

How a Golf Swing Carries Energy

A Book-Length Visual Journey Through Motion, Forces, Timing, Models, and Uncertainty

Dieter Olson

Table of Contents

How to Read This Book	5
1 Follow the Energy	6
1.1 A Concrete Picture	6
1.2 How the Mechanism Works	6
1.3 Where the Picture Breaks	11
2 Choose the System Before Counting	12
2.1 A Concrete Picture	12
2.2 How the Mechanism Works	12
2.3 Where the Picture Breaks	16
3 The State Is the Complete Snapshot	17
3.1 A Concrete Picture	17
3.2 How the Mechanism Works	17
3.3 Where the Picture Breaks	22
4 Geometry Is Part of the Machine	24
4.1 A Concrete Picture	24
4.2 How the Mechanism Works	25
4.3 Where the Picture Breaks	29
5 Speed, Acceleration, Energy, and Power	30
5.1 A Concrete Picture	30
5.2 How the Mechanism Works	30
5.3 Where the Picture Breaks	35
6 Constraints Push Back	37
6.1 A Concrete Picture	37
6.2 How the Mechanism Works	37
6.3 Where the Picture Breaks	42
7 From One Pendulum to Two	43
7.1 A Concrete Picture	43
7.2 How the Mechanism Works	43
7.3 Where the Picture Breaks	47
8 The Summation-of-Speed Story and Its Limits	48
8.1 A Concrete Picture	48

8.2	How the Mechanism Works	48
8.3	Where the Picture Breaks	53
9	Force Direction Changes the Job	54
9.1	A Concrete Picture	54
9.2	How the Mechanism Works	54
9.3	Where the Picture Breaks	61
10	Force Is Not Work Until Something Moves	62
10.1	A Concrete Picture	62
10.2	How the Mechanism Works	62
10.3	Where the Picture Breaks	69
11	Counterfactuals Are Scientific Scissors	70
11.1	A Concrete Picture	70
11.2	How the Mechanism Works	71
11.3	Where the Picture Breaks	74
12	Two Hands Make One Visible Wrench	76
12.1	A Concrete Picture	76
12.2	How the Mechanism Works	77
12.3	Where the Picture Breaks	84
13	Negative Torque Can Help a Positive Outcome	85
13.1	A Concrete Picture	85
13.2	How the Mechanism Works	85
13.3	Where the Picture Breaks	90
14	Arms, Wrists, Preload, and the Meaning of Slack	91
14.1	A Concrete Picture	91
14.2	How the Mechanism Works	91
14.3	Where the Picture Breaks	98
15	The Shaft Is a Spring With a Memory	99
15.1	A Concrete Picture	99
15.2	How the Mechanism Works	99
15.3	Where the Picture Breaks	106
16	The Ground Is Part of the Conversation	107
16.1	A Concrete Picture	107
16.2	How the Mechanism Works	107
16.3	Where the Picture Breaks	115
17	When the Base Moves, Every Shortcut Changes	116
17.1	A Concrete Picture	116
17.2	How the Mechanism Works	116
17.3	Where the Picture Breaks	121
18	How a Constrained Forward Model Solves the Next Instant	122
18.1	A Concrete Picture	122
18.2	How the Mechanism Works	122

18.3 Where the Picture Breaks	127
19 A Ladder of Models, Not One Final Model	128
19.1 A Concrete Picture	128
19.2 How the Mechanism Works	128
19.3 Where the Picture Breaks	133
20 From a Plane to Three Dimensions and Independent Engines	134
20.1 A Concrete Picture	134
20.2 How the Mechanism Works	134
20.3 Where the Picture Breaks	142
21 Sensitivity and Identifiability	143
21.1 A Concrete Picture	143
21.2 How the Mechanism Works	144
21.3 Where the Picture Breaks	148
22 Timing Is a State Question, Not Just a Clock Question	150
22.1 A Concrete Picture	150
22.2 How the Mechanism Works	151
22.3 Where the Picture Breaks	155
23 Fast Once Is Not the Same as Robustly Fast	156
23.1 A Concrete Picture	156
23.2 How the Mechanism Works	156
23.3 Where the Picture Breaks	162
24 What Has Actually Been Measured in Golfers	163
24.1 A Concrete Picture	163
24.2 How the Mechanism Works	163
24.3 Where the Picture Breaks	169
25 Muscles, Scapulae, Wrists, and Biological Ambiguity	170
25.1 A Concrete Picture	170
25.2 How the Mechanism Works	170
25.3 Where the Picture Breaks	175
26 Design an Experiment That Can Say No	176
26.1 A Concrete Picture	176
26.2 How the Mechanism Works	177
26.3 Where the Picture Breaks	181
27 Open Evidence and the Reviewer Path	183
27.1 A Concrete Picture	183
27.2 How the Mechanism Works	184
27.3 Where the Picture Breaks	188
28 A Practical Synthesis Without a Swing Prescription	190
28.1 A Concrete Picture	190
28.2 How the Mechanism Works	190
28.3 Where the Picture Breaks	195

29 Walk Through a Whole Swing Without Losing the Ledger	196
29.1 A Concrete Picture	196
29.2 How the Mechanism Works	196
29.3 Where the Picture Breaks	203
30 A Curious Golfer and a Skeptical Reviewer Ask the Hard Questions	204
30.1 A Concrete Picture	204
30.2 How the Mechanism Works	204
30.3 Where the Picture Breaks	216
Glossary	217

This is a patient exploration of a moving system. It follows forces, motion and energy accounting through ground contact, the body and hands, a flexible club, and the ball. An ideal stationary ground contact can supply impulse without doing work; it is not a universal energy source. Ordinary scenes make the mechanics visible; equations and experiments keep the scenes honest. No model result becomes swing instruction merely because it is easy to picture. The purpose is to make the scientific questions understandable enough that a reader can examine, challenge, and improve them.

Follow the Interactions and the Energy Ledger



Ground contact can redirect momentum without supplying work.
Other interfaces may transfer, store or dissipate mechanical energy.

Figure 1: A Five-Stage Map Connects Ground Contact, Body, Hands, Club, and Ball Without Treating Every Interaction as an Energy Supply.

How to Read This Book

The book is organized as a long walk rather than a stack of definitions. Part I builds the language of motion. Part II follows forces through a double pendulum. Part III opens the two-hand and flexible-shaft mechanisms. Part IV shows how models grow without pretending that the largest model is automatically the best. Part V asks what is identifiable, robust, measured, and falsifiable. The final part turns the argument into a reviewer path and a neutral practical synthesis.

Every chapter has four recurring landmarks:

- **A Concrete Picture** begins with an ordinary situation that can be held in the mind while the mechanics develops.
- **How the Mechanism Works** supplies the sustained explanation, equations, figures, and worked reasoning.
- **Where the Picture Breaks** states exactly where the analogy or reduced model stops earning conclusions.
- **Go Deeper** links to the corresponding technical chapter, executable evidence, and real literature.

Claim labels recur throughout. **Model Result** means an executable mathematical system produced a result under declared assumptions. **Human Evidence** means people were measured. **Hypothesis** means a proposed explanation awaits a decisive test. **Practical Interpretation** translates a question into ordinary language without declaring a universal technique.

Readers who want to change the models can use the [interactive proximal-distal workbench](#). Readers who want every equation and numerical contract can open the [governed scientific monograph](#) or [download its exact PDF](#). Its models, claims, and evidence are pinned to [UpstreamDrift commit 85cce4d3](#).

This book is **not a coaching instruction**. It explains testable mechanics, evidence boundaries, and open questions. The preload discussion also links to the machine-readable [transmission robustness study](#) used by the technical analysis.

Chapter 1

Follow the Energy

1.1 A Concrete Picture

Imagine standing above a busy railway yard at dusk. A locomotive enters with fuel in its tanks and motion in its wheels. It couples to a line of cars. Some cars begin moving, one spring-loaded coupling compresses, brakes warm on another car, and the last car eventually rolls faster than the first. If we watched only the peak speed of each car, we would miss nearly everything that made the event interesting. We would not know which engine supplied work, which coupling stored energy for a moment, which brake removed it, or which force merely kept a car on the curved track.

A golf swing is smaller and faster, but the accounting problem is similar. The pelvis, trunk, arms, hands, shaft, and clubhead do not behave as a row of rigid blocks passing a package along a conveyor. They are linked bodies whose geometry changes while forces act. A connection can transmit power in either direction. A segment can slow while still doing positive work on its neighbor. A large force can redirect motion while transferring almost no energy. A flexible shaft can accept energy, keep it briefly, and return some later. Gravity can help one part of the path and oppose another. Damping quietly turns organized mechanical energy into heat.

The first discipline of this book is therefore simple to state and surprisingly hard to maintain: follow the complete energy ledger, not just the visible order of speed peaks.

The familiar phrase *proximal to distal* means, roughly, from nearer the center of the body toward farther-away segments. In golf, researchers often describe a sequence in which pelvis speed peaks, then trunk speed, then arm or hand speed, and finally club speed. Ordered peaks have been reported, but the reported order depends on which segments and angular-velocity quantities are measured ([Tinmark et al. 2010](#); [Marsan et al. 2019](#)). It is an observation about motion. It is not yet an explanation of energy flow. The explanation requires forces, moments, velocities, and the boundaries across which power is computed.

1.2 How the Mechanism Works

Begin with a modest definition. Mechanical energy is a bookkeeping quantity for a chosen system. A moving rigid body has translational and rotational kinetic energy. A raised body has gravitational potential energy. A bent shaft or stretched tendon may have elastic potential energy. When external

Follow the Interactions and the Energy Ledger



Ground contact can redirect momentum without supplying work.
Other interfaces may transfer, store or dissipate mechanical energy.

Figure 1.1: A Five-Stage Map Connects Ground Contact, Body, Hands, Club, and Ball Without Treating Every Interaction as an Energy Supply.

forces do work, the system's mechanical energy can change. Internal interactions can redistribute mechanical energy, convert elastic storage to kinetic energy, or dissipate mechanical energy into heat. Internal muscles can convert chemical energy into mechanical work. Conservation of total energy does not imply that the mechanical energy of a golfer-plus-club system can increase only through external work.

That last sentence is the heart of the matter. Suppose we draw one boundary around the golfer and club together. The forces between a hand and the grip are internal to that combined system. The forces are equal and opposite. At an ideal shared no-slip contact their material-point velocities agree, so the two contact powers cancel in the combined ledger. If slip, deformation or an intervening compliant element is retained, its power, storage and dissipation require explicit accounting. Merely reporting the same physical contact in two ways cannot create an unexplained power gap. Now draw a smaller boundary around the club alone. The same hand forces become external inputs to the club. They can change its kinetic energy, bend its shaft, or redirect its motion. Neither boundary is more correct. They answer different questions.

For a force $\mathbf{F}$ acting at a point moving with velocity $\mathbf{v}$, the instantaneous power delivered through that point is

$$P_F = \mathbf{F} \cdot \mathbf{v}.$$

For an applied free moment τ acting on a rigid body with angular velocity ω in the chosen observer frame, moment power is

$$P_\tau = \tau \cdot \omega.$$

For several forces on one rigid body, reduce their resultant force and moment to one declared point. Pair the force with the rigid velocity field evaluated at that point and the moment with the body's angular velocity. The total power is the sum of the two terms; a moment arm must not be counted again. A deforming club instead needs its contact/deformation velocities. At a rotational joint, paired moment power uses relative angular velocity between the bodies. Net joint power is not identified individual-muscle power.

These dot products are more informative than force or torque magnitude alone. When force points generally with velocity, power is positive. When it points against velocity, power is negative. When it points perpendicular to velocity, it may bend the path strongly while doing no instantaneous work. A centripetal force in uniform circular motion is the clean textbook example: it is essential to the curved path, yet its dot product with tangential velocity is zero.

This gives us several distinct verbs for the swing. An actuator can **supply** energy by positive work. An interaction can **transport** energy across a chosen boundary. A spring can **store** and later **return** energy. Damping can **dissipate** it. A perpendicular force can **redirect** motion. Constraint reactions can **redirect** motion or exchange power between subsystems. A prescribed moving constraint can also supply external power; its actuator must be included when the accounting boundary includes the mechanism that moves it. Saying only that energy “flows down the chain” compresses all those verbs into one and can hide the mechanism we most need to understand.

Consider a relay race, another useful picture. The order in which runners reach their top speed resembles a kinematic sequence. The baton resembles a physical object crossing a boundary. But a golf swing has no single baton of energy. Energy is a scalar assigned to states, and power is the rate at which work crosses a chosen boundary. What passes through the wrist is not a glowing substance. It is a calculable mechanical interaction whose sign and magnitude depend on the force, moment, motion, reference point, and frame.

The technical study uses an energy ledger precisely because several mechanisms can overlap. In a double-pendulum model, shoulder torque can add energy while the arm-club interaction redistributes it. In the two-hand model, a pair of opposed hand forces can create a club moment even when their net force is small. In the flexible-shaft model, interface power can move into elastic storage and return later. In an ideal stationary no-slip ground contact, reactions constrain motion and provide impulse while doing zero boundary work. Deforming, slipping or moving contacts require a different power calculation. The ledger asks each contribution to identify itself rather than accepting one story for all of them.

1.2.1 A Small Numerical Ledger

Suppose a simplified club receives 500 watts of positive hand power for 0.10 seconds. If the power were constant, the hands would deliver 50 joules of work. During the same interval, imagine gravity contributes 3 joules, damping removes 5 joules, and the shaft finishes with 8 joules more elastic energy than it began. Treat these as synthetic values and include all retained club kinetic modes in K . Count gravity as external work here, rather than also including its potential-energy change. The remaining increase in total kinetic energy is

$$\Delta K = 50 + 3 - 5 - 8 = 40 \text{ J.}$$

Nothing here says the shaft “lost” eight joules. At the chosen final time, those joules are stored rather than expressed as total kinetic energy. If the shaft subsequently loses eight joules of storage, two are dissipated and there is no other work or storage change, total kinetic energy rises by six joules. This does not say that all six reach the clubhead. Timing and interface powers decide which part holds the energy near impact.



Figure 1.2: A Worked Energy Ledger Separates Hand Work, Gravity, Shaft Storage, Damping Loss, and the Remaining Kinetic-Energy Gain.

The same care is needed when a proximal segment slows. A slowing arm may be losing kinetic energy, but that loss is not automatically equal to the club’s gain. The shoulder may still be doing work. Gravity and damping may contribute. Interaction power may move energy between arm and club. Their kinetic energies also depend on configuration because the inertia matrix of a linked system is not a set of independent constants. The only safe claim comes from a closed balance in a declared model or from measurements rich enough to estimate one.

Model Result: The [archived forward two-hand study](#) reports decreasing, nonzero work-energy residuals under timestep refinement. Those numerical checks support bounded internal consistency, not exact trajectories or biological validation. Chapter 29 distinguishes these residuals from tighter algebraic wrench-power identities.

Human Evidence: Tinmark and colleagues reported ordered pelvis, torso and hand peak angular-speed magnitudes in their 45-golfer study ([Tinmark et al. 2010](#)). Marsan and colleagues found strong dependence of the identified sequence on angular-velocity definition ([Marsan et al. 2019](#)). These kinematic results do not measure joint work or identify a causal energy-transfer route.

Hypothesis: A useful late-downswing strategy may depend on arranging the state so passive and interaction terms assist the club while direct distal actuation is delayed or even briefly opposing. The rest of the book makes this proposal precise enough to challenge.

Practical Interpretation: Instead of asking which body part “creates all the speed,” ask what each boundary is doing during each phase: supplying, carrying, storing, returning, dissipating, or redirecting

mechanical energy.

1.2.2 A Swing-Phase Walkthrough

At address, the club already has a state and a preparation history. Its gravitational potential energy is defined relative to a chosen reference height. Raising it requires work against gravity; gravity has not supplied that positive energy change. Holding a pose may require muscle activity even when the net mechanical work is zero. Every later claim must name its ledger. During the backswing, muscles do work to raise and rotate body segments and the club. Some of that work becomes gravitational potential energy, some becomes kinetic energy, some is stored in deformed tissue or the shaft, and some is dissipated. Saying that the backswing “loads” the swing is therefore reasonable only after “load” is divided into measurable forms.

Around transition, club speed may be low while forces and internal activity are not. The direction of motion is changing, the golfer is reorganizing the state, and compliant elements may carry load. This is a useful antidote to the idea that low visible speed means nothing mechanical is happening. Early in the downswing, proximal work can increase the energy of the whole system while the club is still largely being carried. The club’s own speed need not immediately mirror the rate at which energy enters elsewhere.

In the middle and late downswing, the ledger becomes especially interesting. For the whole club, hand power and gravity are external entries, while shaft return is internal conversion from stored energy. Coupling to the body shapes the hand power and trajectory; it is not a third external power to add to the same ledger. At the same time, one generalized torque can have negative power and still participate in a state evolution that produces greater later club speed. Near impact, energy in translation, rotation, and shaft deformation all affect the collision. After impact, substantial energy remains in the body and club or is dissipated; the ball never receives the entire pre-impact system energy.

1.2.3 A Numerical Thought Experiment

Suppose two simulated swings both deliver a clubhead at 45 metres per second. For a deliberately reduced example, represent each head as a point mass moving at that speed: 0.20 kg in Swing A and 0.22 kg in Swing B. Their kinetic energies are $\frac{1}{2}mv^2 = 202.5$ J and 222.75 J. These are not whole-club energy estimates. A rigid head requires center-of-mass speed and a separate rotational term; a flexible club also carries distributed motion. Impact effective mass is a directional contact-response property and must not be substituted for physical mass in this energy calculation. Equal speed need not mean equal energy. Now suppose Swing A reaches that speed with 400 watts crossing an interface for half a second, while Swing B has a 1,000-watt peak lasting only a few hundredths of a second. The 200 J of interface work in Swing A is only one ledger entry; initial energy, gravity, storage changes and losses must close its complete balance. Peak power alone still does not say which boundary supplied more work. Integrating the complete histories does.

The numbers are illustrative, not measurements of a golfer. Their job is to force the vocabulary apart. Speed is a state variable, energy is a ledger quantity, power is a rate, and work is an accumulation. When a result is surprising, returning to those four definitions usually reveals whether the surprise is physical or merely verbal.

1.2.4 Questions for a Skeptical Reader

Before accepting any energy-transfer plot, ask whether all external powers were included; whether the sign convention is stated; whether the shaft was modeled as rigid or flexible; whether joint power was assigned consistently; and whether numerical energy residuals are small relative to the reported effect. Then ask the separate experimental questions: Were inertial parameters measured or assumed? Were force and motion synchronized? Does uncertainty in those inputs change the conclusion? A closed ledger earns the right to continue the argument. It does not finish it.

1.3 Where the Picture Breaks

Energy is not a liquid, electricity in a wire, traffic on a road, or a baton. Those images help us remember conservation and boundaries, but they can mislead us into imagining a material substance with a unique route. In a multibody system, the decomposition itself depends on coordinates and system boundaries. Some internal transfers can be described in more than one mathematically valid way. Equivalent coordinate descriptions must recover the same physical motion and consistent whole-system balance in the same observer frame. Moving to another observer can change kinetic energy and power values without violating mechanics. Individual coordinate-term labels are not invariant physical transfers.

The railway picture also makes couplings look sequential. In a swing, forces act throughout the system at the same time. The distal club can influence the proximal arms immediately through action-reaction. A shaft mode can affect grip loads before its stored energy is returned. The correct picture is a coupled network with time-varying geometry, not a one-way bucket brigade.

Finally, a closed simulated ledger is necessary but not sufficient. A wrong model can conserve its own energy perfectly. Numerical closure shows that the implemented equations agree with their accounting. Experimental agreement is a separate question requiring measured motion, wrenches, shaft deformation, ground reactions, and uncertainty.

Go Deeper: Read the scientific monograph’s [mechanics chapter](#), the detailed [interaction-force decomposition](#), and Robertson and Winter’s segment-energy framework ([Robertson and Winter 1980](#)). The [open evidence directory](#) contains the machine-readable ledgers used by the technical study.

Chapter 2

Choose the System Before Counting

2.1 A Concrete Picture

Picture a household checking its monthly finances. If the “system” is one person’s wallet, a payment to a spouse is money leaving the system. If the system is the whole household, the same payment is an internal transfer: one account falls while another rises, but household wealth does not change. Rent paid to a landlord is external under either household boundary. A loan adds cash now and an obligation later. The accounting only becomes meaningful after we say what is inside the boundary and what quantities the ledger tracks.

Mechanical accounting works the same way. A statement such as “the arms gave energy to the club” quietly draws a boundary around the club. A statement such as “the golfer-club system gained energy from the ground” draws a larger boundary. Both can be useful. Trouble begins when numbers computed for one boundary are interpreted as though they belonged to another.

The choice is not philosophical decoration. It decides which forces are external, which interactions are internal, which energies appear in the ledger, and what conservation statement should close. Before counting transfer, we must choose the system.

2.2 How the Mechanism Works

Start with the club alone. Its rigid-body kinetic energy includes translation of its center of mass and rotation about that center. If the shaft is flexible, the club system also has elastic strain energy and possibly energy in several vibration modes. The hands exert external forces and moments on this chosen system. Gravity is external. Aerodynamic drag is external. If the club strikes the ball, contact with the ball is external. A club-only balance can therefore answer: how much mechanical work entered through the grip, how much gravity added, how much drag removed, how much remained stored in flex, and how much appeared as club kinetic energy?

Now enlarge the boundary to include the golfer and club. Hand-grip forces are internal action-reaction pairs. They still matter because they redistribute energy among body segments and the club, but they do not appear as net external work on the combined idealized system. Ground reactions, gravity, air drag, and metabolic-to-mechanical conversion remain relevant at or beyond the boundary. This

Choose the Boundary Before Naming the Transfer

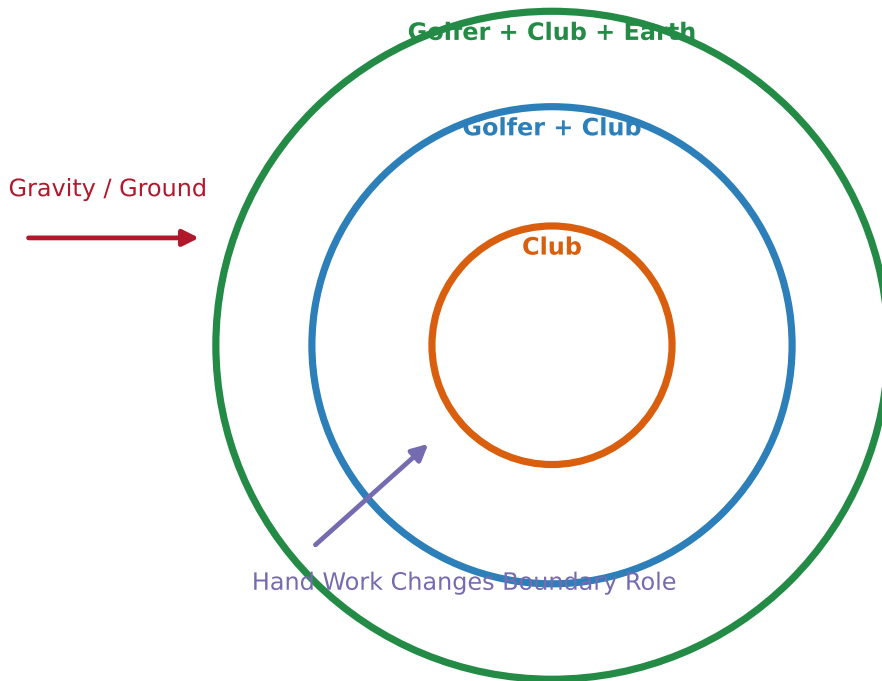


Figure 2.1: Nested Boundaries Separate Club, Golfer-Club, and Golfer-Club-Earth Energy Accounts.

larger balance can answer a different question: how did external work and internal actuation change the mechanical energy of the entire moving system?

Enlarge the boundary again to include Earth. Gravitational potential energy is now stored within the chosen system rather than supplied by an external gravity force. Ground contact becomes an internal interaction with Earth, although the Earth's change in kinetic energy is ordinarily negligible because its mass is enormous. The equations have not contradicted one another. The labels have changed because the boundary changed.

2.2.1 Internal Forces Can Still Transfer Energy Between Parts

Readers sometimes hear that internal forces “do no work.” That phrase is only safe for particular idealized whole-system sums. An internal force can do positive work on one subsystem and negative work on another. Consider two skaters pushing apart. The contact forces are internal to the pair, but each skater gains kinetic energy because biological chemical energy is converted by muscle action during the push. If we examine one skater, the other's contact force is external and plainly does work.

For the arm and club, the wrist reaction force can carry power across the arm-club boundary. Let $\mathbf{F}_{H \rightarrow C}$ be the hand force on the club and $\mathbf{v}_H$ the velocity of the grip point on the club. The club-side power through that contact is $\mathbf{F}_{H \rightarrow C} \cdot \mathbf{v}_H$ plus any direct moment power. The equal and opposite club force on the hand participates in the body-side balance. With perfectly coincident rigid contact points, the pair's translational contact powers cancel in the combined-system sum. With compliance, relative motion, or a different partition, energy can temporarily enter stored interface states.

This is why words like *source*, *transfer*, and *storage* need care. Muscles are sources of mechanical work only by converting biochemical energy. The hand-club interface is usually a transfer boundary, not an independent energy source. A shaft spring stores energy but does not create it. Damping removes recoverable mechanical energy from the modeled degrees of freedom. Gravity exchanges kinetic and potential energy. Each verb belongs to a declared boundary and model.

2.2.2 Reference Points Matter for Moments and Power

A force is a vector that can be expressed in different coordinate frames without changing the physical interaction. A moment is always about a point. Move the reference point, and the force moment changes according to

$$\mathbf{M}_B = \mathbf{M}_A + (\mathbf{r}_A - \mathbf{r}_B) \times \mathbf{F}.$$

The combined force-moment pair is called a **wrench**. Its power is invariant when the force, moment, translational velocity, and angular velocity are all transformed consistently. But comparing a moment about the lead hand with a moment about the club center without performing the transformation can create a false difference. The two-hand chapters will return to this point because a pair of separated forces may have nearly zero resultant force yet a substantial moment about the club.

Reference levels also matter for potential energy. We may choose the ground as zero gravitational potential or choose the club’s address height. Only changes in potential energy affect the balance, so either reference is valid if applied consistently. Absolute numbers without a declared zero are not comparable.

2.2.3 The Boundary Decides the Question, Not the Answer

Suppose a simulation reports that the club gains 40 joules during a phase, while the arm loses 12 joules. It would be wrong to conclude that exactly 12 joules were “transferred from arm to club” and the remaining 28 appeared mysteriously. Shoulder and wrist actuators may do work, gravity may contribute, the shaft may release stored energy, and other body segments may exchange power through the arm. A valid attribution needs interface powers and the complete ledger for the chosen partition.

Conversely, a whole-system balance can hide the internal route. If total mechanical energy rises by 50 joules and the numerical residual is tiny, we still do not know whether hip, shoulder, wrist, or elastic mechanisms delivered the work. Conservation is a consistency condition, not an anatomical diagnosis. To locate pathways, we deliberately draw subsystem boundaries and compute power through each interface.

Model Result: The technical work reports grip power, shaft-interface power, actuator work, gravitational work, damping loss, stored elastic energy, and whole-system residuals for multiple model tiers. Agreement of those ledgers is evidence that the implemented partitions are mechanically consistent.

Human Evidence: Inverse-dynamics studies can estimate net joint moments and powers from measured kinematics and external forces, but results depend on body segment parameters, filtering, joint definitions, and the treatment of two-hand contact. Net joint power is not a direct measurement of a particular muscle’s work (Nesbit and Serrano 2005; Winter 2009).

Hypothesis: A boundary placed around the club and shaft may expose transfer pathways that are invisible in segment-speed plots, particularly when a negative grip couple coexists with positive power arriving through another channel.

Practical Interpretation: Whenever someone says energy was “generated,” “stored,” or “transferred,” ask three quiet questions: what system was chosen, what crossed its boundary, and did the ledger close?

2.2.4 The Same Swing Under Four Boundaries

Imagine drawing four outlines on the same video frame. The smallest outline contains only the club. Every force and moment applied by the hands crosses that boundary, so hand power is external to this system. Shaft strain energy is internal if the flexible shaft remains inside the outline. Aerodynamic drag and the impact force are external. This boundary is excellent for asking how the club acquires and loses mechanical energy, but it says almost nothing about how the body produced the hand wrench.

The next outline contains both hands and the club. Grip contact becomes an internal exchange; wrist and forearm loads cross the new boundary. This view is useful for separating a two-hand internal couple from the net wrench delivered by the arms. The third outline adds the upper body. Shoulder, trunk, and lower- body interfaces now matter. The final outline contains golfer, club, and ground contact. Muscle work is internal, while gravity and environmental contacts are external. Each outline describes the same physical event, yet the words “source” and “transfer” refer to different terms.

This explains why two apparently contradictory papers can both balance. One may call power at a joint a transfer between rigid segments; another may treat the same joint and its spanning muscles as internal to a larger system. A useful comparison first translates both analyses to a common boundary and reference point. Without that translation, disagreement may be bookkeeping rather than mechanics.

2.2.5 A Boundary Audit in Plain Language

For any claimed transfer, write a one-line sentence: “The system is ; *energy crosses through* ; positive points ____.” Next list the stored forms. For a rigid club those might be translational and rotational kinetic energy plus gravitational potential. For a flexible club, add modal or strain energy. Then list external ports: hands, gravity, air, and ball contact. The balance should read like a bank statement in which the change in account value equals deposits minus withdrawals, with a small numerical residual.

Reference points deserve equal care. A force applied away from the club’s center of mass creates both translation and rotation. Moving the reporting point changes the force–moment pair, although the physical wrench is the same. Power remains consistent only when force, moment, linear velocity, and angular velocity are transformed together. Mixing a force from one point with a moment from another is the mechanical equivalent of combining dollars and euros without an exchange rate.

2.2.6 Why the Choice Matters for the Two-Hands Question

When the hands exert equal and opposite forces, their net force can be nearly zero while their couple is not. For the club-only system, that couple is an external moment and can do rotational work. For a hands-plus-club system, the same contact forces are internal, and the energetic question moves to the

wrist/arm boundary. Neither description makes the couple disappear. It changes where the accounting line is drawn.

The distinction is essential when comparing an arm-dominant allocation with a wrist-dominant allocation. If both produce the same net club wrench, club-level motion cannot identify which internal route was used. Expanding the system may make the hidden route explicit, but it also demands more measurements and more anatomical assumptions. A good model chooses the smallest boundary capable of answering its declared question and refuses to imply more.

2.3 Where the Picture Breaks

The household-budget analogy suggests that every transfer has a unique payer and recipient. Multibody mechanics is less tidy. Coordinate choices can move terms between what is called inertial coupling and what is called a generalized force. Segmental energy methods can partition joint power in more than one way, especially at multi-degree-of-freedom joints. Soft tissues span several joints, and biarticular muscles can move energy between segments without fitting a single-joint wallet.

Rigid-body boundaries also cut through biological structures. A modeled “wrist joint” may stand in for carpal motion, forearm rotation, grip deformation, and small translations. A “shoulder torque” may aggregate glenohumeral action, scapulothoracic motion, and trunk coupling. The boundary remains useful if its meaning is explicit, but it should not be mistaken for a direct anatomical measurement.

Finally, work and energy say nothing by themselves about comfort, accuracy, injury risk, neural control effort, or repeatability. Two allocations can have the same club energy and very different internal loads. The later chapters add those objectives rather than letting one ledger pretend to answer them all.

Go Deeper: The scientific monograph’s [whole-system and interface-power treatment](#) shows the balance for rigid and flexible cases. Its [two-hand wrench chapter](#) defines reference points and action-reaction explicitly. For the broader biomechanics convention, see Winter’s treatment of segment and joint energetics ([Winter 2009](#)).

Chapter 3

The State Is the Complete Snapshot

3.1 A Concrete Picture

Two photographs can look identical and conceal opposite futures. In the first, a playground swing has reached its highest point and is about to fall. In the second, it is moving upward through the same angle and is about to slow. The seat, chains, and child occupy nearly the same places in both photographs. Yet one moment leads downward and the other upward because velocity is missing from the picture.

Now photograph a flexible beam only at its two ends. Different bends inside the span can leave those endpoints in the same places while storing different elastic energy. The photograph has missed deformation. Changing the material or rest shape is a different comparison: it changes the model parameters. For a fixed elastic law, complete deformation determines strain energy; energy is not an independently adjustable hidden quantity. Two muscles at the same joint angle can likewise have different activation or internal tissue states and respond differently to a disturbance.

A useful dynamics model gathers everything it needs to continue the motion into a **state**. For a rigid double pendulum, the state may consist of two angles and two angular velocities. For a flexible club, it also includes shaft deflection and deflection rate. For an actuator with first-order lag, it may include an activation state. A pure time delay instead requires relevant past history, or a declared finite-dimensional approximation. A compliant contact may require interface deformation, friction memory, and a contact mode. The complete snapshot is more than a pose.

3.2 How the Mechanism Works

The formal state is commonly written $\mathbf{x}$. For a two-link planar model, one convenient choice is

$$\mathbf{x} = [q_1, q_2, \dot{q}_1, \dot{q}_2]^T,$$

where q_1 and q_2 describe configuration and the dotted variables describe angular velocity. The equations of motion define a rule

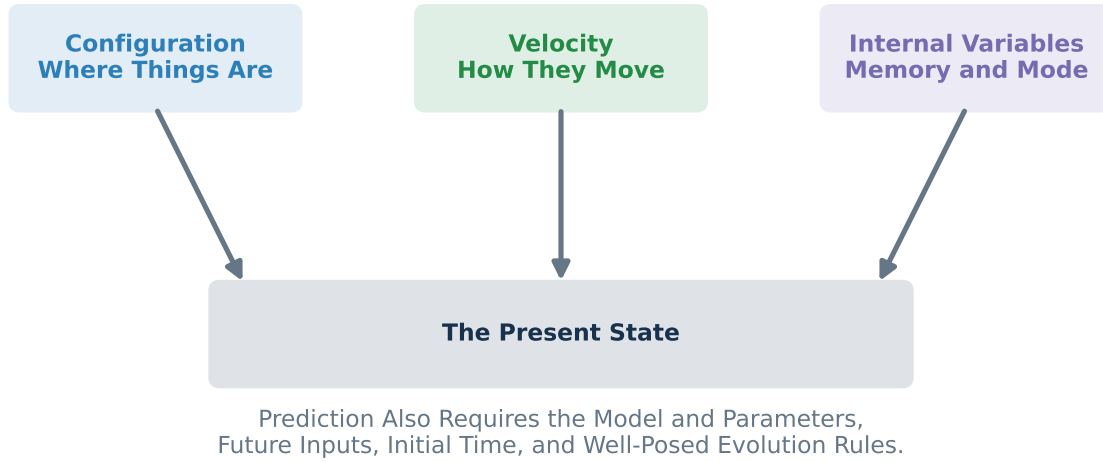


Figure 3.1: State Components and the Additional Conditions Required for Prediction.

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}, t),$$

which says: given the present state, the present control $\mathbf{u}$, and any explicit time dependence, compute the rate at which the state changes. A numerical integrator advances that rule through many small steps.

The word *state* has a strict purpose. Within a declared model, it contains the information about the past needed to evolve from the present. Identical states produce identical futures **when the same model, parameters, initial time, future inputs, and well-posed evolution rules are used**. State alone does not supply tomorrow's commands. In a nonautonomous model, starting the same state at a different clock time can also change the future.

Well-posedness matters. A single-valued deterministic right-hand side does not by itself guarantee uniqueness: $\dot{x} = \sqrt{x}$ for $x \geq 0$, starting at zero, admits both a permanently zero solution and a solution that waits before growing. A common sufficient uniqueness condition is local Lipschitz continuity in state, with appropriate regularity of the inputs; it is not a necessary condition in every model. The claim concerns times for which the solution exists in the model's valid domain, not an unconditional forecast forever.

Contacts and impacts may require more than the displayed smooth ODE. Specify constraint consistency, discrete modes, event guards, and reset rules, and verify that they define a unique admissible evolution for the comparison. An impact can reset velocity while leaving configuration continuous. Numerical time stepping and tolerances approximate this declared evolution; disagreement between simulations can reveal numerical error or ill-posedness as well as missing state.

3.2.1 Configuration Is More Than a Photograph

Configuration describes where the modeled bodies are within their allowed geometry. In a planar double pendulum, two angles may be enough. In a spatial golfer model, configuration can include pelvis position and orientation, spinal and shoulder rotations, elbow and wrist coordinates, club pose, and perhaps shaft modes. Rotations require care because three-dimensional orientation is not just three independent slider positions. Different angle sequences can describe the same physical orientation, and some parameterizations contain singularities. With redundant orientation coordinates, coordinate rates and physical angular velocity are different objects: a kinematic map can relate $\dot{q}$ to a velocity vector v . Normalization and closed-chain constraints restrict admissible states; arbitrary coordinate and velocity arrays need not describe a possible motion.

Configuration also determines mechanical advantage. The mass matrix, moment arms, constraint Jacobian, generalized gravity forces, and mapping from joint torques to club acceleration can change with posture. Gravity need not change direction in an inertial frame; its body-frame components and generalized projections depend on configuration. The same torque vector applied at two configurations can produce different accelerations. State-dependent geometry is therefore not scenery around the mechanism; it is part of the mechanism.

3.2.2 Velocity Gives the Snapshot a Direction

Velocity tells how configuration is changing. For ordinary rigid-body mechanics in independent coordinates, write the mechanical part as

$$M(q)\ddot{q} = \tau - c(q, \dot{q}) - g(q) - D\dot{q},$$

Here q and τ are coordinate and generalized-torque vectors. With positive inertias and a regular independent-coordinate description, M is positive definite and can be inverted to obtain acceleration. The vector c collects quadratic Coriolis and centrifugal terms, g is the potential-force term, and $D\dot{q}$ illustrates viscous resistance. At the same configuration, $M(q)$ is the **same matrix**. Reversing every velocity reverses momentum $M\dot{q}$ and, for a fixed torque, power $\tau^T\dot{q}$, but leaves quadratic $c(q, \dot{q})$ unchanged: each product contains two velocity factors. The matrix factorization $C(q, \dot{q})$ can change sign while its product $c = C(q, \dot{q})\dot{q}$ stays the same. Viscous resistance changes sign instead. These statements apply to this model class; rotating-frame, magnetic, or other odd-in-velocity forces need separate treatment.

Consequently, in the undamped model, opposite velocities can yield **equal instantaneous accelerations** under the same posture and torque. Their futures still differ because their initial velocities differ. Acceleration is only the velocity derivative, not the full state derivative. Reversing just one joint's velocity, or changing speed magnitudes, can change quadratic coupling terms. Large velocity-dependent effects therefore do not imply that every reversal changes every term.

Velocity is also needed to evaluate power. A 100-newton force at a stationary point delivers zero instantaneous translational power. The same force at a point moving 5 metres per second in its direction delivers 500 watts. Reverse the velocity and the power is -500 watts. Force without velocity cannot tell the energy story.

3.2.3 Stored Variables Carry History Into the Present

Rigid-body coordinates and velocities are not always enough. A flexible shaft has modal displacement and velocity. A series-elastic actuator may have tendon stretch. A first-order activation model has a state that lags its neural command. A transmission model with an evolving deflection can use that state to determine engagement. A memoryless dead-zone map alone does not acquire an independent state merely because it is nonlinear. These variables are **history summarized**: they contain the past information needed to continue under specified future inputs. Once two histories lead to the same sufficient state, their earlier histories need not remain separate predictors in that model.

This matters directly to preload. In the finite-preparation experiment, both programs begin relaxed, build their own transmission deflections for 180 ms, and carry those states continuously through transition. Continuity holds **within each branch**; the two strategies need not have the same internal state at transition. Comparing their futures tests the effect of preparation and its resulting state under the declared later commands, not the isolated effect of a new command from identical states. Chapter 14 of this companion gives the model and its limits. Finite preparation also need not reach equilibrium to avoid a state reset.

Pure delay is a different kind of memory. For a scalar channel $y(t) = u(t - d)$, two past input histories can have the same current value $u(0)$ and identical future commands, but yield different outputs during the next d seconds. One must retain the relevant history over $[-d, 0]$, not only the current command. Replacing that history with a first-order lag is a modeling approximation with different transient behavior; assess it over the frequencies and time scales used in the claim.

3.2.4 State and Input Must Not Be Confused

An input is something the model treats as externally prescribed for the future: a joint torque command, a desired actuator level, or a measured base trajectory. A state is something the model evolves. The distinction can change between model tiers. A prescribed hand path can be a kinematic input to an inverse-dynamics model; it does not determine every joint motion or internal force without additional measurements and constraints. In a forward model driven by joint torques, the hand path can instead be an outcome. A prescribed pelvis motion can hide the reaction of the club on the body; a dynamically coupled moving-base model can let that feedback alter the state. A base trajectory prescribed in advance still prevents that response.

If a controller generates commands from measurements, its memory also belongs in the closed-loop state, or its complete future input must be prescribed. Calling a signal an input does not mean it is independent of state in the physical experiment.

This difference explains why two models can answer different questions even if they draw the same stick figure. An inverse model asks what forces are compatible with measured motion. A forward model asks what motion results from forces and an initial state. Neither is automatically superior. They have different estimands and different opportunities for circular reasoning.

3.2.5 A Matched-State Intervention

Suppose we want to ask what the current shoulder torque contributes to club acceleration. One useful experiment freezes the state at an instant and solves the acceleration twice: once with the recorded torque and once with that torque set to zero. Hold the entire relevant state, time, parameters,

constraints, and other inputs fixed. Matching configuration and velocity is sufficient for the simplest rigid model, but not for models with additional actuator or contact states. The acceleration difference is then a pointwise intervention contrast. If acceleration is affine in the selected torque at that state and contact mode, the contrast equals its linear control contribution. For nonlinear control maps, the contrast remains defined but need not form an additive partition across actuators.

If we then integrate the zero-torque branch forward, its state begins to diverge. After a few milliseconds, force and power differences include both the missing torque and the changed trajectory it caused. That forward counterfactual answers a persistence question rather than a pure instantaneous attribution question. Both are legitimate if their contracts are stated. An instantaneous torque step changes acceleration without itself causing a velocity jump; an impulse or a contact reset is a different intervention. Zeroing a neural or actuator command also need not instantly remove transmitted torque when activation or transmission states persist.

The initial state difference is zero by construction. Fix other inputs, parameters, disturbances, and either the future open-loop program or the feedback policy being compared. Numerical divergence beyond tolerance then represents the declared intervention and its later interactions within that model. Reoptimizing each future or resetting an internal state changes the question and requires a new contract.

3.2.6 Phase Landmarks Need an Event Contract

Golf language uses transition, early downswing, delivery, and impact. A clock time alone may not align these phases across participants or simulations. An event can use a position or velocity crossing, a peak criterion, a contact guard, or a measurement history. Specify the quantity, direction, admissible time window, which crossing counts, and whether the event must be detected causally. A velocity peak requires a sign change or comparison over a neighborhood; retrospective peak labeling differs from real-time detection. Freezing a criterion in advance limits outcome-driven selection but does not guarantee that it exists, is unique, or is robust to noise.

Comparing two runs at their respective impact events also differs from comparing them at a common elapsed time: the intervention may shift the event itself. Report that timing difference and handle missing events explicitly. Otherwise, aligning curves can hide part of the effect one intended to measure.

3.2.7 State-Based Timing Versus Clock-Based Timing

A command labeled 120 milliseconds before impact is not necessarily equivalent across golfers or clubs. If impact time is known only after the swing, this is retrospective labeling; a real-time controller needs an estimator or a causally detectable trigger. One system may already have a small arm–club angle and high relative speed; another may still be carried with substantial lag. A state-based rule might refer to relative angle, angular velocity, shaft mode, or the sign of an interaction term. It may adapt across tempo changes if that variable remains observable and its threshold predicts the desired response. This advantage needs testing: a noisy or poorly estimated state trigger can be less reliable than a clock rule.

The backswing-preload proposal belongs here. Its plausible benefit is not that one direction of effort is inherently superior. It is that continuous loading might deliver a transition state with less force-development delay than a full role reversal. To test that claim, the state must include the modeled transmission or activation variable. Without those variables, it cannot attribute an effect to that particular transmission-memory pathway. Other modeled pathways, such as changed velocity or

geometry, could still produce an allocation difference and must be separated.

3.2.8 What an Incomplete Observation Can Miss

Now give the club a flexible shaft. The grip and clubhead may occupy the same visible positions in two trials while unresolved shaft modes or modal rates differ. For example, a scalar observation $y = a + b$ cannot distinguish modal amplitudes $(a, b) = (0, 0)$ from $(1, -1)$. With unit modal stiffnesses, their elastic energies $\frac{1}{2}(a^2 + b^2)$ differ despite equal y . This is an abstract observability example, not a calibrated shaft. Complete deformation and modal rates, together with the constitutive law and loading, determine storage and exchange; visible endpoints alone do not. Add muscle activation and short-range stiffness, and two outwardly similar states can also resist perturbation differently. “Same position” is therefore far weaker than “same state.”

3.2.9 What Must Be Measured

Motion capture provides observations from which model coordinates and velocities are estimated. Filtering and differentiation affect noise, phase, and uncertainty near transition. Strain gauges observe strain at selected locations; recovering shaft modes requires a model and enough independent measurements. Bilateral grip sensors measure the wrench components they are designed and calibrated to resolve. EMG records electrical activity; inferred activation timing and force require additional dynamics and assumptions.

Neither a plausible fit nor a larger sensor count guarantees observability. For a linear model, observability depends on the state-to-output map and dynamics; for a nonlinear model it can also depend on trajectory, excitation, operating point, and parameters. Different states can produce indistinguishable outputs within measurement precision. A single measured output at one instant is also different from a time series under known excitation.

Represent uncertainty in states, parameters, sensor calibration, and model choice as far as the evidence allows. A distribution is conditional on the estimator and its assumptions, not automatically a calibrated statement of truth. Repeating counterfactuals across plausible reconstructions tests sensitivity to those assumptions; it cannot establish a causal human effect or cure a structurally unidentifiable state by itself.

3.3 Where the Picture Breaks

The “complete snapshot” phrase can suggest that a finite model state contains all physical reality. It does not. A model state is complete only relative to the equations chosen. A two-link rigid model omits shaft vibrations, soft tissue, muscle activation, grip deformation, aerodynamics, and neural feedback. Its four-number state, specified future inputs, and well-posed rules can determine a unique future of that reduced model while remaining a sparse description of a human swing.

Biological systems may also contain memory that simple first-order states do not capture: muscle history dependence, reflex delays, thixotropy, fatigue, and learning. Adding every possible state would make estimation impossible. The scientific task is to add a state when it changes the question or survives a falsification test, not because complexity sounds realistic.

Finally, measurements never reveal state without uncertainty. Velocities are often derived from noisy

positions. Internal force and activation states may be only indirectly observed. A responsible model carries those uncertainties into its conclusions instead of treating one reconstructed trajectory as exact.

Model Result: In the registered rigid double-pendulum counterfactual study, each commanded/kill-switch pair begins with the same four-component angle/velocity state, at the same cut time. The commanded branch continues its declared program; the other sets both generalized torques to zero. The finite-preparation study instead preserves each transmission branch’s state through its command switch and deliberately allows preparation to create different states. These are different experimental contracts, with different causal questions.

Measurement Boundary: Motion capture estimates visible kinematics, not a complete biological state. Inferring velocities, activation, deformation modes, or contact memory requires a measurement model and uncertainty assumptions. McPhee’s review surveys prescribed-motion, torque-driven, predictive, and contact models in golf ([McPhee 2022](#)); that survey does not certify a particular sensor set as a complete state reconstruction.

Hypothesis: A golfer may use backswing loading and early-downswing motion to arrive at a state from which interaction dynamics assist late club acceleration. That is a state-arrangement hypothesis, not merely a claim about waiting a fixed number of milliseconds.

Practical Interpretation: A pose is the cover of the book, not the page you are on. To understand what comes next, ask how everything is moving and what internal elements are already loaded.

Go Deeper: See the monograph’s [matched-state persistence chapter](#), the [continuous preparation experiment](#), and the [interactive workbench](#) for guided state-versus-clock interventions.

Chapter 4

Geometry Is Part of the Machine

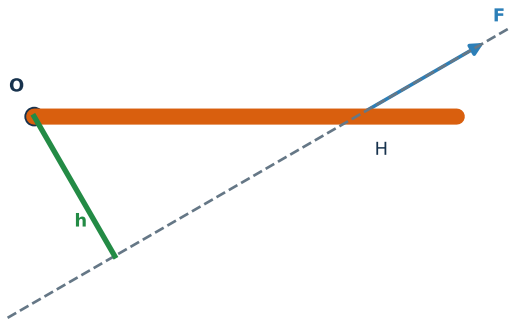
4.1 A Concrete Picture

Push a door near its hinge, then with the same force near its handle. The turning effect changes because the force has a different moment arm. Push directly toward the hinge and the moment about that hinge is zero in the ideal model. Your hand can still exert force and load the structure. Moment, force, motion, and effort are related, but they are not interchangeable.

A swing has several moving versions of this geometry. Joint axes, hand positions, club orientation, and shaft deformation change how forces map into moments and how commanded torques produce acceleration. The useful question is not whether one posture has more “leverage” in general. It is leverage for which motion or force, about which reference, under which constraints.

Moment About O: Force Magnitude $\times$ Perpendicular Distance h

Shorter Moment Arm



Perpendicular Force

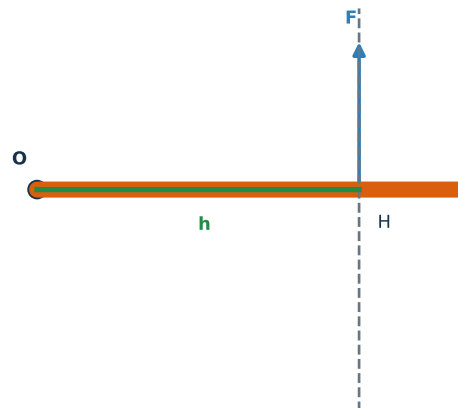


Figure 4.1: The Moment Arm Is the Perpendicular Distance From O to the Force's Line of Action.

4.2 How the Mechanism Works

4.2.1 Name the Point, Plane, and Direction

For a force on the club applied at point H , its moment about O is

$$\mathbf{M}_O = (\mathbf{r}_H - \mathbf{r}_O) \times \mathbf{F}.$$

This instantaneous definition does not require O to be fixed or inertial. Using that moment in an angular-momentum balance about a moving point requires the appropriate transport terms; the definition alone is not that balance. An additional applied free moment must be included separately.

The magnitude of a point-force moment about O is force magnitude times the perpendicular distance from O to its line of action. It is not generally the distance to the arrow tip, the contact point, or the end of the club. A 100 N force with a 0.04 m perpendicular moment arm gives 4 N m. Doubling that distance at the same force gives 8 N m. Passing the line through O gives exactly zero.

For a planar idealization, let $\mathbf{n}$ be the declared unit normal to the plane, let the unit vector $\mathbf{e}_r$ point along the straight club, and define the **in-plane** transverse direction $\mathbf{e}_t = \mathbf{n} \times \mathbf{e}_r$. The transverse direction is not the normal to the swing plane. If the force and club lie in the plane, write $\mathbf{F} = F_r \mathbf{e}_r + F_t \mathbf{e}_t$. When $\mathbf{r}_H - \mathbf{r}_O = \ell \mathbf{e}_r$, the signed moment about $\mathbf{n}$ is ℓF_t . The signed distance ℓ matters: changing which side of O contains the contact can change the sign.

The radial term has zero moment under those collinearity assumptions. It can nevertheless accelerate the club center, load the shaft, or transfer power when the contact moves along its force. A curved shaft, an offset reference, or an out-of-plane force needs the full geometry. “Radial” alone does not specify a zero moment about every point.

4.2.2 Changing the Reference Preserves a Contract, Not Every Number

A wrench combines resultant force $\mathbf{R}$ and total moment about a declared point. Moving the reference from A to B , with $\mathbf{d} = \mathbf{r}_B - \mathbf{r}_A$, gives

$$\mathbf{M}_B = \mathbf{M}_A - \mathbf{d} \times \mathbf{R}.$$

A pure couple has $\mathbf{R} = 0$ and the same moment about every point. For a nearly opposed force pair, the moment change is bounded by $\|\mathbf{d}\|\|\mathbf{R}\|$. A small resultant supports a small-change claim only with a bounded reference displacement and a stated tolerance. It does not make a nonzero-resultant wrench a pure couple.

Power provides a second check. On a rigid body, use the velocity of the same reference point as the moment:

$$\begin{aligned} P &= \mathbf{R} \cdot \mathbf{v}_A + \mathbf{M}_A \cdot \boldsymbol{\omega}, \\ \mathbf{v}_B &= \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{d}. \end{aligned}$$

Transporting both moment and velocity leaves P unchanged. Transporting only one creates a false power

difference. For a deforming shaft or distributed contact, a single rigid-body twist need not describe every loaded point; retain the individual force–velocity pairings or the corresponding distributed power integral.

Expressing the same vectors in another orthonormal basis changes their components while preserving dot products. Changing the physical observer is a different operation: an observer translating at velocity $\mathbf{U}$ measures force power shifted by $-\mathbf{R} \cdot \mathbf{U}$ when the same physical forces are used. Wrenches and trajectories have transformation rules; their component lists are not universally frame-invariant observables. The [Modern Robotics wrench discussion](#) develops the paired force/motion representation.

4.2.3 The Jacobian Maps Velocities and Their Dual Forces

Suppose a hand position is $\mathbf{x}_H = \mathbf{f}(\mathbf{q}, t)$. Differentiating gives a velocity contribution from the modeled coordinates and another from any explicit prescribed motion:

$$\begin{aligned}\mathbf{v}_H &= \mathbf{J}_H \dot{\mathbf{q}} + \mathbf{b}_H, \\ \mathbf{J}_H &= \partial \mathbf{f} / \partial \mathbf{q}, \quad \mathbf{b}_H = \partial \mathbf{f} / \partial t, \\ \mathbf{Q}_H &= \mathbf{J}_H^\top \mathbf{F}_H.\end{aligned}$$

Here $\mathbf{F}_H$ acts on the modeled hand. The force that hand exerts on the club has the opposite sign; choose the system before applying the mapping. The last line maps this applied force into generalized force; it does not assert static equilibrium or identify the total actuator torque. Force and Jacobian must use compatible components and the same point. Generalized force is a torque for an angular coordinate and a force for a length coordinate; its units and components change with coordinate scaling.

The power identity is $\mathbf{F}_H \cdot \mathbf{v}_H = \mathbf{Q}_H \cdot \dot{\mathbf{q}} + \mathbf{F}_H \cdot \mathbf{b}_H$. Ignoring the last term can hide the power associated with prescribed base motion. If base coordinates are included in the dynamical system instead, their rates and generalized forces belong in the larger mapping. A force-plus-free-moment contact uses the corresponding linear and angular velocity Jacobians.

Coordinate derivatives are not always the software’s generalized speeds. Quaternion orientation, for example, uses four constrained coordinates but three rotational speed components. Write $\dot{\mathbf{q}} = \mathbf{N}(\mathbf{q})\nu$ and use $\mathbf{J}_H \mathbf{N}$ when mapping those speeds. The dual generalized force is then $\mathbf{N}^\top \mathbf{Q}_H$. [Tedrake’s multibody notes](#) explain this distinction. Neither a particular coordinate count nor a matrix entry by itself identifies an anatomical actuator.

4.2.4 Near a Singularity, Motion and Force Have Different Scalings

A Jacobian becomes singular for its declared task when its rank falls below the maximum attainable rank. A robot with fewer joints than task dimensions can already lack some instantaneous motions without being at a special singular posture. Instantaneous inability to move in one direction also does not prove that a finite path to another configuration is impossible.

Consider a dimensionless, consistently scaled example $\mathbf{J} = \text{diag}(1, \varepsilon)$ with $\varepsilon > 0$. Unit task velocity in the second direction requires joint rate $1/\varepsilon$ in that direction. But a specified task-force component of 10 gives generalized force 10ε , not $10/\varepsilon$. Conversely, solving for task force from a fixed generalized force gives $1/\varepsilon$ scaling under this ideal invertible mapping.

At $\varepsilon = 0$, no finite joint rate gives the missing task velocity, while that task-force component contributes zero generalized force. Its magnitude is therefore not determined by joint torque through this mapping. Structure, contact, compliance, strength limits, and dynamics still matter; the equation does not grant unlimited physical force or independent force control. The [Modern Robotics singularity examples](#) illustrate the difference between active motion and structural force support.

For multiple coordinates, the analogous statements apply along singular-vector directions with declared units and scaling. Distinguish motions in the Jacobian's nullspace from force components in the transpose nullspace. Neither algebraic ambiguity alone proves that a biological force is unidentifiable from every possible sensor, dynamical model, and experiment.

4.2.5 Inertia Couples the Response Through the Equations of Motion

In independent coordinates, a reduced equation can be written $\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{h} = \mathbf{Q}$. At fixed state and fixed non-control terms, the acceleration change is $\delta\ddot{\mathbf{q}} = \mathbf{M}^{-1}\delta\mathbf{Q}$. It is the inverse mapping, not one off-diagonal mass entry alone, that answers which coordinate accelerates under an added generalized force.

For example, in consistent normalized units, let $\mathbf{M} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. An added force vector $(1, 0)$ gives acceleration change $(2/3, -1/3)$. The off-diagonal mass entry is positive while the cross-acceleration response is negative. A linear coordinate transformation can diagonalize this matrix; transforming the force and reconstructing the motion gives the same physical response. Removing an off-diagonal entry by relabeling coordinates does not remove the original endpoint response.

In a constrained two-hand model, admissible acceleration changes also satisfy the linearized acceleration constraint at the fixed state. Reactions change with the command, so solve the coupled constrained equations or use a valid reduced-coordinate model. Applying an unconstrained inverse mass matrix and ignoring the changed reactions answers another question. Near a constraint-rank change, check the solve's validity before interpreting large numerical values.

Posture affects these mappings, while velocity affects the remaining dynamics. A same-state control comparison holds both fixed. Comparing club angles is a comparison across different configurations, each with its own same-state allocation experiment; it is not one unchanged state with several angles.

4.2.6 A Two-Hand Couple Has Exact, Conditional Controls

Let the contact separation be $\mathbf{d} = \mathbf{r}_T - \mathbf{r}_L$. If the trail contact applies $+\mathbf{F}$ and the lead contact applies $-\mathbf{F}$, the resultant is zero and the couple is exactly $\mathbf{d} \times \mathbf{F}$. Its magnitude is dF only when the vectors are perpendicular. Collapsing separation or making $\mathbf{F}$ parallel to $\mathbf{d}$ removes this point-force couple. Applied free moments can remain.

Reversing the force with the separation held fixed reverses the couple. Reversing the separation while retaining the same force assignment also reverses it. Rigidly rotating the entire force system rotates its moment; it does not necessarily reverse a consistently transformed scalar component. Merely swapping contact labels changes no physical force system at all.

For arbitrary forces, define the common component $\mathbf{F}_c = (\mathbf{F}_L + \mathbf{F}_T)/2$ and differential component $\mathbf{F}_d = (\mathbf{F}_L - \mathbf{F}_T)/2$. The resultant is $2\mathbf{F}_c$, and the point-force moment about the midpoint is $-\mathbf{d} \times \mathbf{F}_d$. For the pure pair above, $\mathbf{F}_d = -\mathbf{F}$, recovering the same positive $\mathbf{d} \times \mathbf{F}$ expression. A common component

can contribute a moment about another reference. Direct free moments must still be added. This decomposition is useful bookkeeping, not a measurement of separate muscles.

These geometric controls require declared held-fixed forces. If changing grip separation also changes solved reactions, the resulting couple need not scale linearly with separation. A measured or simulated reversal test must report both the changed geometry and force response. Otherwise a failed simple scaling law need not be a failed cross-product implementation.

4.2.7 What the Allocation Study Actually Holds Fixed

The pinned provider uses a planar rigid-club model with two constrained arms. At each of 19 club angles from -12° to 24° , it constructs a constraint-consistent configuration with club center at $(0, -0.50)$ m and zero generalized velocity. This is an instantaneous dynamical comparison; zero initial velocity does not imply static equilibrium.

For each configuration it evaluates commanded and zero-command dynamics at the same state, then subtracts them. The fixed task is the club’s rotational inertia times the **control-only angular-acceleration difference**, set to 8 N m. It is not the full net wrench, every acceleration component, or a complete trajectory. The force resultant is allowed to change.

One control subspace uses four shoulder/elbow torques with both wrist commands zero; another uses the two wrist torques with proximal commands zero. In each, the implementation chooses the minimum unweighted Euclidean norm of actuator commands that satisfies the scalar task. It blends these two normalized controls, scaled to the common task, at 21 fractions from zero to one. Linearity of the fixed-state constrained solve preserves the scalar task across that blend; it is not a general superposition claim for finite nonlinear swings.

The blending fraction is **not** the fraction of net club moment supplied by direct wrist moments. At the wrist-only endpoint, direct moments sum to about 8.13–10.02 N m across the angles; opposing contact-force moments bring the control-only total back to 8 N m. The archive’s variable called a grip-force “couple” is the contact-force moment about the club center. With a nonzero resultant, it need not be a pure couple of equal opposite forces.

The reported hand-force RMS is the root mean square of the two **control-only force-vector norms at that instant**. It is not a time RMS or the norm of total physical contact loads. A control-only force is the difference between two same-state solutions; the norm of that difference is not the difference of their norms. The torque metric is the unweighted norm of six actuator commands, not muscle force, metabolic cost, tissue loading, work, or a coordinate-free measure of effort.

Independent recomputation of the archived arrays reproduces the scalar task and moment closure with residuals below 6×10^{-15} N m. The hand-force metric ranges from 7.58 to 91.51 N and the torque norm from 5.74 to 25.45 N m. Among the sampled blend fractions, hand-force minima occur from 0.75 to 0.90, whereas torque-norm minima occur from 0.85 to 0.95, depending on angle. These are grid minima along one restricted blend of two subspace solutions. They do not establish global optima over every admissible allocation, a physiological optimum, or an optimal human grip. Archived summaries were recomputed; the constrained model was not rerun for this chapter review.

4.2.8 Connect Geometry to the Question Being Tested

Geometry, inertia, actuation, constraints, and objectives enter the same chain of reasoning. Geometry maps contact motion and forces; inertia and constraints determine acceleration response; a chosen objective prices the resulting delivery and internal demands. Changing one of these can change a preferred allocation without proving that the original physical mechanism was absent.

For a geometry hypothesis, first declare the system, reference, coordinates, contact model, and task. Then distinguish a bookkeeping transformation from a physical intervention. Check algebraic identities at fixed forces, and report the dynamically recomputed forces for changed geometry. Finally evaluate the outcomes and constraints that matter to the proposed use. Chapter 3 explains state matching, Chapters 9–10 connect force to work, and Chapter 23 treats uncertainty and competing objectives.

Human motion and contact are more complicated than this planar illustration. Kinematic measurements can describe geometry, but they do not alone select the bilateral force allocation that produced it. McPhee’s review provides context for the range and limitations of golf models (McPhee 2022); it is not direct validation of this allocation surface or a technique prescription.

4.3 Where the Picture Breaks

The door has a fixed hinge; the golfer, compliant grip, and shaft do not reduce to that single joint. A contact’s moment about one point is not automatically its contribution to every generalized acceleration. Likewise, favorable force transmission in one direction can coincide with poor velocity authority, limited motion range, or unacceptable loading.

Model Result: The archived reduced model admits different actuator blends with the same scalar control-moment task and different internal metrics. Geometry changes the preferred sampled blend for each declared metric.

Measurement Boundary: Coordinate-consistent wrenches and motion are needed to compare physical actions. Independent information is needed to constrain individual hand forces and biological actuator allocation; small numerical closure residuals supply no human validation.

Hypothesis: An opposing club-moment component can arise from two-hand force geometry. Whether it persists or improves delivery under a changed grip must be tested with the changed reactions and full dynamical state, not inferred from one negative projection.

Practical Interpretation: Ask where the force acts, which direction it points, what motion it accompanies, and which task it helps accomplish. Those questions turn a vague claim about leverage into a comparison that can fail.

Go Deeper: Read the pinned [two-hand wrench derivation](#), [allocation implementation](#), [archived allocation record](#), and [wrench–power frame audit](#).

Chapter 5

Speed, Acceleration, Energy, and Power

5.1 A Concrete Picture

A car dashboard separates quantities that casual speech blends together. The speedometer reports how fast the car is moving now. Acceleration describes how its velocity changes, including a change of direction. The fuel gauge concerns stored chemical energy, whereas power describes a rate of energy conversion or transfer. A heavy truck and a bicycle can have the same speed and very different kinetic energies. These quantities answer different questions even when they rise together.

The same distinction matters in a swing. Clubhead speed describes one point, while the club's kinetic energy includes motion throughout the club. Pelvis angular speed alone does not determine pelvis energy, and a slowing pelvis does not identify where energy went. A large power peak may deliver little work if it is brief or followed by negative power. Before interpreting a transfer, name the body or collection of bodies, the point whose motion is reported, the time interval, and the observer frame. In this chapter, velocities are measured in a declared inertial laboratory frame unless a relative velocity is explicitly named. Changing the direction of coordinate axes in that frame does not change energy; changing to a physically moving observer can.

5.2 How the Mechanism Works

5.2.1 The Point, the Body, and the System

Velocity includes direction; speed is its magnitude. For a rigid club, the velocity of a head point H is

$$\mathbf{v}_H = \mathbf{v}_C + \omega \times \mathbf{r}_{H/C},$$

Here C is the club's center of mass, $\mathbf{v}_C$ its velocity, ω the club's absolute angular velocity, and $\mathbf{r}_{H/C}$ the vector from C to H. Translation and rotation combine as vectors. A wrist rate multiplied by club length omits motion of the wrist and may also confuse a relative joint rate with the club's absolute angular velocity. A flexible club requires additional deformation velocities; one rigid-body formula cannot describe every material point exactly.

Acceleration is the time derivative of velocity. When speed is nonzero, its rate of change is

A Speedometer Is Not an Energy Meter

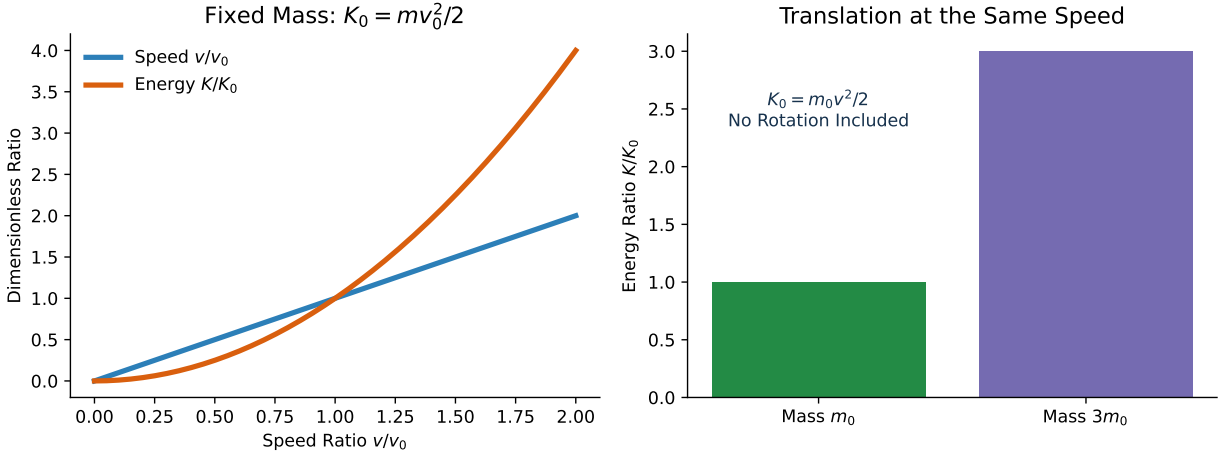


Figure 5.1: Normalized Speed and Kinetic Energy, and the Effect of Mass at a Fixed Translational Speed.

$$\frac{d|\mathbf{v}_H|}{dt} = \frac{\mathbf{v}_H \cdot \mathbf{a}_H}{|\mathbf{v}_H|}.$$

A Cartesian acceleration component may change sign while speed rises; the tangential component in this equation determines the speed change. Normal acceleration bends the path and is present even at constant speed. For a particle moving around a fixed circle, an exactly radial force is perpendicular to velocity and has zero instantaneous power. A force directed along a club toward moving hands need not be perpendicular to the head's laboratory-frame velocity. Calling it radial does not establish zero work.

For a translating particle of mass m , $K = \frac{1}{2}m|\mathbf{v}|^2$. For a rigid segment with arbitrary three-dimensional motion,

$$K = \frac{1}{2}m|\mathbf{v}_C|^2 + \frac{1}{2}\boldsymbol{\omega}^T \mathbf{I}_C \boldsymbol{\omega}.$$

The inertia tensor $\mathbf{I}_C$ and angular velocity must be represented in the same axes. The familiar $\frac{1}{2}I\omega^2$ is the fixed-axis specialization, with I taken about that axis. For a translating club, adding a rotational term about the moving grip to a center-of-mass translational term is not generally this decomposition; the reference shift needs its accompanying cross-term. Using center-of-mass translation and center-of-mass rotational inertia avoids that particular double count.

In generalized coordinates for a system with time-independent position maps,

$$K = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}}.$$

Here $\mathbf{q}$ contains the generalized coordinates, $\dot{\mathbf{q}}$ their rates, and $\mathbf{M}(\mathbf{q})$ is the symmetric generalized mass matrix. Configuration-dependent cross-terms describe shared physical motion. They do not make

the energy of a specified rigid segment arbitrary: its center-of-mass and rotational terms still define it in the chosen observer frame. What can be arbitrary is assigning each generalized coordinate its own independent energy share while distributing the cross-terms by convention. Prescribed moving supports add explicit time dependence unless their motion is included among the coordinates; then the simple quadratic expression needs additional terms. The moving-base chapter returns to that distinction.

5.2.2 What a Speed Increase Costs

Compare equal-mass idealized clubheads moving at 40 and 44 m/s. The second is 10% faster and has 21% more **translational** kinetic energy because $(44/40)^2 = 1.21$. For a 0.20 kg point-mass representation, those energies are 160 and 193.6 J. The extra 33.6 J is a useful accounting requirement for that model, not a measurement of extra muscular work. It could reflect different external work, release of stored energy, lower dissipation, or different initial energy. A full club also has rotational and possibly elastic energy.

Likewise, a 0.20 kg head at 50 m/s has 250 J of translational energy, while a 70 kg golfer's center-of-mass translation at 1 m/s contributes 35 J. The latter calculation omits all motion relative to the golfer's center of mass. It therefore does not say that the golfer's entire kinetic energy is 35 J or that the club contains more energy than the body. Comparisons are useful only when their included motions are explicit.

A point can speed up while its extended body loses kinetic energy. As a manufactured example, take a uniform 2 m rod of mass 3 kg with center-of-mass planar inertia 1 kg m^2 . At one instant let its center move at $(1, 0) \text{ m/s}$ with acceleration $(-3, 0) \text{ m/s}^2$, its angular velocity be 2 rad/s , and its angular acceleration be 4 rad/s^2 . For the endpoint 1 m below the center at that instant, rigid-body kinematics gives velocity $(3, 0) \text{ m/s}$ and acceleration $(1, 4) \text{ m/s}^2$. Its speed increases at 1 m/s^2 , but the rod's kinetic-energy rate is $3(1)(-3) + 1(2)(4) = -1 \text{ W}$. The center's translational energy falls faster than rotational energy rises. This is a consistent instantaneous mechanics example, not measured golf behavior. For a single particle, by contrast, increasing speed at fixed mass requires positive **net** power $m\mathbf{v} \cdot \mathbf{a}$. Changing coordinates cannot evade that identity.

5.2.3 Power Needs a Named Interaction

Power is work per unit time. A force acting at a material point supplies $P = \mathbf{F} \cdot \mathbf{v}$ using that point's velocity. For a rigid body, a resultant force and moment represented at C supply

$$P = \mathbf{F} \cdot \mathbf{v}_C + \mathbf{M}_C \cdot \boldsymbol{\omega}.$$

Moment and point velocity must be shifted together if another rigid-body reference is used. For several deforming contacts, retain their separate force-velocity and moment-angular-velocity pairings unless a common rigid motion has been justified. One interaction's positive power does not require positive net power after all other interactions are included. Nor does power at an external boundary always equal total mechanical-energy rate: internal elastic storage, dissipation and, for a biological system, chemical-to-mechanical conversion must be accounted for.

A constant 1,000 W input for 0.05 s supplies 50 J. A 5,000 W input for 0.002 s supplies 10 J. For a variable signal the work is its signed integral, $W = \int P dt$. Peak height alone supplies neither that

integral nor its sign. Positive and negative portions can nearly cancel even when both peaks are large.

5.2.4 Joint Actuator Power and Segment Power

Torque sign is a direction statement, not an energy verdict. In a planar model, a moment τ acting on a segment contributes $\tau\omega$ using that segment's absolute angular velocity in the declared frame. If τ is instead the actuator moment conjugate to a **relative** joint angle q , its actuator power is $\tau\dot{q}$ using relative angular velocity. These are different quantities. In three dimensions, Euler-angle rates are not generally angular-velocity components; their generalized moments must be transformed to preserve power.

For an ideal joint joining two bodies, an actuator applies equal and opposite moments. If the distal and proximal angular velocities are ω_d and ω_p , their moment powers are $\tau\omega_d$ and $-\tau\omega_p$. Their sum is actuator power $\tau(\omega_d - \omega_p)$. For example, $\tau = 2$ N m, $\omega_d = 1$ rad/s and $\omega_p = 3$ rad/s give moment-power contributions of +2 W on the distal body and -6 W on the proximal body. Of the 6 W removed from the proximal body through this moment channel, 2 W enters the distal body and 4 W is absorbed by the actuator. This example isolates the moment channel; any joint-force power must be included separately when finding the total power on either segment. This does not imply anything about muscle activation or metabolic cost. It shows why a joint-power label needs its velocity convention.

For a negative moment acting along a declared swing-plane normal, the sign of its contribution also depends on angular velocity along that normal. Even when that moment channel removes energy from the club, hand forces or other channels can supply more energy than it removes. Do not add “wrist power” to already complete hand-wrench power if both represent the same transmitted moment. The final speed depends on the coupled trajectory, rather than the sign of one moment channel.

5.2.5 A Worked Whole-Club Ledger

Choose the **whole flexible club**, including its shaft, as the system. Write its mechanical storage as $E = K + V + U_s$, where V includes gravitational potential and U_s is shaft strain energy. Assume no internal actuator, no shaft damping, and no other omitted work channels for this illustration. Then

$$\Delta(K + V + U_s) = W_{\text{hands}} + W_{\text{aero}}.$$

Gravity is represented through V and must not also be added as gravitational work on the right. Shaft release changes internal storage; it is not another external supply. Real calculations must add their actual losses and other interactions instead of inheriting these illustrative exclusions.

Consider two intervals of 0.05 s. In the first, total hand-wrench power averages 800 W, supplying 40 J. In the second it peaks at 1,500 W but averages only 300 W, supplying 15 J. Let aerodynamic work be -2 J in each, and let shaft strain energy remain unchanged in the first interval but fall by 8 J in the second. The ledgers are:

Quantity	First Interval	Second Interval
Hand work	40 J	15 J
Aerodynamic work	-2 J	-2 J
Change in shaft strain energy	0 J	-8 J

Quantity	First Interval	Second Interval
Change in total club mechanical energy	38 J	13 J
Change in club $K + V$	38 J	21 J

The second whole-club energy gain is **13 joules**, while its $K + V$ gain is 21 J because strain energy decreases. Adding the 8 J to the whole-club gain would count an internal conversion as new energy. If instead the system is just a rigid head, excluding the shaft, shaft-to-head interface work is external to that subsystem. To split the second interval consistently, add two illustrative assumptions: the shaft's own $K + V$ rises by 5 J, and all -2 J of aerodynamic work acts on the head. The shaft's mechanical-energy change is then $5 - 8 = -3$ J, so 15 J entering from the hands and 18 J leaving through the head interface close its ledger. The head receives those 18 J and loses 2 J aerodynamically, gaining 16 J. The head's 16 J and shaft's -3 J sum to the whole club's 13 J gain. These assigned storage and drag partitions are not measurements: without them or corresponding observations, whole-club hand work alone does not determine head-interface work. Adding proximal hand work again to that head ledger would count a transmission path twice.

Equal terminal head speeds therefore do not establish equal work histories or equal total club energy. The trials may begin with different storage, distribute energy differently among translation, rotation and shaft deformation, or produce different face and path outcomes. For a specified rigid club, mass and body-frame inertia do not change just because its orientation changes; changing orientation alters projections and the relation between endpoint speed and the rest of its motion.

5.2.6 Sequence, Acceleration, and Impulse

An ordered sequence of pelvis, trunk, arm and club speed peaks describes coordination. It does not demonstrate that one energy package passed along those peaks. A lower angular speed does not alone prove lower segment energy: translation, axis direction and the inertia tensor still matter. Track the segment's actual energy change, the work through its boundaries, and any storage or conversion within the selected system (Robertson and Winter 1980; Winter 2009). Ideal joint reactions can transmit power between individual segments while their net power cancels for the combined system. Zero net work at that larger boundary does not make a force dynamically irrelevant.

For a fixed collection of material bodies, net external force impulse equals the change in total linear momentum in an inertial frame. Net external angular impulse equals the angular-momentum change about a fixed point in that frame; moving-reference formulas require their appropriate transport terms. Work instead pairs forces with displacement and moments with angular motion. A perpendicular force may redirect momentum substantially while doing no instantaneous work. Momentum, energy and speed therefore need separate graphs.

At a fixed state and fixed contact/constraint mode, suitable equations can split generalized acceleration into named force contributions. These are not automatically separate physical energy supplies. Power requires the appropriate force-velocity pairing, and coordinate velocity-bias terms must be interpreted together with the changing mass matrix. Comparing instantaneous contributions also differs from changing an input and following the resulting trajectory. The force/work and counterfactual chapters develop those distinctions.

5.2.7 What the Evidence Supports

Model Result: In the pinned double-pendulum timing sweep, selected later-drive schedules produce more delivery speed than early drive or the passive baseline under the stated input bounds and event rules. A separate coupled uncertainty study reports higher held-out 10th-percentile delivery speed for early restraint than late drive, but worse mean planar face/path error. These are different model comparisons, not one universal optimum or a demonstration of human preactivation. The [timing results](#) and [uncertainty/control study](#) specify the respective assumptions and outcomes.

Human and Methodological Evidence: Tinmark and colleagues measured pelvis, upper-torso and hand kinematics in 45 skilled golfers, using full and partial shots; their reported peak sequence is kinematic evidence, not a direct measurement of the segment energy route ([Tinmark et al. 2010](#)). Robertson and Winter’s segment-power study used a sagittal linked-body model and data from two walking participants; it supports an accounting methodology rather than a golf-specific transfer magnitude ([Robertson and Winter 1980](#)). McPhee’s narrative review is a map of golf models and measurement methods, not an additional intervention experiment ([McPhee 2022](#)). Keep those evidence roles distinct when reading a claim about how a golfer generated speed.

5.2.8 Reader Lab: Diagnose a Misleading Result

Suppose a report favors one strategy because head speed rises more steeply late in the swing and wrist power has a larger peak. First identify the speed point, observer frame, joint-rate convention, system boundary and integration interval. Compare initial storage and integrate distinct work channels exactly once. For a whole-club ledger include strain-energy change inside storage; for a head-only ledger use shaft-to-head work at the boundary. Check the closure residual against the size of the proposed effect, with consistent event detection and numerical resolution.

Then compare the actual task outputs: face orientation, path, impact location, variability and modeled loads. A speed improvement may be real while “better strategy” remains unsupported. A large force with little mechanical work may still matter for tissue loading, and mechanical work is not metabolic effort.

Finally, compare schedules from the same admissible state, including controller or actuator memory where the model has it. Hold parameter choices, event rules and other inputs fixed for that comparison. If the ranking survives timestep refinement and specified uncertainty tests, it has support within those tested conditions, rather than universal robustness. If it disappears, investigate numerical error, changed event selection, initial preparation, parameter choices or other altered conditions; disappearance alone does not identify which explanation is responsible. A simulation intervention does not by itself establish what a human can execute.

5.3 Where the Picture Breaks

The dashboard analogy separates the questions, but the swing couples their answers. Head velocity combines translation and rotation; joint power depends on relative motion; physical segment energies differ from arbitrary coordinate energy shares. Changing the system boundary moves an interaction between internal conversion and external transfer without creating or destroying energy. Those relationships are why apparently contradictory graphs can coexist.

Biological systems also contain chemical energy, active muscle processes and losses that a rigid mechanical model does not resolve. An internally driven system can increase mechanical energy without positive external wrench work. Conversely, high muscular effort may coexist with little external mechanical work. Do not infer muscle action or a coaching prescription merely from a closed rigid-body ledger.

Use “head speed increased,” “club mechanical energy increased,” and “the hands supplied positive work” as separate statements until the accounting connects them. Avoid “power was stored,” because power is a rate, and “speed was transferred,” because speed is not a conserved substance. A useful explanation lets readers follow the connection without substituting one quantity for another.

Go Deeper: See the monograph’s [mechanics and operational energy definitions](#), [phase-resolved results](#), and [coupled uncertainty study](#). The manufactured examples here test mathematical consistency; they are not measurements of golfers.

Chapter 6

Constraints Push Back

6.1 A Concrete Picture

Swing a key on a cord. Your hand does not continuously command the key's inward acceleration. The cord supplies tension because it prevents the key from flying away. Cut the cord and the key leaves along its instantaneous tangent. The constraint is not a passive absence of freedom; it is an interaction that pushes or pulls as needed to keep the allowed geometry.

Human joints, two hands on one grip, and feet on the ground all impose versions of this idea. A wrist reaction can be large even when commanded wrist torque is zero. Two arms can develop internal forces that cancel in the club's net force. The ground can exert a reaction determined by the whole body's state and constraints. "No motor command here" does not mean "no force here."

6.2 How the Mechanism Works

A holonomic constraint can be written $\phi(\mathbf{q}) = 0$. It says which configurations are allowed. Differentiating gives the velocity condition $\mathbf{J}\dot{\mathbf{q}} = 0$, where $\mathbf{J}$ is the constraint Jacobian. Differentiating again produces an acceleration condition involving $\mathbf{J}\ddot{\mathbf{q}}$ and a velocity term. The constraint force must be whatever value makes the equations of motion and this allowed acceleration agree.

One common system is

$$\begin{bmatrix} \mathbf{M} & -\mathbf{J}^T \\ \mathbf{J} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}} \\ \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{Q} - \mathbf{h} \\ -\dot{\mathbf{J}}\dot{\mathbf{q}} \end{bmatrix}.$$

λ contains constraint reactions. They are solved together with acceleration, not pasted on afterward. Change configuration, velocity, or control and the required reactions change.

6.2.1 Zero Torque Does Not Mean Zero Reaction

Set one selected control torque to zero at a fixed moving state. Gravity remains. Momentum remains. Other actuators remain. Most importantly, constraints remain. The wrist joint must still keep arm and club connected, so it transmits force. The resulting **zero-torque counterfactual force** is a

A Constraint Pushes Back to Preserve the Allowed Path

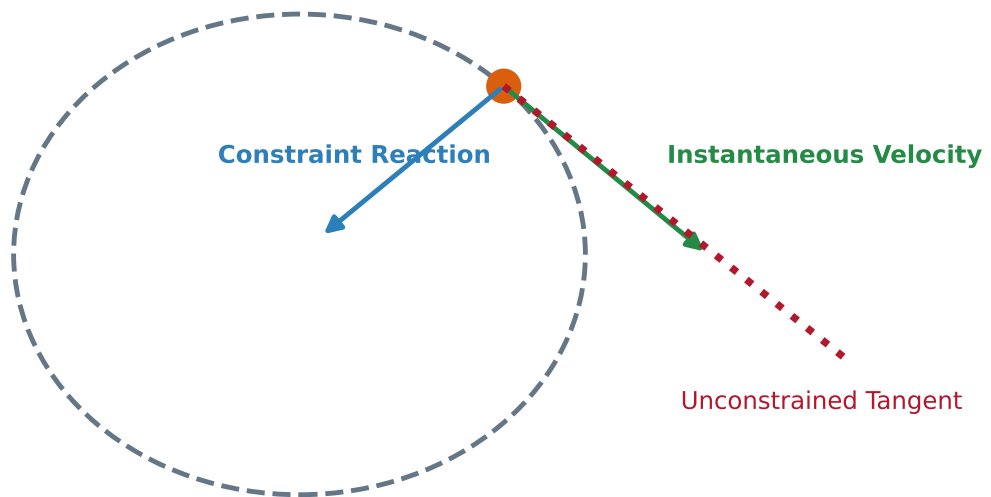


Figure 6.1: A Constraint Force Holds a Moving Mass to Its Allowed Path While an Unconstrained Counterfactual Leaves Tangentially.

model-defined answer to a narrow question: what reaction does this model require now if that selected torque is zero while the state is held fixed?

It is not “the passive force in the human wrist.” It depends on which control was removed, which terms remain, and how the joint is modeled. Yet it reveals an important mechanism: interaction forces can arise from linked motion itself and can contribute substantially to the club’s acceleration or hand-path work.

6.2.2 Ideal Constraints Can Carry Force Without Work

An ideal constraint force is often perpendicular to the allowed virtual motion, so it does no work on the complete constrained system. That does not make it unimportant. A rail pushes a train sideways around a curve. The normal force changes momentum direction while doing little work. At a subsystem boundary, the same interaction can have nonzero power on one part and opposite power on another.

Real biological contacts are not perfectly rigid or frictionless. Grip compliance, joint deformation, soft tissue, and shaft flex permit relative motion and energy storage. The ideal constraint is a reference case that separates geometric enforcement from compliance; it is not a claim that the interface has infinite stiffness.

6.2.3 Constraints Create Hidden Internal Modes

With two hands attached to one club, more contact-force components may exist than the club’s net wrench can identify. Add equal and opposite forces to the two hands and the resultant can remain unchanged. That internal squeeze can alter joint loading and stability without appearing in club motion. Closed-chain inverse dynamics therefore cannot uniquely recover individual hand forces unless extra measurements or allocation rules are supplied.

The same issue appears at the feet. A net ground-reaction wrench may not uniquely identify left and right foot forces or pressure distributions. A force plate per foot, pressure insoles, and careful calibration reduce ambiguity, but model assumptions remain.

6.2.4 Reaction Forces Are Not Free Energy

Constraint reactions can accelerate one coordinate and oppose another. This can look like energy creation if only one segment is inspected. The whole-system work-energy balance restores perspective. The reaction redistributes momentum and energy under the constraint; active work, gravity, and stored energy still fund the ledger.

This is particularly important for the phrase “interaction forces increase transfer.” They may increase club acceleration relative to a counterfactual by redirecting momentum or mediating power from proximal actuation. They do not manufacture energy. The scientific question is which state and geometry make their contribution favorable, for how long, and at what internal cost.

Model Result: In the double-pendulum decomposition, zero selected wrist torque leaves a substantial wrist reaction generated by gravity, velocity terms, proximal actuation, and the connection. Pointwise and forward tests quantify where its contribution persists.

Human Evidence: Joint reactions and individual hand forces are difficult to measure directly. Instrumented grips can constrain the two-hand problem, while motion and external-force data support

inverse dynamics with stated uncertainty (Choi and Park 2020; Koike 2016).

Hypothesis: The late downswing may exploit constraint and inertial reactions so the club receives favorable acceleration even while a direct distal moment is small or opposing.

6.2.5 What a Constraint Actually Does

A constraint removes a motion that would otherwise be possible. A rigid pin keeps two points together while allowing rotation. An ideal no-slip contact keeps a foot point stationary relative to the ground. A bilateral grip model keeps specified hand and handle frames related. The equations must generate reaction forces large enough to satisfy those conditions while all masses, applied forces, and velocities evolve.

Those reactions are not corrections added after the motion. They are part of the simultaneous solution. Accelerations and constraint multipliers are solved together. Reaction at one joint therefore depends on applied torque elsewhere, gravity, configuration, velocity, inertia, and every other active constraint. This dependence is the mechanical foundation for interaction-force transfer.

6.2.6 A Wagon-and-Hinge Example

Imagine a light wagon connected by a hinge to a heavier powered wagon. Accelerate the powered wagon. The hinge pulls on the light wagon even though it has no motor. That force can increase its kinetic energy because the attachment point moves along the force direction. If the hinge instead forces a curved path, part of the reaction may redirect velocity with little direct work. The light wagon’s motion is still caused in part by the connection.

Now brake the light wagon. The hinge reaction changes and affects the powered wagon. “Proximal causes distal” is therefore a useful phase description, not a one-way causal law. Linked dynamics is reciprocal. Distal speed often rises late because of evolving state and control, not because forces pass through a valve in only one direction.

6.2.7 Radial and Tangential Jobs

In a planar arm–club model, wrist reaction can be resolved along and across the club. The radial component helps enforce connection and supplies inward acceleration for curved motion. The tangential component more directly changes relative angular motion. Yet their labels rotate with the club, and a radial force at the wrist can contribute to club-center translation and system-level work. It is safer to calculate power and generalized acceleration than infer function from component names.

During early downswing, rapid arm motion can produce a reaction that carries the club while relative release remains limited. Later, changed geometry can project reactions more strongly into club rotation. A direct wrist command may supplement, oppose, or reshape that interaction. The model separates these terms at fixed state, then tests their persistence forward.

6.2.8 Pointwise Removal Versus Forward Removal

For a pointwise zero-torque counterfactual, copy one state, set the selected torque to zero, and resolve acceleration and reactions. This answers “what is this input contributing now under these equations?”

It does not say what a person would do after losing the input and does not allow state to change before comparison.

For a forward killswitch, both trajectories begin at the same state but one input is removed over an interval. They then occupy different states, so later differences include direct removal and nonlinear consequences of divergence. This is closer to causal intervention within the model but depends on the chosen continuation of all other controls. Agreement between local and forward stories is stronger than either alone.

6.2.9 When Reactions Become Ambiguous

If several contacts enforce the same net restriction, individual reactions may not be unique. Two hands can share a net force in infinitely many ways; two feet can share a ground wrench in many ways. A solver can choose a minimum-norm allocation, but that is a convention rather than a measurement. The null space should be reported or extra constitutive rules and sensors introduced.

Compliance changes the picture too. Real grip and shaft deformation turn an exact geometric constraint into a high-stiffness force law with state. That may smooth reactions, introduce delay, and store energy. A mechanism that survives both rigid and compliant representations is more credible than one existing only because an ideal constraint is infinitely stiff.

6.2.10 What Would Refute the Transfer Claim

The claim weakens if removing proximal drive at matched state produces no meaningful change in distal reaction or acceleration; reversing geometry does not reverse the predicted projection; work-energy closure fails; or the effect vanishes under finer integration and an independent solver. In human data, predicted reaction direction and timing must agree with synchronized hand wrenches and motion within uncertainty. Calling the reaction “passive” does not lower that evidentiary standard.

6.2.11 Reader Lab: Separate Motor and Connection

Run a two-link model from one copied state under four conditions: all inputs; proximal input only; distal input only; and neither input. At the copied instant, record direct generalized accelerations, constraint reactions, and endpoint acceleration. Over a short forward window, record power, work, and trajectory. The proximal-only branch should make visible how the connection moves the distal link without an explicit distal motor.

Now alter link angle while holding masses and instantaneous inputs. The reaction and its distal projection should change. Reverse a coordinate sign and transform all related quantities; physical endpoint motion and power should not change. Replace the ideal joint with a stiff compliant element and test convergence as stiffness rises. These operations distinguish a real coupling effect from one created by notation or constraint implementation.

The exercise also demonstrates cancellation. A direct distal input may oppose the proximal-generated reaction at one state. Total acceleration can be smaller than either component. That is why components are not percentages and why a negative contribution can still be part of a later-beneficial schedule.

6.2.12 The Everyday Meaning of Passive

In ordinary speech, passive means doing nothing. In mechanics, it often means a force follows from state, geometry, material law, or constraint rather than an explicit selected control. A spring force is passive even if it is large; a joint reaction is passive in that narrow channel even while motors elsewhere work hard. Every use in this book carries that technical meaning. Human intent and muscle effort require separate evidence.

6.2.13 One Sentence to Retain

A connection can strongly affect motion without being an independent energy source. Its force depends on the whole state and can transmit, redirect, store, or constrain the consequences of work performed elsewhere. That statement is both more accurate and more testable than saying the club accelerates “by itself.”

6.3 Where the Picture Breaks

The cord-and-key picture has one exact constraint and no anatomy. A wrist has several degrees of freedom, ligaments, muscle stiffness, and distributed contact. The shoulder girdle moves. Grip friction can slip. A rigid planar hinge can test a coupling mechanism but cannot estimate tissue loading or injury risk.

Constraint-force decomposition can also depend on the mathematical partition. At redundant contacts, λ may be nonunique. Reporting one minimum-norm solution does not make it the biological allocation. The model must fail closed or show the null space.

Go Deeper: Study the exact [wrist-reaction decomposition](#), the [forward two-hand constraints](#), and the [ground-reaction attribution](#).

Chapter 7

From One Pendulum to Two

7.1 A Concrete Picture

A single playground swing has one dominant rhythm. Add a loose second link at the seat and the motion changes character. The outer link can trail, fold, accelerate relative to the first, and feed reactions back into it. There is no moment when the first link acts alone and then politely hands control to the second. Both influence one another throughout.

The golf double pendulum replaces the inner link with an arm-like segment and the outer link with a club-like segment. It is deliberately spare. That is its strength: it is the smallest familiar model that contains changing geometry, inertial coupling, interaction force, relative-angle retention, and distal release.

The Distal Segment Is First Carried, Then Accelerated Relative to Its Base

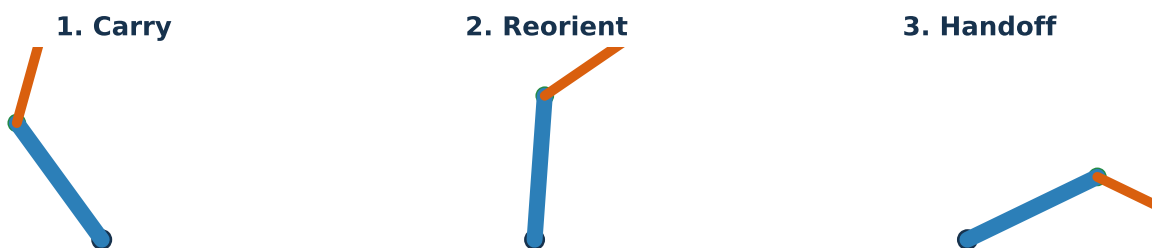


Figure 7.1: Three Snapshots Show the Club First Carried by the Arm, Then Reoriented, Then Accelerated Relative to the Moving Hand.

7.2 How the Mechanism Works

Let q_1 describe the arm angle and q_2 the club angle. The equations have the standard multibody form

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}) = \boldsymbol{\tau}.$$

The diagonal mass-matrix terms resemble each link’s own inertia. The off-diagonal terms couple them. $\mathbf{h}$ collects gravity and velocity-dependent effects. Even with zero direct club torque, shoulder torque and the moving state can produce club angular acceleration because $\mathbf{M}^{-1}$ is generally not diagonal.

7.2.1 Carrying Is Not Wasted Motion

Early in many downswing patterns, the arm and club rotate more nearly together while the wrist angle remains relatively closed. The hand translates rapidly, and the club’s center of mass gains motion even before large relative club rotation. Calling this only “holding lag” misses the transport: the distal body is being carried by a moving base.

As geometry changes, coupling can favor rapid relative rotation. The club’s angular speed about the hand then rises and adds to hand velocity at the clubhead. The visible release is not energy appearing at the wrist. It is the outcome of earlier work, current geometry, interaction forces, direct torque, gravity, and possibly shaft storage.

7.2.2 Relative Angle Changes the Coupling

When links are nearly aligned, their combined effective inertia and moment arms differ from when they form a right angle. The same shoulder torque can produce different club acceleration at those configurations. Velocity-dependent terms also grow with angular speed. A late release window emerges from the moving state, not from a universal clock time.

This helps explain why early positive wrist torque can be an inefficient tool in some optimized double-pendulum studies. It opens the relative angle before the system reaches a geometry where distal rotation produces a large clubhead-speed benefit. The result is model- and objective-dependent, but the mechanism is clear enough to test.

7.2.3 The Club Pushes Back

Action-reaction runs both ways. Accelerating the arm affects the club, and the club’s inertia affects shoulder and wrist reactions. As the club swings outward, it can slow the arm or demand additional proximal torque. A description that calls the club “passive” must not imply it is dynamically invisible.

The wrist reaction contains radial and tangential components. Radial force keeps the club attached to the moving hand and supports curved motion. Tangential force contributes more directly to angular acceleration about the hand. Their signs and powers change through the downswing.

7.2.4 A Limiting-Case Test

If the club mass tends toward zero, its reaction on the arm should vanish. If the club length tends toward zero, its distal leverage disappears. If the links are decoupled, off-diagonal mass terms vanish and proximal torque no longer accelerates the club coordinate through inertial coupling. These limiting cases are more than classroom exercises; they are implementation tests.

Likewise, setting direct wrist torque to zero does not freeze the wrist angle. The unactuated coordinate can move because coupling and gravity remain. An underactuated system is not an unmoving system.

7.2.5 Why the Model Remains Valuable

A model earns its place by isolating a mechanism, not by resembling every surface detail. The double pendulum lets us compute exact pointwise contributions, integrate counterfactual branches, close work-energy balances, and sweep timing without the identifiability burden of hundreds of muscles. If a proposed mechanism fails here, adding anatomical detail will not rescue its simplest claim. If it succeeds, higher tiers must test whether it survives added degrees of freedom and measured constraints.

Model Result: The declared double-pendulum schedules reproduce a familiar finding: late distal torque can improve club speed, while early drive can reduce the favorable release window. Zero-torque branches still show substantial coupled club motion.

Human Evidence: The model’s arm and club coordinates are abstractions. Human golfers show multi-planar shoulder, elbow, forearm, wrist, trunk, and lower-body motion. The double pendulum is not an anatomical fit, although it has a long and useful history as a mechanism model ([Jorgensen 1999](#); [Sharp 2009](#); [McPhee 2022](#)).

Practical Interpretation: The club can be carried before it spins rapidly relative to the hands. Delayed visible release need not mean delayed energy input; it may reflect when stored and transported energy becomes distal speed.

7.2.6 Building the Model One Assumption at a Time

The single pendulum begins with one angle, angular velocity, inertia, gravity, and perhaps a driving torque. It already teaches an essential lesson: torque, acceleration, and velocity are different. The same torque produces different acceleration as inertia or configuration changes, and speed reflects everything that happened earlier.

Add a second link and the state doubles. The club angle can be absolute or relative to the arm; either convention works if equations and signs are consistent. Kinetic energy now contains cross terms because both coordinate rates contribute to distal-mass motion. Those terms change with the angle between links. The club therefore responds even when its own applied torque is zero.

The fixed base is deliberate. It makes coupling visible without pelvis translation, balance, or ground contact. Lengths, masses, centers of mass, and inertias are declared rather than inferred from one golfer. The model’s value comes from traceable causal experiments, not from looking anatomically complete.

7.2.7 A Slow-Motion Downswing

At downswing start, proximal torque accelerates the first link. The club initially lags because inertia and geometry resist immediate relative rotation. The wrist connection transmits force, so the club is already moving through space before a large release. This carried motion is lost if the swing is described only by the club’s relative angle.

As arm angular velocity increases, velocity-dependent interaction terms grow. Changing arm–club angle changes effective coupling. Later, relative club rotation can accelerate rapidly. Depending on

schedule, distal torque can add positive work, regulate release, or temporarily oppose it. There is no single instant at which stored energy is handed over like an object; mechanisms overlap.

Near impact, available time is short. A command with a physiological or actuator rise time may arrive too late even if its ideal step version looks effective. An earlier command can alter geometry so as to reduce later transfer. Preferred onset in this model is consequently a state-and-dynamics problem, not simply “as early as possible” or “as late as possible.”

7.2.8 Tests That Make a Toy Model Useful

The mass matrix should remain symmetric and positive definite across the tested range. Zero-input conservative trials should maintain mechanical energy within declared numerical tolerance. Reducing timestep should converge outcomes. Independent formulations—for example Lagrangian and constrained Cartesian or spatial implementations—should agree. Known limiting cases should be recovered when a mass, link, or coupling is removed.

Then come causal controls. At matched state, remove proximal torque, remove distal torque, set velocities to zero for a pointwise velocity counterfactual, reverse relevant geometry, and integrate killswitch branches. If the claimed interaction survives only one plotting convention or one solver, it cannot support a broader argument.

7.2.9 What Carries Into Higher-Order Models

An exact numerical optimum rarely survives. A signed mechanism can: proximal acceleration changes distal reaction; geometry changes its projection; a late opposing moment can coexist with later speed benefit; and force allocation can be hidden by a net wrench. These become invariants to test on a moving base, in three dimensions, with shaft flexibility, and under two-hand constraints.

This ladder avoids opposite mistakes. One is rejecting a simple model because it lacks anatomy even though it isolates a real mechanism. The other is treating that mechanism as proof of a human strategy. Each rung asks which conclusion remains and which was an artifact of earlier assumptions.

7.2.10 A Reader’s Replication Exercise

Choose one parameter set and torque schedule. Record the complete initial state, coordinate definitions, solver, timestep, tolerances, and output window. Reproduce baseline, then halve timestep. Set distal torque to zero only at a copied state and compare instantaneous acceleration with a short forward branch. Finally perturb mass and length. If the sign changes easily, the honest result is a sensitivity map rather than a law.

7.2.11 Reader Lab: Discover Coupling Yourself

Begin with both links hanging and zero velocity. Apply a brief proximal pulse while distal torque stays zero. Plot both angles, speeds, wrist reaction, and energies. Repeat with the club initially aligned, lagged, and partly released. The distal response changes because geometry changes even though its motor input does not.

Next copy one moving state and set all torques to zero. The links continue and exchange energy under inertia and gravity. Set velocities to zero in a separate pointwise calculation while retaining angles. The

difference isolates the velocity-dependent part at that instant. Do not forward-integrate the artificial zero-velocity state as though a golfer stopped instantaneously; it is an attribution construct.

Finally, sweep distal-torque onset and amplitude. Plot a surface rather than one best trace. Add finite rise time and mass uncertainty. Mark regions where speed improves, where face/path proxy degrades, and where the result reverses. The exercise turns “delayed release” from a slogan into a conditional map.

7.2.12 The Most Important Limiting Cases

If distal mass tends to zero, its reaction on the arm should diminish. If the second link length tends to zero, the model approaches a different one-link inertia. If both links lock, they behave like one compound rigid body. If gravity is removed and no work enters, energy should remain constant. If initial velocity and inputs are zero, nothing should spontaneously accelerate except under gravity. These checks are simple enough that failure cannot be excused by biological complexity.

7.2.13 What a Lay Reader Can Safely Carry Forward

The useful insight is that distal motion depends on the entire linked state, not only distal effort. Early proximal motion can create conditions for rapid later club motion, and direct distal action can regulate that evolution. The model does not tell a golfer which muscle to activate or prove one delay is ideal. It teaches what measurements and counterfactuals a stronger model must preserve.

7.2.14 A Bridge to the Rest of the Book

The remaining models do not discard the double pendulum. They ask whether its coupling story survives separated hands, a moving base, spatial motion, and a flexible shaft. Whenever a later result looks complicated, returning to the two-link state, force, power, and counterfactual definitions provides a useful check. Complexity should enrich the mechanism without changing the meanings of its basic observables.

The checkpoint is simple: if a claim cannot be reproduced in the transparent two-link benchmark when that benchmark contains the claimed ingredients, debug it before adding anatomy.

7.3 Where the Picture Breaks

The two-link picture encourages the “whip” metaphor. A whip is flexible, distributed, and driven through waves; a rigid double pendulum has discrete links and joints. Both show distal speed amplification, but the mechanisms are not identical. Golf shafts are flexible enough to matter yet far from cloth whips.

The model also collapses two arms and two wrists into one connection, fixes a plane, and often prescribes the proximal base. It cannot identify scapular muscle action, grip-force allocation, clubface control, or injury load. These questions require later tiers.

Go Deeper: Read the monograph’s [double-pendulum mechanics](#), [numerical mechanism audit](#), and Putnam’s linked-segment synthesis ([Putnam 1993](#)).

Chapter 8

The Summation-of-Speed Story and Its Limits

8.1 A Concrete Picture

Walk forward inside a moving train. Relative to the carriage you may move at 2 m/s; the carriage moves at 20 m/s relative to the ground; your ground speed is roughly 22 m/s when directions align. Endpoint speed in a linked chain has a similar geometric addition. The clubhead moves because the hands translate, the club rotates about them, and all upstream joints contribute to the hand motion.

This gives the summation-of-speed story its appeal. Proximal motion carries the distal base, and distal rotation adds speed farther from the center. But unlike the straight train aisle, the velocity vectors rotate, segment lengths matter, and the same motions can cancel when directions oppose.

8.2 How the Mechanism Works

For a planar arm and club, clubhead velocity can be written as hand velocity plus velocity from club rotation about the hand:

$$\mathbf{v}_{head} = \mathbf{v}_{hand} + \omega_{club} \times \mathbf{r}_{head/hand}.$$

Hand velocity already contains contributions from proximal joint rates. This is a kinematic identity. It tells how motions combine at an instant, not which forces caused those motions or where their energy came from.

8.2.1 Why Distal Rotation Has Large Leverage

Linear speed from rotation grows with radius: $v = r\omega$. A given angular velocity near the club can contribute strongly at the clubhead because the radius is long. Yet creating that angular velocity requires energy and control under the system's inertia. The benefit of distal speed is not free leverage; it is a mapping between angular and endpoint motion.

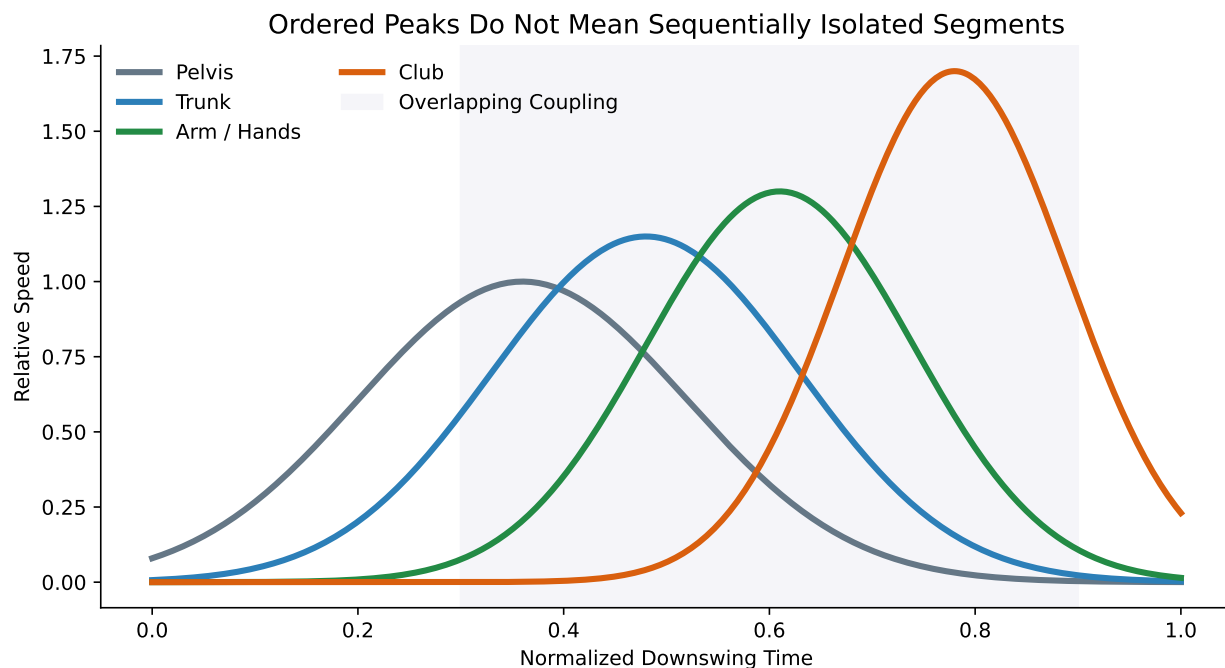


Figure 8.1: Segment-Speed Peaks Occur in an Ordered Sequence While Their Contributions to Clubhead Velocity Overlap in Time.

Timing matters because velocity vectors can align or oppose. If hand velocity and club-relative velocity point similarly near impact, their magnitudes add. If the club releases too early, its relative velocity may point less favorably later or change hand dynamics through reaction. The optimal schedule depends on the full trajectory and outcome definition.

8.2.2 Peaks Do Not Hand Off Like Relay Batons

When pelvis speed peaks and falls before trunk speed peaks, it is tempting to say pelvis speed was transferred into trunk speed. But speed is not conserved. Momentum and energy have conservation laws under stated conditions; angular speed does not. A declining pelvis speed may coexist with continued positive hip power, changing effective inertia, or work transferred through several joints.

Peak order is therefore a coordination descriptor. To claim transfer, pair it with joint powers, interface powers, segment energies, and external forces. Even then, the partition must be declared.

8.2.3 Sequence Can Vary While Outcome Remains Similar

Human movement is redundant. Different joint-rate combinations can produce a similar clubhead path and speed. One golfer may use more pelvis rotation and another more thoracic or arm motion. Averages can show a common sequence while hiding participant-specific solutions.

The uncertainty study reinforces this point in model space. Eight preselected control programs are all nondominated across held-out outcomes: improving one metric costs another. A single “ideal sequence” does not emerge from speed alone.

8.2.4 Phase-by-Phase Reading

During transition, the system changes direction and builds load. Early in the downswing, proximal motion can raise hand speed while the club remains lagged. In mid-downswing, geometry and velocity coupling change rapidly. Late, club rotation becomes a dominant contributor to clubhead speed. These labels are useful, but continuous powers and states are more informative than hard phase boxes.

A meaningful plot aligns events such as transition, peak hand speed, onset of rapid relative club rotation, and impact criterion. It then overlays segment speeds with the forces and powers proposed to cause them. The sequence becomes an entry point to mechanics rather than the conclusion.

Human Evidence: Proximal-to-distal peak ordering has been observed in many skilled striking and throwing tasks, including golf, but methods and definitions vary (Milburn 1982; Putnam 1991, 1993; Tinmark et al. 2010).

Model Result: The double-pendulum and higher tiers reproduce distal speed amplification through overlapping proximal transport, coupling, and late relative rotation. Their control programs are not unique human solutions.

Hypothesis: A useful strategy arranges geometry and state so the distal velocity contribution aligns with hand velocity near the chosen impact event without paying unacceptable face, path, load, or robustness costs.

8.2.5 Why the Relay-Race Story Is Attractive

In many skilled striking and throwing motions, proximal segment angular velocities peak before distal ones. Plot the curves and they resemble runners passing a baton. The picture captures the broad fact that distal speed is built on motion already present proximally. Problems begin when peak order becomes a mechanism by itself or a command to stop one segment before starting the next.

Endpoint velocity is a vector sum. Clubhead velocity includes proximal-base translation, contributions from trunk and arm rotation, relative club rotation, and deformation. These contributions overlap and change direction. A proximal angular velocity can decline while its current motion still contributes strongly to distal linear velocity. A distal segment can accelerate because of interaction forces before its own peak.

8.2.6 Peak Time Does Not Equal Transfer Time

A velocity peak occurs when that coordinate's acceleration crosses zero. An energy-transfer event is identified by power and work across a boundary. They need not coincide. Proximal speed can peak because distal reaction loads it, because drive declines, or because geometry changes—not necessarily because energy was deliberately “passed.” Distal speed may keep rising after one joint- power term turns negative because other channels remain positive.

Filtering and coordinates can move peaks. Global segment angular velocity, relative joint angular velocity, one anatomical component, and the magnitude of a three-dimensional vector produce different event times. A sequence claim must state quantity, frame, filter, event definition, and uncertainty.

8.2.7 A Three-Link Thought Experiment

Imagine pelvis, arm, and club speed peaks at 300, 220, and 80 milliseconds before impact. Ordering alone does not reveal whether pelvis work caused club acceleration. Inspect power: pelvis-to-trunk transfer may be positive over a broad interval; hand power may alternate; shaft return may peak late. If a filter shifts the pelvis peak while power integrals remain stable, the causal story should follow the ledgers rather than the prettiest order.

Hold final club speed fixed while altering torque schedule. A model can produce the same peak sequence through different work allocations. Conversely, two schedules can have different sequences but similar final energy. Sequencing is an observable constraint, not an identifying fingerprint.

8.2.8 Overlap Is the Rule

Muscles do not operate as isolated switches, and segments never become mechanically disconnected. Proximal and distal torques coexist. Constraint forces act bidirectionally. The nervous system may prepare stiffness before visible motion and regulate objectives simultaneously. A better phrase is “a wave of changing emphasis across a continuously coupled system.”

This matters for the arm-versus-wrist proposal. In one extreme, arms establish the dominant handle wrench while wrists resist and regulate. In the opposite, wrists create more club moment while arm action supplies another allocation. Both can generate familiar-looking kinematics. Bilateral wrenches, power histories, and activation measures are needed to separate them.

8.2.9 How to Use Sequence Evidence Well

Combine sequence data with kinetics. Report segment and joint kinematics, external and grip wrenches, interface power, work by phase, and uncertainty in event timing. Test whether a mechanism predicts not only peak order but also force direction, couple sign, and changes under geometry or equipment manipulation.

For an individual, repeatability matters. A mean peak time with wide within- person variability may be a weak control target. A state relation—such as a force-direction change over a consistent arm-club-angle range—may generalize better across tempo. That possibility must be tested rather than assumed.

8.2.10 What Would Change the Interpretation

If a participant preserves club speed while systematically reversing expected peak order, the order is not necessary. If the order appears but predicted transfer powers and hand-force signs do not, it is insufficient evidence for the mechanism. If order changes with coordinate definition, conclusions should be framed at the level that remains invariant. These productive failures turn a slogan into a bounded empirical statement.

8.2.11 Reader Lab: Rebuild an Endpoint Velocity

Take a planar three-link chain and compute endpoint velocity as the sum of each joint-rate contribution through its Jacobian column. Draw each contribution as an arrow tip-to-tail. Some arrows can partially

cancel even when every angular speed is large. Change posture without changing joint rates and watch the arrows rotate. Geometry is part of the summation.

Now plot each segment's angular speed and mark peaks. Compare those marks with endpoint speed, joint power, and cumulative work. They will not generally align. Repeat after applying a low-pass filter or switching from relative to absolute angular velocity. If the reported peak order changes, record that methodological sensitivity rather than choosing the preferred plot.

For a model intervention, remove proximal torque at a copied state while keeping current proximal velocity. Immediate club velocity is unchanged because state is unchanged, but acceleration and future trajectory differ. This shows why an existing proximal velocity contribution is not the same as continuing proximal power.

8.2.12 Alternative Explanations for the Same Sequence

A proximal peak can precede a distal peak because proximal drive declines, because distal reaction brakes the proximal segment, because joint geometry changes effective inertia, or because the chosen coordinate reaches an extremum. Several explanations can coexist. The sequence alone does not select among them.

Competing mechanisms make different predictions for forces, powers, and response to perturbation. A transfer explanation predicts particular interface-power and reaction patterns. A purely kinematic constraint may reproduce peak order without the same work exchange. Measuring and modeling those discriminators makes sequence evidence useful.

8.2.13 Responsible Practical Translation

A golfer need not consciously pause each body segment. Skilled motion can show overlapping peaks as the coupled system and control change emphasis. Practical analysis should avoid chasing exact population-average peak times. It can instead ask whether an individual's motion, wrench, and outcome are repeatable and whether a proposed change improves the full tradeoff under uncertainty.

8.2.14 What the Sequence Can Still Tell Us

After all these cautions, sequence is not discarded. It is a compact description of how motion emphasis changes, a useful check on model realism, and a way to locate phases for kinetic analysis. It becomes much stronger when paired with repeatability and mechanism predictions.

For example, a model that predicts a late negative couple should also predict where that interval lies relative to segment-speed peaks and arm-club state. Measured disagreement can refute or relocate the mechanism. Sequence is thus most useful as one layer in a synchronized evidence stack.

The responsible conclusion is neither “sequence proves transfer” nor “sequence does not matter.” It is that peak order constrains candidate explanations but does not identify one. Forces, powers, counterfactuals, and uncertainty complete the test.

8.2.15 Chapter Checkpoint

The observed sequence is one set of constraints on an explanation. The stronger explanation also closes forces and energy, predicts what changes under an intervention, and survives alternate coordinate

definitions. If those pieces are absent, describe peak order without assigning a unique cause.

This standard still leaves sequence plots valuable for communication. Their role is to orient the reader in time and state, then point toward the forces and powers that require deeper analysis.

8.3 Where the Picture Breaks

The train analogy adds velocities in a straight line and ignores reaction. In a swing, walking “inside the train” pushes back on the carriage. Segment rotations alter one another through inertia and constraints. Three-dimensional vectors can cancel even when scalar angular speeds look large.

The phrase summation of speed can also imply that more speed at every segment is always better. A proximal segment may need to slow or reorient so distal motion aligns. Peak magnitude, timing, and direction all matter, and the best pattern depends on the task.

Go Deeper: See the monograph’s [summation-of-speed discussion](#), [empirical evidence chapter](#), and the review of kinematic heterogeneity ([McPhee 2022](#)).

Chapter 9

Force Direction Changes the Job

9.1 A Concrete Picture

Push a shopping cart along its rolling direction and it can speed up. A sideways push may turn it or be largely balanced by wheel reactions. A downward push can change wheel loading without much forward acceleration. Direction changes the interaction, but the motion follows the net force, moments, and constraints—not one applied arrow in isolation.

At the grip, resolve a hand force along the shaft and along two transverse directions. These components can load the shaft, accelerate the club center of mass, and create a moment about that center. The moment depends on the force's line of action and chosen reference; a force applied at the hand has zero moment about that same point. An independent grip moment is a separate load. As the club rotates, a force fixed in the room changes its club-frame components. Direction is a moving relationship, not a permanent assignment to translation or rotation.

9.2 How the Mechanism Works

Power begins with projection. The dot product $\mathbf{F} \cdot \mathbf{v}$ selects the component of force along point velocity. Moment begins with a cross product, $\mathbf{r} \times \mathbf{F}$, which selects the component perpendicular to the moment arm. The same vector can therefore do little translational work yet create a large moment, or deliver power through a moving base while its moment about a particular point is small.

For the planar illustration, let H be the hand point, G the club center of mass, $\hat{e}_s$ the shaft direction from grip toward head, and $\hat{n}$ a chosen unit normal to the illustration plane. Define $\hat{e}_\perp = \hat{n} \times \hat{e}_s$. Then $F_s = F \cdot \hat{e}_s$ and $F_\perp = F \cdot \hat{e}_\perp$; retain $F_n = F \cdot \hat{n}$ in a spatial model. These are shaft-based components, not automatically the normal and tangent of the hand or clubhead trajectory.

The force's moment about G is

$$M_G^{(F)} = (r_H - r_G) \times F.$$

In the straight-link idealization $r_G = r_H + \ell \hat{e}_s$, its projected moment is $M_G^{(F)} \cdot \hat{n} = -\ell F_\perp$. Here ℓ is the hand-to-center-of-mass distance, not necessarily the full club length. The figure labels the

Direction and Reference Point Set the Moment

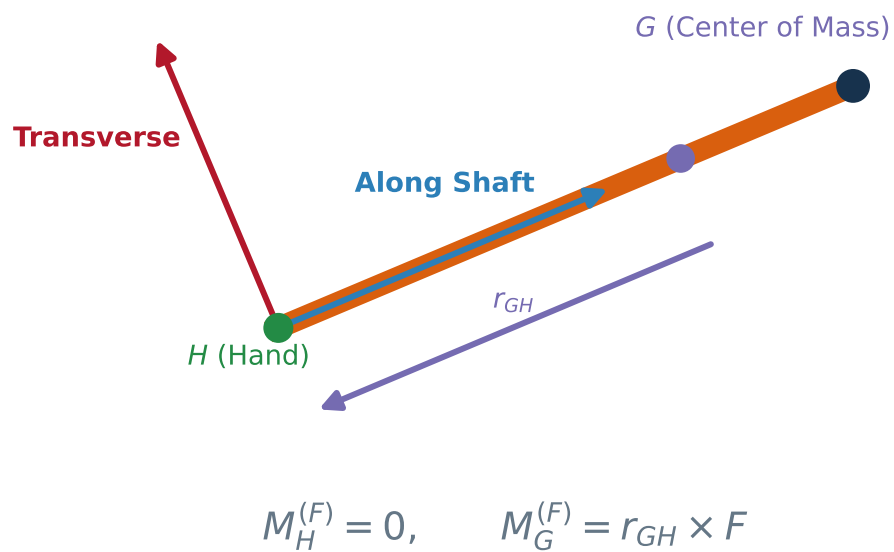


Figure 9.1: Hand-Force Components and the Moment Arm From the Club Center of Mass.

center-to-hand lever arm $r_{GH} = r_H - r_G$. The minus sign follows the stated basis. Its moment about H is zero. Add any intrinsic grip moment separately. A real club's center of mass may lie off the shaft line, so even an axial force can have a moment about G ; use the actual vector lever arm.

Nor does an axial force automatically equal a centripetal requirement. For a rigid offset $r_{HG} = r_G - r_H$, with angular velocity ω and angular acceleration $\dot{\omega}$, the center acceleration is

$$a_G = a_H + \dot{\omega} \times r_{HG} + \omega \times (\omega \times r_{HG}).$$

Hand acceleration and all external loads enter the balance. Only in a declared fixed-center circular case does the inward net-force component reduce directly to mv^2/R . An axial grip force can deliver positive power when the hand moves along it, even when its moment about an on-axis center is zero.

9.2.1 Vectors Rotate While Labels Stay Put

Suppose the laboratory vertical force remains upward through a short interval. When the club is nearly horizontal, upward may be largely transverse to the shaft. When the club is nearly vertical, upward may be largely axial. A plot of global vertical force can remain positive while the signed shaft-transverse component $F_{\perp}$ crosses zero. Mechanistic claims should use the frame tied to the proposed job and provide the transform back to measured coordinates.

Figure 9.1 identifies the application point and moment arm. The technical figures draw vectors on model snapshots because time traces alone hide this geometry. At each selected phase, the reader can see force origin, direction, scale, reference point, club orientation, and sign convention. A good vector figure is an argument, not decoration.

9.2.2 A Force Can Accelerate One Coordinate and Decelerate Another

Generalized coordinates need not be independent Cartesian directions. For a specified force on a modeled body, virtual work gives $Q_H = J_H^T F_H$, where J_H maps generalized velocity to its application-point velocity. Include the opposite interface load when the other body is also inside the modeled system. Generalized forces are moments only for the corresponding angular coordinates. After resolving ideal constraints, a contribution Q_k produces $\ddot{q}^{(k)} = M^{-1}Q_k$ at the same state in a nonsingular model. For example, $M = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $Q_k = (1, 0)^T$ give $\ddot{q}^{(k)} = (2/3, -1/3)^T$. Coupled response can have opposite signs without violating action–reaction; it does not identify the source of energy or muscle effort. Unreduced constrained equations require the associated reaction solve, not an unconstrained inverse mass matrix.

The mass matrix then mixes generalized forces into accelerations. Looking only at the sign of $F_{\perp}$ does not guarantee the sign of the relative wrist acceleration $\ddot{q}_2$, where q_2 is the wrist angle relative to the arm in the two-link example, when other terms are present. Exact decomposition solves the coupled equations at the same state and reports the contribution rather than inferring it from one arrow.

9.2.3 Direction Through the Downswing

Early in the downswing, the hand path and club orientation can make a large reaction mostly along the shaft. Its path-radial role depends on the declared center of curvature. Later, as relative angle

changes, a similar interaction can have a stronger tangential projection. Near the impact zone, two separated hand forces may form a negative swing-normal couple even while other pathways increase club kinetic energy.

This phase dependence is why the scientific question is not “is hand force good or bad?” It is: which component, in which frame, over which interval, about which reference point, with what power and impulse?

9.2.4 Three Names for Parts of One Motion Equation

The double-pendulum model gives a more precise vocabulary for the forces that appear as the arm turns and the wrist angle opens. It calls one contribution the **Coriolis Cross Term**. This part contains the arm’s rotation rate multiplied by the wrist’s relative opening rate. If either motion stops, the product is zero. It is therefore a compact way to ask how simultaneous proximal rotation and distal unfolding reshape the joint-coordinate balance.

A second contribution is the **Squared-Speed Term**: the collected terms proportional to individual coordinate-rate squares in this registered model. Curvature contributes to such terms, but this bin is not a measured inward contact force. Centripetal describes an inward acceleration requirement in a declared motion; centrifugal is an inertial term introduced in a rotating observer’s equations. Those are different balance conventions. Do not add a rotating-frame pseudo-force to an inertial free-body diagram that already accounts for the motion.

Gravity supplies a third, configuration-dependent contribution. Unlike the other two, it does not require motion. It can help or oppose the declared coordinate depending on club orientation. Damping and applied moments complete the model’s account. Adding every declared contribution reproduces the modeled acceleration; inventing a separate extra “centrifugal force” on top would count the same mechanism twice.

These categories are coordinate-dependent. In this study the arm angle is absolute and the wrist angle is relative to the arm. A different coordinate choice can move algebra between cross-speed and squared-speed bins without changing the physical trajectory. The full equations must transform together. Even the complete velocity-bias vector is not independently invariant under a nonlinear coordinate change: a free Cartesian particle $x = q^2$ has $\ddot{x} = 0$, yet $\ddot{q} = -\dot{q}^2/q$ for $q > 0$. Its curved coordinate description introduces a velocity term. Compare reconstructed physical accelerations and power, and carry the coordinate and decomposition convention with every named contribution.

9.2.5 Why an Outward Energy Story Needs More Than a Force Name

It is tempting to picture proximal energy as a parcel pushed outward by a Coriolis force. The safer picture is a state being prepared. Earlier applied moments and external forces build configuration and velocity. Once both links are moving, geometry redirects the consequences of that stored motion through the hinge. A Coriolis term can then make a large contribution even if the instantaneous distal command is small. It did not create energy from nothing; it describes how the moving system routes the effects of work already done and work still entering.

The sign also needs care. The code reports each left-hand equation term and its equal-and-opposite generalized drive. A positive equation term is not automatically positive acceleration, and a large force impulse is not automatically positive work. The drive must be paired with the appropriate velocity to

obtain power. This is why the source display shows impulse and work side by side instead of giving one arrow a permanent “speed-producing” label.

9.2.6 Normalization Can Hide Magnitude

Force arrows are often normalized for visibility. That communicates direction but not magnitude. Other figures use one scale across all phases so arrow length can be compared. Reports should say which convention is used. A tiny vector and a huge vector drawn at equal length can otherwise invite a false mechanism.

Likewise, unit-body-mass normalization aids participant comparison but does not restore missing contact allocation. Directional averages can cancel when participants use different frames or handedness conventions. Preprocessing is part of the interpretation.

Model Result: The archived planar double-pendulum audit reconstructs wrist reaction from the links’ tangential/centripetal accelerations and gravity support. That kinematic force balance is distinct from the coordinate-equation Coriolis/squared-speed split. Both require the state and declared conventions; neither alone identifies biological effort. The two-hand geometric reconstructions test the separated-force couple under frozen loads. They reverse that component under the stated transformations; they do not necessarily reverse the entire grip moment or describe a new forward trajectory.

Human Evidence: Net club motion alone does not uniquely identify both hand wrenches. Additional measurements or explicit allocation assumptions are needed. Koike’s single-instructor example used a strain-gauge handle with static calibration, fixed contact points, and neglected handle-piece inertia (Koike 2016). Choi and Park combined a six-axis internal-grip sensor with inverse dynamics for nine right-handed male professionals. Their estimates depend on the sensor/segment model; the heavier grip, hand overlap, and foam-ball impact limit interpretation (Choi and Park 2020). Neither study proves a passive human control strategy. Rigid-body wrench conventions provide the mathematical background (Featherstone 2008).

Hypothesis: A late negative projected force couple can occur without an instantaneous commanded free moment in a specified model. “Passive” must name which commands and dissipative terms were removed, the retained state and stored energy, and whether the result is a frozen-state evaluation or a new rollout. Its sign alone proves neither passive muscles nor whole-system passivity.

Practical Interpretation: Replace “push harder” with a more scientific question: where does the force point relative to the club, hand path, and desired turning axis during that phase?

9.2.7 The Same Force Can Do Different Jobs

Draw a force arrow at the grip. If its point of application is moving in the same direction, the force delivers positive power. If motion is opposite, it absorbs power. If motion is perpendicular, instantaneous translational power is zero even though the force changes momentum direction. Rotate the club and the same laboratory-frame force obtains different shaft-axis and transverse components. No arrow has a permanent mechanical job independent of state.

This is why phrases such as “pulling on the handle” need coordinates. Pulling toward the golfer, along the shaft, along the hand path, or opposite clubhead motion are different vectors. A force can be inward relative to the clubhead path but tangential relative to a joint. Resolve each vector in declared

axes while retaining the same observer and physical velocity. A common orthogonal axis rotation preserves a dot product. Changing to an observer translating with velocity U instead changes force power by $-F \cdot U$. Distinguish a coordinate representation from a new observer; the scalar work and energy balances must use matching conventions.

9.2.8 Force at a Point Becomes a Wrench

A rigid body's external loading is summarized by a force and a moment at a reference point. Move the reference point and the force remains the same while the reported moment changes by the cross product of offset and force. This does not change the physics. It changes the point about which the moment is reported.

Use net external force for center-of-mass translation. In body axes, spatial rotation about the center obeys $\sum M_G = I_G \dot{\omega} + \omega \times (I_G \omega)$, with constant body inertia I_G . An arbitrary reference-point moment is not simply $I \dot{\omega}$; a moving-point balance needs its transport terms.

For two contact forces F_1, F_2 at $r_1 = c + d/2$ and $r_2 = c - d/2$, define resultant $R = F_1 + F_2$ and differential force $D = (F_1 - F_2)/2$. Include intrinsic contact moments C_1, C_2 . About a point with position r_O ,

$$M_O = (c - r_O) \times R + d \times D + C_1 + C_2.$$

The separated-force couple is $d \times D$. The common-force moment vanishes at the midpoint, but generally not at another reference point. Intrinsic moments can survive when $d = 0$. An axial differential load D parallel to d changes internal loading while leaving both resultant force and force couple unchanged. Thus net club loading cannot generally recover individual hand loading; Section 12 examines the additional constraints and measurements needed.

9.2.9 Following a Vector Through the Downswing

Early in the downswing, a large grip reaction may point largely along the club, maintaining the curved path and transmitting proximal motion. Its direct tangential acceleration contribution can be modest. As the club changes orientation, the force's shaft-axis components can change even if its laboratory-frame components change smoothly. Late in the downswing, a smaller force with a larger component perpendicular to the center-to-hand lever arm, or a longer lever arm, may contribute a larger moment about the club center.

At the same time, hand-point velocity changes. Positive grip force power can coexist with a negative swing-normal couple power because the translational and rotational pieces differ. A complete display therefore shows the force vector, moment vector, contact-point velocity, angular velocity, and the paired power terms $F \cdot v$ and $M \cdot \omega$ at the same reference point. One scalar cannot represent this structure.

9.2.10 A Geometry-Control Experiment

Take a simulated state and freeze its resultant R , differential force D , midpoint c , and intrinsic moments. Reconstruct the force couple $d \times D$ under four controls: retain d ; replace d by $-d$ by swapping contact positions; replace D by $-D$; and shrink d to zero. The force couple respectively stays the same, reverses,

reverses, and tends to zero **if the differential force D remains bounded**. Allowing D to grow inversely with separation can keep the couple finite. Neither the common-force moment about another reference point nor intrinsic contact moments must vanish or reverse. Test the declared component, not a mislabeled total wrench.

These controls are more informative than simply changing a force magnitude. They challenge the proposed geometric pathway. If a nonzero separated-force couple survives zero separation while D remains bounded, the calculation or labeling is wrong. A surviving intrinsic moment is not such an error. If the control behaves correctly, the model has passed a necessary mechanical test but has not shown that golfers use that allocation.

9.2.11 Measurement Frames Are Part of the Experiment

Instrumented grips report forces in sensor axes. Motion capture reports markers in laboratory axes. The club carries a moving frame, while anatomical analysis may use segment or joint frames. A calibration must locate each sensor and its orientation relative to the club. Rotate force and moment components together, and also transport moments when the reference point changes. Correct inertial loads of sensor-side hardware when needed; a raw internal sensor wrench is not automatically either hand's complete wrench. Synchronize kinematics and loads before combining them, then propagate both rotation and position uncertainty.

Small orientation errors can mix large axial forces into smaller transverse components. Let δ be the sensor-axis misalignment angle. For one sign convention, a true axial/transverse pair of 100 N and 1 N becomes $1 \cos \delta - 100 \sin \delta$ N transversely: at $\delta = 1^\circ$ it is about -0.745 N. The inferred sign reverses without a physical force reversal. Such leakage can alter a moment's swing-normal projection when that moment depends on a small transverse force and its lever arm. Calibration uncertainty should therefore be propagated into couple sign and timing, and reference transforms should be tested with known loads before any biological interpretation.

9.2.12 What a Useful Figure Should Show

A publication-grade force-vector figure should include coordinate axes, reference point, hand locations, arrow scale, moment-arm line, phase or timestamp, and sign convention. If arrows from different trials use automatic scaling, the caption must say so. A sequence of frames should retain scale where comparison matters. Accompanying plots should show components and resultant, not just one selected direction.

For reviewers, the underlying numeric table should be linked. That makes it possible to recalculate moments, transform frames, and verify that plotted arrows match reported values. Visual intuition is valuable precisely when it is anchored to auditable quantities.

9.2.13 Questions That Keep the Claim Honest

Where is the force applied? About which point is moment calculated? In which frame is “negative” defined? Is the vector measured, inferred, or simulated? Does it do positive or negative power at that instant? Does reversing geometry reverse the effect? Can another hand-force allocation produce the same net club wrench? Answering these questions turns an arrow from decoration into evidence.

9.2.14 Reader Lab: Verify a Two-Hand Diagram

Choose two three-dimensional hand positions and force vectors. Select a club reference point. Compute net force and moment directly. Translate the reference point and recompute; then verify total wrench power is unchanged for points on the same rigid body, seen by the same observer, when the new point velocity is transformed consistently. A cut through a deforming shaft or slipping interface requires its own local velocity and energy balance. This catches the common mistake of moving the moment without moving the velocity term.

Resolve each force in laboratory, club, and swing frames. Reconstruct the original vector from every component set. Perturb sensor orientation by one or two degrees and observe how much a large axial force leaks into the smaller transverse component. Recompute the moment's swing-normal projection; the resulting change becomes part of sign uncertainty.

Animate the arrows through a simulated late downswing using one fixed scale. Overlay moment arms and hand velocities. The visual should explain the calculated couple and power rather than merely make forces look dramatic. Release the numeric array so every arrow can be audited.

9.2.15 Competing Force-Direction Stories

One hypothesis may say inward handle force changes club acceleration through parametric geometry. Another may say a differential hand force supplies a negative couple. A third may say shaft reaction explains the observed wrench. They can coexist, but each predicts a different relationship among force direction, hand motion, deformation, and state.

Separate algebraic checks from feasible dynamical interventions. Frozen-force separation changes test the couple formula; alternative velocities paired with the same force illustrate the power projection. Neither automatically supplies a trajectory satisfying Newton's laws and contact constraints. For a causal comparison, reintegrate a declared model after changing geometry, compliance, actuation, or controller while matching the initial state and reporting new reactions. Removing shaft flexibility changes stored energy and often other dynamics; specify what was matched. Compare those trajectories separately from reconstructions so a correct cross product is not mistaken for a demonstrated human mechanism.

9.3 Where the Picture Breaks

The cart and planar arrows ignore distributed contact, three-dimensional axes, and deformable tissue. The real grip force acts across an area; a six-axis sensor reduces it to an equivalent wrench at a reference point. That reduction preserves resultant loading and, for a common rigid velocity field, its power. It does not preserve local deformation work, contact pressure, discomfort or tissue load without additional information.

Force direction alone also cannot identify muscle intent. The same hand wrench can arise from many shoulder, elbow, wrist, and co-contraction patterns. Electromyography (EMG) adds activation timing but not a unique force mapping.

Go Deeper: Use the monograph's [force-vector mechanism chapter](#), [coordinate force-source chapter](#), [two-hand local geometry results](#), and the instrumented-grip studies (Choi and Park 2020; Koike 2016).

Chapter 10

Force Is Not Work Until Something Moves

10.1 A Concrete Picture

Hold a heavy suitcase motionless for a minute. Your muscles fatigue, but the suitcase has not moved and the external mechanical work on it is zero. Carry it horizontally at constant height and the upward hand force is nearly perpendicular to horizontal velocity, so that force again does little mechanical work on the suitcase. Lift it onto a shelf and force aligns with displacement; positive work by the hand raises the suitcase–Earth gravitational potential energy. During a slow lift, the hand and gravity do nearly equal and opposite work on the suitcase, so its kinetic energy changes little. Hand work and net work are different ledgers.

This example feels unfair because biological effort is real in all three cases. That is exactly the lesson. Mechanical work, metabolic cost, muscle activation, and perceived effort are different quantities. A large grip force may stabilize the club with little club-level work. A modest force aligned with fast hand motion may transfer large power.

10.2 How the Mechanism Works

Over a path, translational work is

$$W = \int \mathbf{F} \cdot d\mathbf{r}.$$

Here $\mathbf{r}$ is the position of the material point where the force acts, measured in a declared inertial observer frame; $\mathbf{v} = \dot{\mathbf{r}}$. Over time, $W = \int \mathbf{F} \cdot \mathbf{v} dt$. For a rigid body, rotational power at the chosen reference is $\mathbf{M} \cdot \boldsymbol{\omega}$, so its work channel is $\int \mathbf{M} \cdot \boldsymbol{\omega} dt$. Only for rotation about a fixed axis does this reduce directly to $\int M_{\text{axis}} d\theta$. In general three-dimensional motion, Euler-angle rates are not the components of angular velocity. Generalized moments must be transformed to be power conjugate to those rates before taking their dot product.

Impulse instead integrates force over time, $\mathbf{J} = \int \mathbf{F} dt$. For a fixed collection of bodies, the **net**

Power Depends on the Projection of Force Along Velocity

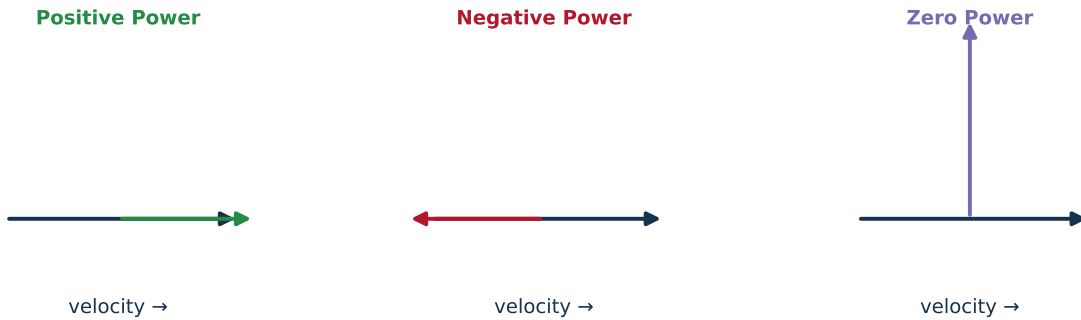


Figure 10.1: Force, Velocity, and Their Projection Distinguish Large Load From Large Mechanical Power.

external impulse equals the change in total linear momentum in that inertial frame. A single force contributes to that balance; other forces may cancel it. These answer different questions.

A force can have large impulse and small work when it acts mostly perpendicular to motion. It can have small impulse and meaningful work if it aligns with rapid motion over a short interval. Reporting magnitude, impulse, power, and work together exposes the difference.

10.2.1 Hand-Path Force Needs a Declared Path

Some golf analyses project net hand force along the hand path and integrate over path length. This can be a useful transfer observable. But a large value does not prove large biological effort. A chosen model may assign much of the instantaneous force to its drift/constraint part and an opposing contribution to its input part. This is a decomposition at the same achieved state, not a finding that muscle activity was unnecessary. Removing an input changes later states and reactions; its causal effect requires a new trajectory under a declared intervention or controller.

The pinned hand-path study evaluates this estimand in a planar double pendulum, a one-arm three-link point-mass model, and a two-arm floating-club closed loop. In two forward cases, drift supplies slightly more than 100% of signed hand-path force work because the contemporaneous control contribution is opposing. A percentage above 100 does not violate conservation; positive and negative contributions cancel to the total. In a prescribed two-arm kinematic sweep, rather than a forward rollout, drift and control nearly cancel, making fraction language unstable.

10.2.2 What the Coriolis-Impulse Search Actually Optimized

A separate archived pendulum example asks a deliberately narrow question: among 135 frozen torque programs, which one produces the greatest *absolute* along-hand-path impulse from the coordinate-defined Coriolis cross term before qualified impact? Absolute impulse counts both signs, so a force that reverses direction does not disappear through cancellation. Signed impulse is published beside it to retain sign relative to the changing path tangent. This scalar projected impulse is not the vector

impulse that appears in the momentum balance. The registered search uses shoulder-absolute and wrist-relative coordinates; its Coriolis label denotes their cross-speed equation terms. It is not an additional force acting on a free-body diagram.

The grid combines three proximal torques (60, 80, 100 N m), three restraint magnitudes (0, 5, 10 N m), and fifteen onset times from 0 to 0.350 s. After onset, the wrist drive is 15 N m. Ninety-one programs reach the registered event: the first upward crossing of the club angle through downward vertical, with arm angle no greater than 2 radians. The other 44 are excluded. This is a geometric pre-impact score interpolated from sampled history, not a simulated ball collision. The maximum absolute Coriolis impulse is 13.817 N s with a 100 N m proximal drive, 10 N m of early wrist restraint, and wrist-drive onset at 0.100 s. That program reaches 46.780 m/s. The highest-speed program also uses 100 N m proximally, but 5 N m of early restraint and a 0.125 s onset; it reaches 46.913 m/s with 13.701 N s of absolute Coriolis impulse. The optima are close, but they are not the same program. Maximizing a named force component is therefore not identical to maximizing clubhead speed, even inside this small deterministic model. These are discrete grid maxima, not continuous optimal-control solutions or human swing prescriptions.

The signed values deepen the lesson. The maximum-absolute-impulse program has a signed Coriolis impulse of -6.825 N s and Coriolis generalized work of -100.69 J. A large absolute impulse can coexist with negative net work because the source reverses and because force, velocity, and generalized-coordinate signs matter. Squared-speed work is -33.73 J, gravity work is +25.56 J, and applied-drive work is +358.20 J in the same impact-truncated interval. These figures are an equation-term ledger, not a claim that a Coriolis agent supplied or removed a measured amount of biological energy. Nor does their sum, by itself, equal the change in total kinetic energy: the configuration dependence of the mass matrix matters.

For example, in a smooth ideal model with time-independent constraints, let q be the generalized coordinates, $M(q)$ the mass matrix, $U(q)$ the potential, and τ the applied generalized forces. Define kinetic energy $K = \frac{1}{2}\dot{q}^T M(q)\dot{q}$ and write $M\ddot{q} + h + \nabla U = \tau$, with h the full velocity bias and dots denoting time derivatives. The kinetic-energy rate contains $\frac{1}{2}\dot{q}^T \dot{M}\dot{q}$ as well as $\dot{q}^T M\ddot{q}$. For the usual consistent rigid-body equations, $\dot{q}^T h = \frac{1}{2}\dot{q}^T \dot{M}\dot{q}$; these terms cancel to give $d(K+U)/dt = \tau^T \dot{q}$. Coriolis and squared-speed terms redistribute the coordinate acceleration balance without becoming extra energy sources. Include dissipation, moving constraints and explicit time dependence when they are present.

10.2.3 The Hand-Path Arrow Does Not Carry the Whole Wrench

The modeled wrist point moves with the arm coordinate but not independently with the relative wrist coordinate. Its task Jacobian is therefore rank-deficient for mapping a two-coordinate generalized drive into one force-only hand-path vector. The software computes the minimum-norm force that can be represented at the wrist and keeps the leftover generalized couple as a residual. It never hides that remainder inside the displayed hand force. For component k , use generalized drive Q_k , point Jacobian $J_H = \partial x_H / \partial q$, mapped force $F_{H,k}$, residual r_k , and point velocity $v_H = J_H \dot{q}$. Then $Q_k = J_H^T F_{H,k} + r_k$. Consequently, $Q_k^T \dot{q} = F_{H,k}^T v_H + r_k^T \dot{q}$. The generalized work totals above include the residual-power channel; they are not just force work along the hand path. A least-squares force equivalent is not automatically a measured contact force.

The archived maximum residual norm is 54.44 N m for the impulse-selected program and 52.81 N m

for the speed-selected program. Each maximum is taken over time **and all attributed components**, not solely the Coriolis component or the net physical wrist moment. This is a structural mapping diagnostic, not a measured joint load or a small numerical rounding error. It tells the reader that an along-path force plot answers a useful but incomplete question. A complete grip description needs a wrench, or an endpoint-and-couple mapping with enough independent coordinates and measurements.

At the top of the simulated swing, hand speed begins at zero, so no path tangent exists. The integration marks that interval undefined and uses only adjacent samples with a defined direction (the archived speed threshold is 10^{-12} m/s). Generalized power remains defined at zero speed and its work is integrated over the full recorded interval. More than 99.6% of the impact-truncated time is valid in the highlighted runs. Reporting that fraction prevents an arbitrary initial direction from contaminating the impulse. It does not bound the missing impulse: even a short excluded interval can matter if its force is large. Report the masking threshold and convergence when interpreting a small difference between programs.

10.2.4 Fractions Become Dangerous Near Cancellation

If drift contributes +10 J and control contributes −9 J, total signed work is +1 J. Calling drift “1,000% of total” is arithmetically correct and scientifically awkward. If total crosses zero, the fraction is undefined.

Better reports show signed contributions, absolute magnitudes, a cancellation index, and intervals separately. Ratios are reserved for denominators safely away from zero. This is not statistical fussiness; it prevents a small residual from turning two large opposing mechanisms into a sensational percentage.

10.2.5 Power Reveals Timing Hidden by Work

Two strategies can deliver the same integrated work with different power histories. One may load the shaft early and receive energy back late. Another may deliver direct power near impact. Their club energies at one instant can match while internal loads and robustness differ.

Plotting cumulative work beside instantaneous power is useful. The power curve shows when exchange occurs; the cumulative curve shows what remains after positive and negative intervals cancel. A phase-integrated table connects both to transition, early downswing, delivery, and impact.

10.2.6 Joint Power Is Not Muscle Power

A measured or inferred net joint power is a mechanical constraint on a model; it is not a muscle-energy meter. Tendon storage, biarticular transfer and opposing muscle forces can produce different biological allocations with the same net result. The segment-work section below specifies the required relative angular velocity and explains what inverse dynamics identifies.

This distinction is central to the arm-versus-wrist proposal. A net club wrench can match while direct wrist moments, two-hand force couples, internal squeeze, and proximal joint loads vary. Club-level work is an outcome constraint, not a biological allocation measurement.

Model Result: The hand-path attribution separates pointwise drift and control contributions, then reports impulse, power, work, force vectors, joint effects, and cancellation rather than one percentage.

Recomputing the selections from the archived 135 outcome records reproduces the stated grid winners and counts. That check does not independently reproduce their ODE trajectories or quadrature.

Human Evidence: Nesbit and Serrano used kinematically driven body/club models to examine four selected amateur golfers; the outputs are model-based work estimates, not direct muscle-energy measurements (Nesbit and Serrano 2005). MacKenzie and colleagues analyzed 76 right-handed golfers with handicaps below 15 using the same driver, inverse dynamics and between-golfer regressions. Their strong force/work associations do not measure the effect of instructing a golfer to pull harder (MacKenzie et al. 2020). Model boundaries, differentiation, inertial parameters and shaft treatment affect interpretation. Contact measurements and muscle-informed models can narrow the ambiguity without eliminating it.

Practical Interpretation: Force tells how strongly the system interacts; work tells how that force aligns with motion over distance; power tells how fast the exchange happens; effort belongs to a separate biological account.

10.2.7 From Force History to Mechanical Consequence

Start with a force vector through time. Its integral is that force’s impulse; only the sum of external impulses fixes the system’s momentum change. Pair each force sample with contact-point velocity and take the dot product to obtain power. Integrate power for work. Pair a moment with angular velocity for rotational power. These operations answer related but distinct questions; none can be safely inferred from the appearance of the force curve.

A high force applied while a point barely moves may create little work. A lower force acting through a long displacement can do more. A force perpendicular to velocity can bend a trajectory without changing kinetic energy at that instant. In a golf swing, all three patterns occur while the club follows a curved, rotating path.

10.2.8 The Hand Interface Has Two Power Channels

Choose a club reference point. Translational power is net force dotted with reference-point velocity. Rotational power is net moment dotted with angular velocity, all expressed in the same observer frame. Their sum is wrench power. For two material points on the same rigid body, let r_{AB} run from A to B ; F is the resultant force and M_A, M_B its equivalent moments. Then $v_B = v_A + \omega \times r_{AB}$ and $M_B = M_A - r_{AB} \times F$, so $F \cdot v_B + M_B \cdot \omega = F \cdot v_A + M_A \cdot \omega$. The separate force and moment powers can change even though the sum does not. A change of observer is different: kinetic energy and power are frame dependent. A point across a bending shaft need not satisfy this rigid-body velocity relation. Reporting only the couple may miss positive translational power; reporting only net force may miss a large rotational exchange.

For two hands, individual contact powers can also be calculated. Their sum must match net wrench power after consistent transformations and inclusion of any contact moments, provided the contacts share the rigid-body motion used in the reduction. With slip, local deformation or distinct hand/club velocities, retain those contact powers and losses explicitly. That closure is a strong audit. Yet individual hand power does not identify which muscle supplied it, because work can cross wrist, elbow, shoulder, and trunk pathways and because co-contraction can add internal cost without changing net external power.

10.2.9 A Late Negative-Couple Example

Suppose the net force supplies +900 watts of translational club power while the two-hand couple supplies -150 watts of rotational power over a short interval. Net hand power remains +750 watts. The negative couple is real and locally absorptive, but the hands as a whole still add club energy. If the couple changes orientation so its power later becomes positive, its integrated work may be near zero while its angular impulse is nonzero. A change in angular momentum requires the sum of all external moments about a consistently chosen inertial origin (or a separately justified center-of-mass balance).

Now run a matched-state branch without that couple. The club may release sooner, changing geometry and reducing later positive power from another channel. The baseline could then finish faster despite the early negative work. That result would not mean negative work created energy; it would mean the intervention changed the route and timing by which other work acted.

10.2.10 Segment Work Is Not Muscle Work

Inverse dynamics produces net joint moments consistent with measured motion and external loads under a model. For a rotational joint, dot the net moment acting on the distal segment with its angular velocity **relative to the proximal segment**, expressed in a common frame. This gives the joint's net rotational power; moments dotted with absolute segment velocities instead describe the separate segment power flows. Include translational joint power if relative translation is allowed. Several muscles, passive tissues, contact loads, and model residuals can contribute to that moment. Biarticular muscles can absorb energy at one joint and generate it at another. Assigning net joint work to one named muscle is therefore not warranted.

Musculoskeletal optimization can propose allocations, and EMG can constrain activation timing, but redundancy remains. A strong report keeps rigid-body joint work, muscle-tendon work, and metabolic energy on separate evidence levels. This distinction is especially important for claims about wrists “resisting passively” or scapular muscles “creating” a club couple.

10.2.11 Numerical Closure Before Storytelling

First declare the resolved mechanical energy: kinetic energy plus the chosen gravitational and elastic potentials. A club-only boundary receives hand work; a golfer-plus-club boundary instead contains actuators or muscles that convert chemical energy into mechanical energy internally. “Closed” to mass does not mean mechanically isolated or unable to perform internal work.

A useful smooth-phase ledger is

$$\begin{aligned}\Delta E_{\text{mech}} = & W_{\text{boundary,nc}} \\ & + W_{\text{internal actuation}} \\ & - E_{\text{dissipated}}.\end{aligned}$$

where “nc” denotes the nonconservative boundary channel, and $E_{\text{dissipated}} \geq 0$ is mechanical energy converted into unresolved heat or other dissipative modes. Boundary work excludes forces already represented by the included potentials, and internal actuation excludes elastic storage already counted in E_{mech} . This equation assumes no unlisted mass flow, time-varying potential or other energy conversion. Use a separate jump balance for impacts rather than integrating across an unresolved impulse. For a

passive full club including its shaft, hand and aerodynamic work feed kinetic, gravitational and shaft elastic energy plus dissipative losses. For the combined body and club, hand-interface work cancels internally when contact motion is compatible, but muscle/actuator conversion remains. Adding gravity work while also treating gravitational potential U_g as a destination would double-count it. Adding shaft storage to an energy that already includes it would do the same.

Calculate closure over the whole trial and important phases. Report absolute residual and residual relative to the effect being discussed. A 2 J residual may look small compared with total swing energy yet be fatal to a claimed 1 J mechanism.

Timestep refinement, alternate differentiation/filtering, and independent power calculations should preserve the conclusion. Agreement from two quantities derived from the same filtered motion is an internal consistency check; it is not an independent validation of the assumed mass properties or contact model. At contact interfaces, action-reaction and power consistency provide additional checks. A model that cannot close its ledger should not explain an unexpected sign by inventing a new transfer narrative.

10.2.12 What to Compare Across Strategies

For the arm-dominant and wrist-dominant extremes, compare more than terminal speed. Plot bilateral forces and moments; common and differential hand modes; interface power; cumulative work; club and body energy; shaft strain energy; joint load proxies; and face/path sensitivity. Align comparisons by both clock and state. Match the final club task when asking about internal allocation, and match the initial state when asking about causal input contribution.

A role reversal may preserve net club work while changing peak force, internal co-contraction, or timing sensitivity. Those are scientifically interesting differences even if clubhead speed barely changes. Conversely, a small speed gain may be unattractive if it depends on fragile timing or unmeasured internal load.

10.2.13 A Reviewer Can Recompute the Claim

The most useful release contains time, state, forces, moments, velocities, powers, cumulative works, energy terms, and residuals in machine-readable form. Units and frames live in a schema, and each figure links to its source. A reviewer can then verify dot products and integrals instead of accepting a caption. This is how a casual explanation remains connected to a scientific product.

10.2.14 Reader Lab: Recompute a Power Ledger

Download one time series of force, moment, reference-point linear velocity, and angular velocity. Check units and frames, then calculate translational power, rotational power, and their sum sample by sample. Numerically integrate each with at least two reasonable quadrature rules and compare. A difference larger than the reported mechanism needs explanation.

Next transform the wrench and velocity to another point on the same rigid club segment, keeping the observer fixed, and verify summed power is invariant. Reverse the declared swing-normal axis and confirm only the coordinate label changes, not the physical power. Compare cumulative hand work with the declared club energy and losses: count gravity and shaft storage once, include any power crossing a cut through a flexible shaft, and retain the residual with its sign.

Finally, shorten and lengthen the integration window. If a claimed positive contribution depends entirely on an unreported boundary choice, the conclusion must name that interval. This exercise is simple enough for a reviewer and powerful enough to expose many sign and synchronization errors.

10.2.15 The Difference Between Necessary and Sufficient

For a constant-mass particle in an inertial frame, increasing kinetic energy requires positive net force work. For an articulated system's total mechanical energy, the required positive balance may come from boundary work or internal chemical/actuator conversion, after accounting for potentials and losses. Neither statement makes one particular joint-power peak necessary: other routes can substitute. A negative couple may be sufficient to alter release in one model but not necessary for all high-speed trajectories. Counterfactuals and alternative allocations establish these logical distinctions.

Use cautious language: “contributed under this decomposition,” “improved the declared outcome in this branch,” or “was associated in this sample.” Avoid “caused all transfer” unless interventions and closure exclude competing paths. The strength of the claim should match the design.

10.2.16 A Compact Reporting Template

Name the system, boundary, reference point, frame, sign, and interval. Report force and moment, point and angular velocity, power, cumulative work, energy change, residual, and uncertainty. State whether each quantity is measured, inferred, or simulated. Then link the machine-readable series. This template is short enough for every mechanism result and prevents most category errors.

10.3 Where the Picture Breaks

The suitcase treats the hand as one contact and the object as rigid. In golf, two hands, a flexible shaft, and moving joint centers complicate the boundary. Work at one reference must be transformed consistently, and distributed contact may store energy locally.

Mechanical work also does not judge usefulness. Negative work can be essential for control, accuracy, or preparing a favorable state. Positive work at the wrong time can worsen the final task.

Go Deeper: Compare the [pinned hand-path attribution snapshot](#) (provider revision [69eb7e9d](#)) with the separate [coordinate force-source chapter](#) and its [135 archived outcome records](#). The latter archive is at revision [f70f4620](#); its recorded generating revision is [942c4e32](#). Archive inspection is distinct from rerunning that environment. For methodological background, Robertson and Winter examined segment power balance in a sagittal-plane walking model; that is not golf-specific validation ([Robertson and Winter 1980](#)). Winter's textbook supplies broader background ([Winter 2009](#)).

Chapter 11

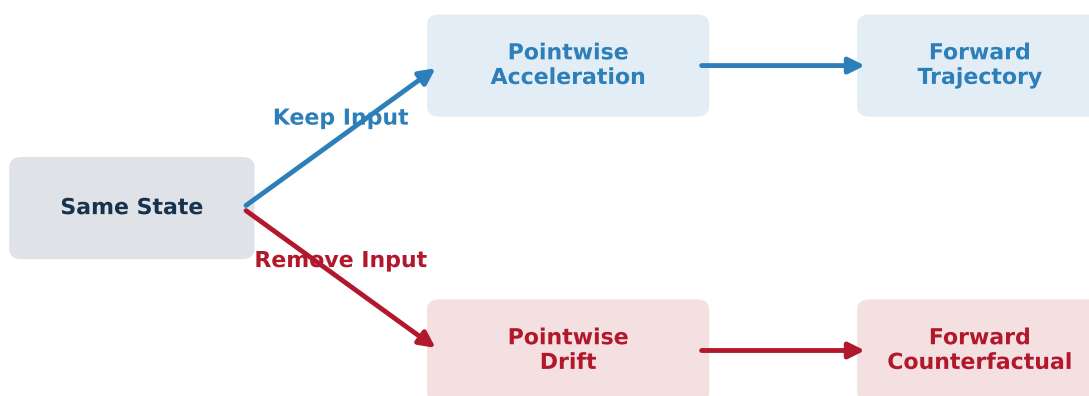
Counterfactuals Are Scientific Scissors

11.1 A Concrete Picture

A baker wants to know what yeast contributed to a loaf. Removing yeast from the finished loaf is impossible. Baking a second loaf without yeast changes the rise, temperature history, moisture, and structure. Comparing one instant of the two doughs answers a different question from comparing the finished loaves.

Models let us make cleaner cuts, but each cut has a meaning. We can remove a torque at one frozen state and recompute acceleration. We can continue that altered branch forward. We can rerun the entire swing with a different schedule. Or we can remove gravity, damping, shaft flex, or a contact moment arm. These are not interchangeable “what ifs.” They are scientific scissors with different blade shapes.

Two Questions That Must Not Be Confused



Pointwise: what is the acceleration now? Forward: where does the changed system go?

Figure 11.1: Four Counterfactual Branches Separate Instantaneous Attribution, Short-Horizon Persistence, Full Resimulation, and Mechanism Ablation.

11.2 How the Mechanism Works

A pointwise counterfactual holds configuration and velocity fixed, changes one input or term, and resolves acceleration or reaction force. Because the state is identical, the difference isolates an instantaneous modeled contribution. It does not say what a complete no-input trajectory would have looked like.

A matched-state forward counterfactual begins both branches at the same state and integrates with different future inputs. At the cut, attribution is clean. As time passes, states diverge. The difference now includes direct input effect, changed geometry, changed velocity coupling, and changed impact time. This answers whether the consequence persists over a declared horizon.

A full resimulation changes an input schedule from the beginning. It answers a strategy question: what trajectory and outcome follow if the entire command program changes? It cannot by itself attribute the outcome to one instant or mechanism because every later state changes.

An ablation removes a modeled mechanism such as gravity, damping, compliance, or contact separation. If the claimed effect remains when the proposed mechanism is absent, that mechanism is not necessary within the model. If it disappears, necessity is supported only for the declared family; another unmodeled mechanism could still produce a similar observation in humans.

11.2.1 Cut Time and Horizon Are Part of the Experiment

Choosing a cut after seeing the largest difference can exaggerate a result. The technical work therefore registers cut times, event definitions, and horizons. It repeats tests across nearby cuts and timesteps. A result that exists for one numerical sample but vanishes under refinement is not a mechanism.

Impact creates another trap. One branch may hit the impact criterion earlier. Comparing both at the baseline impact clock time may evaluate one branch after its own impact. Comparing each at its own first valid impact changes the state being compared. Both variants can be reported, but the distinction must be visible.

11.2.2 Decomposition and Intervention

An algebraic decomposition divides the acceleration at one state among gravity, velocity terms, controls, and constraints. An intervention modifies the model or input and observes the outcome. Decomposition says how the equation balances now; intervention says what changes under a defined alteration. Confusing them turns a bookkeeping term into a causal claim.

The zero-torque counterfactual force is a decomposition tool when computed at a fixed state. Integrating a zero-torque branch is an intervention. The technical paper reports both so readers can see where the instantaneous contribution continues and where trajectory divergence takes over.

11.2.3 Negative Controls Challenge the Story

Good experiments include cases where the mechanism should vanish or reverse. Coincident hand contacts remove a force-generated couple. Reversing moment arms should reverse its sign. Setting a dead zone to zero should remove any advantage attributed specifically to crossing that dead zone. Equalizing shaft stiffness should remove differences attributed only to flexibility.

Negative controls are more persuasive than a baseline-versus-favored comparison because they ask the mechanism to predict its own failure. A model that always produces the preferred result, regardless of geometry or removed mechanism, is describing an outcome rather than explaining it.

11.2.4 Counterfactuals Need Common Observables

Every branch should report the same quantities in the same frames: club speed, face/path proxy, forces, moments, powers, work, constraint residual, energy closure, and event timing. Otherwise one model can appear better merely because it reports a convenient metric.

This common-observable contract also connects model tiers. A double pendulum and a spatial contact model have different coordinates, but both can report an equivalent club wrench and interface power. Mechanisms that survive this ladder earn broader confidence.

Model Result: Pointwise wrist-force attribution, matched-state persistence, driver killswitches, geometry controls, zero-dead-zone controls, and stiffness ablations form a linked counterfactual program in the technical release.

Hypothesis: If delayed transfer depends on a particular preload or interaction mechanism, its advantage should disappear when that mechanism is removed while the task and state are appropriately matched.

Practical Interpretation: A fair “what if” must say what stayed identical, what changed, when the change occurred, and how long the altered branch was allowed to become a different motion.

11.2.5 Four Different Meanings of “Take It Away”

The phrase “remove wrist torque” can hide several experiments. A pointwise removal changes the torque only in the acceleration calculation at one copied state. A forward killswitch removes it from that state onward while other controls follow their original schedules. A reoptimized branch removes it but lets other controls compensate. A human perturbation asks a person to change a strategy and inevitably changes feedback, intent, and perhaps the starting state. These are not interchangeable.

Pointwise removal is best for decomposition. Forward removal is best for persistence within the declared model. Reoptimization asks whether another control route can substitute. Human manipulation tests whether the mechanism appears in an adaptive biological system. A complete research program uses the first three to sharpen predictions before attempting the fourth.

11.2.6 The Matched-State Rule

At the branch instant, copy every modeled state: positions, velocities, shaft modes, activation or transmission states, and controller memory. Change only the declared intervention. If one branch begins with a different club angle or force history, later divergence cannot be assigned cleanly to the removed input.

This rule is particularly important for preload. Comparing a continuously loaded backswing with a role-reversed downswing naturally produces different transition states. That comparison answers a strategy-level question but not the narrow causal question “what does the transmission state do?” A second matched-state experiment should equalize geometry and motion while manipulating only the declared gap or activation state. Both comparisons are informative when labeled correctly.

11.2.7 Pointwise Decomposition Does Not Add Like Percentages

At a fixed state, the acceleration equation may be decomposed into gravity, velocity, applied-control, external-load, and constraint contributions. These terms can oppose. One positive component can exceed total acceleration because another is negative. Dividing each by total and calling the result a percentage creates values below zero or above one hundred, which are mathematically possible but easy to misread.

Report signed components in physical units, cancellation indices, and residual closure instead. For nonlinear forward branches, effects generally do not add because each intervention changes the future state. A combined removal can differ from the sum of separate removals. That interaction is part of the result, not an accounting error.

11.2.8 A Counterfactual Sequence for the Negative-Couple Claim

Begin with baseline forward simulation. At selected late-downswing states, reconstruct bilateral contact and compute swing-normal couple. Next perform a pointwise removal of the relevant actuator or driver and observe the changed constraint reaction. Then run a short forward killswitch. Reverse hand geometry and force direction in separate negative controls. Finally repeat across model rungs and parameter samples.

The mechanism is strengthened if sign, timing, and geometric response remain coherent while magnitude varies. It is weakened if the couple persists when hand separation collapses, changes sign under a mere coordinate relabeling, or depends on one integration step. Human data must then test the predicted wrench, not merely reproduce the club trajectory.

11.2.9 Counterfactuals About Timing

To study onset, branch from several candidate states and apply identical command shapes shifted in state or time. Include finite rise time. Compare terminal speed, face/path proxies, peak loads, and sensitivity. The preferred command by one objective may be poor by another. Repeat under parameter uncertainty and on held-out samples.

A clock-shift experiment asks how a command schedule performs at different times. A state-trigger experiment asks whether activation at a geometric or dynamic event is more robust to tempo. These interventions can produce different answers. The study should expose both rather than translate one into a universal instruction.

11.2.10 Compensation Is a Separate Scientific Question

If wrist torque is removed and an optimizer increases arm torque to recover the same club motion, the result demonstrates redundancy. It does not show wrist torque was irrelevant in baseline. Compare internal loads, power routes, and robustness between baseline and compensated solutions. Multiple feasible routes are expected in a redundant biological system.

Conversely, failure to compensate within a restricted controller does not prove humans cannot. It establishes a limit of that control family. Expanding the family is a model-ladder step and should be preregistered enough to prevent the optimizer from becoming an explanation that can fit anything.

11.2.11 From Model Intervention to Human Falsifier

A safe human experiment cannot instantaneously remove a muscle torque. It can manipulate grip spacing, club inertia or stiffness, instruction, or perturbation while measuring the predicted mediators. The model must say which observable should change and in what direction. If a grip-spacing reduction should shrink a differential couple, that is more testable than asking a participant to “use less wrist.”

The final evidentiary bridge is therefore indirect but disciplined: model intervention, robust observable prediction, physical manipulation, synchronized measurement, and a predeclared result that counts against the mechanism.

11.2.12 Reader Lab: Build a Branch Manifest

For one intervention, write a small manifest containing source run, copied state hash, branch time, removed input, retained inputs, controller memory, solver, tolerance, horizon, and primary outcomes. Generate baseline and branch from that manifest. A third party should be able to confirm that no hidden state changed.

Plot zero difference at the branch instant for all states and the expected nonzero difference in acceleration or reaction. Then show divergence through time. Repeat with half timestep and a second engine. Store both successes and any failed closure. This turns a counterfactual from a narrative choice into a reproducible intervention.

Run the same design pointwise, forward, and with compensation. The pointwise effect identifies local contribution; forward effect identifies persistence; compensation identifies redundancy. If conclusions differ, report why rather than choosing the most favorable version.

11.2.13 Counterfactual Etiquette

Never describe an artificial zero-velocity calculation as a motion a golfer could perform. Never call a reoptimized trajectory “the effect of removal” without noting compensation. Never compare branches with different initial states as though only control changed. Never turn nonlinear component differences into percentages assumed to sum.

These rules do not make causal modeling timid. They allow strong statements within scope: “At this exact modeled state, removing this driver changed this reaction by this amount,” or “Over this forward branch, retaining the negative couple produced this later outcome.” Precision makes the result easier to challenge and reuse.

11.3 Where the Picture Breaks

The baking analogy suggests one ingredient can be removed without changing the meaning of the recipe. In nonlinear dynamics, interventions change the state trajectory, and mechanisms interact. Removing damping can alter phase and impact time, not just energy loss. Counterfactual design reduces ambiguity but does not make coupled systems separable.

Human counterfactuals are harder. A golfer cannot repeat an identical internal state while one muscle command is magically removed. Randomized cues, perturbations, and participant holdout approximate

interventions but include adaptation and measurement noise. Model results should guide experimental design, not substitute for it.

Go Deeper: Read the [matched-state counterfactual chapter](#), the [same-state spatial killswitch](#), and the registered [experimental negative controls](#).

Chapter 12

Two Hands Make One Visible Wrench

12.1 A Concrete Picture

Turn a steering wheel by pushing down with one hand and pulling up with the other. The two forces can nearly cancel as a net force on the wheel's center, yet together they create a clear turning effect. Now squeeze the wheel equally from both sides. The internal load in your hands rises while the wheel's motion may not change at all.

Two hands on a golf grip create the same distinction. The club responds to the combined **wrench**: net force plus net moment about a chosen point. The arms and wrists may experience very different individual forces while producing the same club wrench. Club motion alone cannot reveal the hidden allocation.

Opposing Hand Forces Can Form a Couple Without a Net Push Narrow Grip Wider Separation

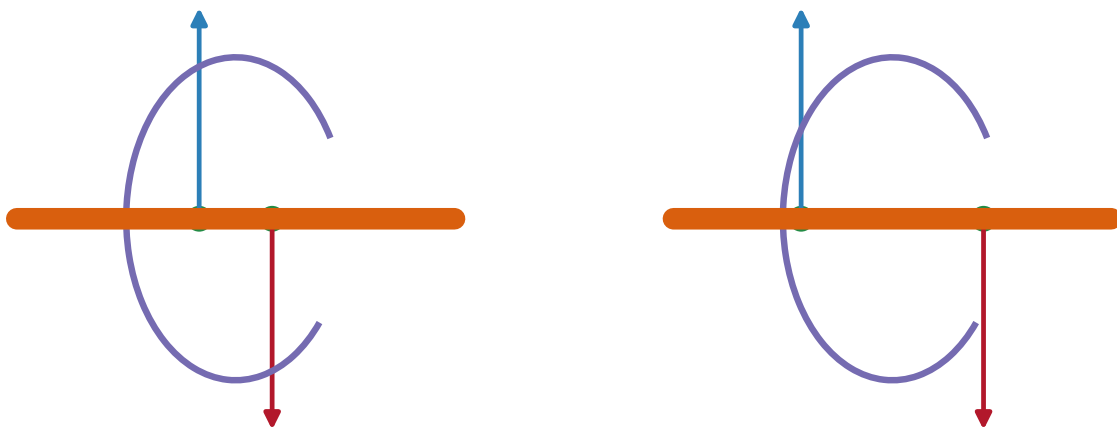


Figure 12.1: Equal and Opposite Hand Forces Produce a Turning Couple Whose Magnitude Grows With Grip Separation.

12.2 How the Mechanism Works

Let hand forces be $\mathbf{F}_L$ and $\mathbf{F}_R$ at positions $\mathbf{r}_L$ and $\mathbf{r}_R$ relative to club reference O . Include direct hand or wrist moments $\mathbf{M}_L$ and $\mathbf{M}_R$. The equivalent club wrench is

$$\mathbf{F}_O = \mathbf{F}_L + \mathbf{F}_R,$$

$$\mathbf{M}_O = \mathbf{M}_L + \mathbf{M}_R + \mathbf{r}_L \times \mathbf{F}_L + \mathbf{r}_R \times \mathbf{F}_R.$$

All components use the same orthonormal axes at the same instant; each position vector points from O to its contact. The moments are contact moments exerted on the club, not automatically anatomical wrist joint torques. For a rigid club with known inertia and all other loads specified, the combined wrench determines the hands' contribution to acceleration. Many allocations produce the same sum. A flexible club also has deformation modes that a single resultant does not describe.

12.2.1 Common and Differential Modes

Define $\mathbf{F}_c = (\mathbf{F}_L + \mathbf{F}_R)/2$ and $\mathbf{F}_d = (\mathbf{F}_L - \mathbf{F}_R)/2$. Let $\mathbf{r}_c$ locate the contact midpoint relative to O , and $\mathbf{d} = \mathbf{r}_L - \mathbf{r}_R$ point from the right contact to the left. Then, exactly,

$$\mathbf{F}_O = 2\mathbf{F}_c,$$

$$\mathbf{M}_O = \mathbf{M}_L + \mathbf{M}_R + 2\mathbf{r}_c \times \mathbf{F}_c + \mathbf{d} \times \mathbf{F}_d.$$

About the midpoint, the common force has no moment. About another point it usually does. Only the part of $\mathbf{F}_d$ perpendicular to $\mathbf{d}$ creates the differential force couple. Its axial part changes internal loading without changing the wrench.

If we add $+\eta$ to one hand and $-\eta$ to the other, resultant force stays unchanged. Depending on contact geometry, moment may change or remain unchanged along certain null-space directions. This is the mathematical source of hidden internal loading.

12.2.2 What Inverse Dynamics Can Identify

If full club motion and inertia are known well enough, inverse dynamics can estimate the required net external wrench. Isolating the hands requires subtracting gravity, aerodynamic loads and any ball-contact load using the same boundary and reference point. Impact may require an impulse balance rather than a smoothly differentiated acceleration. Clubhead position alone does not supply the complete rigid-body state or flexible-shaft motion. Even a perfectly known hand resultant cannot identify its split without additional measurements or assumptions.

A minimum-norm allocation is a reproducible mathematical choice. It is not a measurement of how a golfer used the arms. A muscle-effort optimization is still model-dependent. Instrumented grips with force and moment sensing provide the most direct route to individual hand wrenches.

12.2.3 The Invisible Axial Push–Pull Mode

For two point forces at distinct locations, with no free contact moments, the map is

$$G = \begin{bmatrix} I & I \\ [\mathbf{r}_L]_{\times} & [\mathbf{r}_R]_{\times} \end{bmatrix}, \quad [\mathbf{r}]_{\times} \mathbf{f} = \mathbf{r} \times \mathbf{f}.$$

Zero resultant force requires opposite force increments. Zero moment then requires $\mathbf{d} \times \boldsymbol{\eta} = 0$: the increment lies along the contact line. The null space has dimension one, so the map has rank five. Equivalently, the moment about the midpoint cannot have a component along $\mathbf{d}$ in this point-force model. If the locations coincide, the rank drops to three and every opposite force pair is invisible.

For distinct known contacts, an added scalar measurement sensitive to the axial null direction closes the structural rank gap. A redundant measurement does not. Short separation also amplifies force errors associated with transverse moment noise, so full rank does not ensure reliable estimation. One scalar is insufficient if each hand can also apply its own three-dimensional contact moment: the full 12-to-6 bilateral-wrench map retains a six-dimensional invisible family. The calculation is a sensor-design result, not an anatomical result. It does not identify wrist action, scapular action, muscle recruitment, or intentionality, and calibrated noise, cross-talk, and moving contact centers still require separate qualification.

These dimensions concern unconstrained linear maps. Friction, unilateral contact, finite grip strength and allowable moments restrict physically possible loads. Their intersection with an algebraic solution family need not have the same dimension or contain any solution.

12.2.4 Internal Force Can Matter Without Moving the Club

Hidden loading may change grip stability, friction margin, joint compression, muscle activation and effective stiffness. The axial point-force null mode is not itself a measurement of normal grip pressure. At a fixed state, a zero-wrench load increment does zero power against a rigid velocity field, exactly. Its metabolic cost and tissue loading can still be nonzero. With deformation or slip, equal resultants can also conceal different mechanical power. There is no single internal metric called “effort” that settles these trades. Reducing an axial null load does not algebraically require increasing a contact moment.

This is why the technical article refuses to infer scapular recruitment from a club couple. A proximal generalized-torque subspace can be labeled and tested in the model, but the mapping to muscles, scapular motion, and hand pressure is not unique.

12.2.5 Action-Reaction Must Close

At a common physical interface, each force has an equal and opposite reaction. Transform moments to the same point and axes before summing both sides. An ideal rigid grasp uses closure constraints; a compliant grasp needs a constitutive law with stored-energy and dissipation terms. Opposite forces do not make contact-pair power vanish when the two sides have different velocities.

For loads on one rigid club,

$$P_{\text{hands}} = \sum_i (\mathbf{F}_i \cdot \mathbf{v}_i + \mathbf{M}_i \cdot \boldsymbol{\omega}) = \mathbf{F}_O \cdot \mathbf{v}_O + \mathbf{M}_O \cdot \boldsymbol{\omega}.$$

Here $\mathbf{v}_i = \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}_i$ is the club’s velocity field in one inertial observer. Changing the reference point transports both wrench and velocity; changing observer is a different operation. For a deforming grip or shaft, use actual local velocities and rotations. A single rigid-body power formula is insufficient. Power and energy residuals test these ledgers separately from action-reaction closure.

The [archived spatial contact test](#) uses MuJoCo and Pinocchio for independent kinematics and forward dynamics, but shares the contact law and time-stepping scheme. Agreement tests those engine implementations under shared assumptions; it cannot independently validate the shared law. The hands are finite-mass translational carriages, not anatomical arms.

12.2.6 Why Two Hands Can Create Negative Coupling

Near the late downswing, separated hand forces can produce a couple with a negative projection onto a declared swing-plane normal. The forces need not themselves point along that normal. Couple power is the full dot product with angular velocity: one negative component alone does not establish negative power. Even negative moment power can coexist with increasing club kinetic energy when force power and other external powers outweigh it. Inertial coordinate terms are not an extra energy source. The model asks where the couple comes from and whether the mechanism persists in a specified forward intervention.

At fixed bounded forces, collapsing hand separation eliminates the differential force couple; reversing $\mathbf{d}$ with $\mathbf{F}_d$ fixed reverses its sign. Free contact moments do not vanish in either control. These geometric checks are not predictions for a newly integrated swing: moving contacts can change forces, constraints, commands and motion. Keeping a nonzero force couple as separation tends to zero would require unbounded perpendicular differential force.

Model Result: Wrench reconstruction reveals nonunique internal modes. In the pinned reduced spatial test, switching off the declared driver at 0.18 s retains a negative swing-normal force couple for 0.0375 s, with minimum about -0.4093 N m in both engines. This is a model-time interval, not a measured phase of a golfer’s downswing. Coincident and reversed geometry checks transport the same achieved forces; they are algebraic controls, distinct from the integrated driver-switch experiment. Stored energy and initial motion remain after the switch.

Human Evidence: Instrumented grips add information that club motion lacks, but sensor signals and inferred hand loads must be distinguished. Koike’s example used one professional instructor and statically calibrated strain gauges, with fixed hand reference points and neglected handle-piece inertia ([Koike 2016](#)). Choi and Park studied nine right-handed male professional golfers using a six-axis sensor between two grip segments and inverse dynamics. Their heavier grip, possible hand-to-hand load bypass and foam-ball impacts limit interpretation ([Choi and Park 2020](#)). Neither study identifies muscle recruitment or validates the present synthetic model’s hand allocation.

Hypothesis: Opposed arm-chain actions can generate a useful club couple while wrists supply resistance or impedance, but only synchronized bilateral wrench and biological measurements can test that allocation.

Practical Interpretation: The club can feel one net push and twist while the two hands experience a much richer internal negotiation.

12.2.7 From Two Contacts to One Club Load

Each hand can apply three force components and three moment components at its contact reference frame. Transform both six-dimensional wrenches to one club frame, including the moment created by each position offset, then add them. The resultant hand wrench contributes to rigid-body acceleration together with all other external loads and inertia. Shaft forces are internal to a whole-club boundary; they become boundary loads when a shaft or clubhead subsystem is cut out. Count them once at the chosen boundary. This sum is estimable in principle, but its split between hands is not determined by club motion alone.

In a planar simplification, the arithmetic is easier: two two-dimensional forces at separated points create a net force and an out-of-plane moment. Even there, multiple pairs yield the same resultant. Adding equal and opposite force to the two hands changes internal loading while leaving net force unchanged; depending on direction, it can change the couple. Distinct point-force contacts have one axial null direction in either two or three dimensions. Allowing full contact wrenches increases the nullity from three in the planar model to six in space.

12.2.8 Interpreting Push–Pull

The exact midpoint decomposition separates resultant force from the differential force couple and free contact moments. This coordinate change does not claim the nervous system controls those modes explicitly. It is a clean way to see which parts of bilateral loading are visible at the club and which can cancel.

The distinction also clarifies “push–pull.” Equal opposition is not required. One hand may supply a larger force, both may point partly in the same direction, and contact moments may matter. A measured push in grip-local coordinates can coexist with a laboratory-frame component that looks like a pull. The complete wrench should replace verbal labels wherever data permit.

12.2.9 How a Negative Couple Can Emerge

Near impact, proximal motion, club inertia, grip constraints, and contact geometry determine the reactions required at the hands. A differential force component acting through hand separation may project onto the swing-plane normal with negative sign even when neither hand is commanded with an explicit negative club torque. Calling a reaction “passive” requires a declared partition: the same contact force can depend on active proximal commands, stored elastic energy and inertia. State dependence alone does not establish muscular passivity.

That statement is narrower than “the scapulae passively torque the club.” The generalized drivers can stand for many body actions. Reconstructing a negative contact couple does not uniquely map it to shoulder-girdle motion, wrist action, or muscle activation. The model supplies an observable prediction for bilateral wrenches; anatomy remains a separate inference problem.

12.2.10 A Worked Wrench Example

Let the hands be separated by 0.20 metres. Suppose they apply opposite 50-newton forces perpendicular to that separation. Net force is zero and couple magnitude is 10 newton-metres. Add the same 100-newton vector to each hand: net force becomes 200 newtons and moment about the contact midpoint remains 10 newton-metres when free contact moments are zero. About an offset reference, include the additional moment of the common force. If separation drops to 0.10 metres, the same opposing forces make 5 newton-metres.

Real force vectors are not perfectly perpendicular, contact points are distributed, and hand moments may contribute. The arithmetic nonetheless gives testable dependencies: couple scales with perpendicular separation and force; reversing either reverses sign; collapsing either collapses the couple. A model or measurement pipeline should pass those controls exactly within tolerance.

12.2.11 Matching the Club Task, Varying the Allocation

To compare arm-dominant and wrist-dominant strategies fairly, first specify a club-level task: perhaps the same net wrench history or a set of terminal club states. Solve for distinct internal allocations that meet it. Then compare joint demands, individual hand power, co-contraction proxy, transmission state, peak load, and sensitivity. The matched task prevents one strategy from “winning” merely because it was asked to produce a larger club moment.

Next relax the task and optimize performance under common constraints. This second experiment asks whether internal freedom can improve speed or robustness. Keeping matched-task and optimized comparisons separate avoids confusing mechanical equivalence with performance preference.

12.2.12 What Bilateral Sensors Add

Separate six-axis grip sensors directly constrain each contact wrench. Pressure mapping can reveal distributed contact and shifts in effective application point. Shaft strain helps close the club load balance. Synchronized motion supplies velocities for contact power and transforms sensor axes to club and laboratory frames. Force plates and body motion support whole-system closure.

Calibration needs known loads, crosstalk characterization, thermal drift checks, and frame registration. Grip modification can itself change behavior, so the instrumented club should be compared with a standard club. Uncertainty in hand position and sensor orientation should propagate to couple sign and magnitude.

12.2.13 Why Closure Is Not Calibration

A [pinned synthetic sensor qualification](#) makes the distinction quantitative. Its net-wrench-only inverse closes the resultant to numerical tolerance while giving 11.86 N component-wise RMSE across the six hand-force components and 29.05 N RMSE for its declared axial scalar. These are errors against synthetic ground truth. Closure tests consistency with the supplied resultant; it does not establish that either the resultant or allocation is physically correct.

The same controlled study separates practical measurement errors. Correcting a declared cross-talk matrix reduces allocation RMSE from 0.94 N to 0.15 N; a 10% error in the cross-talk correction amplitude raises it to 0.18 N; this is not ten-percent device accuracy. Holding contact centers fixed

with the migration amplitude parameter set to 8 mm causes 2.02 N RMSE; this parameter scales prescribed sinusoidal coordinates, not total travel. Tracking those centers removes that modeled bias in the idealized migration case. The combined case, with noise, residual cross-talk error and untracked migration amplitude set to 4 mm, gives 1.02 N component-wise RMSE and 3.87 N at the 95th percentile of the six-component error-vector norm over 32 seeded trials with 301 samples each. The percentile and RMSE use different aggregations; they are not two statistics of one scalar hand-force error. The 8-mm and 4-mm settings are different cases.

These values come from a synthetic point-force sensor qualification, not a physical transducer, distributed-pressure experiment, or participant study. They define pre-acquisition acceptance tests: verify the augmented rank, report allocation error in addition to wrench closure, calibrate cross-talk, track effective contact centers, propagate residual calibration uncertainty, and fail closed when any required channel or registration is absent. Governed bilateral six-axis human data are still required to test the proposed hand, wrist, arm, or scapular mechanisms.

12.2.14 Internal Force Is Not Wasted Force

An equal-and-opposite hand mode may cancel at the club's center-of-mass force balance while increasing grip compression, joint load, or stiffness. It may help stabilize the clubface or may be needless co-contraction. Net club motion cannot decide. Outcomes and internal-load measurements are required.

This is where the question of “slack” becomes scientific. If continuous bilateral loading reduces a measurable force-development delay or perturbation response time without excessive internal cost, preload has operational value. If the same club task occurs with no gap or robustness difference, the proposed benefit is unsupported. A feeling of connection is not yet a measured state.

12.2.15 One Point per Hand Is Itself a Modeling Choice

Treating each hand as a single contact point is convenient, and the natural worry is that the convenience is doing the work. A [registered reduced-model comparison](#) examines one part of that question.

Each hand was replaced by one, three, or five tension fibers spread across a 30 mm band, while **total per-hand** stiffness and damping were held fixed at 1800 N/m and 18 N·s/m, divided equally among that hand's stations. That control matters: adding contact points without holding the total constant would silently stiffen the grip, and any change could then be blamed on stiffness rather than on distribution.

The archived design contains 288 trajectories: twelve initial states, three station counts, two velocity branches, two integration steps and two engines. The 4, 10, 25 and 50 ms outputs are nested observation horizons within each 50 ms trajectory, not integration timesteps or separate trials. The steps are 1 and 0.5 ms. Initial motion supplies the starting condition; this experiment does not use active joint drivers.

Peak single-station force reached 4.03 N. The maximum concentration of 0.77 is the largest station-force norm divided by the sum of all station-force norms at an instant. It is neither pressure nor a named finger's share. The summary also reports zero transitions of whether **any station is active**. That counter does not track every station's engagement. Indeed, the maximum fraction of samples with fewer than all stations active is 1.0. Zero recorded global on/off transitions therefore cannot establish that every contact stayed closed.

The maximum three-to-five-station comparison at the fine step is 0.00355. Its implementation compares final coordinate arrays using maximum absolute component difference divided by the larger of one and the two maximum coordinate magnitudes. This is a registered coordinate-dependent diagnostic, not a unit-invariant physical error or proof of convergence of forces, energy and contact events. The stored coordinates mix translations and rotations; units and parameterization affect the comparison. The worst normalized work-energy residual also falls from 0.03119 to 0.01560 on timestep refinement. Passing these particular checks supports a bounded comparison, not universal discretization convergence.

Holding total stiffness and damping fixed makes the comparison interpretable. It does not measure grip pressure, identify finger anatomy, or model friction or soft tissue. A number of stations is not a set of fingers. These statements describe the pinned archive, not a new solver run or current-human validation.

12.2.16 Minimum Evidence for a Human Claim

The model prediction should specify phase, frame, couple sign, magnitude range, geometry dependence, and competing explanations. A human test should measure bilateral wrench, motion, shaft response, and relevant activation, with participant-held-out analysis. Only then can researchers ask whether a modeled contact mechanism is commonly used, individually variable, or absent. Until then, it remains a falsifiable mechanical hypothesis—not a prescription.

12.2.17 Reader Lab: Explore the Hidden Mode

Start with one pair of hand forces that produces a target club wrench. Add a small vector to one hand and subtract it from the other. Recompute the resultant. Depending on the vector and hand separation, net force stays fixed while moment changes. Add compensating contact moments so the full target wrench is restored. You have constructed a distinct internal allocation with identical rigid club loading.

Repeat while minimizing different costs: total force squared, peak wrist moment, lead-hand load, or deviation from a measured pressure pattern. Each criterion selects a different solution. This makes the role of regularization visible. Without bilateral measurement, none should be called the observed allocation.

Now add simulated sensor noise and contact-position uncertainty. Estimate the confidence interval on swing-normal couple and on each hand's contribution. A negative mean with an interval spanning zero is not firm evidence of sign. The lab links elementary wrench geometry to realistic qualification.

12.2.18 How This Guides Product Design

The software should display net wrench and the family of compatible allocations when individual forces are unavailable. When bilateral data exist, it should show measured allocation with uncertainty and closure residual. Any optimized allocation should be labeled as a model solution with its cost function.

This prevents the interface from converting hidden assumptions into anatomical facts. It also makes the tool useful for experiment design: users can see which sensor or manipulation would shrink the ambiguity most.

12.3 Where the Picture Breaks

The steering wheel is rigid, symmetric, and fixed to one axis. A golf grip moves in space, the shaft bends, wrist axes rotate, pressure is distributed, and both arms attach to a moving body. Two point contacts are a reduction.

The common/differential transformation also does not identify comfort or intent. The same internal wrench can arise from different muscles and co-contraction. Technique claims need participant-level evidence.

Go Deeper: Read the complete [two-hand wrench mechanics](#), the [forward constrained model](#), and the instrumented-grip literature ([Choi and Park 2020](#); [Koike 2016](#)). The technical chapter links follow upstream development; numerical claims above use the explicitly pinned artifacts so their definitions remain inspectable.

Chapter 13

Negative Torque Can Help a Positive Outcome

13.1 A Concrete Picture

When you throw a ball, muscles often brake one segment while another continues accelerating. When you drive through a bend, turning the steering wheel left can be correct even while the car's velocity still points partly right. Signs belong to axes and moments; words such as *helpful* belong to objectives.

In a golf model, positive club rotation is defined by a chosen swing-normal axis. A negative torque points the other way. It may reduce club angular speed, retain a relative angle, preload an interface, stabilize face orientation, or arise as a passive reaction. None of those interpretations follows from the minus sign alone.

13.2 How the Mechanism Works

In one rotational coordinate, power is $P = \tau\omega$. If the club rotates in the positive direction and torque is negative, power is negative: that channel removes club energy at that instant. But total club energy can still rise because hand-force power, shaft release, gravity, and inertial coupling contribute. A negative component in a sum need not make the sum negative.

13.2.1 Braking Can Prepare a Better State

An early opposing wrist command can retain the wrist-cock angle against opening. It may do negative work locally while allowing proximal work to carry the club into a geometry where later release has greater endpoint leverage. Whether the trade helps depends on later work, timing, actuator limits, and outcome metrics.

This resembles holding a spring-loaded door briefly before letting it open, but the club is not powered by one spring. Retention changes the entire coupled state. It can increase proximal demand and alter hand path. The benefit must be shown by forward simulation and work accounting, not assumed from the appearance of lag.

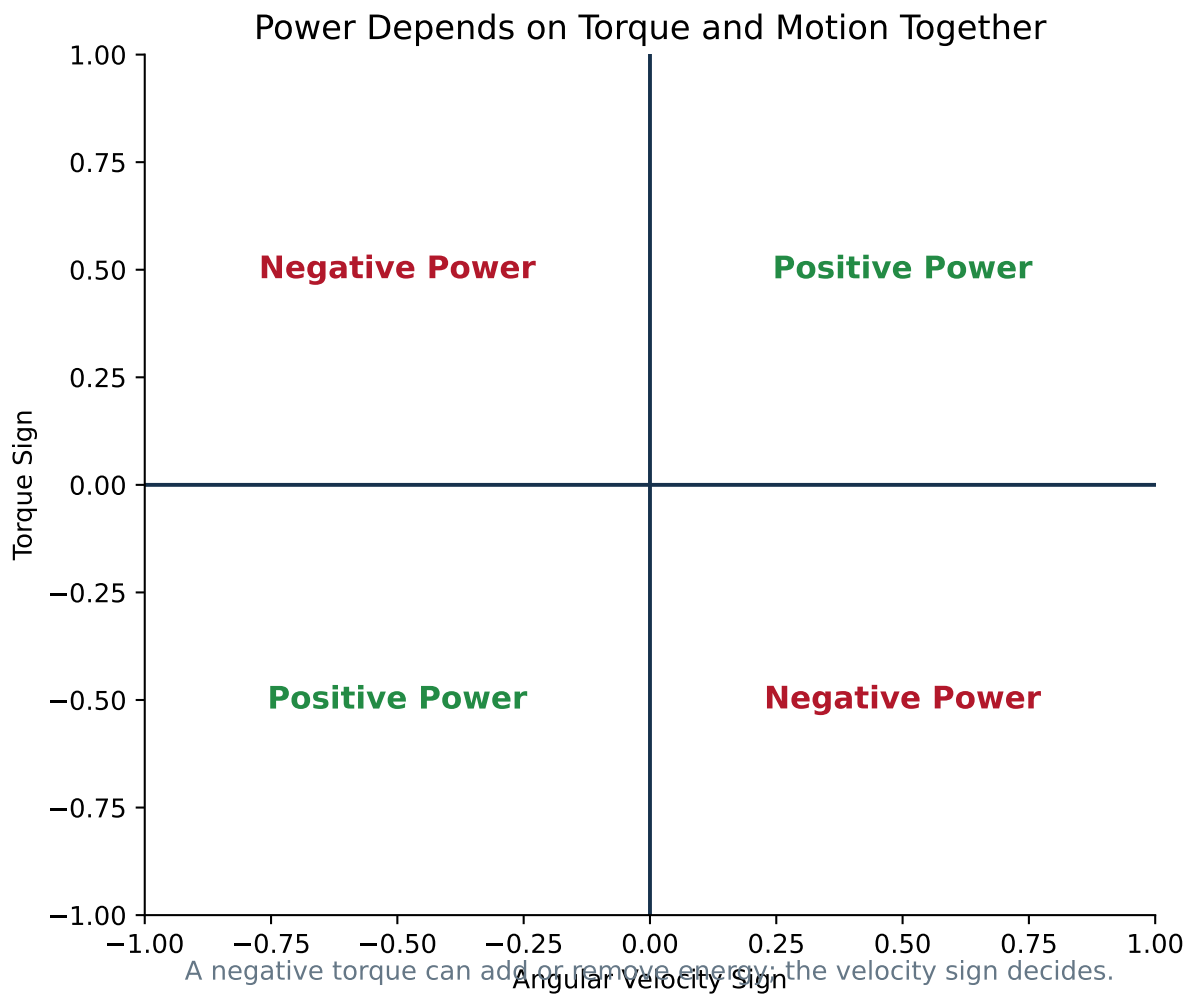


Figure 13.1: Torque and Angular-Velocity Signs Form Four Power Quadrants, Showing That Negative Torque Can Supply or Remove Energy.

13.2.2 A Passive Negative Couple

The WSCG-inspired two-hand question is different. It asks whether constraint and interaction forces can create a negative club couple even without commanding a matching direct negative wrist moment. In the reduced model, separated hand reactions project into a swing-normal couple whose late sign can be negative. That is a reaction mechanism, not proof of passive human wrists.

The technical work separates direct wrist moments from force-generated grip couple. It reconstructs the equivalent wrench and applies geometry controls. The negative couple persists in a declared forward interval, vanishes when the moment arm is collapsed, and reverses with reversed geometry.

13.2.3 Negative Power Is Not Lost by Definition

Negative work at one interface may transfer energy out of the club coordinate and into another body or elastic state. Damping turns energy into heat, but a conservative spring can return it. A negative wrist-power interval therefore needs a destination in the ledger.

At human joints, negative net power is often called absorption. Muscles may act eccentrically, tendons may store energy, or opposing muscle groups may coexist. Net joint power does not identify which tissue performed the work.

13.2.4 Timing Changes the Verdict

The same negative torque applied early, late, or after impact can have different effects. Early retention may preserve geometry; late braking may reduce speed; brief negative coupling near impact may coexist with a still-positive net club power. Strategy comparisons must report onset, duration, impulse, work, and the state at application.

The uncertainty experiment adds another caution. An early opposing command improves lower-tail speed relative to one late-drive program but worsens a planar face/path proxy. Both remain nondominated. “Helpful” requires a declared vector of outcomes.

Model Result: Negative direct or interaction torque can coexist with a positive final speed effect in declared schedules, but the mechanism and trade differ between retention, passive coupling, and robust-control tests.

Human Evidence: Wrist EMG and inverse dynamics show phase-dependent activity, but available measurements do not uniquely identify a passive two-hand negative couple or its muscle source ([Morrison et al. 2016](#); [Nesbit 2005](#)).

Hypothesis: Late negative club torque may be produced partly by two-hand interaction geometry while energy continues arriving through proximal and elastic pathways.

Practical Interpretation: Read a torque sign as an arrow, not a grade. Ask which axis, which velocity, which power, which phase, and which final objective.

13.2.5 First Decide What “Negative” Means

A sign has meaning only after a coordinate is declared. In a planar model, positive club rotation may be defined in the downswing direction relative to the arm. In a spatial model, “negative swing-normal

moment” refers to projection onto a time-varying axis. An anatomical flexion moment, a club-frame moment, and a generalized coordinate torque are not automatically the same quantity.

Next distinguish torque from power. A negative torque acting while angular velocity is positive has negative power. It removes energy through that coordinate at that instant. A negative interaction contribution to generalized acceleration is different again: it describes how state-dependent dynamics affect motion under a decomposition. Careless writing collapses these meanings and makes an ordinary sign look paradoxical.

13.2.6 Braking Can Improve a Later Outcome Without Creating Energy

Consider steering a loaded cart around a corner. A brief braking or inward force can reduce one velocity component, improve orientation, and permit more useful acceleration afterward. The brake does not supply energy. It changes state so other forces act more effectively. In a linked swing, an opposing distal moment can delay relative release, alter coupling geometry, or regulate face/path while proximal and hand forces continue to do work.

A matched-state simulation can make that precise. Compare baseline with a branch where the opposing moment is removed. If the branch releases earlier and ends slower, baseline’s negative interval contributed to the later outcome through state evolution. The energy ledger must still show where positive work came from. Language such as “negative torque creates speed” should be replaced by “the schedule containing negative torque produced a faster later state through the following measured pathway.”

13.2.7 Direct, Interaction, and Contact Contributions

Direct generalized torque is an input term specified by the controller. Interaction torque is a decomposition of dynamics arising from configuration, velocity, inertia, and other driven coordinates. Contact couple is reconstructed from forces and moments at hand interfaces. All three can be negative over the same interval, but they are not synonyms.

A passive negative contact contribution is especially easy to overstate. It means that, given the modeled state and drivers, enforcing contact produces a couple with negative projection. Muscles are still active elsewhere; maintaining geometry and stiffness may require work. “Passive” describes the absence of an explicit torque in that selected model channel, not the absence of biological effort.

13.2.8 The Role-Reversal Spectrum

At one extreme, arm and shoulder-girdle action establishes most of the handle wrench while wrists maintain alignment, stiffness, or a smaller regulating moment. Continuous loading from backswing through early downswing may keep a modeled transmission variable engaged. At the other extreme, wrists establish more of the backswing club moment while arm action resists; during downswing the allocation reverses, with wrists opposing while arms drive.

These extremes organize experiments; real golfers can occupy mixtures that change by phase. To compare them, match club-level task and initial state where appropriate. Measure sign changes in bilateral wrench, activation timing, force-rise delay, stiffness proxy, shaft response, and resulting work. A strategy name cannot substitute for these observables.

13.2.9 When Retention Is Useful

Retention is valuable only relative to an objective. Delaying release can allow proximal motion and geometry to develop before rapid relative club rotation. It may preserve a later positive-power window or improve face/path regulation. Too much or too-late retention can leave insufficient time to accelerate, increase load, or deliver poor impact orientation.

This produces an optimum region rather than a maxim: some opposing contribution, over some state interval, under some parameter distribution. The region shifts with club inertia, shaft stiffness, tempo, strength, and objective weights. A model should map that region and its uncertainty, not canonize one torque curve.

13.2.10 A Simple Schedule Comparison

Imagine three commands applied from a copied early-downswing state. Schedule A uses positive distal torque throughout. Schedule B uses a short negative phase then positive drive. Schedule C keeps the negative phase too long. A plausible model can make B fastest, A earlier-releasing, and C too late. But change arm speed, inertia, or rise time and the ranking can change.

The scientific result is not that B is universally best. It is the response surface and mechanism: when B wins, which power channels and geometry explain it; where it loses; and how readily the ranking reverses. Held-out parameter samples and independent solvers test whether that explanation is robust.

13.2.11 Human Evidence Needed

Kinematics can show relative release and angular acceleration. Inverse dynamics can estimate net generalized moments with assumptions. Bilateral grip sensors can measure the contact couple. Shaft strain can show concurrent storage and return. EMG can show activation timing, but force and passive tissue contribution remain uncertain. Perturbation or system identification may estimate effective stiffness and force-development delay.

A convincing dataset synchronizes these signals and preregisters the late interval and sign conventions. It tests whether predicted negative couple precedes or accompanies the expected state change, whether geometry manipulations alter it, and whether the effect generalizes across held-out participants. A motion trace alone cannot establish passivity.

13.2.12 Failure Conditions

The proposed benefit is weakened if an opposing interval never improves any declared speed-control tradeoff; if apparent benefit disappears after matching initial state; if energy accounting assigns positive work to the negative channel through a sign error; if contact couple does not respond to geometry; or if human wrenches show the opposite sign in the preregistered window. Each failure would refine the theory rather than threaten the value of testing it.

13.2.13 Reader Lab: Audit a “Negative Helps” Claim

Locate the coordinate and sign definition. Plot torque, angular velocity, power, and cumulative work for the same channel. Mark the proposed negative interval. Next plot all other power ports and total energy residual. Identify the positive work source supporting any later speed increase.

Copy state at interval start and remove only the negative input or contact mode. Compare immediate acceleration and forward outcomes. Repeat after matching terminal task through reoptimization. The first comparison asks causal contribution; the second asks whether another route substitutes. They should not be blended.

Reverse geometric controls, alter interval boundaries, and sample parameters. If benefit is robust only for one hand-picked trace, report a demonstration. If the mechanism retains sign and improves a defined tradeoff over a broad region, report a conditional strategy result. Neither supports an anatomical instruction without human measurements.

13.2.14 Three Statements That Sound Similar but Are Not

“The wrist moment was negative” is a kinematic/dynamic sign statement. “The wrist removed club energy” is a power/work statement requiring angular velocity and integration. “The negative wrist action improved final speed” is a causal counterfactual statement. A paper should provide separate evidence for each.

Likewise, “passive negative torque” can mean a model interaction term, an elastic material reaction, or absence of measured activation. These meanings must be named. The project uses “passive” only with the producing model channel; it does not infer human intention or zero muscle work.

13.2.15 The Nonparadoxical Summary

Negative torque is ordinary signed mechanics. Its usefulness, if any, comes from how it changes state, timing, and other power pathways—not from producing energy with the wrong sign. Keeping that sentence visible removes the apparent paradox while preserving the interesting causal question.

13.3 Where the Picture Breaks

The brake metaphor suggests negative torque merely slows an independent wheel. In a coupled swing it also changes reactions, hand path, and future geometry. The passive label is especially dangerous: zero selected command in a model does not demonstrate relaxed muscles or absent neural control.

A planar sign also cannot represent all three-dimensional wrist and clubface moments. A favorable swing-normal couple could coexist with an unfavorable face-twisting moment.

Go Deeper: See the [passive couple-reversal chapter](#), the [timing strategies](#), and the wrist-torque simulation literature ([Springs and Neal 2000](#)).

Chapter 14

Arms, Wrists, Preload, and the Meaning of Slack

14.1 A Concrete Picture

Turn a two-hand valve. One strategy uses the arms to create most of the opposing push-pull while the wrists remain firm. Another uses wrist moments more directly while the arms stabilize. The valve can receive the same net turning effect in both cases, but the loads inside the person differ.

Now reverse the motion. Motion reversal need not reverse the applied torque: a continuing torque can first brake one direction and then accelerate the other. A gear train that stays loaded against one tooth face avoids traversing backlash. That does not make torque tracking instantaneous; compliance and actuator dynamics can still delay the response. Reversing the load can instead require crossing a gap before the opposite face engages. Biological tissue is not a gearbox, but the picture creates a testable question: does maintaining preload and load direction reduce a measurable transmission delay compared with complete role reversal?

14.2 How the Mechanism Works

The first experiment matches a **scalar control contribution**, not the entire club task. At a fixed configuration and velocity, the constrained dynamics give an affine acceleration map. Subtracting the zero-control solution leaves a linear control contribution. The planar target is

$$I_C[\ddot{\alpha}(u) - \ddot{\alpha}(0)] = 8 \text{ N m},$$

where I_C is club rotational inertia and α is club angle. The drift acceleration $\ddot{\alpha}(0)$ remains part of the total motion. Proximal-only and wrist-only generalized controls can each attain this increment at the sampled states. Their convex mixtures attain it too, because this fixed-state map is linear. Feasibility under strength, contact, activation, or other constraints requires a separate check; the algebra alone does not establish human capacity.

Equal turning moment does **not** mean equal wrench. A planar wrench includes two force components

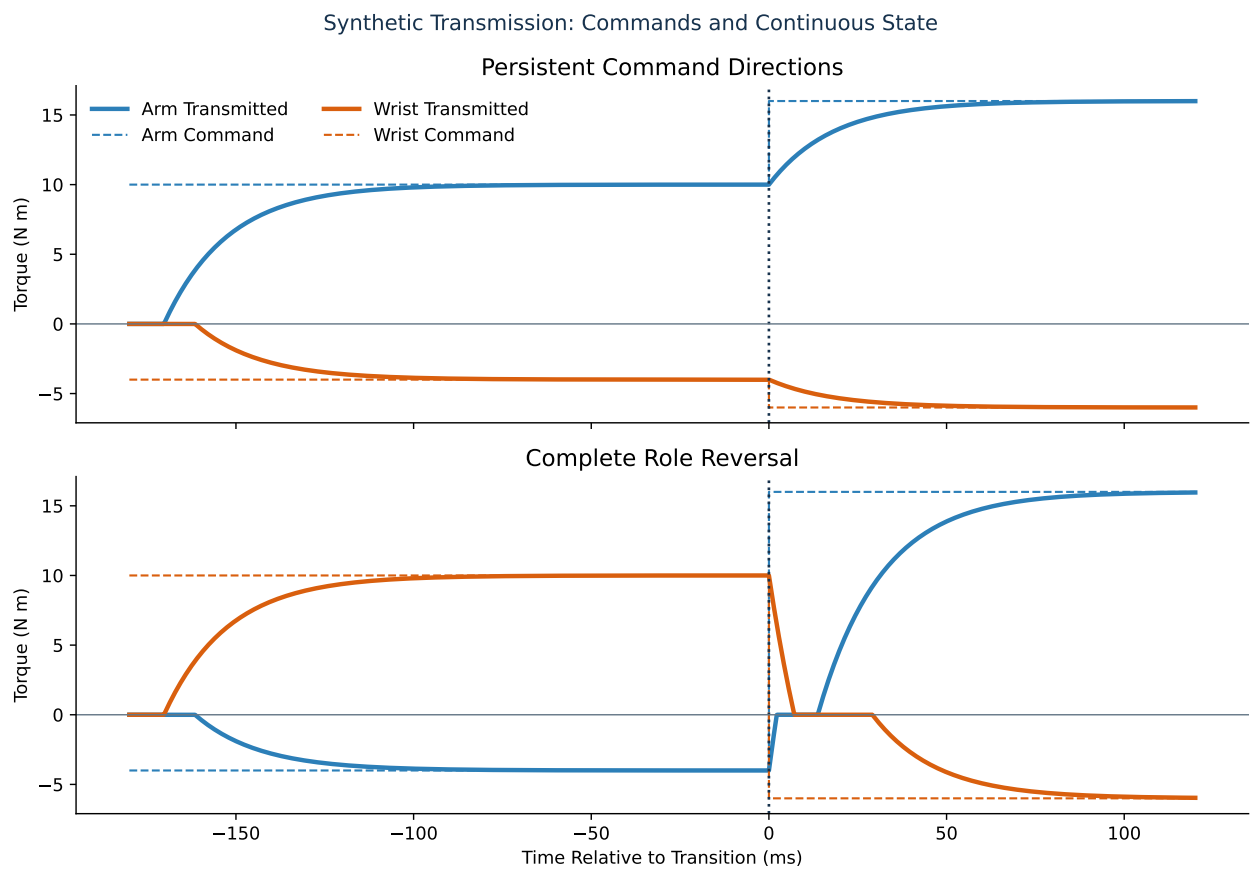


Figure 14.1: Commands and Transmitted Torques in the Declared Synthetic Preload Experiment.

as well as moment about a specified point. Equal moments can accompany different resultant forces, club translation, and work. For example, adding the same transverse force at two symmetrically placed contacts changes the resultant without changing their combined moment about the midpoint. Even matching a complete wrench would leave some internal allocations undetermined.

The study varies 19 club angles and 21 mixture fractions. Its reported hand-force RMS is calculated from the **control-induced contact-force increments**, not complete measured hand loads. Its generalized-torque norm is an unweighted Euclidean norm in the declared actuator coordinates; it is neither muscle effort nor metabolic cost. Changing coordinate scaling or cost weights changes the meaning of a minimum-norm allocation.

The labels *arm channel* and *wrist channel* identify generalized actuator subspaces. A wrist control can also induce contact forces through the closed chain; it does not isolate a pure contact moment from every other effect. Zero direct wrist actuation does not imply zero passive wrist resistance or zero constraint reaction. These results do not identify scapular retraction, muscle activation, or a preferred technique.

14.2.1 Two Extreme Stories

At one end, proximal arm-chain action creates the grip couple while wrists supply resistance or impedance. The same general proximal drive can continue across backswing and early downswing while signs remain loaded. At the other end, wrist moments create more of the backswing couple while arms resist; during downswing, roles reverse so arm-generated couple becomes positive and wrist action opposes release.

These are useful poles, not a binary classification of golfers. Real allocations can lie between them and vary continuously through time. The comparison asks what each pole predicts under matched tasks.

14.2.2 Replace “Slack” With Measured Variables

Slack can mean tendon slack length, grip gap, low muscle activation, low tangent stiffness, joint laxity, series-elastic extension, or a delay in force rise. Those are not synonyms. Having studied instantaneous allocation, we now ask a different question: how does an internal transmission state affect the response to a changing command? This second experiment uses two abstract channels, each with deflection x , commanded torque τ^* , stiffness k , dead-zone half-width δ , and time constant T :

$$x^* = \text{sgn}(\tau^*) \left(\delta + \frac{|\tau^*|}{k} \right),$$

$$T\dot{x} = x^* - x.$$

$$\tau = k \text{sgn}(x) \max(|x| - \delta, 0).$$

For a zero command, $x^* = 0$. The parameters are $k = 600$ N m/rad, $\delta = 0.012$ rad, and $T = 18$ ms in **both** channels. The dead zone has total width 2δ . Deflection follows a first-order lag; torque is obtained through the dead-zone law. This is a phenomenological transmission model, not a measured activation law or a literal anatomical gearbox. It contains no calibrated muscle energetics, and its tracking score cannot establish energetic efficiency.

14.2.3 Continuous Preparation Matters

Starting at a loaded equilibrium would hide how preload arose. The finite-history test starts both channels at zero deflection, applies preparation commands for 180 ms, then changes commands without resetting the state. Each pair is (arm, wrist) in N m:

- **Persistent Directions:** Preparation (10, −4); after transition (16, −6).
- **Complete Role Reversal:** Preparation (−4, 10); after transition (16, −6).

Both desired sums change from 6 to 10 N m. This scalar experiment is distinct from the earlier fixed-state 8 N m allocation experiment; it does not simulate a complete anatomical backswing. Preparation lasts ten time constants, bringing the state close to equilibrium without reaching it exactly. The figure plots the archived transmitted torques alongside the commands, including the initial rise from the relaxed state.

The reported score integrates **absolute net torque tracking error** over the first 120 ms after transition:

$$J = \int_0^{0.120\text{ s}} |\tau_{\Sigma}^*(t) - \tau_{\Sigma}(t)| dt.$$

It has units N m s, but is not signed angular impulse and is not work. Finite preparation gives $J = 0.0717151$ N m s for persistent directions and 0.1954089 N m s for reversal. The calculation uses a 0.1 ms forward-Euler step and a trapezoidal error integral. These are synthetic model outputs, not estimates of human delay.

Persistent directions have no post-transition zero-transmission samples. Reversal has arm and wrist zero-torque occupancies of 11.5 and 22.0 ms, with the study’s one-step brackets [11.4, 11.6] and [21.9, 22.1] ms. These sampling brackets are not total model-error or physiological confidence intervals. Occupancy counts samples with torque magnitude at most 10^{-12} N m; it is distinct from time to useful opposite torque. The corresponding first attainment of 10% of the post-transition target in its new direction is 15.7 and 31.1 ms. From an identical relaxed state at transition, both programs have identical future commands and therefore identical responses ($J = 0.2010714$ N m s).

14.2.4 What the Zero-Dead-Zone Control Establishes

With $\delta = 0$ and the **same T in both channels**, the sum satisfies $T\dot{\tau}_{\Sigma} + \tau_{\Sigma} = \tau_{\Sigma}^*$. Equal initial sums and future net commands then give equal net responses, despite different component histories. The registered 12-case grid combines half-widths 0, 0.004, 0.012, and 0.024 rad with common time constants 8, 18, and 35 ms. Its three zero-width cases are equivalent within 10^{-10} N m s; the other nine favor persistent directions on this score. This is a conditional result within the specified family, not a universal allocation optimum.

Unequal channel time constants break the sum reduction even with no dead zone. For equilibrium initial torques (10, −4) versus (−4, 10) and the common final command (16, −6), the difference in net responses is $14[\exp(-t/T_{\text{arm}}) - \exp(-t/T_{\text{wrist}})]$ N m. It is generally nonzero when the time constants differ. Thus the negative control isolates a mechanism within this model and comparison; it cannot uniquely identify a human mechanism from a similar trace. Removing the dead zone also leaves first-order lag: no gap does not mean immediate command tracking.

14.2.5 Compliance Can Be Useful

Eliminating all compliance is not an implied goal. Elastic elements can store and return energy; damping dissipates it. Compliance can change peak loads, timing, and sensitivity, with benefits or costs depending on the excitation and objective. None of those outcomes is measured by the tracking score alone. The relevant question is which stiffness and preload suit the task, including internal load and physiological cost, rather than how to maximize rigidity.

The wrist-led extreme may reduce some proximal load, use elastic storage, or aid participant-specific accuracy. The current objective does not price those benefits. Its conditional continuity result therefore cannot select a universal winner.

14.2.6 What Must Be Measured

A serious human test needs bilateral six-axis grip wrenches and pressure, closed-chain inverse dynamics constrained by those measurements, wrist and shaft kinematics, shaft strain, transition-aligned force-development delay, and EMG from forearm, shoulder, and scapular muscles. Perturbation or system-identification methods should estimate tangent stiffness at a declared posture, activation state, perturbation amplitude, and time scale. Dynamic impedance can also include damping, inertia, and feedback. Preload is an operating load, not itself a stiffness measurement. Holding out participants tests generalization; it does not remove individual differences or justify prescribing a group average.

14.2.7 Define Slack Operationally

Distinguishing these mechanisms requires different measurements: grip separation can indicate contact loss, bilateral forces resolve load modes, and perturbation response estimates stiffness. EMG and tissue imaging add complementary evidence without directly measuring the net wrench. The transmission experiment alone does not tell us which mechanism generated a measured trace.

A separate synthetic constitutive audit keeps five candidate mechanisms separate: contact disengagement, transmission dead zone, structural preload, biological series compliance, and control deadband. At the selected synthetic operating points, each class has a rank-three local sensitivity matrix for its three fitted parameters under the registered excitations. This concerns parameters **within an assumed class**, not selection of the class itself, global uniqueness, or reliable estimation with noisy data. Contact and biological surrogates have a normalized output RMSE of about 0.0196 under the registered multisine, but that number is not a statistical detection threshold. Their similar traces leave single-channel class identification unestablished. This is a useful negative result: a clean-looking force trace cannot by itself tell a reviewer which kind of “slack” was present. Forward delivery and human tests therefore need independent contact, shaft, tendon, activation, and bilateral wrench measurements.

14.2.8 Why Continuous Loading Might Help

Force does not appear instantaneously in biological tissue or actuators. Electromechanical delay, tendon compliance, muscle short-range stiffness, grip deformation, and neural feedback all introduce history. Maintaining some load through transition might reduce the time required to develop a useful wrench in the early downswing and could improve resistance to perturbation.

Continuous load could also require more co-contraction, raise internal load, or constrain useful reorientation. These are proposed trade-offs to measure; the two-channel model does not estimate

fatigue or physiological cost. “No slack” is therefore not automatically optimal. The correct question is whether a measurable loaded state improves a declared outcome at an acceptable internal cost compared with alternatives.

14.2.9 Why Role Reversal Might Help

Reversal can exploit different muscle groups and mechanical advantages across phases. The backswing has more time and different geometry than the late downswing. A wrist-dominant backswing allocation might position or load the club efficiently, while proximal controls take a larger share of downswing drive and wrist controls resist. That is a proposed allocation, not a measured anatomical hierarchy established by this model. Antagonistic preactivation could also preserve stiffness despite a change in net torque sign.

Therefore role reversal need not mean mechanical disengagement. Two opposing preloaded torsional elements can exert $+5$ and -5 N m at equilibrium while each contributes 100 N m/rad of restoring stiffness: net torque is zero but restoring stiffness is 200 N m/rad. This illustrative construction is not a fitted wrist model. Net moment can reverse while individual muscles or hand contacts remain loaded. The experiment must measure force magnitude, differential modes, and stiffness rather than inferring engagement from net sign.

14.2.10 A Matched-Task Comparison

First state what is matched. Equal instantaneous angular momentum, equal full wrench histories, equal club trajectories, and equal terminal states are increasingly different constraints; none follows automatically from the first. Select a feasible comparison and report its tolerances. Under a matched scalar task, compare the remaining wrench components as outcomes. If matching the full wrench is feasible, compare joint demands, bilateral load distribution, work, and robustness. Account for actuator limits and closed-chain consistency.

A second experiment can compare future evolution from matched configurations **and velocities**, retaining different transmission states. Match other relevant states and disturbances as well, or identify them explicitly as alternative causes. Declare whether future commands or feedback laws are held fixed: freely reoptimizing each program changes the causal question. Matching visible geometry alone leaves momentum and several possible sources of divergence unmatched.

A targeted state intervention can test dependence on preload in a model, but resetting a state may change stored energy, momentum, or constraint consistency. Those changes and any physical preparation needed to realize the reset belong in the accounting. If the entire deterministic state, future control law, and external inputs are identical, uniqueness already implies identical future motion. That equality is not independent proof of mediation. A convincing mechanism test changes the candidate state while controlling competing pathways and checks that the claimed effect survives feasible preparation and measurement uncertainty. Chapter 15 develops the related shaft-memory problem.

14.2.11 Predictions That Separate the Extremes

The continuous arm-dominant hypothesis predicts less force-rise delay after transition, fewer or shorter near-zero-load intervals in the relevant bilateral mode, and a particular sign continuity in arm-level

control. The role-reversal hypothesis predicts a clearer change in net allocation but may retain high co-contraction or contact load. Both can predict similar club motion.

If a verified allocation change produces no practically meaningful difference in the predicted wrench or stiffness mediator within adequately precise bounds, that particular slack explanation loses support. Instructions alone do not establish that an allocation changed. If one allocation improves speed only by increasing peak load or face/path variance, its practical value becomes conditional. If responses differ between participants, report that heterogeneity; subgroup explanations should be prespecified or clearly exploratory and independently tested.

14.2.12 Safety and Coaching Boundary

This comparison is not an instruction to apply maximal opposing force or to manipulate the wrists aggressively. The model does not estimate tissue safety, and the linked EMG evidence does not identify a preferred strategy. Human experiments require bounded tasks, clinical and ethics review where appropriate, and monitoring of discomfort and load.

For a reader or coach, the useful principle is narrower: distinguish continuous mechanical engagement from continuous net torque, and distinguish a felt cue from the physical variable it is supposed to change. A cue becomes scientifically useful only when it reliably changes the measured mediator and outcome.

14.2.13 What Would Count Against Preload

The specific dead-zone account is weakened if its predicted contrast fails when the declared model assumptions are reproduced; if an apparent advantage instead comes from unequal channel dynamics, unmatched states, or a different control law; if measured transition load or stiffness does not mediate the outcome; if continuous loading raises internal cost without a robust task benefit; or if held-out participants show no repeatable relation. These conditions prevent “connection” from becoming an explanation that fits every result.

14.2.14 Reader Lab: Make the Opposite Case Strong

Construct the strongest case for each allocation using the mechanisms above. Separate what the declared transmission model predicts from the anatomical assumptions that would need human measurements.

Write predictions that differ. Continuous loading should reduce a defined gap in a particular force mode. Role reversal may show a sign change in allocation without reduced stiffness. Both could yield equal club motion. Specify how bilateral wrench, perturbation response, EMG, and shaft state distinguish them.

Now test a middle family rather than only endpoints. Map allocation continuously from arm-dominant to wrist-dominant and allow phase-specific mixture. Plot speed, face/path, load, robustness, and transmission measures. If an interior region dominates both extremes, the spectrum has done its job by revealing that the original dichotomy was too coarse.

14.2.15 Interpreting a Null Result

If instructions produce no detectable allocation change, the experiment has not established the intended contrast; inadequate measurement sensitivity is also a possibility. It has not shown strategy equivalence. If allocation changes but a well-resolved preload measure does not, the proposed pathway loses support. If preload changes without a precisely estimated practical outcome, distinguish an uncertain effect from one bounded below a prespecified useful size. A nonsignificant result alone establishes neither absence nor equivalence. If the outcome changes without the predicted mediator, examine competing pathways and mediator observability before drawing a causal conclusion.

This staged interpretation prevents every outcome from supporting the same story. It also respects participant differences: one subgroup may maintain stiffness during reversal while another develops a measurable gap.

14.3 Where the Picture Breaks

The gearbox picture is intentionally simplified. Muscle-tendon systems can have slack-to-taut transitions and history-dependent behavior, but the chosen torque dead zone and scalar deflection lag do not identify those biological mechanisms. Surface EMG cannot measure force or stiffness directly. Scapular retraction does not map uniquely to one hand force.

The model also treats two abstract channels, not dozens of muscles and joint axes. Its numerical advantage is a conditional prediction awaiting data, not a coaching instruction.

Model Result: Persistent load direction outperforms complete role reversal on the chosen tracking score in the registered transmission family. Its zero-dead-zone equivalence also depends on equal channel time constants and matched initial net states and commands.

Human Evidence: [Kao et al. \(1995\)](#) recorded scapular-muscle EMG in 15 competitive male golfers and described phase-dependent activity. [Robinson et al. \(2023\)](#) studied 15 sub-elite male golfers and found side-dependent wrist kinematics and extensor carpi ulnaris EMG. Their reported lack of a significant relationship between downswing EMG and impact clubhead kinematics is not an equivalence test. Neither study identifies the bilateral wrench allocation proposed here. EMG is an electrical measurement, not a direct measure of muscle force or stiffness.

Hypothesis: Proximal push-pull during preparation may preserve useful transmission continuity while wrists resist, but the opposite allocation could win on load, accuracy, or participant-specific objectives.

Practical Interpretation: “No slack” should become a measurement question: was there a force gap, low tangent stiffness, delayed transmission, or merely a different internal allocation?

Go Deeper: Read the full [arm-wrist allocation and preload chapter](#), the [dynamic constitutive audit](#), the administrative [epic record](#), and the linked scapular and wrist activation studies above.

Chapter 15

The Shaft Is a Spring With a Memory

15.1 A Concrete Picture

Bend a flexible ruler and release it. The ruler does not respond only to the force you apply now. Its present shape and motion carry a short history of earlier loading. It can return energy after your hand force has fallen, overshoot, and vibrate in several patterns.

A golf shaft behaves far more subtly than the ruler, but the central point survives: flexibility adds state. Treating the club as perfectly rigid may be adequate for one question and misleading for another, especially when late timing, grip moment, and impact-window energy are under study.

The illustration prescribes $x = A \sin^2(\pi t/T)$ with $A = 0.025$ m, $T = 0.6$ s, $m = 0.2$ kg, $k = 80$ N/m and $c = 0.8$ N s/m. The required drive force is $m\ddot{x} + c\dot{x} + kx$, so input power minus $c\dot{x}^2$ equals the rate of change of kinetic plus elastic energy. Negative drive power means energy returns to the actuator; it does not label energy delivered to a clubhead.

15.2 How the Mechanism Works

The simplest flexible model adds one coordinate s for shaft deflection. Its stored elastic energy might be $U_s = \frac{1}{2}ks^2$, and damping force might be proportional to $\dot{s}$. The state now includes s and $\dot{s}$. Grip work can increase U_s ; later shaft motion can return energy to rigid-body coordinates.

15.2.1 Timing, Not Free Speed

An ideal passive spring stores and releases energy without creating it. Its strain energy is not constant while it exchanges work with the rest of the system. For a fixed linear stiffness $k > 0$, $\dot{U}_s = ks\dot{s}$: the same nonzero deformation can be storing or releasing energy. Mechanical energy is conserved only for the complete conservative system with no net boundary work. Damping converts mechanical energy to heat; contact can transfer energy to the ball or other bodies.

The potentially useful feature is timing: accepting work and returning it when the geometry and task make the transfer useful. That usefulness is an outcome to test, not a property guaranteed by elasticity.

In a rigid-flexible comparison, the work-energy ledger must track boundary input, gravity, damping and all retained kinetic and strain energy so that any change in the specified delivery outcome has an

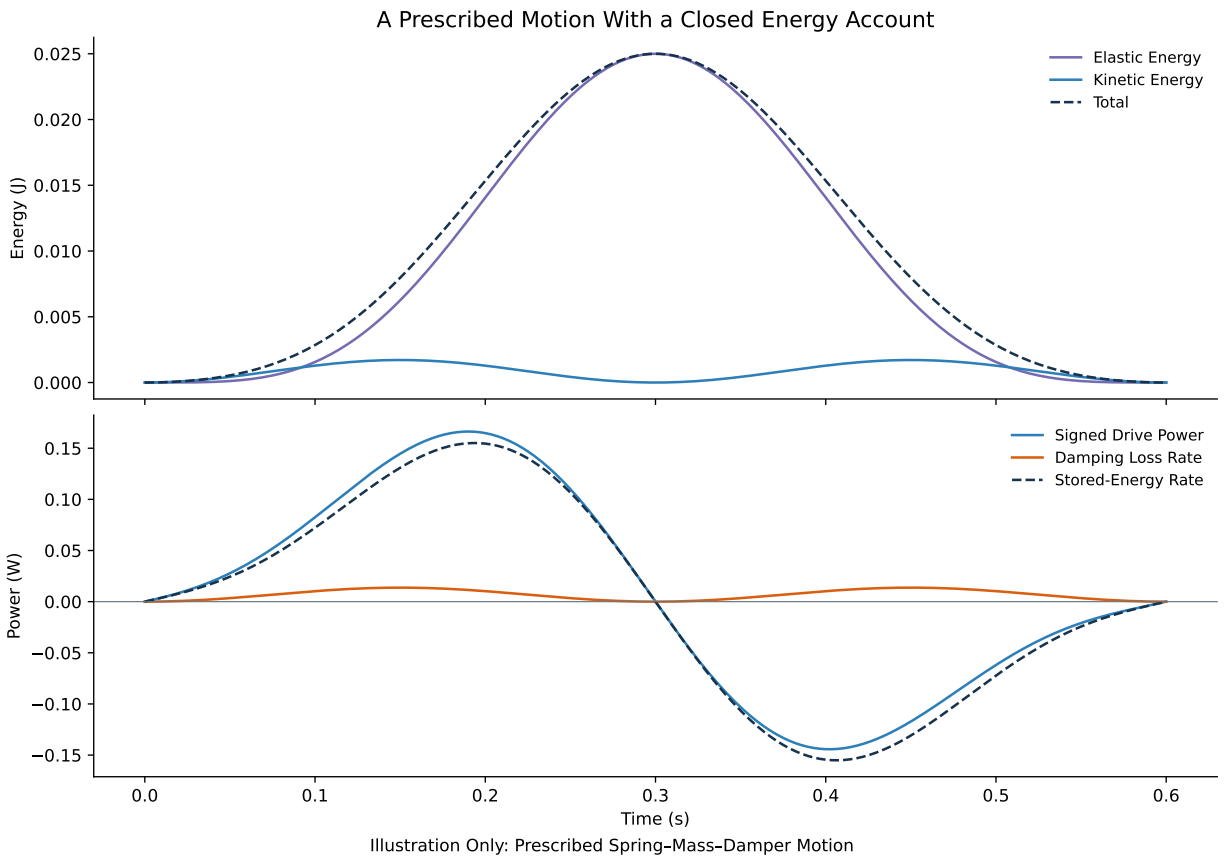


Figure 15.1: A Prescribed Spring–Mass–Damper Motion Separates Stored Energy From Input Power and Damping Loss. This Is an Illustrative Energy Ledger, Not a Shaft Simulation.

accounted source.

15.2.2 One Mode Versus a Distributed Shaft

A real shaft bends continuously. Modal analysis represents its shape as a sum of vibration patterns. The first mode may capture most low-frequency deformation; higher modes matter when excitation is rapid or spatially complex. A one-mode model is efficient but should be compared with a higher-order beam reference.

An isolated structural study assembles an Euler–Bernoulli beam with tip mass and rotary inertia, computes modal frequencies, identifies synthetic stiffness and mass, and compares reduced responses. A separate study transports bending modes into forward two-hand dynamics. Its baseline fails its declared small-deflection screen, as discussed below; numerical closure does not remove that failure. Mode truncation should be judged by specified load, power, energy and trajectory errors, not by appearance or frequency agreement alone.

15.2.3 Excitation Decides Required Complexity

Slow loading may barely excite high modes, making a rigid or one-mode model adequate. A sharp command, impact, or rapid contact change can excite them. The right model therefore depends on the frequency content of the question.

This is analogous to describing ocean swell with one broad wave versus describing a splash. More modes are not inherently more truthful; they are needed when the input and observable are sensitive to them.

15.2.4 Interface Power Connects Hand and Shaft

For an ideal point load F at a deforming shaft station, write its material-point velocity as $v = V + \omega \times r + B\dot{\eta}$. Here V and ω describe the reference motion, r is the current offset, and $B\dot{\eta}$ is the relative deformation velocity. In one observer frame the same power can be written as

$$\begin{aligned} P &= F \cdot v \\ &= F \cdot V + (r \times F) \cdot \omega \\ &\quad + (B^T F) \cdot \dot{\eta}. \end{aligned}$$

These are two representations of one port. Adding the last modal term to $F \cdot v$ would count deformation power twice. Applied contact moments require their own compatible angular-velocity decomposition. A resultant wrench alone represents all contact power only when the interface motion is rigid, or the unrepresented deformation power vanishes. Distributed tractions can do work on deformation even when their net wrench is zero.

The flexible model separates grip moment from internal shaft moment. A negative grip couple can coexist with shaft energy release and positive club kinetic-energy growth. This makes the shaft relevant to the negative-torque discussion without turning it into the sole explanation.

15.2.5 Impact Is a Severe Boundary

Near ball contact, high-frequency response and contact duration matter. A model that stops at a pre-impact criterion can discuss delivery but not claim a full impact prediction. A distributed shaft needs measured stiffness, damping, geometry, clubhead inertia, grip conditions, and timestep resolution before it becomes equipment calibration.

Model Result: The isolated beam and articulated first-mode atlas report bounded synthetic checks. The separate planar forward-modal baseline retains numerical coupling evidence but fails its small-deflection screen. These are different models and acceptance results, not interchangeable validation claims.

Human Evidence: Betzler and colleagues tested 20 golfers with two clubs designed to differ in shaft bending stiffness. The less stiff club produced a small average clubhead-speed increase, alongside changes in shaft recovery and grip motion; the authors judged the overall effects small for the shafts tested (Betzler et al. 2012). This does not identify a universal fitting rule or calibrate the synthetic models here. Strain or optical measurements need their own calibration and uncertainty treatment.

Hypothesis: Shaft state may mediate how a late negative grip couple and proximal power coexist, returning stored energy while the grip channel opposes rotation.

Practical Interpretation: Flex behaves like short-term mechanical memory: what the shaft does now depends on how it was loaded and how fast that loading changed.

15.2.6 What the Shaft Adds to the State

A rigid club needs pose, including orientation, and compatible linear and angular velocity. A flexible shaft needs additional deformation coordinates and their rates. A one-mode model might use one bending amplitude; a distributed beam uses many modes or elements. These states carry memory because current shaft force depends not only on grip and clubhead position but also on how the shaft is bent and moving.

That memory complicates matched-state comparisons. Two swings with identical grip and clubhead pose can have different shaft curvature or modal velocity. Removing a command from those apparently identical frames can produce different responses. Within the same flexible model, a same-state intervention must match the complete state, including activation or contact-memory variables if present. Between rigid and flexible models, the state spaces differ: declare a mapping of common pose and velocity, the elastic initial state, mass properties, contact geometry and energy. Equal rigid coordinates alone do not establish equal total energy or a compatible constrained state.

15.2.7 Loading, Storage, and Return

When relative motion bends or twists the shaft, load power can increase strain energy. Inertial coupling redistributes energy; it is not a separate external energy supply. Damping turns some mechanical energy into heat. As deformation changes, the shaft can return energy to the rigid motion of grip and head. Return is not guaranteed to occur in the useful direction or at the useful time; phase matters.

It is tempting to say the shaft “kicks” the clubhead. A better account plots strain energy, damping loss, power at grip and head interfaces, and modal phase. If shaft energy decreases while clubhead

kinetic energy increases, return may be part of the explanation, but whole-system closure and other power ports still matter.

15.2.8 A Simple Spring Thought Experiment

Take a spring with stiffness k and displacement x . Stored energy is $\frac{1}{2}kx^2$. Doubling deformation quadruples energy, but a real shaft's distributed bending and torsion do not reduce exactly to one scalar spring. Add damping and a changing boundary, and release timing depends on natural frequencies, mode shapes, and excitation history.

If the downswing duration aligns poorly with a mode, extra bending can return too late or increase presentation variability. A linear spring in static equilibrium stores $F^2/(2k)$ under a prescribed force, so increasing k reduces storage in that comparison. At fixed displacement it stores $kx^2/2$, so increasing k increases storage instead. Dynamic loading also depends on inertia, damping and drive frequency: resonance can reverse a simple stiffness trend. Increasing stiffness raises $\sqrt{k/m}$ only with the same effective mass and boundary model; it does not guarantee a faster useful response. There is no universal “more flex equals more speed” rule.

15.2.9 Interaction With the Negative-Couple Mechanism

A two-hand couple loads the handle and can excite shaft bending or torsion. Shaft reaction then changes the wrench at the hands and the clubhead's future state. A negative contact couple may partly regulate deformation, or it may be a reaction to deformation generated elsewhere. Causal separation requires killswitches and energy accounting in the flexible model.

Compare rigid, one-mode, and distributed-shaft rungs with an explicit initial-state mapping and the same endpoint definition. Does the late couple retain sign? Does its timing shift with modal frequency? Does a speed benefit remain after shaft strain energy and damping are accounted for? If a mechanism vanishes in the flexible rung, that is important evidence about the rigid approximation.

15.2.10 Interaction With Preload and Slack

Continuous hand loading can maintain shaft deformation through transition, while a role reversal can unload and reload one mode. Yet “shaft preload” is not the same as muscle or grip preload. The shaft may remain bent while a hand-force component crosses zero, or vice versa. Each state requires its own measurement.

The phenomenological dead zone should not absorb shaft dynamics implicitly. If shaft deformation is modeled, transmission parameters and modal states must be identifiable enough to avoid two mechanisms explaining the same delay. Ablation tests—rigid shaft, zero dead zone, altered damping—show which declared model mechanism changes the result. They do not by themselves identify the mechanism in a golfer; distinct models can still fit the same observations.

15.2.11 Model Validation Ladder

Begin with static deflection under known loads. Continue with free-vibration frequency and damping tests, then dynamic boundary excitation. Verify modal convergence and timestep stability. Compare strain, tip displacement, and energy balance with an independent beam or finite-element reference.

These are verification steps. After coupling to the swing, repeat the power, convergence and deformation-domain checks: a sound isolated beam can be driven outside its valid range by the coupled model. Experimental validation is a separate step.

Human validation uses instrumented clubs with calibrated strain or optical shape measures. Club properties should be measured, not selected merely from a label such as “stiff.” Repeated trials reveal whether deformation phase and contact wrench are consistent within a golfer. Equipment changes should be bounded and randomized to reduce learning and expectation effects.

15.2.12 What a Shaft Result Can Support

A validated model can say that, under declared properties and motion, certain modal forces, powers, and energy exchanges occurred. It can predict how a controlled property change shifts them. It cannot infer a golfer’s preferred feel, injury risk, or universal fitting recommendation without additional data.

For the proximal-distal framework, the shaft is most valuable as a test of mechanism survival. If the interaction and allocation conclusions remain under realistic flexibility and uncertainty, confidence grows. If not, the model ladder has found the boundary where the simpler explanation must be revised.

15.2.13 Reader Lab: Watch Phase Change the Answer

Excite a declared one-mode shaft model with the same peak handle force at three different times. Plot deformation, modal velocity, strain energy, grip power, head power, and damping loss. The same force magnitude can store or return different energy because phase and boundary motion differ. Equal peaks do not match force history, impulse or work; report those differences too.

Copy two states with identical rigid pose and velocity but opposite modal velocity. Apply the same future hand wrench and test whether the chosen clubhead observable differs. Symmetry, mode coupling or observation timing can hide a state difference; a different internal state does not guarantee a different measured endpoint.

Next, reset shaft deformation and modal velocity to zero while retaining the rigid coordinates and rates. This is a model intervention with a new initial condition. Record its energy change, momentum change and any constraint or contact projection. Removing strain energy does not necessarily reduce total energy: rigid-modal cross terms in kinetic energy can change either way. The reset requires explicit work and impulse bookkeeping before interpreting the subsequent trajectory. It is not an available human command or a measured causal contribution in a golfer.

Repeat the negative-couple branch across stiffness and damping. Report whether couple sign, onset, work, and terminal speed remain. A result that survives realistic property uncertainty deserves more confidence than one tuned to a single resonance.

15.2.14 Avoiding Double Explanations

If both a dead-zone transmission and flexible shaft can produce delay, fit or vary them jointly. Look for observables that separate them: grip force may change before shaft strain under one mechanism, while modal phase dominates another. Without such information, their parameters can compensate and neither is identifiable.

The release should therefore include ablations and joint sensitivity, not one best-fit flexible trace. When two explanations remain equivalent, the honest output is an experiment recommendation—usually synchronized wrench and shaft state—rather than a preferred story.

15.2.15 A Minimum Flexible-Club Result

At minimum, publish the measured or declared club properties, modal frequencies and damping, deformation and modal velocity, grip and head power, strain-energy history, damping loss, and work-energy residual. Show timestep and modal convergence. Declare the rigid-to-flexible state mapping and distinguish fixed-input comparisons from comparisons selected to match an outcome.

Then qualify the conclusion. A synthetic property sweep shows mechanical possibility and sensitivity. A calibrated participant club permits predictions for that equipment; held-out observations test their accuracy. Replication can extend support only across the equipment, golfers and conditions actually tested. Skipping these levels turns shaft “memory” into a metaphor too broad to fail.

15.2.16 The Registered Test Says the Benefit Is Not Universal

The [articulated shaft atlas](#) tests a passive synthetic shaft with the first bending shape in each of two orthogonal planes, plus torsion. These are not the first and second bending frequencies in one plane. Bending is anchored to a 24-element reference at 5.2399 Hz; torsion is declared at 70.1260 Hz. Twelve initial states, four activations (rigid, bending, torsion, coupled), two velocity branches, two timesteps and two engines give **384 trajectories**. The motion starts from initial conditions, with no active driver or joint torque. Zero modal coordinates and rates give a common initial energy in this declared comparison. The velocity branches add ± 0.05 m/s to club-frame x translation; the timesteps are 0.25 and 0.125 ms, using MuJoCo and Pinocchio. Five tension-only spring-damper stations per hand supply grip loads and grip dissipation. These are synthetic contact elements, not measured fingers or pressure distributions.

The observed maxima are 1.696 mm of tip bend and 0.00186 rad of twist; these are results, not the acceptance thresholds. The configured limits are 5% of the 1.08 m shaft length (54 mm) and 10 degrees of twist (about 0.175 rad). Two additional coarse-timestep probes that left the linear domain were excluded and reported separately; they do not reduce the accepted 384-trajectory grid.

The coupled-versus-rigid comparison uses four horizons: 4, 10, 25 and 50 ms. Its **384 comparison cells** count twelve states, two velocity branches, two timesteps, two engines and four horizons. This happens to equal the trajectory count, but counts something different. Repeated horizons, engines and timesteps are not independent golfer trials.

Only **126 of 384** comparison cells pass the post-registration load-and-work matching rule: 28 at 4 ms, zero at 10 ms, 40 at 25 ms and 58 at 50 ms. For each quantity a, b , the rule is

$$\frac{|a - b|}{\max\{f, (|a| + |b|)/2\}} \leq 0.05.$$

The floor f is 1 N for the largest station-force norm over the horizon and 10^{-6} J for grip-plus-shaft damping loss. When the mean magnitude falls below its floor, the absolute tolerances are 0.05 N and 5×10^{-8} J, respectively. These compare dissipation, not total grip boundary work. Selecting cells by

these two outcomes is a descriptive comparison; it does not match every load history or identify a causal shaft effect in a golfer.

Across those selected cells, the reported coupled-minus-rigid speed difference is -0.0285 to $+0.0212$ m/s. Both signs appear. Crucially, the [recording code](#) uses the norm of the club-frame translation rates, with origin at the modeled grip mass. This omits the rotational and deformation contributions to clubhead-point velocity. It is measured at fixed horizons, not at impact. Sometimes the flexible shaft is slower **in this translation-speed observable**; the quoted interval is not an impact clubhead-speed interval. At 50 ms alone, the matched range is -0.0208 to $+0.00634$ m/s.

A positive effect on this observable is therefore **rejected as a universal rule** within the registered synthetic comparison. A human clubhead-speed or equipment-fitting conclusion requires a different endpoint and calibrated data. The atlas also omits ground reaction and free moment, so it cannot establish a contribution from the feet.

Two other studies answer different questions. In the isolated beam reference, one-mode versus six-mode tip-displacement RMS disagreement rises from about 0.69 micrometres under an 80 ms pulse to 41 micrometres under a 4 ms pulse with a different force-and-moment input. That comparison motivates checking excitation and truncation; it is not a universal frequency cutoff. In the separate planar forward-modal baseline, maximum absolute modal tip displacement is **13.48% of shaft length**, above its **5%** screen. Its small contact-power residual verifies a coupling identity but cannot establish valid small-deflection dynamics outside the screen. The articulated atlas passing its own checks does not repair this distinct baseline.

15.2.17 Chapter Checkpoint

Before crediting shaft return, check the complete state, energy-transfer direction, damping and ledger closure. Test the specified endpoint across property uncertainty and independent models. Distinguish verification, domain screens and held-out validation. The current translation-speed result supports both signs of change; clubhead-speed and individual control/load tradeoffs still require calibrated human data.

15.3 Where the Picture Breaks

The ruler bends in one obvious plane. Shafts bend and twist in space, taper, vary in material, and interact with a distributed grip and clubhead. Classical Euler–Bernoulli bending neglects shear deformation and distributed beam rotary inertia; a tip body’s rotary inertia can still be included, as in this reference. Torsion requires its own model. Synthetic parameter recovery checks an inverse pipeline under its generating assumptions, not equipment validity.

The phrase “kick” can also imply a free late boost. Any returned energy was previously supplied and survived damping. A fair comparison counts the complete cycle.

Go Deeper: Read the [gravity, damping, and shaft contribution chapter](#), the [beam reference](#), and the [forward modal coupling](#).

Chapter 16

The Ground Is Part of the Conversation

16.1 A Concrete Picture

Jump from a small boat and the boat slides backward. Push against a wall and the wall returns a force through your body. Walking, throwing, and swinging all rely on external contact with the environment. The ground is not merely a stage; it is one boundary through which force and sometimes power enter the body.

Force plates make this interaction visible, but their curves are easy to overinterpret. A vertical-force peak does not identify which muscle created it. A center-of-pressure shift does not prove a unique segment sequence. The ground reaction is a measured boundary load. A model must reproduce it consistently with the measured motion, inertias, gravity, and other external loads; that consistency does not identify the muscles that produced the motion.

16.2 How the Mechanism Works

In an ideal constrained-contact model, ground reactions appear as multipliers enforcing the selected contact constraints. Solving the equations can decompose reaction at a fixed state into configuration/gravity, velocity-dependent, control, and external-load terms. The technical study defines ground-reaction versions of zero-torque and zero-velocity counterfactual forces.

16.2.1 The Reaction Reflects the Whole System

Accelerating the club changes body reactions. Body motion changes required foot forces. Gravity contributes. If the feet are constrained, the solver finds a reaction satisfying both motion and contact. A large ground force is not a separate motor delivering energy directly to the club; it is part of the coupled external interaction.

Ground power depends on motion at the physical boundary. For a rigid contact wrench referred to point O , in one observer frame,

$$P_g = \mathbf{F}_g \cdot \mathbf{v}_O + \mathbf{M}_{g,O} \cdot \boldsymbol{\omega}.$$

Modeled Net Reaction Does Not Uniquely Specify Foot Allocation

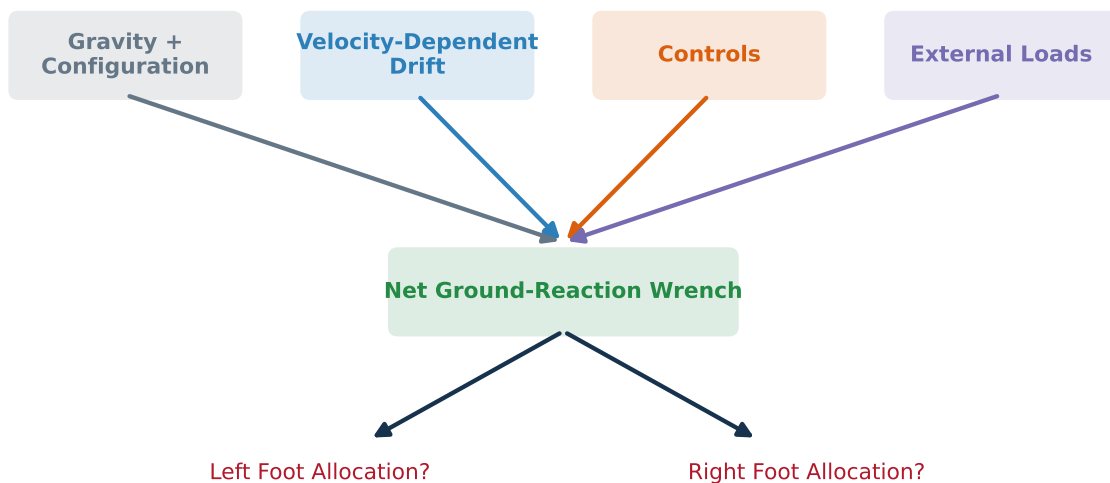


Figure 16.1: A Model Decomposes Net Reaction; a Summed Wrench Alone Leaves Foot Allocation Unresolved.

The moment and velocity must refer to the same point on the same rigid body. An ideal nonslipping rigid contact at rest in the chosen observer frame has zero boundary power despite large reactions. For a deforming foot, use traction dotted with local material-point velocity over the contact surface, including any applied contact couples in the chosen continuum model. A center-of-pressure path is a changing force-resultant location, not a material trajectory: its migration alone is not ground work. Internal muscle, joint, and tissue work can occur while the external ground port does zero work. A force plate measures wrench, not metabolic source.

16.2.2 ZTCF Family and ZVCF Ask Narrow Questions

At one achieved state, the ground-reaction ZTCF sets declared controls to zero and recomputes reaction. In the pinned technical study, ZVCF sets **both velocity and declared control to zero**, retaining configuration and declared other loads. Keeping control while zeroing velocity is the separate **zero-velocity control-preserved reaction**. These pointwise calculations do not advance a new trajectory or describe a person instantaneously stopping all motion.

In a fixed-support double-pendulum benchmark, pointwise ZTCF predicts the general reaction waveform but retains substantial amplitude error. Opposing control can make one ZTCF impulse exceed the total because contributions cancel. This is a useful warning against treating a component as a measured fraction.

16.2.3 Two Feet Add Allocation Ambiguity

A single net wrench under the body may be distributed across left and right feet in many ways, especially if both contact regions can support moments. Separate plates reduce ambiguity. Pressure insoles add spatial information. Yet foot models, friction, and coordinate transforms remain.

The same common/differential logic used for hands applies to a summed wrench: its net value does not specify its split. Separate six-axis plates directly resolve each foot’s aggregate wrench within measurement uncertainty, though neither plate uniquely resolves within-foot pressure or muscle co-contraction. Two feet do not automatically make every forward dynamics problem indeterminate; constraint rank and the contact constitutive law determine that question.

16.2.4 What Ground Data Can Test

Synchronized whole-body kinematics, separate force plates, club motion, and bilateral grip wrenches can test the predicted **total** reaction in timing, direction, and amplitude. Individual counterfactual components are not separately measured by a plate. Their attribution requires additional model assumptions and perturbation predictions. Residual forces and moments can expose inconsistencies in segment parameters or filtering. Participant holdout tests whether a fitted pattern generalizes.

Model Result: The fixed-support benchmark establishes a reproducible reaction decomposition with partial waveform agreement and explicit amplitude limitations.

Human Evidence: Han and colleagues studied 63 skilled men, with handicap at most 3, using optical motion capture and two force plates for driver, 5-iron, and pitching-wedge shots. Their abstract reports associations between selected force/moment peaks and maximum clubhead speed, not a muscle-level causal decomposition (Han et al. 2019). Ball and Best identified two major direction-of-hit COP clusters in 62 golfers; this does not establish two exhaustive patterns of two-dimensional foot pressure (Ball and Best 2007). A 2026 systematic review retained 24 studies, mostly cross-sectional, and identified inconsistent methods and remaining reliability and intervention questions (Watson et al. 2026). These studies measure different quantities and populations; none validates the present counterfactual decomposition in humans.

Hypothesis: State-dependent reaction components may explain part of observed ground-force timing without requiring each peak to be directly commanded, but full-body human data must test the prediction.

Practical Interpretation: The ground tells us what net external conversation occurred at the feet. It does not provide a transcript assigning every sentence to a muscle.

16.2.5 Force Plates Measure a Net External Wrench

A six-axis force plate reports three force components and three moments at its declared origin. Transform these to the support surface before deriving COP. For a planar surface $z = 0$, reaction on the golfer, and $F_z \neq 0$, $x_{\text{COP}} = -M_y/F_z$ and $y_{\text{COP}} = M_x/F_z$; the remaining $M_z - (x_{\text{COP}}F_y - y_{\text{COP}}F_x)$ is the surface-normal free moment. Declare axes and signs, and mask low-load intervals where the ratios are unstable. Separate plates preserve left–right aggregate allocation. Pressure insoles usually measure normal pressure; shear and a full wrench require additional sensing or a calibrated model.

The plate does not report muscle force, joint torque, or energy “from the ground.” It reports the reaction required at the contact boundary. To understand how that reaction relates to the club, synchronize it with body and club motion and solve the dynamics with uncertainty.

16.2.6 Large Force Does Not Necessarily Mean Large Power

At an ideal stationary rigid support, both terms in the wrench-power expression vanish. Body kinetic energy can still increase through active muscle work, gravitational potential decrease, or release of elastic storage. Which changes count as new mechanical input depends on the declared energy boundary. Force magnitude is not an energy-flow rate, and a kinetic-energy increase alone does not establish new active work.

This is similar to pushing a locked wall during a squat. The wall or ground provides a reaction that permits body acceleration, while chemical energy in muscle supplies work. Saying the golfer “gets energy from the ground” can be a useful cue, but it is not a precise mechanical ledger unless contact work is actually calculated.

16.2.7 A Whole-Body Reaction Walkthrough

Early in the downswing, changes in pelvis and trunk motion require horizontal, vertical, and rotational ground reactions. As arms and club accelerate, their momentum changes alter the reaction needed by the rest of the system. Late club acceleration can therefore appear in foot-force features even when no new foot command specifically targets the club.

The influence is reciprocal. Ground reaction enables changes in whole-body momentum that shape proximal motion, and distal acceleration changes the reaction required for balance and constraint. Cross-correlation of peaks cannot determine direction of causation. A forward model and perturbations are needed.

16.2.8 Pointwise Ground-Reaction Decomposition

For an autonomous holonomic contact model, write

$$\begin{aligned} M\ddot{q} + h_0 + h_v &= Bu + Q_e + J^\top \lambda, \\ J\ddot{q} + \gamma &= 0. \end{aligned}$$

Here h_0 is the declared velocity-independent bias, h_v the velocity bias, Bu the selected control, and Q_e other loads. With positive-definite M and J of full row rank, let $A = JM^{-1}J^\top$. A well-conditioned linear solve yields

$$A\lambda = JM^{-1}(h_0 + h_v - Bu - Q_e) - \gamma.$$

Holding configuration, contact rows, and the declared load conventions fixed gives an additive split $\lambda = \lambda_0 + \lambda_v + \lambda_u + \lambda_e$. ZTCF retains $\lambda_0 + \lambda_v + \lambda_e$; ZVCF retains $\lambda_0 + \lambda_e$ when $h_v(q, 0) = \gamma(q, 0) = 0$. Thus total minus ZTCF isolates the declared control term, and ZTCF minus ZVCF isolates the velocity term. **ZTCF and ZVCF overlap and are not additive parts of the total.**

Time-dependent constraints, velocity-dependent controllers or other loads, slip, and contact opening require recomputation under the actual intervention. They cannot inherit this fixed-operator shortcut without qualification. Compute COP from each reconstructed wrench; dividing each component separately does not produce an additive COP decomposition.

Components can cancel, so a counterfactual reaction may exceed total in one direction. That does not mean it contributed more than one hundred percent of a measured force. It means another component opposed it. Report signed waveforms, impulses, cancellation, and residual rather than forcing an additive percentage story.

16.2.9 Two Feet Create an Allocation Problem

The same net body wrench may be supported by different left and right contact wrenches within friction and balance constraints. Separate plates observe much of this allocation. If both feet occupy one plate, a model or pressure data must supply the split, and nonuniqueness should be shown.

Foot models also need contact assumptions. A fixed point, rolling contact, distributed pressure, and compliant shoe produce different reaction paths. The choice should match the measurement question. A mechanism that depends on an unrealistic fixed foot may not survive a richer contact rung.

16.2.10 Connecting Ground and Hand Ledgers

For the **body-plus-club** system before ball impact, ground, gravity, and any other external contacts or aerodynamic loads determine total momentum change. Hand-club interface loads are internal: include both sides with consistent points and signs, so their complete wrench cancels. For the body alone, the club-on-hand wrench is external; for the club alone, the opposite wrench is external. Do not add hand loads again to the combined-system external ledger.

About a fixed inertial origin O , $\dot{\mathbf{H}}_O$ equals the total external moment about O . If the reporting origin moves, include the transport term appropriate to the chosen angular-momentum definition. Segment powers and internal elastic/dissipative storage then connect the system ledger to the hand interface; action-reaction alone does not make internal work vanish.

This creates powerful error checks. If measured hand and ground loads combined with motion leave large residuals, segment inertias, filtering, synchronization, or unmeasured contacts may be wrong. A polished transfer story should wait until those residuals are bounded.

16.2.11 What Ground Manipulations Could Test

Researchers can compare stance width, lead/trail pressure constraints, surface friction within safe limits, or externally paced tempo while measuring the predicted hand and club mediators. The mechanism should specify whether changing a ground variable shifts proximal state, hand wrench, negative-couple interval, or only balance strategy.

A finding that ground-force peaks change without the predicted hand or club change would weaken a simple ground-to-club transfer narrative. A finding that hand-wrench timing mediates the relation would strengthen the chain. Participant holdout is essential because people may adopt different compensations.

16.2.12 Practical Boundary

Force-plate patterns can characterize an individual and expose constraints, but they should not become universal target traces. Similar club outcomes can arise from different foot allocations, and larger peaks can reflect braking or balance as readily as useful propulsion. The valuable question is how the measured external wrench fits the complete system ledger and whether a proposed change improves robust outcomes without unacceptable load.

16.2.13 Reader Lab: Close the External Wrench

Take synchronized body and club kinematics, segment parameters, separate foot wrenches, gravity, and any other external loads. Compute body-plus-club linear and angular momentum change and compare with integrated external wrench about a fixed inertial origin. Use hand measurements to close the body and club subsystems separately, without double-counting them in the combined system. Plot residual by direction and phase. A large residual near peak club acceleration may reflect synchronization, filtering, inertia, or omitted contact—not a new force.

Next sum the two foot plates and verify the result against a single combined wrench calculation. Shift reference point consistently. Compute center of pressure only when vertical force and moment assumptions make it well defined. Flag low-force intervals rather than allowing unstable ratios.

Finally, validate the total modeled reaction against the plates and run the declared pointwise counterfactuals. Show component timing as model attribution. A component-to-total difference is not itself a validation error: the other components are expected to contribute. Test counterfactual predictions through separately specified perturbations and held-out trials.

16.2.14 Competing Explanations for a Force Peak

A vertical peak can reflect accelerating the body center of mass, resisting downward motion, club reaction, balance, or a combination. A horizontal peak can turn or translate the system. A free moment can relate to rotational balance but does not map uniquely to pelvis or club torque.

Test proposed pathways with mediation and intervention. If a stance manipulation changes ground force but neither proximal state nor hand wrench as predicted, the simple transfer chain is unsupported. If all links change in order and a forward model predicts them on held-out trials, the explanation strengthens.

16.2.15 From an Observation to a Causal Claim

Suppose higher club speed is associated with a larger lead-side vertical impulse. That is an observation. A causal hypothesis might say the impulse changes pelvis and trunk state, which changes hand wrench and later club acceleration. The hypothesis requires each mediator and its timing. A model can predict the chain; a manipulation and synchronized data can test it.

Alternative explanations include stronger golfers producing both variables, different stance or club, balance response to the faster club, or processing choices. Randomized safe manipulations, within-person contrasts, and held-out prediction reduce these alternatives. The ground trace alone cannot.

16.2.16 What the Product Should Display

Show separate and summed foot wrenches, center of pressure only in valid intervals, whole-system momentum residual, predicted reaction components, and uncertainty. Link peaks to concurrent body, hand, and club state. Let readers toggle pointwise ZTCF and ZVCF components while explaining that they are model counterfactuals.

Do not rank a golfer from force magnitude alone. Normalize only when the scientific comparison justifies it and retain dimensional values. A transparent display should make cancellation, allocation ambiguity, and residuals as visible as the headline pattern.

16.2.17 A Final Ground Check

For a closed accounting of kinetic, gravitational, and elastic mechanical energy, any increase must be supplied by declared active work or external ports, with dissipation also recorded. Release of already-counted elastic storage redistributes that energy; it does not create it. If the support is compliant, specify whether its stored energy belongs inside the system or enters through an external port. This keeps everyday “use the ground” language tied to an explicit ledger.

16.2.18 What Happened When the Ground Was Actually Added

The archived finite-base experiment adds horizontal/vertical translation, pitch about world y , or both to an articulated model. Each pathway is paired with the fixed-base version at the same registered initial state. Its intrinsic **pitch moment** differs from a force plate’s usual surface-normal free moment. The support law is linear spring-damper mechanics, without two feet, unilateral ground contact, Coulomb friction, or measured plantar pressure.

The design comprises 12 states, four primary pathways, two signed initial velocity perturbations, two timesteps, and two native dynamics backends: $12 \times 4 \times 2 \times 2 \times 2 = 384$ primary trajectories. Two control families add 192 trajectories. Each is summarized at 4, 10, 25, and 50 ms, giving 1536 primary pathway summaries. The coupled-versus-fixed subset contains 96 trajectory pairs, each compared at four horizons. Consequently, it has **384 summary comparisons**, $12 \times 2 \times 2 \times 2 \times 4$, rather than 384 independent human trials or 384 distinct trajectory pairs. Engine and timestep variants are numerical replications of the modeled conditions.

MuJoCo and Pinocchio supply independent native dynamics operators but use a **shared contact law** and semi-implicit integration in this experiment. The maximum archived trajectory relative discrepancy is about 1.77×10^{-10} . The worst primary normalized energy residual decreases from about 0.01986 to 0.00995 when the timestep halves. These results support implementation agreement and refinement over the sampled domain; they are not independent validation of all contact physics. All primary and control numerical gates pass, while the separate load/work matching criterion fails.

16.2.19 What the Matching Criterion Actually Measures

The archive’s **declared primary criterion** requires both peak grip load and total dissipated work to match within 5%. Here load is the maximum single-station force norm over the interval, not the summed hand force. A separate timestamped prospective registration was not verified here. The work comparison is

$$\delta_W = \frac{|W_c - W_f|}{\max(10^{-6} \text{ J}, (|W_c| + |W_f|)/2)},$$

where c and f denote coupled and fixed support and W is nonnegative dissipated work. Peak load uses the same symmetric normalization with a 1 N floor. **Zero of 384** comparisons meet both criteria; 202 meet the load criterion alone. The work discrepancy ranges from 1.723 to 1.999. This does **not** mean twice as much energy. With positive work and no active floor, the symmetric discrepancy approaches 2 as the ratio diverges. The actual coupled/fixed work ratios range from **13.44 to 3519.40**. Small fixed-base dissipation makes a large ratio possible; the archived fixed and coupled ranges are approximately 4.46×10^{-7} –0.00290 J and 0.00157–1.872 J.

The sampled tolerance sweep finds no matches through 100% and all 384 at 200%. For nonnegative quantities this symmetric discrepancy cannot exceed 200%, so that last threshold imposes no meaningful matching requirement. It is not an estimated minimum tolerance, nor proof that unequal work is unavoidable in every possible ground model. Adding support damping changes dissipation directly and also changes motion, hence grip and shaft dissipation.

The supported conclusion is narrow: **no eligible comparison exists to estimate a speed benefit under the joint load-and-total-work matching criterion**. An empty matched cohort does not estimate a zero effect. That criterion must not be replaced silently after seeing the result.

16.2.20 A Total Simulated Contrast Is a Different Question

At the finer timestep and 50 ms, the coupled pathway has higher terminal club translation speed in all 48 summaries, by approximately 0.028–0.177 m/s. These are 12 states, two velocity signs, and two backends. The atlas has no active driver or joint torque; motion begins from the assigned state and perturbation, with initial mechanical energy matched across pathways.

Changing support alters the equations, energy redistribution, and losses. The resulting paired trajectory difference is a **total effect of that declared model intervention**. Dissipation is partly downstream of the support change, so equalizing it would ask a different question. Failing the matching screen does not erase the total simulated contrast; conversely, the contrast does not establish an equal-load/equal-work advantage, optimal coordination, energy supplied by stationary ground, or a causal benefit in golfers. The modeled endpoint is 50 ms translation speed, not impact speed or whole-swing performance.

A post-hoc sensitivity excludes ground damping from the matched-work quantity while retaining grip and shaft dissipation. It finds 60 matches: **48, 8, 4, and 0** at 4, 10, 25, and 50 ms respectively, across both steps and backends. Their speed differences range from approximately -0.00075 to $+0.01394$ m/s, with 20 positive and 40 negative. These are not 60 successful 50 ms comparisons. This alternative estimand remains exploratory and does not replace the primary result.

16.2.21 Initialization Is Part of the Experiment

A separate 4 ms diagnostic uses seven branches, three steps, and two backends: 42 trajectories. At the 0.125 ms step, three coupled-support initializations give the following peak ground forces and terminal club translation speeds:

Initialization	Force (N)	Speed (m/s)
Natural-Zero	32.8	0.264
Gravity-Only	565.5	1.908
Conditional Base	510.3	0.946

Natural-zero matches the registered initial coordinates and energy but is not static equilibrium. Gravity-only balance omits the grip preload contribution. Conditional equilibrium balances the three base coordinates while holding the remaining posture and club fixed; it is not equilibrium of every coordinate. The resulting initial storage and transients differ, so these numbers demonstrate initialization sensitivity, not three human standing strategies.

This archive also uses distributed central tension-only grip fibers. Their equal-and-opposite endpoint forces have zero combined pair moment. The noncentral vector-dashpot limitation discussed in Section 20 belongs to a different archived pathway and must not be attributed to these fibers.

16.2.22 Chapter Checkpoint

Ground reaction is external force evidence, not a muscle transcript or automatic energy source. Use it to close momentum, test contact models, and constrain the causal chain through body and hands. Report left/right allocation and model components without converting them into additive percentages.

And when someone shows you a speed difference between a moving base and a fixed one, ask what was held equal before the comparison was made.

A practical pattern becomes mechanistically meaningful only when its predicted mediators and outcome survive within-person manipulation and participant holdout.

16.3 Where the Picture Breaks

The wall and boat pictures use simple rigid contacts. Feet roll, deform, slip microscopically, and distribute pressure. Fixed-support pendulums omit balance, pelvis translation, and bilateral contact.

Correlation between force peaks and segment motion cannot establish causation. Timing can reflect shared state, feedback, or filtering. Reaction decomposition must be validated against synchronized data before becoming a human claim.

Go Deeper: Read the pinned monograph's [ground-reaction attribution](#), [finite-ground experiment](#), and [human falsification protocol](#).

Chapter 17

When the Base Moves, Every Shortcut Changes

17.1 A Concrete Picture

Stand on a stable floor and swing a pendulum from your hand. Then repeat while standing on a rolling board. The pendulum pulls you; your motion changes its path; the board moves beneath both. Prescribing your hand path from the first experiment and replaying it on the board would erase this feedback.

Many simple golf models prescribe a hub, shoulder, or hand trajectory. That is useful for asking what club forces are compatible with a measured path. It is less useful for asking how club reactions alter the body. A moving-base forward model turns the prescribed support into a dynamical participant.

17.2 How the Mechanism Works

In the simplest double pendulum, the shoulder may be fixed. In a prescribed-motion model, shoulder or hub kinematics are supplied as known functions of time. The solver computes joint reactions needed to realize them. In a coupled model, base coordinates join $\mathbf{q}$ and their accelerations emerge from force and inertia.

17.2.1 Prescribed Motion Is an Infinite-Authority Driver

A perfectly prescribed trajectory occurs regardless of reaction. Mathematically, the driver supplies whatever force and power are needed. This can hide a large back-reaction from the club. The computed wrench is still useful, but the model cannot predict how the body would deviate under finite strength or mass.

Calling prescribed motion “measured” does not remove this limitation. It makes the model an inverse or hybrid analysis of that trial. Causal strategy comparison needs a forward tier where changes in control can change the base state.

Back-Reaction Is Hidden When the Driver Cannot Move

Prescribed Base Coupled Finite-Mass Base



Figure 17.1: A Prescribed Base Follows a Fixed Path While a Coupled Moving Base Responds to the Club Reaction.

17.2.2 Coupled Inertia Creates Feedback

When base and club coordinates share one mass matrix, off-diagonal terms connect their accelerations. A club reaction can slow, redirect, or translate the base. Base motion changes hand velocity and therefore club power. The whole-system balance must include both.

The coupled base-flex model separates grip and shaft moments, includes convective terms, and uses a constrained solve. Same-state interventions remove selected drivers or reverse geometry while preserving the instant. Sensitivity sweeps ask whether the negative-couple mechanism survives plausible parameters.

17.2.3 Finite Mass Matters

An infinite-mass base barely responds; a light base moves strongly. Human body segments have finite, distributed mass and active control. A reduced moving hub does not become a torso, but it tests whether a mechanism depends on an immovable support.

The independent spatial test uses finite-mass hand carriages rather than direct club actuation. Paired compliant contacts transmit action-reaction. The late negative couple survives for a bounded interval in both engine realizations. This broadens the mechanism beyond one prescribed planar driver.

17.2.4 Whole-System Power Prevents Hidden Work

A prescribed base can inject or remove power implicitly. A forward base requires explicit drivers or external forces. Summing actuator work, gravity, damping, contact power, shaft storage, and kinetic-energy change exposes hidden sources. Timestep refinement distinguishes physical residual from numerical error.

17.2.5 A Hierarchy of Questions

Use fixed-base models for clean local coupling. Use prescribed motion to attribute forces along a measured path. Use a moving reduced base to test back-reaction. Use full-body models to represent spatial mass distribution and joint structure. Each tier should inherit common observables so the mechanism can be compared.

Model Result: The negative-couple and transfer findings survive a finite-base, forward constrained tier under registered controls, while quantitative values change.

Human Evidence: Measured hub paths and the modeled effect of inward hand-path motion show why base and handle translation cannot be treated as decorative details ([Nesbit and McGinnis 2009](#); [Miura 2001](#)).

Hypothesis: Human proximal control may arrange a moving base whose reaction with the club supports late transfer, but the reduced hub cannot identify the specific body coordination.

Practical Interpretation: If the club can push back on the golfer, a model that forces the hands along an unchangeable path answers only part of the story.

17.2.6 Why a Fixed Pivot Is Both Useful and Dangerous

Fixing the proximal joint makes the double pendulum easy to interpret. It removes translational base work and exposes arm-club coupling. Yet a golfer's hands travel because pelvis, trunk, shoulder girdle, arms, and wrists all move. The hand path has curvature, inward motion, elevation change, and out-of-plane components. Once those are important to the question, a fixed pivot can distort both force direction and energy flow.

The danger is not that every fixed-base result is false. It is that a mechanism may rely on compensation for an omitted base motion. The correct response is a model ladder: preserve the same observable definitions and controls while releasing the base in stages.

17.2.7 Prescribed Motion Versus Dynamic Motion

A prescribed base follows a measured or designed path regardless of reaction from the club. This is computationally convenient and can test how hand-path geometry influences club motion. Work needed to enforce the path is an external input and must be included in the energy ledger. The club cannot feed back to change the prescribed body trajectory.

A dynamic base has mass and equations of motion. Applied forces and torques, ground constraints, and club reactions determine its acceleration. Energy and momentum exchange are reciprocal. This is more physical but introduces control, contact, and parameter questions. Comparing prescribed and dynamic versions reveals which findings depend on one-way forcing.

17.2.8 The Moving-Handle Contribution

Clubhead velocity equals handle velocity plus the contribution of club rotation and deformation. Accelerating the handle can raise clubhead speed even before relative release. Pulling the handle inward can change angular momentum and effective geometry, a mechanism explored in golf models as parametric acceleration. Whether it helps depends on timing and the rest of the state.

Handle power can be calculated from the grip wrench and handle twist. A moving base adds translational power channels that a fixed hinge cannot represent. The negative-couple question must therefore be revisited: a locally negative rotational channel can coexist with substantial positive translational hand power.

17.2.9 Releasing the Base in Rungs

Rung one prescribes planar pivot translation. Rung two adds a finite-mass planar base driven by declared controls. Rung three adds pelvis and trunk rotation with ground constraints. Rung four introduces three-dimensional shoulder-girdle and arm motion. At each rung, calculate the same hand wrench, swing-normal couple, club power, work, and terminal outcomes.

The sign of a physical wrench projection should transform consistently. Exact magnitude and preferred onset can change. If the mechanism reverses under a small plausible base motion, the fixed-base conclusion is fragile. If it persists across a broad ensemble, it becomes a stronger candidate for human testing.

17.2.10 A Counterfactual Base Test

Copy one state from a dynamic simulation. In one branch retain proximal driver; in another remove it while retaining distal control. Observe immediate changes in base reaction, handle acceleration, club constraint force, and club angular acceleration. Then integrate both branches briefly. This traces a causal path from proximal input through moving base to distal response within the model.

Repeat with a prescribed base using the baseline path. If the club effect remains despite removal of dynamic feedback, it is mostly kinematic forcing. If it changes, reciprocity matters. Neither result is “more true” automatically; the comparison identifies the assumption carrying the conclusion.

17.2.11 Moving Bases and Timing

A clock-based wrist command can meet different geometry when base motion changes. A state trigger based on arm-club angle, handle radial velocity, or a coupling term may be more portable. But a trigger chosen after inspecting results can overfit. Candidate triggers and outcome windows should be declared before held- out evaluation.

Base motion also changes the location and orientation of the late impact zone. Alignment by impact alone can hide meaningful differences earlier. Reports should show both absolute time and state-space trajectories, with uncertainty in event detection.

17.2.12 Human Measurements Required

Whole-body motion should include pelvis, trunk, scapular/shoulder proxies, arms, hands, and club. Separate force plates and instrumented grips constrain external and interface wrenches. Segment inertias and joint centers carry uncertainty. High-speed club tracking and shaft strain add the distal state.

Soft-tissue artifact is especially important at the shoulder girdle. A moving- base model should not claim anatomical precision beyond its measurement. It can use a generalized handle driver for mechanism testing while explicitly reserving muscle and scapular interpretation.

17.2.13 Practical Meaning

The useful lesson is not a command to pull the handle along a particular path. It is that club release cannot be optimized independently of how the hand base moves. Geometry, speed, and force at the handle jointly determine club response. A strategy should be evaluated across realistic base variations and control objectives rather than on one fixed-pivot animation.

17.2.14 Reader Lab: Prescribe, Then Release

Record or design one handle path and drive a club model with it as a prescribed boundary. Calculate the external work needed to enforce that motion. Then build a finite-mass base driven to approximate the path. Compare club trajectory, handle wrench, body reaction, and energy. Differences reveal the effect of dynamic reciprocity.

Run an inward-handle-path perturbation at matched state. Hold distal torque fixed and observe radial/tangential reaction, relative release, and clubhead speed. Repeat with changed club inertia. If the outcome reverses, the proposed path effect is parameter-dependent and should be mapped, not prescribed.

Finally compare a clock-scheduled distal command with a state-triggered one across varied base tempo. A state policy may retain performance better, or it may overfit an unobservable model variable. Report both the performance and whether the trigger has a measurable proxy.

17.2.15 Common Modeling Trap

Prescribing a measured body path while also assigning body torques can double-count control: the path is already enforced by whatever unreported force is needed. Decide whether motion is input or outcome. If prescribed, report its constraint wrench and work. If dynamic, show how controls and ground contact produce it.

This distinction is crucial when attributing energy to proximal drivers. A prescribed path can inject exactly the energy required and make distal transfer look inevitable. A dynamic base exposes the energetic cost and feedback.

17.2.16 A Minimal Moving-Base Benchmark

Use one finite-mass base translating in a plane, one arm link, and one club link. Publish mass and inertia, applied base force, joint torques, initial state, and constraints. Verify momentum and energy. Compare with a prescribed trajectory that matches baseline base motion. This small benchmark can be implemented in multiple engines and isolates reciprocity before a whole human model is added.

Run the two-hand geometry controls on both versions. The physical contact-couple sign should respond consistently even if reaction magnitudes differ. Run proximal killswitches and show how base, hand, and club variables diverge. These tests create a clear bridge from the fixed double pendulum to spatial whole-body dynamics.

17.2.17 Why This Matters for Strategy

An arm or scapular action changes both force and the moving base from which the club is carried. A wrist-focused model that prescribes hand path may hide the work needed to create that path; an

arm-focused model may underrepresent distal regulation. A balanced comparison includes both controls and their reaction on the base.

The useful strategy question becomes: which allocation produces a favorable handle state and club interaction across realistic body variability, with acceptable load and control? The moving-base rung makes that question computable without pretending it is already answered for a golfer.

17.2.18 Chapter Checkpoint

Ask whether the base motion is prescribed or dynamically produced, where its work enters, and whether club reaction can feed back. Compare both formulations on shared observables. A mechanism that survives this comparison has moved beyond the most consequential fixed-pivot shortcut.

17.3 Where the Picture Breaks

The rolling board has one translation. A human base has pelvis and trunk motion, joint limits, balance, ground contact, and active feedback. A finite carriage is a falsification bridge, not anatomy.

Forward models also require control choices. A wrong controller can make a good mechanism look poor. Comparisons must separate mechanism failure from controller failure and report both.

Go Deeper: See the [coupled base and compliance chapter](#) and the [independent spatial contact test](#).

Chapter 18

How a Constrained Forward Model Solves the Next Instant

18.1 A Concrete Picture

A navigation app knows where a car is, how fast it moves, road rules, engine commands, and map geometry. To predict the next second, it cannot choose position directly. It computes how forces and constraints change acceleration, then advances velocity and position. If the predicted car drifts through a barrier, the model violated a constraint or took an unreliable step.

A forward swing model performs a more demanding version of that calculation. At every tiny step it balances inertia, gravity, velocity effects, controls, contact reactions, and flexible states, then checks that joints and contacts remain closed.

18.2 How the Mechanism Works

The mass matrix $\mathbf{M}(\mathbf{q})$ says how strongly each generalized coordinate resists acceleration and how coordinates are coupled. The vector $\mathbf{h}(\mathbf{q}, \dot{\mathbf{q}})$ collects gravity and velocity-dependent terms. Controls and external loads form generalized force $\mathbf{Q}$. Constraints add reaction $\mathbf{J}^T \lambda$.

The KKT system solves $\ddot{\mathbf{q}}$ and λ together. This is crucial: reaction force is whatever enforces the declared acceleration-level closure, not an independent input selected for a preferred result.

18.2.1 Integration Turns Local Rules Into a Trajectory

Acceleration updates velocity; velocity updates configuration. Numerical methods approximate this continuous evolution with finite steps. A step too large can miss an impact, distort a fast shaft mode, or inject energy. A stable-looking animation is not numerical validation.

The study repeats key cases at smaller timesteps. Physical results should converge while work-energy and constraint residuals decrease. Event detection locates impact consistently rather than accepting the nearest coarse sample.

A Forward Solver Predicts, Advances, and Checks Every Step

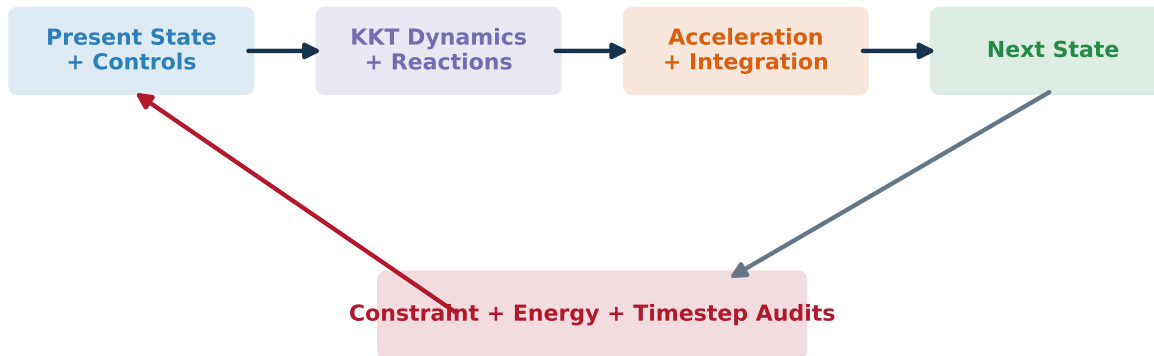


Figure 18.1: A Forward-Dynamics Pipeline Turns State and Control Into Accelerations and Reactions, Integrates the State, and Audits Constraints and Energy.

18.2.2 Constraint Stabilization Is Not Free

Roundoff and integration error can let closed contacts drift apart. Projection or stabilization corrects the state toward the constraint manifold. Corrections must be reported because large projections can hide an unstable solver. The forward tests set tolerances for position closure, velocity closure, and maximum projection.

18.2.3 Inputs Need Dynamics Too

An instantaneous torque step is mathematically convenient and physiologically harsh. First-order command filters, torque-rate limits, and activation states make inputs finite. Sensitivity to rise time tests whether a result depends on a single discontinuous sample.

The preload study goes further by evolving internal transmission state before the transition. The uncertainty study includes delayed actuators. These additions do not create a complete muscle model; they challenge timing claims with more realistic state continuity.

18.2.4 Registered Observables Keep the Solver Honest

The forward two-hand model reports club trajectory, speed, hand reactions, equivalent wrench, common and differential modes, interface powers, actuator work, energy residual, and constraint closure. A result is rejected if one key contract fails even when club speed looks attractive.

Negative controls alter contact separation, moment-arm sign, driver commands, gravity, damping, or stiffness. The solver must reproduce the qualitative and quantitative limiting behavior expected from mechanics.

18.2.5 Inverse and Forward Models Complement Each Other

Inverse dynamics begins with measured motion and estimates required forces. It is strong for trial-specific attribution but vulnerable to noise and contact nonuniqueness. Forward dynamics begins with force/control models and predicts motion. It supports interventions but depends on controller and parameter quality. Agreement between them on common observables is more informative than choosing one as universally superior.

Model Result: The constrained forward tiers pass registered closure, work-energy, negative-control, and timestep tests for their declared synthetic models.

Human Evidence: Existing three-dimensional golf models show both the value and the strong assumption load of forward simulation (MacKenzie and Sprigings 2009; Balzerson et al. 2016).

Hypothesis: Interaction-force mechanisms that persist under forward integration and geometry controls are better candidates for human experiments than effects seen only in inverse replay.

Practical Interpretation: A forward model earns trust by showing its homework: equations, state continuity, small steps, closed constraints, balanced energy, and failed cases where the mechanism predicts failure.

18.2.6 The Solver’s Job at Every Time Step

A constrained forward model begins with state and control. It evaluates mass, gravity, velocity-dependent, elastic, damping, external, and applied-control terms. Constraint equations say which relative motions are permitted. The solver then finds accelerations and reaction multipliers that satisfy dynamics and constraints together. An integrator advances positions, velocities, and internal states to the next instant.

This loop repeats thousands of times. Small sign, frame, or indexing errors can create plausible-looking motion while corrupting forces or energy. Contracts at each layer are therefore as important as the final animation.

18.2.7 Coordinates and Constraints

Minimal generalized coordinates avoid explicit holonomic constraint multipliers for simple joints but can make complex contacts and spatial frames harder to manage. Maximal or Cartesian coordinates describe body poses more uniformly and enforce joints with constraints. Spatial-vector formulations organize rigid-body dynamics efficiently. Independent engines using different formulations provide valuable cross-checks.

No formulation is inherently the truth. They should agree on frame-invariant observables when representing the same physical model. Differences point to implementation, tolerance, or specification mismatch that must be reconciled before interpretation.

18.2.8 Initial Conditions Must Satisfy the Model

Positions should satisfy joint and contact constraints. Velocities should lie in the constraint tangent space. Flexible modes and controller states need declared initial values. Starting from an inconsistent state can create large artificial reaction impulses that resemble interesting interaction forces.

A projection step can place measured or designed states on the constraint manifold, but projection changes them. Record the correction and reject cases whose adjustment exceeds tolerance. For human data, measurement uncertainty should be reconciled with dynamics rather than silently forced into perfect constraints.

18.2.9 Work-Energy and Momentum Audits

The change in total modeled mechanical energy should match integrated external and control power minus declared dissipation. Constraint work should follow the contact model: zero for ideal stationary constraints, nonzero for moving or compliant contacts. Linear and angular momentum balances provide separate checks when external wrench is known.

Audits should run on every major model rung and counterfactual. A branch that removes torque changes power, but it should not suddenly create an unexplained energy residual. Report maximum constraint violation, balance residuals, integration tolerances, and refinement convergence with each result.

18.2.10 Building a Killswitch Correctly

At intervention time, copy the complete state. Modify only selected future input. Do not reset velocities, controller integrators, or shaft modes unless those are the explicit intervention. Use the same solver settings and event handling. Store both trajectories with a branch identifier and intervention manifest.

For pointwise decomposition, do not advance state. Re-evaluate the right-hand side under baseline and counterfactual inputs at the same state. This difference should close against a declared term decomposition. Keeping the two procedures separate prevents a forward divergence from being mislabeled as instantaneous force attribution.

18.2.11 Contact and Friction

Ideal bilateral constraints can transmit whatever reaction is needed. Human hands and feet have finite friction, compliance, and moment capacity. A richer model checks whether reconstructed loads remain within plausible contact cones and pressure limits. If not, the assumed no-slip constraint is infeasible.

Complementarity or compliant contact introduces switches and numerical stiffness. Event detection, regularization, and timestep can affect results. The negative-couple mechanism should be tested across reasonable contact representations rather than tuned to one solver's smoothing.

18.2.12 Verification Before Validation

Verification asks whether code solves the equations intended. Use analytic limits, manufactured solutions, finite-difference derivative checks, timestep studies, conservation cases, and independent engine comparison. Validation asks whether the intended equations represent measured reality. It uses calibrated data, uncertainty, held-out predictions, and residual structure.

A model can be verified and invalid. It can also fit data while being implemented incorrectly through compensating errors. Publication must show both tracks and avoid treating visual resemblance as either.

18.2.13 Reproducible Output Contracts

Every run should record code commit, model version, parameter manifest, control schedule, initial state, solver and tolerance, random seed, and output schema. Derived figures should be reproducible from machine-readable results. Units, frames, and sign conventions belong in the schema, not only the prose.

The open workbench should invoke the same canonical engines used by tests rather than copying equations into a second educational implementation. That reduces drift and lets a reader move from a casual experiment to the exact scientific artifact.

18.2.14 What Forward Simulation Can Establish

It can demonstrate that a mechanism is dynamically possible under declared assumptions, quantify its effects, expose tradeoffs, and make falsifiable predictions. It cannot prove that observed golfers used the simulated controls, that unmeasured muscles produced generalized torques, or that an optimum is safe. Those boundaries make the simulator a stronger scientific resource, not a weaker sales demonstration.

18.2.15 Reader Lab: Break the Solver on Purpose

Reverse one sensor axis, introduce a small initial constraint violation, enlarge timestep, remove damping, and shift one input channel in time. Each defect should trigger a predictable gate: frame closure, projection correction, convergence, energy behavior, or synchronization. If outputs remain silently “valid,” the product is not fail-closed.

Then perform an independent implementation of one benchmark. Share only the specification and input manifest, not internal derivative code. Compare state, reaction, power, and energy residual. Agreement in animation with disagreement in wrench is not enough because this project interprets forces.

Finally replay a stored run from exact commit and seed. Regenerate tables and figures and compare hashes or numeric tolerances. Reproducibility should be a test, not a claim in the methods paragraph.

18.2.16 Interpreting Solver Failure

A failed integration can indicate numerical stiffness, infeasible contact, singular geometry, or a physically extreme control. Retrying with looser tolerances until motion appears is not an acceptable default. Classify failure, retain the case in ensemble statistics, and determine whether the model or numerics needs revision.

Failure regions are scientifically valuable. They reveal where an ideal constraint demands impossible reaction, where contact should switch, or where a control policy lacks robustness. The public tool should visualize these regions rather than omitting them from the success rate.

18.2.17 Model Contracts as Scientific Sentences

“Mass matrix is symmetric positive definite in the qualified state region” is both a test and a statement about kinetic energy. “Constraint violation remains below tolerance” says simulated contacts mean what the prose claims. “Wrench power is invariant under reference-point transform” protects the energy story. “The geometry reversal changes the couple sign” encodes the proposed mechanism.

Writing such contracts before implementation gives the solver an explicit job. Tests then protect scientific meaning as code evolves. A regression that passes visual inspection but violates one contract is caught before publication.

18.2.18 Extending the Engine Safely

When adding a shaft, moving base, or spatial joint, preserve old benchmarks and add new limiting cases. Version schemas rather than silently changing column meaning. Rebuild all derived artifacts and compare qualified observables. Record intentional changes in a release note and evidence manifest.

This discipline supports multiple front ends and repositories. The canonical physics library owns equations and contracts; educational pages consume its versioned outputs. Duplication is minimized, and a casual explanation cannot quietly drift from the tested engine.

18.2.19 What a Reviewer Needs

Provide a small runnable benchmark, exact command, expected tolerances, and human-readable trace. Include failed-case examples. A reviewer should not need the full optimization stack to verify core dynamics. This modularity is one of the most important ways to move the project from an internal modeling package to an open scientific resource.

18.3 Where the Picture Breaks

The navigation picture has known roads. Human joint constraints and controls are uncertain. A numerically impeccable solver can integrate the wrong anatomy or controller. Verification checks code against equations; validation checks model predictions against reality.

Contact models are particularly consequential. Rigid, compliant, frictional, and distributed contacts can change internal forces. Results should be qualified by the contact family tested.

Go Deeper: Read the [forward constrained two-hand model](#), the [solver and platform methods](#), and the open [qualification instructions](#).

Chapter 19

A Ladder of Models, Not One Final Model

19.1 A Concrete Picture

A map can show a continent, a city, a building, or the pipes inside one room. The city map is not false because it omits plumbing. The plumbing drawing is not better for planning a cross-country trip. Scale and detail should match the question.

The proximal-distal project uses a model ladder for the same reason. A double pendulum isolates coupling. A two-hand model exposes wrench redundancy. A flexible shaft adds storage. A moving base adds back-reaction. Spatial and full-body tiers add frames, contacts, and independent engine realizations. Each rung should answer a declared question and preserve common observables.

19.2 How the Mechanism Works

The bottom rung should be analytically transparent. Exact equations, limiting cases, and pointwise decompositions reveal why a mechanism appears. Higher rungs ask whether it survives added realism. If the sign reverses or effect vanishes, the model boundary becomes a scientific result rather than an embarrassment.

19.2.1 Common Observables Are the Handrail

Coordinates differ across tiers. A planar wrist angle has no one-to-one match in a spatial model with two wrists and a flexible shaft. Comparing raw coordinates would confuse model architecture with mechanism.

Instead, every tier reports a club wrench at a declared reference, interface power, club motion, relevant work integrals, and closure residuals. These are the handrail connecting rungs. Added internal detail can be reported without losing the shared comparison.

Confidence Rises Only When the Evidence Changes Kind

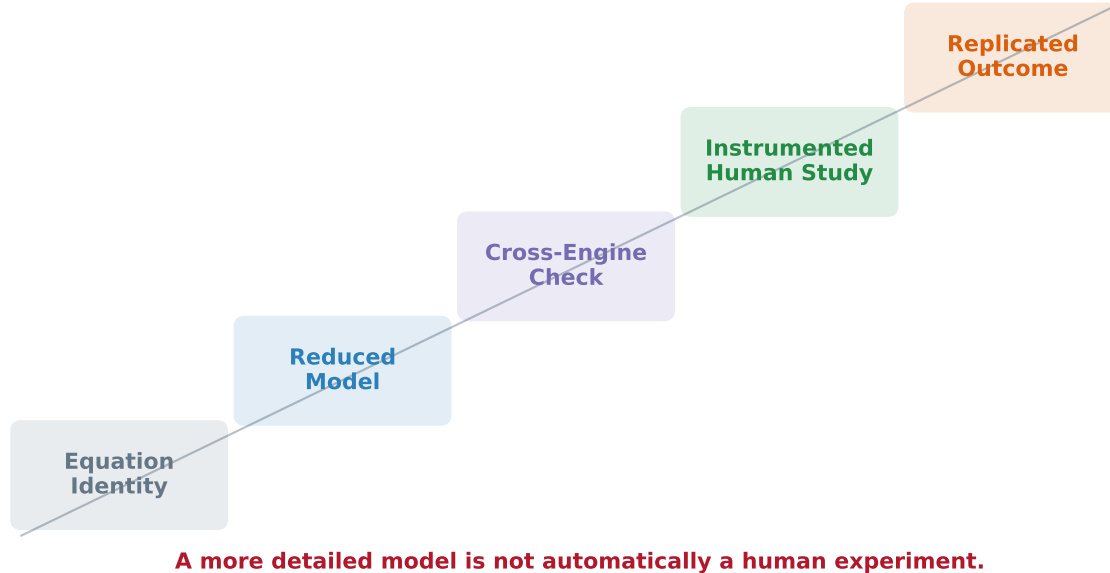


Figure 19.1: A Model Ladder Adds Two Hands, Flexibility, Moving Base, Spatial Motion, and Full-Body Structure While Keeping Common Wrench and Power Observables.

19.2.2 Complexity Must Pay Rent

Adding degrees of freedom increases parameter uncertainty and nonuniqueness. A full-body model can look lifelike while allowing many internal solutions. If it does not improve a relevant observable, falsifier, or uncertainty estimate, its complexity has not paid rent.

The distributed shaft is justified when higher modes change the impact-window wrench or energy beyond tolerance. The moving base is justified when back- reaction changes the mechanism. Spatial contact is justified when out-of-plane geometry or wrench components matter. The question comes first.

19.2.3 Surviving and Changing Claims

Some statements are structural. Action-reaction, wrench transformation, and work-energy closure should survive every tier. Some are model-specific: the exact duration of a negative couple, optimal torque onset, or numerical force magnitude can change with parameters and degrees of freedom.

The release matrix labels what each tier supports, omits, and must not claim. This prevents a result from climbing the ladder merely because a later model exists elsewhere in the repository.

19.2.4 Cross-Formulation Agreement

Inverse, forward, reduced, and spatial formulations can share a common state or observable. Agreement under independently implemented equations reduces the chance that one code path created the effect. Disagreement helps locate hidden assumptions.

The independently executed spatial contact test is especially useful because MuJoCo and Pinocchio

do not share one dynamics implementation. Their agreement on bounded mechanism observables strengthens numerical credibility, though both still represent the same reduced physical idea.

19.2.5 The Ladder Ends at an Open Gate

No model rung becomes a human trial. The top computational tier still needs measured kinematics, bilateral hand wrenches, ground reactions, shaft behavior, and activation. The experimental protocol is not an optional future decoration; it is the next rung.

Model Result: The negative-couple and interaction-force mechanisms survive several rungs with changing quantitative values and explicit failures under geometry controls.

Human Evidence: Available golf measurements constrain motion and some net kinetics but do not yet supply the synchronized dataset needed to validate every common observable.

Optimization with higher-degree-of-freedom golf models illustrates how rapidly the design space grows as anatomical and club freedoms are added ([McNally and McPhee 2018](#)).

Practical Interpretation: Ask for the smallest model that can answer today’s question and the next larger model capable of proving that the answer was not an artifact of the simplification.

19.2.6 Why No Single Model Is “the Model”

Every model answers a question by omitting something. A double pendulum isolates linked inertia. A two-hand constraint model exposes force allocation. A flexible shaft adds stored energy and phase. A whole-body spatial model adds moving base and ground interaction. A musculoskeletal model asks about candidate muscle allocations. None supersedes all earlier models; each changes the balance between interpretability, fidelity, and identifiability.

The ladder works when observables and interventions remain aligned across rungs. If every model uses a different definition of transfer, comparison is impossible. This project carries forward hand wrench, swing-normal couple, interface power, work, club state, energy closure, and declared counterfactuals.

19.2.7 Rung Zero: Algebra and Geometry

Before integrating motion, test wrench arithmetic, sign conventions, moment-arm dependencies, and coordinate transforms. Symbolic or static examples should recover known limits exactly. This rung catches errors that a complex animation can conceal.

It also states propositions clearly: collapsing hand separation collapses a pure differential couple; a consistent reference-point transform preserves wrench power; equal-and-opposite contacts can have zero net force and nonzero moment. These are mechanics identities, not human findings.

19.2.8 Rung One: The Planar Double Pendulum

The fixed-base two-link model tests interaction dynamics, carried motion, relative release, and torque timing. Its small state permits dense parameter sweeps, analytic checks, and transparent pointwise decompositions. It cannot represent bilateral hands, spatial axes, ground contact, or shaft deformation.

A result promoted from this rung should be qualitative and mechanism-focused. Exact speed gains or optimal milliseconds are provisional. Negative controls and sensitivity maps determine which signs or orderings are stable enough to carry upward.

19.2.9 Rung Two: Two Hands and a Moving Base

Add separated contact points and either prescribed or dynamic proximal motion. Now the model can reconstruct common and differential hand modes and test the geometric negative-couple hypothesis. A finite-mass base restores reaction from club to body. Contact remains idealized and anatomy generalized.

This rung asks whether the mechanism survives when the very feature it concerns —two hands—is represented explicitly. It also reveals nonuniqueness: the same club wrench can correspond to many internal contact allocations.

19.2.10 Rung Three: Flexible Club

Add one or more shaft modes, damping, and calibrated boundary coupling. Repeat the same counterfactuals while including modal state and energy. This rung asks whether rigid-club conclusions were artifacts of omitting deformation and whether negative coupling interacts with shaft return.

Modal convergence and independent beam checks matter. A single arbitrary spring can generate an attractive delay without representing a club. Parameters should come from measured or explicitly synthetic sources and remain qualified.

19.2.11 Rung Four: Spatial Whole-Body Dynamics

Add three-dimensional pelvis, trunk, shoulder/arm, hand, and club motion with ground contact. Spatial frames permit out-of-plane moments and full wrenches. The number of parameters and controls grows sharply, so identifiability becomes a central result rather than an inconvenience.

Independent engine agreement is valuable here. Matching trajectories is not enough; constraint reactions, energy, and wrench projections should also agree. A mechanism surviving only one software stack remains suspect.

19.2.12 Rung Five: Biological Allocation

A musculoskeletal or activation-aware model proposes how generalized loads might be shared among muscles and passive tissues. It can study co-contraction, stiffness, and internal load, but it needs EMG, strength, anatomy, and optimization assumptions. Multiple allocations usually fit the same motion.

This rung should not retroactively turn earlier generalized arm and wrist drivers into named muscles. Instead it tests candidate mappings and reports their nonuniqueness. Scapular action, in particular, requires adequate kinematics and measurement-aware shoulder-girdle representation.

19.2.13 Promotion and Demotion Rules

A claim is promoted when it passes implementation verification, negative controls, uncertainty tests, and the next rung's relevant physics while retaining a coherent observable prediction. It is demoted

when sign or mechanism depends on a removed assumption, fails closure, or cannot be distinguished with available measurements.

Demotion is not failure of the project. It locates a boundary: “this effect is specific to a fixed-base rigid planar model” is scientifically useful. The reader can then see what remains established, what is model-conditional, and what human evidence still owes.

19.2.14 Cross-Rung Tables and Figures

For each mechanism, a comparison table should list degrees of freedom, contact model, shaft model, controls, identified parameters, pass/fail gates, and common outcomes. Figures should overlay normalized time histories with consistent signs, not juxtapose unrelated axes. Differences in magnitude should remain visible rather than normalized away when magnitude is the question.

Machine-readable manifests make this comparison reproducible. Each row links to input, output, code commit, and generated figure. Reviewers can rerun one rung or replace it with an independent implementation without adopting the whole software ecosystem.

19.2.15 Deciding When Complexity Is Worth It

Add a degree of freedom or state when it changes a target observable, explains a structured residual, enables a discriminating experiment, or removes an unacceptable assumption. Do not add complexity merely for visual realism. Unidentifiable complexity can make a model less falsifiable by allowing it to fit anything.

The final modeling product should therefore remain modular: a transparent core, optional higher-fidelity rungs, shared contracts, and explicit qualification. That architecture invites criticism and extension as an open resource rather than presenting one branded engine as final authority.

19.2.16 Reader Lab: Write a Promotion Matrix

Choose one claim, such as a late negative swing-normal contact couple. Across rows list algebraic geometry, planar double pendulum, two-hand moving base, flexible shaft, spatial whole-body, and biological allocation. Across columns list sign, onset, duration, magnitude, work, outcome effect, negative controls, uncertainty, and human evidence. Populate every cell with result, not applicable, or unresolved.

The matrix makes a common error obvious: a claim may be well verified in early rungs but unresolved in humans. It also reveals where complexity changes the result. Link each cell to machine-readable evidence and tests.

Now define promotion criteria before looking at the next rung. Perhaps sign must hold in 95 percent of a declared ensemble and geometry controls must pass, while magnitude can vary. If criteria fail, narrow the claim. This avoids silently moving the goalposts as realism increases.

19.2.17 Independent Extension

Invite a second implementation to reproduce one rung using the published contracts. If it disagrees, preserve both outputs and reconcile specification, frames, and numerics. The goal is not to defend an engine. It is to discover which physical conclusion survives independent expression.

A modular open resource makes such extension inexpensive. Shared schemas and benchmarks let a contributor add a contact or anatomical rung without rewriting the manuscript’s conceptual core.

19.2.18 A Claim Can Live at More Than One Rung

The identity that wrench power is reference-point invariant belongs to the mechanics rung and should hold everywhere. The finding that a particular negative-couple interval appears belongs to specific model rungs. The hypothesis that golfers use it belongs to the human-evidence rung. Separating them prevents a human null result from invalidating an identity or a verified identity from being offered as human validation.

Each public statement should therefore carry rung and status. Readers can see which parts are durable foundations and which remain contingent. New evidence updates the appropriate layer without rewriting the entire framework.

19.2.19 Completion Is a Maintained Ladder

The ladder is complete enough for release when every rung has a specification, verification, shared observables, limitations, and a documented route to the next test. It does not require maximal anatomical detail. Future rungs can be added when evidence and questions justify them.

Maintaining simple benchmarks alongside rich models prevents regression and keeps the project teachable. The lay book, technical monograph, simulator, and evidence register should all reference the same ladder identifiers.

19.3 Where the Picture Breaks

The map analogy suggests clean nested scales. Model families are not always nested; a spatial engine may use different contact or damping laws rather than simply adding detail. Common observables and negative controls are needed to compare them fairly.

More agreement also does not guarantee truth if all tiers share a wrong premise. Independent measurement remains decisive.

Go Deeper: Explore the monograph’s [common-observable ladder](#), [model completion matrix](#), and [reviewer workbench](#).

Chapter 20

From a Plane to Three Dimensions and Independent Engines

20.1 A Concrete Picture

A shadow puppet can reveal a hinge mechanism while hiding motion toward the wall. A planar golf model is a carefully chosen shadow. It can expose in-plane forces and swing-normal moments, but it omits motion out of the plane, moments about axes within the plane, changing swing plane, and spatial joint axes.

Moving to three dimensions is not merely drawing the same links with perspective. Finite rotations generally do not commute, moments require declared frames and reference points, and constraints can contain new internal modes. A six-component wrench record does not mean that every contact can exert six independent loads.

20.2 How the Mechanism Works

In a plane, one angle and one signed moment describe each hinge. In space, body orientation belongs to the rotation group $SO(3)$. Angular velocity is a vector, and a moment's components depend on frame. Applying rotations in different orders can yield different orientations.

20.2.1 Spatial Wrenches Preserve the Physical Contract

A three-dimensional wrench records force $\mathbf{F}$ and moment $\mathbf{M}_O$ about a specified point O . For $\mathbf{r} = \overrightarrow{OQ}$, expressing the same loading at another point Q of a rigid body gives

$$\begin{aligned}\mathbf{M}_Q &= \mathbf{M}_O - \mathbf{r} \times \mathbf{F}, \\ \mathbf{v}_Q &= \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}, \\ P &= \mathbf{F} \cdot \mathbf{v}_O + \mathbf{M}_O \cdot \boldsymbol{\omega} \\ &= \mathbf{F} \cdot \mathbf{v}_Q + \mathbf{M}_Q \cdot \boldsymbol{\omega}.\end{aligned}\tag{20.1}$$

All velocities here describe the same rigid motion for the **same observer**. Rotating the components of

A Plane Reveals One Projection and Hides Others

Planar Projection

Spatial Wrench and Changing Axes

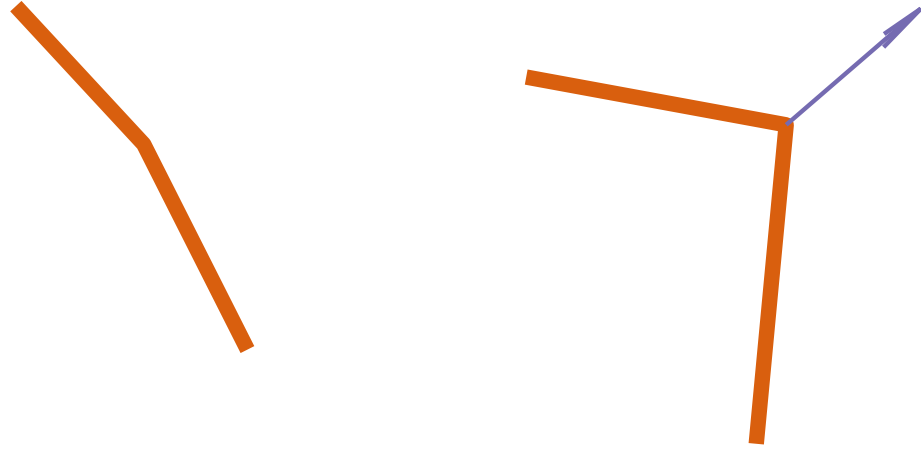


Figure 20.1: A Planar Projection Hides Directions That a Spatial Model Must Resolve.

every vector into another orthonormal basis preserves these dot products. The wrench components themselves change. Likewise, shifting the moment reference requires the matching change in rigid-body velocity; using the old velocity with the new moment generally changes the reported power.

Changing the observer is a different operation. An observer translating with constant velocity $\mathbf{U}$ measures $\mathbf{v}'_O = \mathbf{v}_O - \mathbf{U}$ and $P' = P - \mathbf{F} \cdot \mathbf{U}$. External power and kinetic energy need not have the same value for different observers. Rotating observers require additional relative-motion terms in dynamics. Expressing laboratory-observed velocities in club-aligned axes does not by itself make them club-relative velocities.

A point-force contact has no independent contact moment: its moment about O is fixed by its position and force. Friction, unilateral separation, and contact area further restrict admissible loads. A distributed grip may transmit a resultant moment, but its six recorded components still obey the contact model. For a deforming shaft, one rigid wrench–velocity pairing does not capture all deformation power; sum the actual contact powers and account for internal energy.

This frame-explicit contract is the bridge from the planar negative couple to a spatial test. The swing-normal component can be compared while additional components reveal additional twisting or bending loads.

20.2.2 Full Body Adds Redundancy

A spatial full-body model has many coordinates capable of producing similar club motion. Inverse dynamics may estimate a net generalized-force vector for a prescribed trajectory, but two-hand contact and muscle redundancy remain. More coordinates do not automatically identify more biology.

The common-state spatial model therefore emphasizes action-reaction, wrench closure, and matched

interventions rather than claiming a unique human strategy. It reports what survives and what remains open.

20.2.3 Contact Closure Comes Before Contact Dynamics

A constraint Jacobian can have full row rank even when the points it would constrain are far apart. The project tested that distinction with six deterministic subject-scaled engineering profiles, three grip spans, and 61 prescribed states per case. The anatomical hand points miss the declared grip points by 0.171–0.616 m, and none meets the registered 5 mm tolerance.

The same open configurations still produce a rank-six bilateral contact Jacobian. That is not a contradiction. Rank asks whether six local correction directions are locally available without coordinate bounds; closure asks whether a correction is still required. Rank alone does not guarantee a finite correction within joint limits or without collision. Calling a rank test a contact-feasibility result would therefore hide the most important geometric failure.

This adverse result narrowed the next model rather than weakening the scientific program. A bounded subject-scaled solver then closed both hands on the grip while respecting its declared joint and collision guards. It held the club at 13 prescribed samples over 0–0.24 s, while allowing pelvis, trunk, and arm coordinates to change. All **234 of 234** profile–span–sample cases passed the **0.5 mm** closure gate, with no stored contact error above 1.16×10^{-10} m. That numerical residual is not an anatomical measurement accuracy. A deliberately unreachable 2.0 m grip span failed, preserving a negative control.

That success is a reduced-tree necessary condition, not proof that a person can adopt the poses. The limits are broad engineering guards, collision clearance uses coarse spheres, and the model omits scapular motion, forearm rotation, multi-axis wrists, fingers, and distributed grip contact. The spatial tier still cannot answer whether a passive mechanism reduces timing demand, creates useful slack, improves delivery, or specifies a human technique. Subject-specific anatomy and calibrated compliant forward contact are the next gates.

20.2.4 Scapular Mobility Changes Reachability Without Identifying Strategy

The next screen makes a paired comparison within a different feasible set. It holds the trunk and club pose fixed, then compares fixed shoulder centers with shoulder centers that can move on a declared thoracic ellipsoid. Four screening coordinates per side represent protraction, elevation, upward rotation, and winging. In this surrogate, the first two move the shoulder center; the latter two change joint orientation. This is a reduced geometry experiment, not a validated anatomical shoulder. Its arm-only solve is not a rerun of the 234-case solve that allowed trunk motion.

The fixed branch closes **0 of 54** stored states. The mobile branch reaches the 0.5 mm residual tolerance in **31 of 54**. The optimizer reports successful termination in **39 of 54**, but only **16 of 54** satisfy both criteria: fifteen closed cases lack successful termination, and twenty-three successfully terminated cases remain open. Twenty-eight states touch a declared bound, and an intentionally impossible 2.0 m grip span remains open. Mobility clearly changes modeled reach, but unresolved cases may reflect range choice, omitted coordinates, or numerical termination rather than human infeasibility.

There is a second warning. Both contact Jacobians retain rank six, while local coordinate nullity rises from two to ten. Many scapular and glenohumeral coordinate changes therefore leave the contact residual unchanged to first order in the unconstrained local linearization. At an open pose this is not

yet a closed-contact manifold, and active joint bounds restrict admissible directions. Contact geometry alone cannot tell a reviewer which internal allocation a person used or should use.

The scientific next step is not to rename these coordinates as muscles or coaching actions. It is to repeat the comparison with a validated articulated shoulder, subject-specific bone geometry where governed, calibrated distributed grip contact, and paired forward dynamics. Those additions support tests of force, work, passivity, and delivery with increasing anatomical relevance. Contact geometry alone supplies none of those results or a human strategy.

20.2.5 Independent Engines Test Implementation Dependence

Running the same conceptual experiment in two engines tests the implementation parts that differ, provided observables use a common convention. Report which model construction, contact laws, derivatives, and numerical updates are shared. Matching screenshots prove little. Matching trajectories, wrenches, powers, constraint residuals, and negative controls is stronger.

The archived finite-mass hand-carriage experiment calls native MuJoCo `mj_forward` and Pinocchio articulated-body forward dynamics, both version 3.8.0. It uses a shared Kelvin–Voigt point-interface law and a shared semi-implicit update after native acceleration evaluation. It therefore compares rigid-body dynamics and force mapping; it does not independently test the engines’ native contact solvers or integrators. Neither branch commands the club directly.

The driver killswitch is a genuine forward intervention: starting from the same state at 0.180 s, it removes the grounded hand-carriage driver while retaining gravity, inertia, and interface storage and damping. In contrast, the coincident and reversed moment-arm controls reuse achieved forces **without advancing a new trajectory**. They test the moment calculation, not the later motion of a physically altered grip.

20.2.6 Angular Momentum Is a Separate Check

Equal and opposite forces can still produce a net moment when their application points differ. In the archived vector spring–damper law, let $\mathbf{h}$ be the hand position, $\mathbf{p}$ the club contact point, and $\mathbf{d} = \mathbf{h} - \mathbf{p}$. The club receives $\mathbf{F} = k\mathbf{d} + c\dot{\mathbf{d}}$ and the hand receives $-\mathbf{F}$. With both positions measured from one origin, the pair contributes

$$\begin{aligned}\tau_{\text{pair}} &= \mathbf{p} \times \mathbf{F} + \mathbf{h} \times (-\mathbf{F}) \\ &= -c\mathbf{d} \times \dot{\mathbf{d}}.\end{aligned}\tag{20.2}$$

The spring term cancels, but transverse relative velocity makes the damping forces noncentral. The power ledger can still show nonnegative dissipation. Thus force balance and passivity alone do not establish angular-momentum balance for a closed internal interface. A declared support reaction or contact-couple pathway is needed to account for this moment, or the constitutive model must change.

Direct calculation from the archived traces gives a peak norm of the summed two-interface pair moment of about 0.083 N m. The native adapters apply no additional contact couples, and the recorded ground-pathway proxy accounts for the driver. This is a shared modeling limitation, not a discrepancy between the two engines. Its effect on the proposed club-moment mechanism requires a new, governed comparison; the existing agreement does not quantify that effect. The [provider follow-up](#) tracks this

unresolved moment accounting. The archived sign and power results remain results of the declared engineering model, not validation of an isolated biological grip.

20.2.7 Agreement Has a Boundary

If two engines agree on a reduced spatial mechanism, code-specific error becomes less likely. Shared parameter errors, oversimplified contact, and missing anatomy remain possible. Cross-engine agreement is verification evidence, not human validation.

20.2.8 Three-Dimensional Measurement Is Hard

Marker-based motion capture suffers soft-tissue artifact, especially around the scapula and forearm. Joint centers and axes are estimated. Clubface orientation requires careful rigid-body tracking. Bilateral grip sensors must be registered to the club frame. These uncertainties should enter the model-data comparison.

Model Result: The archived carriage model retains a negative swing-normal projection of its point-force moment after the driver is removed in both native dynamics realizations. The moment-arm identities also pass their algebraic controls. This projection is about the declared club reference; it is not an origin-independent pure couple when net contact force is nonzero.

Human Evidence: Measured trunk and pelvic motion has three-dimensional structure ([Meister et al. 2011](#); [Joyce et al. 2013](#)). Those kinematic studies do not test this specific bilateral compliant-contact mechanism. McPhee reviews the broader modeling and measurement methods ([McPhee 2022](#)); neither that review nor agreement between reduced models supplies participant validation for this mechanism.

Hypothesis: The planar swing-normal couple is one projection of a spatial wrench pattern that should be measurable without requiring a unique muscle allocation.

Practical Interpretation: A planar model can explain a shadow mechanism; a spatial model must show which parts remain when the lights move.

20.2.9 What the Plane Captures

A planar model captures rotation in one chosen swing plane and forces projected into it. It can represent link inertia, relative release, radial/tangential reaction, and an out-of-plane moment. Because signs and geometry are easy to see, it is an excellent place to derive and falsify the basic negative-couple mechanism.

The plane is not a claim that a golf swing is flat. It is a controlled slice. Projecting a spatial trajectory is not a closed planar dynamics model: omitted motion can alter inertia, reactions, and forces within the displayed plane. Imposing planar constraints may itself introduce reactions absent in free space. The key question is whether a spatial extension retains the corresponding physical projection when out-of-plane motion and extra axes are admitted.

20.2.10 What Three Dimensions Add

Each rigid body now has spatial orientation and angular velocity. The club's moment is a vector. Hand loading can be recorded as a six-component wrench at each declared reference, subject to the contact restrictions above. Swing-plane normal can vary with time and must be defined operationally. Forearm

rotation, wrist deviation, shaft droop, torso side bend, and shoulder-girdle motion introduce pathways absent from the plane.

These freedoms can redistribute a planar effect rather than merely add to it. A moment that looked entirely negative in one axis may split across changing axes. Preserve the complete wrench, its reference point, and the observer used for power. A passive coordinate rotation changes components while preserving the physical loading and scalar power; choosing a different physical projection axis changes the reported component. The swing-normal projection remains useful if its construction is reported and tested for sensitivity.

20.2.11 Frames Must Be Registered

Laboratory, pelvis, trunk, arm, hand, club, sensor, and swing frames all move differently. A general spatial wrench transform includes rotation and a moment shift; either part can be trivial when axes or reference points already coincide. Angular velocities require consistent representation; Euler-angle derivatives are not the angular-velocity vector. These are common sources of plausible but wrong signs.

Frame tests use known loads and orientations, action–reaction at contacts, reference-point invariance of power, and round-trip transforms. Coordinate conventions should be machine-readable. A reviewer should be able to reproduce the swing-normal axis from released kinematics.

20.2.12 Defining the Late Impact Zone

“Near impact” can be a clock window, a range of club position, a state-space region, or a phase normalized between events. A spatial study should declare the primary definition and test alternatives. If the negative-couple interval appears only under one post hoc window, confidence is low.

State-based regions may include clubhead height, hand path, shaft orientation, and arm–club geometry. They allow comparisons across tempo but can introduce selection bias if tuned on outcomes. Freeze them before participant holdout.

20.2.13 Independent Engines

Implement the same spatial mechanism in two dynamics engines or formulations. Share model parameters, initial state, control histories, and output schema. Compare trajectories, hand wrenches, constraints, energy, and the projected couple within tolerances. Differences should be investigated, not averaged.

Independence is strongest when implementations do not share derivative code or hidden convention helpers. Agreement then reduces the chance that one algebraic or frame error created the result. The released comparison should include both successful agreement and known deviations.

20.2.14 A Spatial Geometry-Control Suite

These are proposed extension and measurement tests, beyond the two archived same-force moment-arm identities. Reverse the defined swing-normal axis and verify only the reported projection changes sign, not physical power. Swap hand positions in the model and verify the differential couple responds. Collapse separation, align forces with the separation vector, and rotate the swing plane. Perturb sensor-frame orientation to quantify sign confidence.

These tests distinguish coordinate behavior from physical mechanism. If a purported negative couple disappears whenever the reporting plane shifts a few degrees, its interpretation should include that uncertainty rather than a binary label. At a fixed reference point, if two unit normals differ by angle θ ,

$$|\mathbf{M} \cdot (\mathbf{n}' - \mathbf{n})| \leq 2\|\mathbf{M}\| \sin(\theta/2). \quad (20.3)$$

A projection close to zero can reverse sign under a small angular perturbation even when the moment vector is accurately known. For example, a moment $(1, 0, -0.01)$ N m is negative along $(0, 0, 1)$ but positive after that normal tilts one degree toward the positive first axis. A sign probability additionally needs a specified uncertainty distribution; a collection of plausible axes alone gives a sensitivity range, not a probability.

20.2.15 Moving Anatomy Without Overclaiming It

Spatial arm segments permit richer kinematics, but a ball joint at the shoulder does not represent scapulothoracic motion or muscle recruitment automatically. Marker-based shoulder centers and scapular orientation have substantial measurement uncertainty. A generalized shoulder driver remains an abstraction.

The model can predict a required hand wrench or joint-level generalized load. Mapping that to scapular retraction, protraction, or named muscle action requires additional kinematics, physiological measurements, and a biological model. EMG constrains activation hypotheses but does not uniquely identify muscle force. Keeping that boundary explicit protects the very hypothesis the work aims to test.

20.2.16 Quantities That Should Survive the Extension

Whole-system energy balance for a declared observer, contact action–reaction, consistent wrench–velocity power under passive coordinate changes, and geometric control responses should survive exactly or within numerical tolerance. Qualitative timing and sign of the proposed contact contribution may survive over a declared region. Exact optimal controls and magnitudes are expected to change.

The spatial model earns its complexity if it makes new falsifiable predictions: out-of-plane force components, axis-dependent couple timing, or conditions under which the planar mechanism fails. If it merely draws a more realistic golfer while hiding the causal tests, it has moved away from science.

20.2.17 Human Validation Path

Use synchronized full-body and club motion, bilateral grip wrenches, separate force plates, and shaft deformation. Calibrate every frame and report soft-tissue and sensor uncertainty. Evaluate participants held out from parameter tuning. Compare predicted three-dimensional wrench trajectories, not only clubhead speed.

Residuals should be inspected by phase and direction. A systematic out-of-plane residual may identify missing wrist, shaft, or shoulder-girdle physics. Updating the model in response should be declared as a new rung and retested on untouched participants.

20.2.18 Reader Lab: Rotate the World

Take one spatial state and apply one constant proper rotation to every position, orientation, velocity, force, moment, gravity vector, and boundary/input definition. Express inertias and constraints consistently, then recompute dynamics. This is a passive relabeling for the same observer. The rotated trajectory and the same scalar energy and wrench power should match. A quantity tied to the declared swing frame should transform predictably. Failure reveals a frame-dependent implementation error.

Next project the same contact moment onto several plausible swing-plane normals. Report sign confidence and angular sensitivity. If all definitions agree over the late interval, the negative-couple claim is robust to plane construction. If they disagree, publish the dependence and prefer the complete vector result.

Finally compare two independent spatial engines on geometry-reversal and hand- separation controls. Match initial state and input at high precision. Differences in constraint stabilization or quaternion handling should remain below declared observable tolerances.

20.2.19 Spatial Visualization Without Illusion

A three-dimensional rendering can make a model look biologically authoritative. Include visible joint definitions, hand contact frames, force-vector scale, uncertainty, and model-rung label. Let readers rotate the scene and inspect the numeric state. Do not use a detailed human mesh to imply anatomical fidelity not present in the equations.

The best spatial figure pairs an intuitive view with declared observables: wrench power for a named observer, projected moment with axis sensitivity, and closure residual. Beauty should make the mechanism easier to audit, not harder to question.

20.2.20 A Minimal Spatial Report

Publish body and joint frames, orientation representation, state definition, inertias, constraints, controls, solver, and swing-plane construction. Report complete three-dimensional hand wrenches before selected projections. Include energy, constraint, and engine-agreement residuals. Provide geometry-control animations with fixed vector scales and downloadable data.

For the negative couple, show the moment vector in club, laboratory, and swing frames, the angle between vector and plane normal, and sign probability under frame uncertainty. Report the interval by both time and state. This makes a spatial claim reviewable rather than visually persuasive only.

20.2.21 When the Planar Result Fails in Space

Failure can occur because out-of-plane motion redirects the contact wrench, because a different axis dominates, because hand geometry changes, or because the planar constraint created an artificial reaction. Diagnose which term changes. A spatial failure is not proof that coupling is irrelevant; it narrows the specific projection or validity region.

The revised explanation might say that a negative total swing-normal couple is not robust, while one component or state-dependent regulation effect remains. Such refinement is exactly why the model ladder exists.

20.2.22 Accessibility of a Three-Dimensional Result

Pair an interactive scene with two-dimensional explanatory slices and textual alt descriptions. A phone reader should understand hand positions, arrows, and sign without rotating a model. A specialist can open the interactive view and data. Providing both prevents visual sophistication from becoming an access barrier.

20.2.23 Chapter Checkpoint

The spatial extension succeeds when it preserves invariant balances, survives frame and engine tests, and either confirms or precisely narrows the planar mechanism. More degrees of freedom alone are not progress. New, measurable predictions and exposed failure regions are.

For a reviewer, the full wrench vector remains primary. A convenient negative projection should never conceal uncertainty in the axis used to define it.

20.3 Where the Picture Breaks

The shadow analogy makes spatial extension look visually obvious. In fact, coordinate conventions, rotation representations, and contact frames can create subtle sign errors. Every comparison needs automated transformation and power checks.

Engine agreement can also become false reassurance if one result is tuned to the other. Parameters and tolerances should be registered before comparison, with held-out cases and negative controls.

Go Deeper: Read the [spatial common-state formulation](#), the [native forward-dynamics comparison](#), and the project's [reproduction boundary](#). These links identify the reviewed provider revision; proposed validation steps above are not claims that those studies have already been completed.

Chapter 21

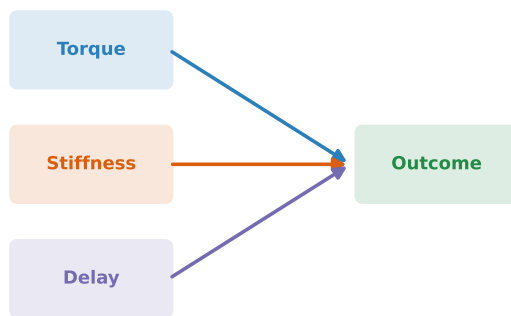
Sensitivity and Identifiability

21.1 A Concrete Picture

Taste a soup and notice it is salty. That observation may be sensitive to the amount of salt, but it cannot tell whether salt was added directly, broth was reduced, or a salty ingredient was included. Several recipes can produce the same taste. Sensitivity asks how the taste changes when an ingredient changes. Identifiability asks whether tasting the soup lets us recover the recipe.

Golf models face the same distinction. Club speed may be sensitive to shoulder torque and wrist timing. Yet one speed trace may not uniquely identify those inputs because shaft stiffness, hand path, damping, contact allocation, and inertial parameters can compensate.

Sensitivity: What Moves the Outcome?



Identifiability: Can We Recover the Recipe?

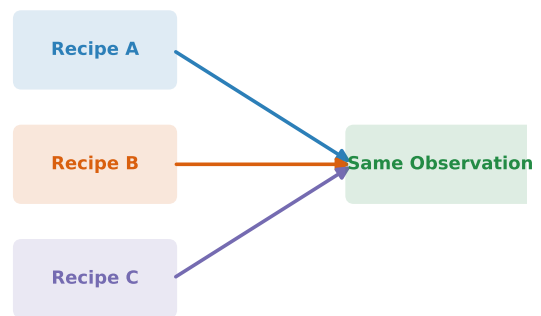


Figure 21.1: Sensitivity Shows Which Inputs Move Each Outcome, While Identifiability Shows Which Inputs Can Be Recovered From the Available Observations.

21.2 How the Mechanism Works

Local sensitivity changes one parameter slightly near a reference and estimates an outcome derivative. Global sensitivity varies many parameters across declared ranges and apportions outcome variance. A parameter can be locally quiet yet important through interactions elsewhere in the range.

21.2.1 Structural and Practical Identifiability

Structural identifiability asks whether perfect, noise-free observations under the model could uniquely recover parameters. Practical identifiability includes finite data, noise, limited excitation, and correlated effects. A parameter may be structurally identifiable and practically hopeless.

The coupled study varies 12 inputs simultaneously but observes six summary quantities. The full parameter vector cannot be identified from that information. This is not a solver failure. It is an information limit.

21.2.2 Hidden Hand-Force Modes

Even perfect net club wrench leaves an individual-hand-force mode unresolved. No amount of optimizing club trajectory can recover a quantity absent from the observations. Bilateral sensors add the missing information; a preferred regularizer merely selects one possible allocation.

21.2.3 Sensitivity Is Outcome-Specific

An input can strongly affect speed and weakly affect face/path, or vice versa. Ranks should therefore be reported by outcome. Combining everything into one score hides tradeoffs and lets arbitrary weights select the answer.

21.2.4 Uncertainty Should Travel to Conclusions

Parameter ranges come from measurement, literature, or explicit synthetic design. Samples propagate those ranges through the model. Reporting medians, tails, and failure rates is more honest than one nominal trajectory. Held-out samples test whether a chosen program generalizes beyond the cases used to select it.

21.2.5 Identifiability Guides Measurement

The value of analysis is not only declaring uncertainty. It identifies which new observations would separate competing explanations: bilateral hand wrenches, shaft strain, activation timing, or a richer time series instead of six summaries. Experiment design can target sensitive and discriminating states.

Model Result: The 12-input coupled design shows practical nonidentifiability, one unresolved two-hand force mode, and outcome-dependent sensitivity.

Human Evidence: Methodological heterogeneity and participant variability in golf research make parameter recovery more difficult, not less ([McPhee 2022](#)).

Hypothesis: Bilateral wrench and shaft measurements will distinguish arm- wrist allocation mechanisms that club motion alone cannot identify.

Different Coordination Histories

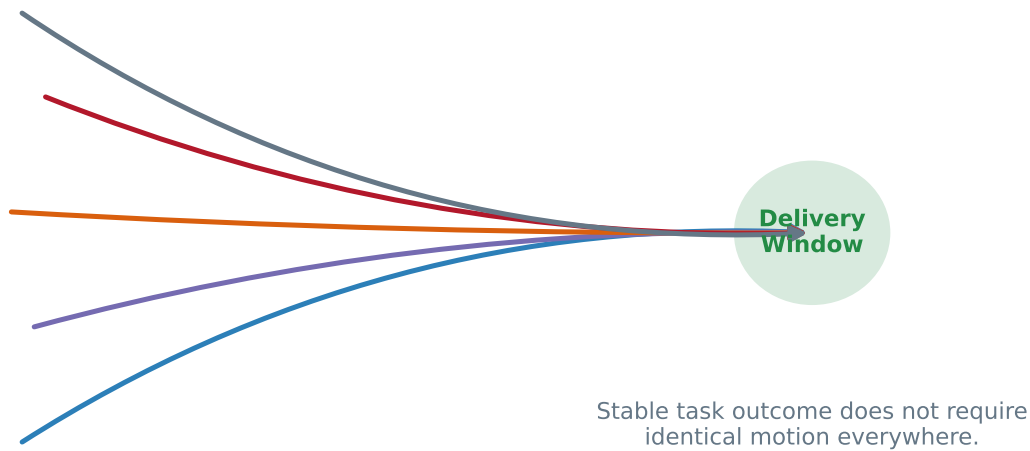


Figure 21.2: Many Internal Paths Can Reach One Club-Level Outcome, Which Is Why Motion Alone Cannot Recover the Hidden Allocation.

Practical Interpretation: A result can change strongly when a parameter changes while the observed result still cannot tell us which parameter changed.

21.2.6 Start With the Question, Not the Method

Sensitivity asks what changes when an input changes. Identifiability asks what inputs can be learned from observations. Optimization asks which inputs best meet an objective. Calibration asks which inputs make a model resemble data. These questions use related mathematics but produce different claims. A parameter can be influential yet unidentifiable, or identifiable yet irrelevant to the output of interest.

For the proximal-distal product, write each question explicitly: Which inputs alter late hand couple? Can bilateral wrench data distinguish arm and wrist allocation? Which timing policy is robust across uncertain inertia? Can shaft stiffness be estimated from strain and motion? This prevents one global ranking from being treated as an answer to all four.

21.2.7 Local Sensitivity

Near a nominal parameter vector, perturb one input by a small amount and estimate the derivative of an outcome. Automatic differentiation, analytic sensitivities, or carefully checked finite differences can compute it. Scaling matters: a derivative per kilogram cannot be compared directly with one per millisecond. Use nondimensional or uncertainty-scaled effects.

Local derivatives reveal sign and steepness around one case. They can miss thresholds, sign reversals, and interactions elsewhere. A dead-zone transmission model is a clear example: sensitivity may be nearly zero while disengaged and large near engagement. Multiple nominal states and directional

perturbations are needed.

21.2.8 Global Sensitivity

Define justified ranges and joint distributions, sample them, and propagate each case through the model. Variance-based indices, elementary effects, or surrogate models can identify main and interaction effects. The result belongs to the chosen range and distribution. A broad arbitrary range can exaggerate a parameter; a narrow convenient range can hide it.

Outcome-specific reporting is essential. Arm inertia might dominate terminal speed variance while hand separation dominates couple magnitude and command rise time dominates face/path sensitivity. A single importance score would erase the design tradeoff.

21.2.9 Structural Nonidentifiability of Two Hands

Suppose only the net club wrench is observed. Adding one internal hand-force mode and subtracting it from the other can preserve the same resultant. No noise-free club trajectory can recover that hidden mode without additional assumptions. This is structural nonidentifiability, not a shortage of optimizer effort.

Regularization can choose a minimum-force or minimum-effort allocation, but the choice is a prior. Report it as one candidate. Bilateral sensors add observables that collapse much of the null space and transform the scientific question from speculation to estimation.

21.2.10 Practical Identifiability in a Rich Model

Even theoretically identifiable parameters can be strongly correlated in finite noisy data. A more flexible shaft can mimic timing delay; a different hand path can mimic torque onset; inertial error can be absorbed by estimated control. Posterior distributions, profile likelihoods, or bootstrap intervals reveal these ridges better than one best fit.

Excitation matters. If all recorded swings occupy nearly the same state region, the data cannot distinguish mechanisms that diverge elsewhere. Safe equipment or task manipulations can increase information by changing hand separation, shaft response, tempo, or load geometry.

21.2.11 A Worked Ambiguity

Imagine terminal speed rises one metre per second when either wrist-torque onset moves five milliseconds or shaft stiffness increases ten percent. One speed number cannot tell which changed. Add a shaft-strain waveform: stiffness affects modal phase differently from command timing. Add bilateral wrench: onset changes contact history. The extra observations make candidate explanations separable.

Now suppose both candidates still fit because motion filtering uncertainty shifts phase. The correct result is a joint confidence region, not an arbitrary winner. Experiment design should target a phase or manipulation where their predictions separate beyond measurement uncertainty.

21.2.12 Sensitivity of a Sign Claim

For the late negative couple, calculate not only mean magnitude but probability of negative sign across parameter and measurement uncertainty. Report onset and duration distributions. Identify the

parameters most associated with sign failure. Geometry reversal and zero-separation controls should remain decisive across the ensemble.

If sign confidence is only slightly above chance, use qualified language. If sign is robust but magnitude uncertain, the mechanism claim can be stronger than the performance claim. Distinguishing those levels is more informative than one binary validation stamp.

21.2.13 Participant-Held-Out Tests

Fit or select policies on one set of participants and freeze them before evaluation on another. Split by person, not swing, because repeated trials from one golfer share anatomy and strategy. Report individual outcomes and population distribution. A model that predicts the average but fails most individuals is not an individualized product.

When data are limited, leave-one-participant-out evaluation and hierarchical models can help, but uncertainty will remain wide. Honest width is preferable to leakage that creates impressive but non-reproducible accuracy.

21.2.14 Measurement Value as a Design Output

Use the sensitivity and identifiability analysis to rank candidate sensors by which hypotheses they discriminate. Bilateral wrench may be essential for hand allocation; shaft strain for elastic state; high-quality shoulder-girdle motion for scapular geometry; EMG for activation timing; perturbation response for effective stiffness.

The best next measurement is not always the most precise. It is the one that separates live hypotheses at acceptable cost and burden. This turns uncertainty analysis into an experimental roadmap rather than a disclaimer.

21.2.15 What to Publish

Release parameter definitions, units, ranges, distributions, correlation assumptions, sample design, convergence diagnostics, sensitivity estimator, outcome definitions, and raw indices. Include failure cases and plots showing nonlinear response surfaces. Link every summary back to a machine-readable table. Reviewers can then test alternate priors without rerunning the entire argument from prose.

21.2.16 Reader Lab: Design One Discriminating Measurement

Select two parameter or mechanism combinations that fit current club motion. Simulate candidate observations: bilateral wrench, shaft strain, ground reaction, or EMG timing. Plot where predictions differ relative to expected sensor noise. Choose the observation and phase with the largest reliable separation and design a bounded trial around it.

After collecting data, evaluate the predeclared discriminator on held-out trials. Do not refit both models until they agree and then claim neither was rejected. If data fall between predictions, quantify whether both are wrong or uncertainty was understated.

Repeat the exercise for the arm/wrist allocation extremes. Club speed may be nearly identical while differential hand force and transition stiffness differ strongly. This demonstrates how identifiability analysis directly determines instrumentation priorities.

21.2.17 Communicating an Unidentifiable Result

Do not say “the parameter could not be estimated” and stop. Show the family of values or combinations consistent with data, the observables they predict, and the measurement that would shrink the family. If a practical conclusion remains constant across it, that conclusion may still be robust. If the conclusion changes, the product should withhold recommendation.

This is one of the clearest ways to expose falsifiability: uncertainty is represented as competing predictions a future experiment can separate.

21.2.18 A Minimum Identifiability Report

List every estimated parameter and observed channel. State structural nonidentifiabilities, priors, bounds, correlations, noise model, calibration data, and holdout design. Plot posterior or profile relationships, not only standard errors around one optimum. Show predictive coverage on untouched data.

For hand allocation, display the null-space dimension with and without bilateral sensing. For preload, show correlation among transmission gap, activation delay, shaft dynamics, and onset. For robustness, show which parameter uncertainty changes policy ranking. These plots reveal what the current product can and cannot personalize.

21.2.19 When More Data Do Not Help

Collecting more repetitions of the same uninformative motion reduces random noise but may not separate structurally equivalent mechanisms. Richer excitation or a different sensor is needed. Conversely, adding many noisy sensors can increase processing assumptions without adding discriminating information.

Experiment design should therefore maximize expected distinction between live hypotheses under realistic noise and burden. The result is a ranked measurement plan tied directly to completion of model claims.

21.2.20 Conservative Product Behavior

When inputs are unidentifiable and conclusions vary across compatible values, the interface should say so and avoid a unique strategy result. It can show the family, sensitivity, and next useful measurement. If all compatible values lead to the same qualitative conclusion, it can report that robustness while retaining magnitude uncertainty.

21.2.21 Chapter Checkpoint

An influential parameter is not necessarily recoverable, and a fitted parameter is not necessarily causally important. Show both maps. When ambiguity affects a decision, withhold the decision and name the measurement that would resolve it.

21.3 Where the Picture Breaks

Soup ingredients are independent items; mechanical parameters interact through equations and may change meaning across models. Sensitivity indices depend on chosen ranges and distributions. Narrow

ranges can make an important mechanism look unimportant.

Identifiability within a wrong model is also not truth. Data may precisely select a parameter that compensates for omitted physics. Residual structure and cross- model validation remain essential.

Go Deeper: Read the monograph's [uncertainty and identifiability chapter](#) and the [machine-readable study outputs](#).

Chapter 22

Timing Is a State Question, Not Just a Clock Question

22.1 A Concrete Picture

A cook does not remove bread from an oven simply because twelve minutes passed. The timer is useful, but color, temperature, and structure tell whether the loaf reached the intended state. Two ovens or loaf sizes reach that state at different times.

Swing instructions often use clock language: start the release later, apply a torque earlier, wait until a position. A model can define these precisely, but the useful trigger may be a club angle, relative velocity, hand-speed event, shaft state, or predicted time to impact rather than elapsed milliseconds.

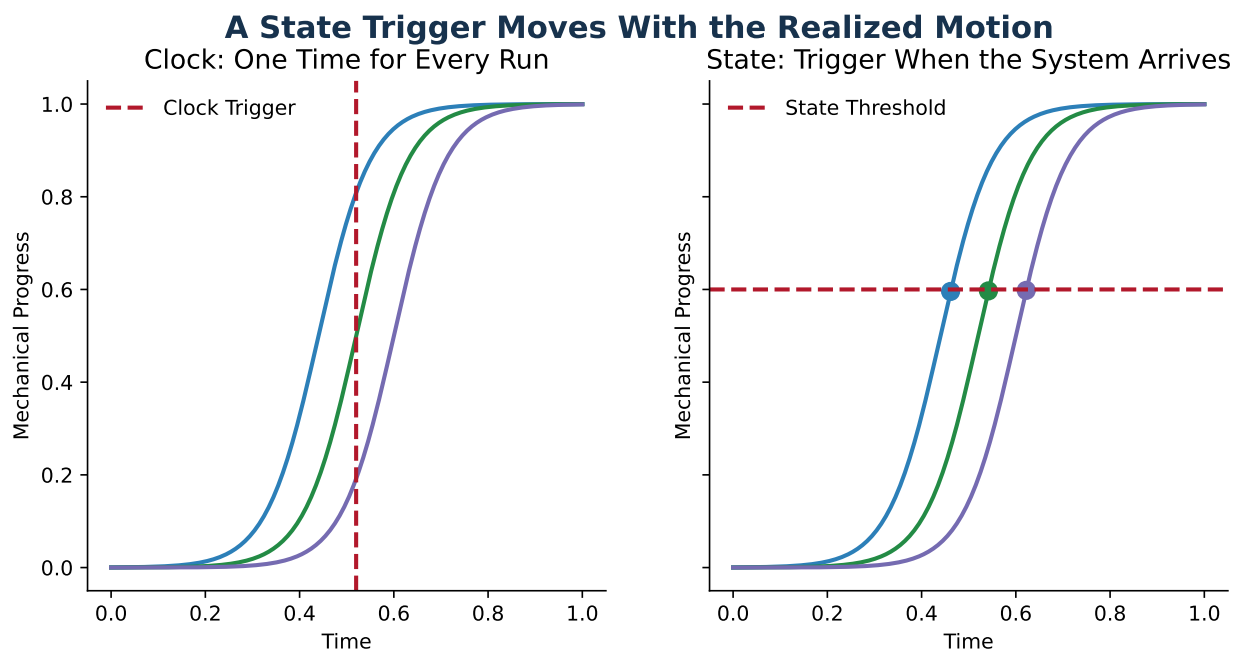


Figure 22.1: Clock-Time Triggers and State-Based Triggers Select Different Command Onsets When Two Swings Progress at Different Rates.

22.2 How the Mechanism Works

A torque program needs amplitude, sign, onset, rise time, duration, and shutdown. An onset can be fixed in time or defined by an event function $g(\mathbf{x}) = 0$. Event-based timing can align mechanisms across parameter variations, but the event must be observable and chosen before outcomes.

22.2.1 Early Drive and Retain-Then-Drive

The simple comparison applies positive wrist torque immediately after transition or uses zero/negative early torque followed by positive late torque. Optimization over onset shows late drive can outperform early drive under declared actuator bounds. This supports a timing mechanism, not a universal delay number.

The early opposing program tests a different idea: retaining state and preload may improve robust speed under delay. In the held-out ensemble it improves one lower-tail speed comparison while worsening face/path proxy. Timing has multiple objectives.

22.2.2 Command Timing Is Not Force Timing

Activation and transmission dynamics delay force after command. A 35 ms rise time means the actuator does not jump to its requested level. A preloaded channel may transmit immediately even when desired magnitude changes. Complete sign reversal may pass through low force. Distinguishing command, internal state, transmitted torque, and resulting motion prevents false timing claims.

22.2.3 Impact Definition Changes Optimization

If impact is defined by a club-angle crossing, a strategy altering trajectory also alters impact time. A fixed terminal clock time can compare different club positions. Robust studies repeat plausible impact criteria and reject schedules that never reach a valid event.

22.2.4 State Feedback Versus Open-Loop Schedules

An open-loop program plays the same command versus time regardless of deviation. A feedback strategy responds to state. Humans use sensory and predictive control, though delays limit rapid correction near impact. The model workbench lets readers compare clock and state triggers without calling either a complete neural controller.

Model Result: Preferred onsets move with parameters, rise time, and objective; the qualitative late-drive ordering survives declared command filters.

Human Evidence: Observed timing varies across golfers and measurement definitions. Kinematic events do not directly reveal command onset.

Computer simulations of delayed release show why timing claims must be stated against a declared model, objective, and state ([Sprigings and MacKenzie 2002](#)).

Hypothesis: Preparation and early downswing arrange a state in which late coupling and distal power become favorable; event-based triggers may generalize better than fixed clock delays.

Practical Interpretation: “Late” should mean late relative to a mechanical state or event, not a mystical timestamp copied across swings.

22.2.5 Clock Time Is Easy to Measure and Easy to Misread

Researchers commonly express events as milliseconds before impact or percentage of downswing. This supports averaging, but impact alignment stretches or compresses mechanically different trajectories onto one axis. Two golfers at the same normalized time may have different arm–club angles, velocities, shaft states, and force directions.

Clock timing remains useful when fully defined. Report event detection, normalization, uncertainty, and absolute duration alongside normalized phase. Do not treat a population mean as the command time used by any participant.

22.2.6 State Timing

A state trigger activates when a measurable condition is met: an arm–club-angle range, sign change in relative angular velocity, handle radial velocity, shaft mode phase, or interaction-term threshold. Such triggers can adapt to tempo and parameter differences. They also risk being difficult to measure or overly tuned.

Candidate triggers should come from mechanism, then be frozen before held-out testing. A trigger based on a hidden model variable may be useful for simulation but cannot guide an experiment until an observable proxy is validated.

22.2.7 Command, Force, Motion, and Outcome Events

Neural command onset, EMG onset, muscle force rise, joint torque, hand wrench, shaft response, angular acceleration, velocity peak, and impact occur at different times. Delay and filtering separate them. Seeing a club event does not reveal when control began.

This chain matters for preload. Continuous loading may shorten the interval from command to useful wrench, but the expected signature is a changed force-rise or stiffness state—not merely an earlier club release. Measuring several links allows mediation tests and prevents outcome timing from being mistaken for cause.

22.2.8 A Command-Shape Experiment

Compare an ideal step, a ramp, and a first-order activation filter with identical nominal onset and steady amplitude. The step supplies force immediately. The ramp and filter delay effective torque and can move the preferred nominal onset earlier. If a result exists only with an impossible step, it has limited human relevance.

Vary rise time across plausible bounds and optimize on training samples. Then freeze schedule or trigger for held-out cases. Report speed, face/path, peak load, and lower-tail outcomes. A robust timing claim survives command dynamics rather than hiding them.

22.2.9 Timing the Negative Couple

Define the couple interval by sign and swing-normal projection, then report its relation to state variables and impact. Test whether it consistently occurs over a state region across tempo changes. Compare direct torque, contact couple, shaft power, and relative release on aligned plots.

If the couple appears after the relevant state change, it cannot be its cause. If its onset precedes the change but varies too widely for prediction, the mechanism may be present without being a useful control target. Causal branches within the model and intervention data are still needed.

22.2.10 Timing Across the Two Allocation Extremes

The continuous arm-dominant strategy predicts continuity in one generalized drive or load mode across transition. The role-reversal strategy predicts a clearer allocation change, perhaps with continued co-contraction. Both may yield similar club event timing if activation and transmission histories differ.

Compare not just onset but duration, rise rate, impulse, and cumulative work. Align by transition, state trigger, and impact. A finding that only appears under one alignment should be described accordingly.

22.2.11 Robust Timing Is a Region

An optimum at 112 milliseconds before impact can be misleading if 108–125 milliseconds perform nearly as well while a narrow peak at 112 collapses under small errors. Map the objective surface and identify a plateau satisfying speed, control, and load constraints. Plateaus are more actionable than isolated maxima.

Uncertainty in event detection, sensor delay, and golfer variability should be shown on the same scale. If experimental timing uncertainty exceeds the width of a simulated optimum, the optimum is not currently testable at that precision.

22.2.12 Feedback Changes the Question

Open-loop schedules prescribe control by time. Humans use sensory feedback, feedforward preparation, and task-level corrections, although feedback delays limit rapid late changes. A state-triggered model approximates one aspect of feedback but is not a neural model.

Add bounded feedback controllers on later model rungs and ask whether the mechanism remains. A strategy that works only open loop may be fragile; a policy that relies on unrealistically instantaneous feedback is also suspect. Control assumptions belong beside mechanical ones.

22.2.13 Experimental Reporting

Publish synchronized raw and processed time bases, latency corrections, event definitions, and confidence intervals. Preregister primary windows and permit exploratory state alignments with clear labels. Provide code that reconstructs events from released signals. This lets reviewers decide whether a timing claim is mechanical, measurement-driven, or an artifact of normalization.

22.2.14 Reader Lab: Compare Three Alignments

Take repeated trials and align them by impact, by transition, and by a preregistered state event such as an arm–club–angle crossing. Plot direct torque, hand couple, shaft power, and club acceleration with uncertainty. A feature that sharpens only under state alignment may be organized by mechanics rather than absolute clock.

Next fit a timing policy on part of the data and freeze it. Evaluate fixed-clock, normalized-phase, and state-trigger versions on held-out participants. Include sensor and command delay. Compare distributional outcomes and trigger failures, not only means.

Finally shift event detection within its confidence interval. If the claimed ordering reverses, the evidence cannot support fine timing language. State that the events overlap within resolution.

22.2.15 Timing as Preparation Plus Opportunity

A useful late action requires preparation: the system must arrive with suitable geometry, velocities, load, and shaft state. It also requires opportunity: enough remaining time and motion for force or work to affect impact. Acting earlier can spoil geometry; acting later can miss the window.

This two-part view unifies delayed transfer and preload. Backswing and early downswing arrange state; later dynamics determine which control channels are effective. It explains why one timestamp cannot be detached from the trajectory that created it.

22.2.16 A Minimum Timing Report

State event definitions, time base, normalization, sampling rate, latency, filtering, and confidence intervals. Show raw-time and normalized-time plots, then one preregistered state alignment. Distinguish command, EMG, force, torque, motion, and outcome events. Report the full response surface around any optimum.

If the preferred region is narrower than measurement or execution uncertainty, say that the current experiment cannot resolve it. If a broad plateau survives held-out testing, report the plateau rather than a false-precision timestamp.

22.2.17 Timing and the Two-Hands Mechanism

The contact couple depends on both state and bilateral force. Its onset may be better described by a region of hand separation, club orientation, and force projection than by a single millisecond. The model can propose this region; instrumented data can test whether it recurs.

Continuous loading and role reversal make different predictions about the events preceding it. One predicts force-mode continuity; the other predicts a sign change with retained or recovered stiffness. Aligning only at impact can blur these transition signatures. The protocol should preserve both transition and state views.

22.2.18 What a Timing Cue Cannot Establish

A cue such as “wait” or “hold the angle” may change several variables and may be implemented differently by each golfer. Its success does not validate a modeled torque schedule unless the predicted mediator changes. Conversely, a mechanism can be real without a conscious cue.

The distinction protects both scientific timing and practical interpretation.

This keeps practical experiments open but bounded. Measure what the cue changed, compare it with the model, and avoid treating subjective intent as a control input known to the equations.

22.2.19 Chapter Checkpoint

Every timing claim should answer five questions: which event, measured how, in which frame or state, with what uncertainty, and related to which later outcome? It should also distinguish command from force and motion. If a state trigger is proposed, the observable proxy and held-out performance must be shown.

The key lesson is not that clock time is useless. Clock and state are two views. Agreement across them is strong; disagreement is information about tempo, preparation, or measurement. Publishing both prevents an exact timestamp from becoming a universal myth.

For the allocation comparison, transition is the critical bridge. Measure load, stiffness, activation, and wrench through it. The timing hypothesis succeeds only if those mediators explain the later difference.

22.3 Where the Picture Breaks

The oven has a small set of observable states. A golfer's relevant state includes uncertain internal loads and activation. A convenient club angle may correlate with, but not cause, the favorable mechanism.

Optimization can exploit unrealistic switches or measurement-perfect feedback. Rate limits, delays, noise, and held-out tests are required before interpreting a schedule biologically.

Go Deeper: Read the [timing counterfactual framework](#), [torque-program methods](#), and [robust delayed-command study](#).

Chapter 23

Fast Once Is Not the Same as Robustly Fast

23.1 A Concrete Picture

A racing setup that wins one qualifying lap may lose its advantage in wind, tire wear, or traffic. Comparing setups therefore requires a task and a declared range of conditions. A probability distribution helps when the question concerns how often an outcome occurs; a worst-case comparison can instead use a specified set of disturbances without assigning probabilities.

The same distinction matters in a swing model. A command that gives high nominal club speed may become sensitive to timing, initial state, shaft properties, or the permitted face error. Robustness asks which conclusions survive specified changes. It does not mean immunity to every disturbance, and a repeatable slow swing is not automatically a useful one.

The points in Figure 23.1 are recomputed from the archived outputs discussed below. They are two projections of a five-objective comparison. A point that looks inferior in one panel can remain nondominated when another objective is included. Lines between programs would not establish that intermediate commands achieve intermediate outcomes.

23.2 How the Mechanism Works

23.2.1 State the Question Before Choosing a Score

Let a policy π specify a command program or a feedback rule, and let θ describe uncertain parameters, initial state, and disturbances. Their modeled outcome can be written $Y(\pi, \theta)$. Name the endpoint as carefully as the policy: speed at a fixed time, speed at a specified crossing, and ball-impact delivery answer different questions. If the event never occurs, the outcome needs an explicit missing-event or failure rule.

A useful comparison fixes the model, admissible policy family, actuator limits, initial-state convention, and endpoint. It then states what varies and what counts as acceptable. The policy itself may react to measurements, but the feedback rule and available information must be fixed before evaluation.

Eight Preselected Programs: Six Held-Out Cases

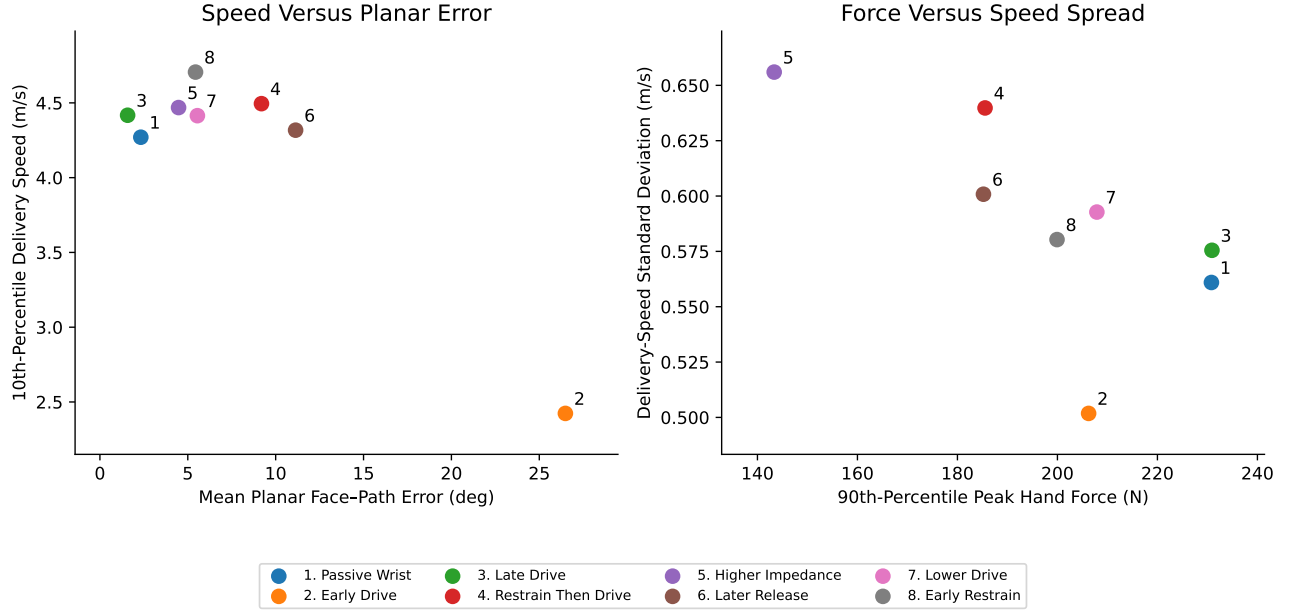


Figure 23.1: Two Projections of the Same Eight Programs on Six Held-Out Cases.

Allowing one program perfect knowledge of a future impact time changes its information advantage, not just its mechanics.

Mean speed, lower-tail speed, accuracy, load, and energy are different objectives. A weighted score requires units or normalization and explicit weights. Changing metres per second to another unit must not silently change the selected policy. A force minimum also does not imply minimum joint torque or muscle effort; those relationships require the relevant mechanical and physiological model.

23.2.2 The Tail Depends on Whether Larger Is Better

For speed treated as a reward, a low percentile summarizes poor-speed cases. For a nonnegative face error treated as a loss, large values are adverse, so an upper percentile describes the difficult cases. Reducing *lower-tail error* could merely improve the already accurate swings while leaving large misses unchanged. A hypothetical speed/error trade must identify the adverse tail and must not be presented as a measured result.

For a scalar outcome Z , define its population quantile at level $0 < \alpha < 1$ by

$$q_\alpha(Z) = \inf\{z : \Pr(Z \leq z) \geq \alpha\}.$$

A quantile is a threshold, not the average severity beyond it. An empirical quantile also needs an interpolation convention. With six speed observations, the linear 10th-percentile estimator used in the provider code is halfway between the smallest and second-smallest values. It is not six observations of a 10-percent tail, nor a demonstrated guarantee for 90 percent of future golfers.

Use the same sampled cases for paired policy comparisons where the experiment allows it. This reveals

whether one policy helps the same difficult cases. The difference between two policy quantiles generally differs from the quantile of their paired differences. In the archived early-restrain/late-drive comparison, the former is about 0.289 m/s and the latter about 0.271 m/s. Neither says that every case improves by that amount.

23.2.3 Severity, Probability, and Worst Case Are Different

For an integrable loss L calculated from the modeled outcomes, conditional value at risk can be defined by the Rockafellar–Uryasev formula (Rockafellar and Uryasev 2002):

$$\begin{aligned}\text{CVaR}_\alpha(L) &= \min_{\eta \in \mathbb{R}} F_\alpha(\eta), \\ F_\alpha(\eta) &= \eta + \frac{\mathbb{E}[(L - \eta)_+]}{1 - \alpha}, \\ (z)_+ &= \max(z, 0).\end{aligned}$$

It measures the average of the worst $1 - \alpha$ fraction of loss probability. At a probability atom, use the appropriate fraction of mass at the threshold; simply averaging observations strictly above the quantile can give another quantity. For four equally weighted losses 0, 0, 0, 10, the 50-percent CVaR is 5, whereas the mean strictly above the median is 10. This example explains the definition, not a golf measurement.

A chance constraint asks for a probability of acceptable outcomes. If all delivery conditions matter together, write a joint condition, such as

$$\Pr(E \leq e_{\max}, F \leq f_{\max}) \geq 1 - \varepsilon,$$

where E is declared delivery error, F is a declared force metric, and the thresholds and distribution are specified. Two separate 90-percent success rates need not give 90-percent joint success: disjoint 10-percent failure sets leave only 80 percent satisfying both. A chosen hand-force threshold is a model constraint, not an established injury threshold.

A set-based worst-case objective instead examines $\sup_{\theta \in \Theta} L(\pi, \theta)$ over a declared uncertainty set Θ . The largest loss among a finite set of sampled cases is generally only a lower bound on that supremum. Dense sampling, local derivatives, and a successful optimizer do not alone certify the unsampled set. Conversely, an implausible range edge can make a worst-case design unnecessarily conservative. Quantiles, CVaR, probability constraints, and worst-case bounds answer different questions; none is the automatic definition of a good swing.

23.2.4 What the Registered Coupled Study Actually Compares

The provider study uses a planar moving-base model with two articulated arms, two solved hand reactions, and one flexible shaft mode. It varies 12 engineering inputs spanning geometry, inertias, base stiffness, damping, grip separation, shaft properties, command delay, lag, and an impedance coefficient. A deterministic Latin-hypercube design supplies 24 global screening cases, six training cases, and six independently generated held-out cases. The seeds are 8426, 8449, and 8450, respectively. Every candidate uses the same cases within a split.

These ranges are not fitted golfer-population distributions. The implementation permutes strata and jitters within the middle portion of each stratum; it is a declared space-filling design, not a collection of independent identically distributed human trials. Delivery is evaluated at 0.24 s with a 4 ms integration step, not at a ball collision. Small constraint and contact-power residuals check particular numerical identities; they do not by themselves bound trajectory integration error or establish convergence of tail estimates.

Eight programs were preselected: passive wrist, early drive, late drive, restrain then drive, higher impedance, later release, lower drive, and early restrain. Their names identify parameter choices, not eight observed human techniques. In particular, early restrain blends its right-wrist command from -2 to $+4$ N m beginning at 0.10 s, while late drive blends from 0 to $+5$ N m beginning at 0.14 s. The command transition lasts 35 ms; the paired left-wrist target is -0.65 times the right-wrist target in the model's signed coordinates. These signs do not alone identify a net club couple or an anatomical muscle action.

The actuator applies pure delay, a discretized first-order command lag with a rate-limited state increment, velocity-dependent resistance, and asymmetric torque-velocity clipping. Its so-called impedance term is proportional to actuator velocity in this implementation; it is not a measured muscle stiffness or a full reflex model. Thus the command schedules are preselected, while delivered torque also depends on velocity. The rate limit on the lag state does not by itself bound every change after velocity-dependent clipping. Even the program named passive wrist can retain modeled velocity resistance.

The five objectives are:

1. Maximize 10th-percentile delivery speed.
2. Minimize mean absolute planar face-path error.
3. Minimize the 90th percentile of peak individual-hand force.
4. Minimize mean integrated squared delivered joint torque.
5. Minimize the sample standard deviation of delivery speed.

The face-path proxy is the absolute wrapped difference between the modeled planar face direction and terminal velocity direction. It is not three-dimensional clubface delivery or ball dispersion. The force statistic first takes the largest force norm over both hands and all recorded times in each rollout, then takes a percentile across cases. The squared-torque integral has units $\text{N}^2 \text{m}^2 \text{s}$; it is not mechanical work, metabolic energy, or measured muscular effort. Standard deviation measures spread, not tracking stability, recovery from disturbance, or probability of meeting a target.

23.2.5 What the Archived Results Establish

Independent recomputation from the stored arrays gives seven nondominated programs on the training cases and eight on the held-out cases. Early restrain has held-out 10th-percentile speed of approximately 4.706 m/s versus 4.417 m/s for late drive. Its mean planar error is approximately 5.44° , compared with 1.58° for late drive. This is a speed/error tradeoff for two specified schedules in six cases. It does not isolate the early opposing interval: the compared schedules also have different onset and late-command values.

The implementation selects programs using **training** objectives. Its balanced, equal-range Chebyshev rule chooses later release, the speed-priority rule chooses early restrain, and the load-priority rule

chooses higher impedance. The balanced rule minimizes the largest objective disadvantage after scaling each objective by its range across the training candidates. Those data-dependent scales encode a choice and can change when candidates change. The held-out outputs evaluate the selected programs and other candidates; they are not the data used by those selection rules.

Omitting one held-out case at a time changes Pareto membership. Passive wrist, early drive, late drive, and higher impedance remain nondominated in all six omissions; restrain then drive, lower drive, and early restrain do so in five; later release does so in four. This is a perturbation check on six cases, not a confidence interval for a population frontier. The complete recomputation and original arrays are linked in the review evidence.

23.2.6 A Pareto Set Is Relative to Its Feasible Comparisons

A feasible policy is dominated if another feasible policy is no worse on every declared objective and strictly better on at least one. Nondominated policies form the Pareto set; their objective vectors form the corresponding front. Restricting the comparison to eight candidates gives a *candidate* Pareto set. Another admissible policy not evaluated here might improve every objective.

For example, the loss vectors $(0, 2)$, $(2, 0)$, and $(1, 1)$ are mutually nondominated. An untested feasible vector $(-1, -1)$ would dominate all three. The archived study therefore leaves a universal optimum unsupported; it does not prove that none exists over every admissible control program. Likewise, nondominance in five sample summaries is neither stochastic dominance nor superiority in every individual case.

A tradeoff plot makes the decision visible but does not assign its values. Speed, accuracy, force exposure, and repeatability can be weighted, constrained, or considered separately. Normalization and weights must be chosen and reported. A weighted sum can miss parts of a nonconvex front, and a continuous line drawn between discrete policies is not evidence of realizable intermediate performance.

23.2.7 Construct Uncertainty Without Hiding the Experiment

Separate execution variability, participant variation, uncertain calibration, unknown physical parameters, and model discrepancy where possible. Declare ranges, distributions, dependence, temporal structure, and their evidence. Independent draws of geometrically related parameters can create impossible bodies or grips; constraint-consistent initial states and admissibility checks belong inside the design. Merely increasing the number of simulations does not repair an omitted mechanism or an unjustified distribution.

Timing, amplitude, direction, and allocation perturbations can expose different weaknesses. Choose temporal correlations and smoothness to match the mechanism under investigation; white or independent noise is not automatically wrong, but a numerical sampling convention is not an execution model. A local derivative describes a neighborhood and can miss distant failures, contact-mode changes, or a missing event. Broad timing plateaus are useful only relative to a performance tolerance, the other disturbances, and the resolution of the search.

Paired simulations should share exogenous cases while retaining the state changes caused by each policy. Resetting transmission state to make two trials look matched changes the intervention. Chapter 3 explains that distinction; Chapter 14 explains finite preparation and role reversal. A comparison of policy families should declare their admissible controls, resource constraints, tuning procedure, and

computational budget. Equal budgets reduce one confound but do not prove equal optimization quality or equal physiological feasibility.

23.2.8 Preserve Failures and the Denominator

Record how many cases were attempted, solved, physically admissible, and evaluated at the required endpoint. A numerical failure, an impossible contact, a missing impact event, and a valid but poor delivery are different outcomes. Investigate numerical failures with refinement or an independent method; do not silently delete them or automatically relabel them as physical failures.

A distribution conditional on successful solves must carry that condition. If a case's loss is unknown, report missingness and bounds or an explicit penalty rule rather than inventing its value. Discarding the worst observed case can improve a reported tail without improving any policy. A failure rate among completed trials also has a different denominator from a failure rate among all attempted trials. Publish both the accounting and the metric rules.

23.2.9 Keep Selection Separate From Evaluation

Use development cases to explore policies, validation cases for tuning choices, and an untouched evaluation set for the final claim. The eight-program archive contains training and held-out splits, not a separate three-way validation exercise. It demonstrates a bounded comparison, not a completed robust-control optimization pipeline. Repeatedly modifying the program after viewing held-out results consumes that independence.

Choose splits for the deployment question. Unseen-person prediction requires participant separation; personalization for a known golfer requires evaluation on genuinely new sessions or conditions and separation from that person's calibration data. Randomly mixing correlated swings between training and test sets can exaggerate transfer. Calibration also has scope: nominal 80-percent coverage should be assessed across enough relevant unseen cases, with sampling uncertainty and subgroup checks. Good average coverage can conceal poor coverage for one participant or regime.

Personalization does not require every parameter to be uniquely identifiable if the target prediction is stable across all plausible parameter sets. Conversely, a precise parameter estimate under the wrong model is not reliable prediction. Show the uncertainty that changes the decision, and withhold confident technique prescriptions when predictive qualification is missing.

23.2.10 Compare Model Tiers and Human Evidence Explicitly

Parameter sensitivity, numerical convergence, structural model changes, and transfer to people are separate tests. A planar fixed-base ranking may change when base motion, shaft memory, spatial contact, or delay is introduced. Define how policies, observations, initial states, and constraints correspond between tiers. A ranking reversal identifies a dependency to investigate; it does not automatically identify the added physical mechanism as its cause.

Likewise, a negative torque component must name an axis, frame, and time window. Persistence of its sign is different from persistence of a performance benefit. A bimodal outcome distribution can arise from nonlinear transformations or sampling choices as well as subgroups or competing mechanisms. A histogram alone cannot decide among those explanations.

Glazier’s discussion of movement variability and McPhee’s review of golf models provide methodological context ([Glazier 2011](#); [McPhee 2022](#)). They are not validation of the eight-program ranking or a certificate of individual robustness. Measured responses to appropriately designed perturbations would test specified predictions, including timing, recovery, and uncertainty. Human adaptation absent from the model can require narrowing the claim or changing the controller model; it cannot be inferred from the synthetic ensemble.

23.2.11 Reader Lab: Make the Robustness Claim Auditable

Start with two policies, a common set of admissible cases, and fixed outcome definitions. Plot their individual results as well as means, adverse tails, and paired differences. State whether uncertainty is sampled probabilistically or bounded as a set. Keep the original sample counts, unsuccessful cases, and the quantile convention alongside the figure.

Then vary one declared assumption: the error tolerance, a parameter dependence, the sampling design, or the model tier. Recompute the comparison without retuning on the evaluation set. Add a simple baseline and report where the ordering reverses. A ranking that survives these tests is evidence within their scope, not proof against every future change.

An open modeling tool should preserve these qualifications in exported tables and figures. It can flag violations of known input bounds but cannot guarantee detection of unknown model inadequacy. Separate an exploratory prediction from a qualified recommendation. A useful report identifies which measurement or experiment would most reduce the uncertainty affecting the choice.

23.3 Where the Picture Breaks

The racing analogy motivates a comparison across conditions; it does not imply that a golf downswing has the same feedback opportunities as a lap. A constructed ensemble also differs from repeated measurements of people. Denser sampling of known variables cannot make an omitted disturbance disappear.

Model Result: The archived eight-program comparison reproduces a conditional speed/error tradeoff and sample-dependent nondominance. It supports no universally best policy and does not exclude better untested policies.

Measurement Boundary: The endpoint, force statistic, and effort metric are declared model quantities. Participant-held-out measurements, appropriate outcome definitions, and calibrated predictive uncertainty are still needed for human robustness claims.

Hypothesis: Preload or an opposing command interval may improve selected outcomes under some delays and uncertainties. Isolating that mechanism requires matched comparisons beyond two schedules that differ in several parameters.

Practical Interpretation: Ask how often the intended delivery succeeds, under which changes, with which error and load consequences, and how much of that answer comes from measurements rather than modeling assumptions.

Go Deeper: Read the monograph’s [coupled uncertainty and control chapter](#), the [archived experiment record](#), and the [small-sample stability audit](#).

Chapter 24

What Has Actually Been Measured in Golfers

24.1 A Concrete Picture

A courtroom separates what a witness saw from the explanation offered for it. Golf science needs the same separation. A marker trajectory, a calibrated grip load, and an electrical signal from a muscle can constrain an explanation of the swing. They do not carry the same information, and collecting them together does not automatically reveal the internal mechanism.

The useful question is specific: what observation would distinguish two plausible explanations? If both predict the same club motion but different hand loads, another camera may add less information than a carefully calibrated grip measurement. If both predict the same hand loads through different muscle combinations, even a perfect grip sensor leaves a biological question open.

24.2 How the Mechanism Works

24.2.1 Separate the Signal From the Explanation

A camera records image information from which marker positions and segment motion are reconstructed. A force plate converts transducer signals into a calibrated external wrench. A grip sensor records a load at its sensing section; the force attributed to a palm can require a further mechanical calculation. EMG records electrical activity associated with selected muscles, not muscle force directly. Every measurement has a location, reference frame, bandwidth, calibration, and uncertainty.

Inverse dynamics combines motion, external loads, and body-segment inertias to estimate net joint loads. Its answer depends on filtering and differentiation, joint definitions, and the assumed contact model. In the closed chain formed by two arms and one club, the net club wrench alone generally does not identify how the hands share the load. Additional sensing or assumptions are required.

Even uniquely determined net joint moments would not uniquely determine muscle forces. Antagonists can increase their opposing forces while leaving a similar net moment. EMG can constrain activation timing and a muscle model, but the conversion to force also depends on muscle length, contraction velocity, activation dynamics, and tendon behavior. Mechanical identification and anatomical identification

Observations, Inferences, and Tests

Calibrated Observations

Motion, external loads,
electrical activity

Mechanical Inference

Frames, inertias,
contact assumptions

Anatomical Inference

Multiple muscle allocations
can remain

Prospective Tests

Frozen predictions;
relevant new data

Measurements constrain explanations.
A unique mechanism is not guaranteed.

Figure 24.1: Observations Constrain Inferences; Each Explanation Still Needs a Test.

are different problems.

24.2.2 A Bounded Register of Human Evidence

The following studies illustrate what the literature can constrain. They are a selected register, not a new systematic search or a pooled dataset. Participant counts refer to people, not independent replications of every recorded swing. The linked sources identify the studies; unspecified trial counts or data access should be checked in the full methods before reanalysis.

- **Kinematics Across Studies.** Bourgain and colleagues’ systematic review included 92 studies and documented heterogeneous definitions and methods ([Bourgain et al. 2022](#)). It supports careful comparison of coordinate systems, clubs, samples, and events; it does not establish one universal sequence. This is distinct from McPhee’s narrative model-and-measurement review ([McPhee 2022](#)).
- **Professional and Amateur Motion.** Zheng and colleagues compared 72 men, with 18 in each of four professional/amateur skill groups, using motion capture at 240 Hz ([Zheng et al. 2008](#)). Group differences describe the sampled swings. They do not establish that copying a professional’s movement causes improvement or prevents injury.
- **Scapular Muscle Activity.** Kao, Pink, Jobe, and Perry studied 15 competitive male golfers with bilateral EMG of four scapular muscles and cinematography ([Kao et al. 1995](#)). Phase- and side-dependent activity supports shoulder-girdle involvement. It does not identify a unique route from a scapular muscle to a handle moment.
- **A Specific Wrist Muscle.** Robinson and colleagues studied bilateral extensor carpi ulnaris activity in 15 subelite, right-handed men, with five shots each using a pitching wedge, 7-iron, and driver ([Robinson et al. 2023](#)). Surface EMG and forearm/hand inertial sensors characterized selected activation and movement variables. These are not measurements of all wrist muscles. The reported nonsignificant downswing-EMG association with impact clubhead kinematics does not establish no effect or equivalence.
- **A Strain-Gauge Handle.** Koike demonstrated reconstructed palm forces and moments for one right-handed professional instructor using a driver ([Koike 2016](#)). Their calibration and mechanical calculation use fixed palm reference points and neglect handle-piece inertia. This is a useful instrument demonstration, not population validation or direct muscle measurement.
- **A Split Grip and Inverse Dynamics.** Choi and Park studied nine right-handed male professionals with five regular-grip driver swings followed by five instrumented-grip swings ([Choi and Park 2020](#)). One six-axis sensor between two grip sections supplied an internal wrench; a nine-segment rigid-body model inferred hand and upper-limb joint loads. Separate hand loads are therefore estimates supported by sensing and a model, not two independent raw sensor outputs.
- **Ground Interaction.** Han and colleagues studied 63 skilled male golfers with handicaps at most three, using driver, 5-iron, and pitching-wedge shots, optical motion capture, and two force plates ([Han et al. 2019](#)). They computed ground-interaction moments and found associations with clubhead speed. Those associations do not identify a unique ground-to-club energy pathway.
- **Shaft Stiffness.** Betzler and colleagues studied 20 golfers with two clubs designed to differ only in shaft bending stiffness ([Betzler et al. 2012](#)). Their examination of shaft strain, wrist motion, club presentation, and ball launch found small speed differences for the tested shafts. This bounds a specific equipment comparison; it does not establish that shaft flexibility is always negligible or that one stiffness suits everyone.

24.2.3 Why Instrument Design Changes the Claim

The Choi–Park instrumented grip weighed about 350 g versus about 100 g for a regular grip; the instrumented club weighed 551 g. Foam/sponge balls, possible hand-to-hand load bypass, and regular-club-first ordering further limit the comparison. Similar average trajectories do not prove unchanged muscle control or mechanical equivalence. The reported roughly threefold right-hand torque relative to the left is a model-derived result in this setting, not a universal hand contribution or muscle-power ratio (Choi and Park 2020).

The central lesson is constructive: record what the instrument closes in the mechanical accounting and what remains assumed. A six-component signal at one section is not automatically a six-component contact wrench at each hand. Loads must be expressed about declared points in compatible frames. Acceleration of material between sensor and contact introduces inertial terms; a path that bypasses the sensor can carry an unrecorded internal load. Static calibration alone does not establish dynamic accuracy over the swing’s frequency range.

24.2.4 Motion, Ground, Hands, and Shaft Belong in One Test

Kinematic peak order is a description of motion. It does not uniquely reveal intersegmental energy transfer, and changing the angular-velocity component or event definition can change the reported sequence (Bourgain et al. 2022; Marsan et al. 2019). Likewise, a large ground force is not itself a large supply of mechanical work: an ideal stationary contact can exchange momentum while its contact power is zero. The body’s muscular work and stored energy must still be accounted for.

At the hands, power pairs a wrench with the corresponding linear and angular velocity; work integrates that power over time. Force magnitude alone is insufficient. At the shaft, deformation can store and return energy and alter clubhead presentation. A head-speed change can therefore accompany changes in body motion, bilateral loading, shaft recovery, or several together. Synchronized measurements let a model predict their joint pattern instead of explaining one curve after the fact. They also require synchronized uncertainty: a timing offset between force and velocity can corrupt a computed power curve.

The EMG result supplies another constraint, not a shortcut around this accounting. Different activation patterns can produce similar net wrenches; similar EMG amplitudes can accompany different forces under different muscle states. Crosstalk, electrode placement, normalization, and limited sampling matter. A nonsignificant association leaves an uncertainty interval to examine; an equivalence claim needs a justified equivalence margin and an appropriate test.

24.2.5 Repeated Swings and Independent Golfers

Extra swings help estimate a person’s repeatability. Extra participants address variation between people. These contributions should not be confused when estimating a population mean or comparing groups. Nested observations require an analysis that accounts for their dependence, such as an appropriate mixed model or participant-level summaries (Lazic 2010).

Consider a **manufactured example**, not measured golfer data. Let y_{ij} be a scalar outcome on swing j from person i . Write

$$y_{ij} = \mu + b_i + e_{ij}.$$

Here μ is the population mean, b_i is a zero-mean person-specific offset with variance σ_b^2 , and e_{ij} is zero-mean within-person variation with variance σ_e^2 . Assume independent participants, errors independent of the offsets and of each other, equal trial counts m , and no additional session or time dependence. For N people and their grand mean $\bar{y}$,

$$\text{Var}(\bar{y}) = \frac{\sigma_b^2}{N} + \frac{\sigma_e^2}{Nm}.$$

The person offset repeats in every one of that person's swings, so its contribution shrinks with the number of people, not the total number of swings. With $N = 15$, $m = 10$, and both variances set to one in squared outcome units, the variance of the grand mean is $11/150$. Falsely treating all 150 swings as independent gives $2/150$: uncertainty variance is understated by a factor of 5.5. The corresponding variance-equivalent independent-observation count is $300/11$, about 27.3. That number is neither 27 independent people nor a universal degrees-of-freedom rule.

More repeats reduce the second variance term but cannot eliminate the first at fixed N . If between-person variance is zero, the formula reduces to the independent-observation case; if each person contributes one swing, there is no repeated-measure inflation. Real analyses may need condition effects, random slopes, unequal trial counts, session effects, or residual autocorrelation. Sample-size calculations must target the actual contrast and dependence model; these invented variances are not a golf-study power calculation.

24.2.6 Prediction, Estimation, and Causation Need Different Designs

An analysis estimating a population association does not automatically need a prediction holdout, but it does need an appropriate sampling and uncertainty model. A claim about predicting a new golfer requires evaluation on participants excluded from fitting and model selection, with preprocessing learned only from training data. Randomly splitting swings can leak a person's characteristics into both sets. A different target, such as predicting a new session for a known golfer, calls for a different split.

Neither a hierarchical model nor successful prediction alone identifies a causal mechanism. A causal claim must state the intervention or contrast, potential confounders, and assumptions that distinguish rival explanations. A model fitted to a mean curve has not passed a held-out test merely because it reproduces that curve. Multiple tested outcomes and exploratory choices also require transparent reporting; select the primary outcome and analysis before inspecting results.

24.2.7 The Specific Gap in This Project

The evidence reviewed here does not provide this project with a single synchronized human dataset that validates the complete proposed passive negative-couple and preload pathway. That is a statement about the evidence assembled for this mechanism, not a claim that no relevant human study exists anywhere. Existing reviews provide context (Hume et al. 2005; McPhee 2022), but combining results from different participants and instruments does not create a joint validation experiment.

Human Evidence: Motion patterns, selected muscle activity, ground reactions, grip measurements, and shaft behavior constrain parts of the explanation within their study designs. **Model Result:** Simulations predict coupled changes in hand loads, club motion, and timing under specified assumptions. **Hypothesis:** The proposed complete biological pathway remains to be tested with measurements and contrasts capable of separating it from alternatives.

For the full pathway, seek synchronized whole-body and club motion, separate foot reactions, sufficiently resolved bilateral hand loading, shaft deformation, selected forearm/shoulder/scapular EMG, and launch outcomes. Contact pressure can help characterize grip location and distribution but does not replace a full wrench. Not every smaller scientific question requires every channel; select channels according to what distinguishes the competing predictions.

24.2.8 Turn Disagreement Into a Diagnostic

Suppose a model predicts a negative hand couple in a specified interval, but the estimated experimental couple is positive. First check sign conventions, reference points, synchronization, and uncertainty. If the disagreement persists beyond the prespecified tolerance under comparable conditions, that prediction has failed. Retuning the model can produce a new hypothesis, but does not erase the failed prospective test.

If hand loads agree while EMG timing differs, examine activation measurement and the proposed anatomical allocation. Agreement in net mechanics need not validate the muscle explanation. If motion agrees while hand or ground loads disagree, the model fails that additional constraint; the discrepancy may come from parameters, missing dynamics, or measurement error. It does not uniquely diagnose one cause. Failure in one observable should remain visible alongside success in another.

24.2.9 A Reproducible Human-Evidence Release

Before testing predictions, specify calibration acceptance, synchronization tolerance, coordinate registration, sensor range, residual checks, exclusions, and missing-data treatment. Relate thresholds to the intended measurement and claim; there is no universal residual limit that validates every model. Freeze the model version and primary comparisons, and disclose later deviations.

Release the protocol, data dictionary, processing code, participant and trial counts, uncertainty, and consent-compatible data. Show individual traces and residuals alongside group summaries. Report raw-data availability explicitly; open access to a paper does not imply that participant-level data are available. Synthetic pipeline examples and digitized published curves must remain labeled as such. Neither substitutes for a new human validation sample.

24.2.10 Reader Lab: Follow One Claim Through the Chain

Take the sentence “elite golfers sequence segments proximally to distally.” Check which people, club, coordinate components, filters, and peak definitions support it. Determine whether the pattern occurs within individuals or only in an average. A kinematic description needs reliable motion measurements; a claim that this sequence transfers energy needs additional kinetic and power evidence.

Now examine “scapular muscles create the handle torque.” Phase-dependent EMG supports selected muscle involvement. Attribution of a particular handle torque requires a closed-chain mechanical and anatomical model constrained by more measurements. Even then, check whether competing muscle

allocations remain possible. Write separate sentences for the observation, the model inference, and the untested causal claim. This makes the next experiment’s purpose concrete.

24.3 Where the Picture Breaks

The courtroom analogy makes observations sound neutral. Markers move relative to bone, electrical signals have artifacts, and force estimates depend on calibration and modeling choices. A richer instrument package improves the constraints only when its uncertainties and load paths are understood.

Small or specialized samples also limit generalization across sex, age, skill, injury history, anatomy, and equipment. Evidence can strengthen a bounded mechanism without becoming a universal technique. The goal is a connected test of motion, contact, energy, and biological allocation whose successes and failures are both available to the reader.

Go Deeper: Read the monograph’s [empirical evidence chapter](#), the kinematics review ([Bourgain et al. 2022](#)), and the distinct model-and-measurement review ([McPhee 2022](#)). The primary grip methods ([Koike 2016](#); [Choi and Park 2020](#)) make the distinction between sensing and inverse dynamics explicit.

Chapter 25

Muscles, Scapulae, Wrists, and Biological Ambiguity

25.1 A Concrete Picture

Several teams can move the same piano through a doorway. One team pushes from behind, another pulls with straps, and a third combines both. The piano follows nearly the same path, but internal effort, stability, and injury risk differ. Watching the piano does not identify the team strategy.

Club motion is the piano. Muscles, joints, two hands, and connective tissues form an overcomplete team. A model may estimate the net wrench required at the grip or the net moment at a joint. Those quantities rarely identify which muscles or scapular actions produced them.

25.2 How the Mechanism Works

Muscles pull through moment arms. Several muscles cross one joint; some cross two. Agonists and antagonists can be active together. A net joint moment is the signed sum of muscle, passive tissue, and contact effects. Infinite or numerous muscle- force combinations can reproduce the same net moment.

25.2.1 Activation Is Not Force

EMG reflects electrical activation near sampled muscle fibers. Muscle force also depends on length, velocity, tendon state, fatigue, architecture, and activation dynamics. Surface amplitude cannot be read directly as torque. Timing evidence is valuable when interpreted at that level.

25.2.2 Co-Contraction Changes Impedance

Equal opposing muscle moments can cancel in net torque while increasing joint stiffness and compression. This hidden activation may stabilize the grip or prepare rapid force transmission. It also costs energy and can raise load.

The arm-wrist comparison therefore needs effort and stiffness measurements, not only net club work. A model that minimizes generalized torque may choose an allocation unlike one that maximizes robustness or comfort.

Club Motion Does Not Uniquely Identify the Biological Allocation

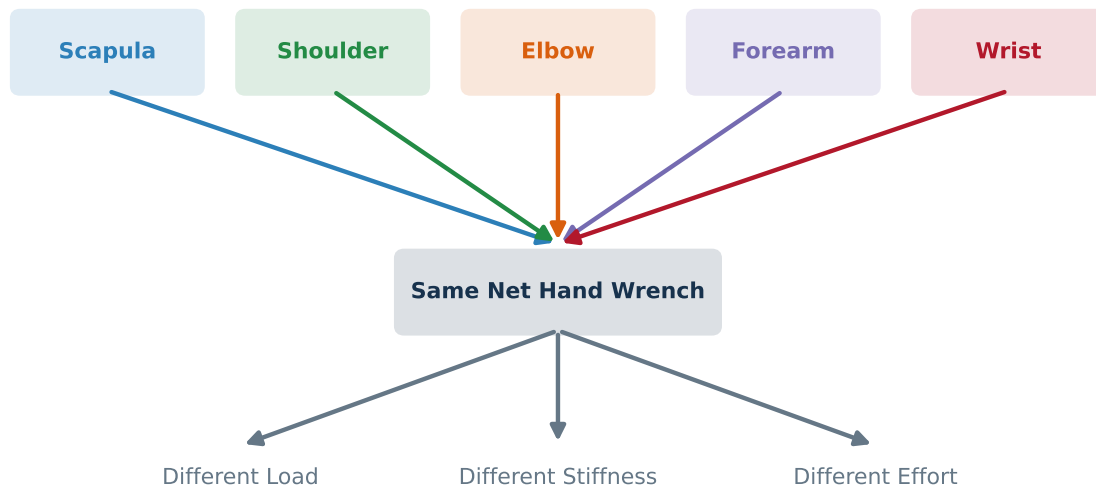


Figure 25.1: Many Muscle and Joint Allocations Converge on the Same Net Hand Wrench While Producing Different Internal Loads and Stiffness.

25.2.3 The Scapula Is Not a Fixed Hinge

Scapulothoracic motion reorients the shoulder socket and changes hand reach, moment arms, and muscle lengths. Retraction and protraction can contribute to a proximal push-pull pattern, but a reduced shoulder torque cannot be relabeled as scapular retraction. Direct scapular kinematics and muscle evidence are required.

Marker measurement over the scapula is difficult because skin moves relative to bone. Specialized clusters, imaging, or validated models may improve estimates. The uncertainty belongs in any mechanism test.

25.2.4 Wrists Are Multi-Axis Transmissions

Flexion-extension, radial-ulnar deviation, forearm rotation, carpal motion, and grip compliance interact. A planar wrist torque is a projection. Direct negative club torque could arise from active wrist moment, force-generated couple, passive stiffness, or a mixture.

25.2.5 Tendon and Muscle History

Tendon slack length is a model parameter, not a casual synonym for a loose swing. Series compliance delays and stores force. Muscle short-range stiffness depends on activation and recent movement. These phenomena may affect preload, but the dead-zone study represents only one falsifiable transmission family.

25.2.6 What a Responsible Biological Model Would Add

A higher-order model would include anatomical joint axes, muscle paths, activation and contraction dynamics, tendon compliance, passive limits, contact pressure, and participant parameters. It would be constrained by synchronized motion, wrench, EMG, and possibly ultrasound. Even then, muscle redundancy requires optimization or probabilistic inference and uncertainty reporting.

Human Evidence: Scapular and wrist EMG establishes phase-dependent activity in small samples, not a unique torque allocation (Jobe et al., 1995; Robinson et al., 2023).

Model Result: Generalized arm and wrist subspaces can produce the same club task with different internal demands. This is a mechanics statement, not muscle identification.

That many-to-one mapping is consistent with the broader multi-articular muscle coordination problem: muscle action depends on the state and on the rest of the mechanism, not on a muscle name alone (Zajac and Gordon 1989).

Hypothesis: Proximal push-pull and wrist resistance may maintain useful preload for some golfers, while wrist-led preparation and arm resistance may be favored for others. Bilateral force and stiffness data must decide.

Practical Interpretation: Similar club motion can be produced by different internal organizations. A visible outcome should not be reverse-engineered into one muscle story.

25.2.7 From Club Wrench Back Toward the Body

The club wrench is a useful boundary observation. Moving inward, each hand contact loads wrist and forearm; elbow and shoulder reactions depend on segment motion; the shoulder girdle transmits load to the trunk; trunk and pelvis couple to the ground. Inverse dynamics can estimate net generalized loads along this chain under a rigid-body model. It does not uniquely allocate those loads to muscles and passive tissues.

At each step, degrees of freedom and uncertainty multiply. Wrist axes are not perfect hinges. Forearm pronation changes orientation. The glenohumeral joint moves relative to a scapula that itself moves relative to the thorax. Skin markers imperfectly track bone. An anatomical story should become more qualified, not more certain, as it moves away from directly measured contact.

25.2.8 Muscle Redundancy

Many muscles can contribute to one joint moment, and muscles spanning several joints can move energy between segments. Agonists and antagonists may activate together, raising stiffness and internal force while leaving net moment nearly unchanged. Passive tissues can contribute state- and history-dependent loads.

An optimization criterion—minimum activation squared, metabolic cost, stress, or fatigue—selects one allocation from many. That solution is conditional on the criterion and muscle model. EMG can reject some allocations and inform timing, but sparse surface electrodes do not fully identify deep muscle forces.

25.2.9 Scapular Action as a Candidate Mechanism

Scapular protraction, retraction, rotation, and tilt reorient the shoulder socket and alter hand reach and leverage. Coordinated lead/trail shoulder-girdle action could plausibly contribute to a push-pull handle pattern. That makes scapular kinematics and activation relevant measurements.

It does not justify replacing a generalized arm driver with the phrase “scapular retraction.” The same hand wrench can arise through different shoulder, elbow, wrist, trunk, and grip combinations. Direct or well-validated scapular tracking, multi-muscle EMG, and a closed-chain model are required to test the specific mapping.

25.2.10 Wrist Resistance Is Not Necessarily Passive

If net wrist torque opposes club rotation, active muscle, passive tissue, contact geometry, and dynamic interaction may all contribute. EMG silence does not prove absence of force, and EMG activity does not prove positive work. Shortening and lengthening contractions, co-contraction, and moment-arm changes complicate the relation.

The word “passive” should be reserved for a declared model channel or measured material behavior. A safer human hypothesis is that some late opposing club- moment contribution may arise without a newly increasing explicit distal drive. The experiment then measures wrench, activation, and state to test candidate sources.

25.2.11 A Many-to-One Example

Suppose the club requires a 12-newton-metre swing-normal moment and a specified net force. Allocation A uses a stronger bilateral differential hand force with modest contact moments. Allocation B uses smaller differential force plus larger wrist contact moments. Allocation C changes arm posture so common force also has a useful moment arm. The club can see nearly the same wrench in all three.

Internally, joint moments, co-contraction, tissue load, and sensitivity differ. Club motion alone cannot identify the allocation. Bilateral six-axis sensing rules out some solutions; EMG and kinematics constrain others; a model proposes remaining possibilities with uncertainty.

25.2.12 Preactivation and Effective Stiffness

Before motion reverses at transition, neural activation and load history may prepare muscles and tissues. Co-contraction can increase resistance to perturbation without a large net joint torque. Tendon and cross-bridge behavior can create history dependence. Grip and shaft deformation add mechanical states.

This gives a plausible biological basis for investigating continuous loading, but the simple dead-zone model is only phenomenological. Effective stiffness should be estimated through perturbation or system identification, not inferred from a feeling or one EMG channel. The benefit must also be weighed against internal load and reduced adaptability.

25.2.13 Lead and Trail Sides Need Separate Treatment

The hands occupy different positions and anatomical orientations. The same club-frame force direction maps to different joint and muscle demands on each side. Roles can change over the swing. Summing

them into one “wrist torque” or one EMG average can erase the differential mode central to the hypothesis.

Report side, hand position, frame, phase, and dominance. A model may use symmetric contacts for mechanism isolation, but human validation should not assume symmetry. Participant handedness and technique further constrain generalization.

25.2.14 How to Compare the Allocation Extremes Biologically

Use matched club tasks and randomized, bounded strategy conditions. Measure bilateral wrenches, grip pressure, whole-arm and shoulder-girdle kinematics, forearm and scapular EMG, shaft response, and perturbation-based stiffness proxy. Track discomfort and peak loads. Verify that instructions actually changed the intended mediator.

Analyze whether force continuity or stiffness mediates outcomes rather than assigning success to the instruction label. Some participants may implement the same cue with opposite internal allocations. Cluster only on measured variables and validate clusters on held-out people.

25.2.15 Claims at Four Levels

Level one is observation: a measured hand force, EMG timing, or motion. Level two is rigid-body inference: net joint or interface load. Level three is biological inference: candidate muscle, stiffness, or tissue allocation. Level four is practical interpretation: a strategy might be worth testing. Every paragraph should make its level apparent.

Moving upward requires new evidence. A beautiful model trajectory cannot skip levels. This discipline makes room for bold hypotheses while keeping them easy to challenge.

25.2.16 What Would Refute the Anatomical Story

If predicted scapular kinematics or activation do not accompany the bilateral wrench, the specific scapular pathway weakens. If wrist activation and measured contact moments contradict the proposed passive resistance, revise the mapping. If multiple biological allocations remain equally supported, report ambiguity rather than choosing the most intuitive one.

The mechanical contact mechanism may survive even when an anatomical hypothesis fails. Conversely, observed activation may exist without the predicted club effect. Separating layers lets evidence refine the right part of the framework.

25.2.17 Reader Lab: Map a Claim Downward and Upward

Start with a measured negative hand couple. Moving downward toward raw evidence, identify sensor voltages, calibration, frame transform, hand positions, and uncertainty. Moving upward, identify inferred wrist moments, candidate muscle allocations, proposed scapular coordination, and practical interpretation. Mark where each new assumption enters.

Now ask which alternate biological allocation produces the same measured wrench. Construct at least two candidates and list observables that separate them: EMG timing, scapular kinematics, joint reaction, stiffness, or fatigue. The model should retain both until data discriminate.

Repeat from an EMG observation. Activation timing alone can support “this sampled muscle was active under this normalization.” It cannot establish sign of club power. Linking both directions teaches why contact mechanics and biology need one another but cannot be collapsed.

25.2.18 Practical Value of Ambiguity

Redundancy can be beneficial: it permits adaptation to anatomy, fatigue, equipment, and perturbation. A modeling product should therefore expose feasible families and costs rather than force every golfer toward one allocation. The most stable individual solution may not resemble the population mean.

Ambiguity also sets a safety boundary. Without identified internal load, a speed-equivalent strategy cannot be assumed equally safe. Clinical and coaching decisions require measurements and expertise beyond the current model.

25.2.19 A Minimum Biological Claim

State the measured activation or kinematics, the net mechanical quantity, the model mapping between them, alternative allocations, and uncertainty. Do not name one muscle or scapular action as the source of a club wrench unless the data and model distinguish it. Report internal-load and safety limitations near any practical interpretation.

This minimum still permits useful conclusions: a candidate allocation is consistent with evidence; a proposed mapping is contradicted; or additional measurement is required. Scientific neutrality is not silence. It is calibrated specificity.

25.3 Where the Picture Breaks

The piano teams choose explicit roles. Human control can be distributed, automatic, and task-dependent. Muscle synergies and reflexes do not map neatly to coaching words such as push, pull, firm, or passive.

Musculoskeletal detail can also create false precision. Unmeasured moment arms or tendon properties may dominate an optimized allocation. More anatomy without better data can increase confidence faster than knowledge.

Go Deeper: Read the technical [biological boundaries in the allocation chapter](#), the [experimental protocol](#), and the linked activation evidence above.

Chapter 26

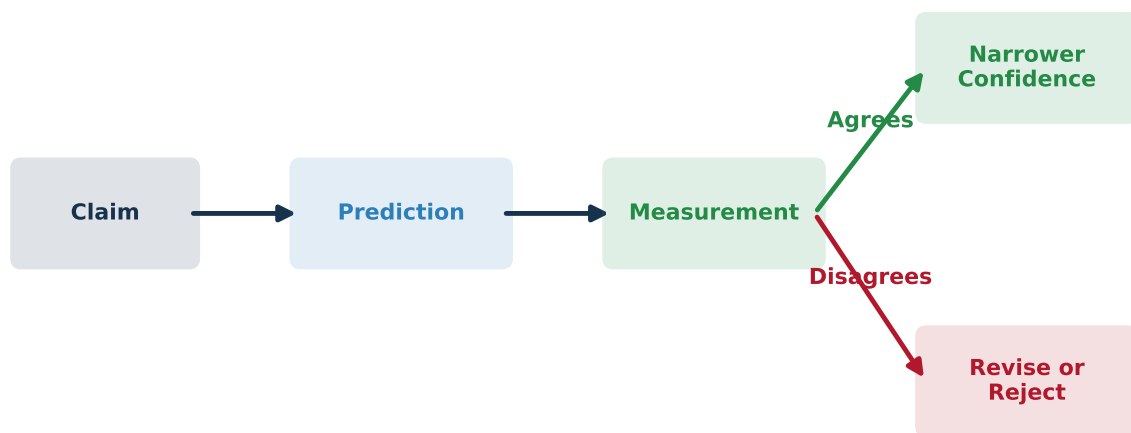
Design an Experiment That Can Say No

26.1 A Concrete Picture

A good smoke alarm is tested with smoke, with clean air, with a weak battery, and with a blocked sensor. A demonstration that rings only when the manufacturer presses a hidden button proves little. Scientific tests should likewise include conditions where the proposed mechanism predicts absence, reversal, or failure.

The proximal-distal hypotheses are valuable only if measurements can count against them. This chapter turns the model's attractive pictures into a protocol that can say no.

A Scientific Story Must Include an Exit



A claim that survives every possible result has not risked enough.

Figure 26.1: A Falsification Map Connects Each Mechanism Claim to Required Measurements, Negative Controls, and a Predeclared Failure Condition.

26.2 How the Mechanism Works

Preregistration freezes primary hypotheses, outcomes, windows, exclusions, filtering, coordinate conventions, and analyses before outcome inspection. It does not guarantee truth; it prevents the test from quietly changing to fit the data.

26.2.1 Synchronized Measurements

The protocol calls for whole-body and club kinematics, separate ground-reaction wrenches, bilateral six-axis grip wrenches and pressure, shaft strain or deformation, launch outcomes, and forearm/shoulder/scapular EMG. Shared timing is essential because millisecond phase differences are part of the mechanism.

26.2.2 Calibration Before Interpretation

Force sensors need zeroing, axis checks, and reference transforms. Motion and force coordinate systems must be registered. Club and body inertias need uncertainty. Known-load trials and action-reaction checks reveal sign or scale errors before a biological narrative is attached.

26.2.3 Filtering and Residual Gates

Differentiating marker positions amplifies noise. Filtering force and motion with incompatible cutoffs creates artificial phase shifts. The protocol freezes methods and checks inverse-dynamics residual force and moment. Trials failing closure are not used to support a mechanism.

26.2.4 Primary Falsifiers

The interaction-force claim is weakened if measured force direction, power, and counterfactual predictions do not match. The two-hand couple claim fails if bilateral wrenches lack the predicted late sign or geometry dependence. The preload claim fails if role reversal produces no measurable force/stiffness gap or if any advantage persists after gap-related variables are equalized. The robust-control claim fails if held-out participants do not preserve the declared tradeoff.

26.2.5 Negative Controls

Hardware or task manipulations should alter grip separation, shaft properties, or load direction within safe limits. Analysis controls scramble event timing, reverse sign conventions, or use impossible reference transforms to confirm the pipeline rejects them. Synthetic dry runs test code paths but never count as human outcomes.

26.2.6 Participant Holdout and Hierarchy

Repeated swings from one person are not independent people. Training and test splits occur at participant level. Hierarchical models report individual and population variation. A mechanism may be supported for a subgroup without becoming universal.

26.2.7 Identity and Governance

Human data require consent, privacy, controlled access, provenance, and an identity-safe analysis plan. Session or row order must not be used to infer player identity. Public releases can contain deidentified derived observables and qualified synthetic examples while raw authority remains governed.

Hypothesis: The predicted force-couple, preload, and timing signatures will survive preregistered participant-held-out testing.

Human Evidence: No such complete dataset has yet been analyzed by this project. The protocol and synthetic dry run establish readiness, not evidence.

Practical Interpretation: The strongest version of a claim names the result that would make its authors change their minds.

26.2.8 Turn Each Mechanism Into a Prediction Table

For every claim, list the model rung, intervention, primary observable, expected sign or ordering, uncertainty window, negative control, and failure condition. The table should distinguish direct measurement from derived quantity. This prevents a broad theory from retreating to a different outcome whenever one prediction fails.

For interaction-force transfer, predict hand-force direction, acceleration contribution, and forward persistence. For the negative couple, predict bilateral wrench projection and geometry response. For preload, predict force-rise or stiffness mediation. For robustness, predict held-out distributions rather than one optimum.

26.2.9 Phase One: Bench and Instrument Qualification

Before recruiting participants, calibrate force plates, grip sensors, pressure mapping, shaft strain, motion system, and timing hardware with known loads and motions. Characterize crosstalk, drift, hysteresis, range, orientation, latency, and repeatability. Verify action–reaction and wrench transforms on a rigid test fixture.

Run synthetic and hardware-in-the-loop signals through the complete pipeline. Intentionally swap axes, shift time, and exceed ranges to confirm gates fail. Qualification records belong in the release and should be tied to hardware and software versions.

26.2.10 Phase Two: Feasibility and Manipulation Checks

Use a small pilot to test comfort, sensor durability, event synchronization, and whether bounded instructions or equipment changes alter the intended mediator. Pilot outcomes do not confirm the primary hypothesis. They refine protocol and effect-size uncertainty before preregistration.

If an “arm-dominant” instruction does not change measured allocation, comparing its club speed with a “wrist-dominant” label is meaningless. Manipulation checks use bilateral wrench and activation/stiffness observables, not participant self-report alone.

26.2.11 Phase Three: Confirmatory Study

Freeze inclusion, exclusion, conditions, randomization, primary windows, outcomes, model versions, filters, transforms, residual gates, and statistical analysis. Power the study at participant level. Separate discovery and holdout participants and prevent identity leakage across splits.

Randomize condition order and allow familiarization. Blind data processing to condition where possible. Track adverse or uncomfortable trials. Preserve raw authority and generate deidentified derived releases under consent and governance.

26.2.12 Phase Four: External Replication

An independent group should receive the protocol, schemas, calibration fixtures or specifications, analysis container, and preregistered predictions. Replication with different hardware and participants tests whether findings belong to the mechanism rather than one laboratory pipeline.

Discrepancies should be decomposed by measurement, population, model, and intervention. A multi-lab result need not be identical to be informative; it should make heterogeneity explicit.

26.2.13 A Negative-Control Catalog

Geometric controls reverse hand positions or force directions and collapse separation. Temporal controls shift or scramble event alignment. Computational controls remove dead zone, shaft flexibility, or selected drivers. Sensor controls apply known loads and impossible transforms. Statistical controls test null data and label permutation.

Each control has an expected failure signature. If the pipeline still reports a negative couple after separation is mathematically zero, stop. If shuffled timing produces the same mediation result, timing evidence is not specific.

26.2.14 Primary Analysis for the Two Extremes

Compare continuous arm-dominant and role-reversed wrist-dominant families only after confirming measured allocation differs. Primary endpoints might include transition force-gap duration, effective stiffness, late swing-normal couple, club speed, and a face/path variability metric. Internal load and discomfort are safety outcomes.

Use hierarchical models with participant-specific effects. Test whether preload metrics mediate outcome differences. Report the full distribution and participants whose response reverses. Avoid declaring one universal strategy if heterogeneity is substantial.

26.2.15 Analysis of Slack

Predefine literal contact loss, near-zero differential load, force-development delay, and low stiffness as separate variables. Choose thresholds from sensor qualification or physiology, not outcome inspection. Report duration and phase, not only whether a gap occurred.

Test whether each variable predicts or mediates speed, control, and load after accounting for initial state. A finding that a brief low-load interval is common and beneficial would directly challenge a simplistic “never allow slack” claim.

26.2.16 Model-to-Data Comparison

Do not fit every parameter to each trial and then call the fitted trajectory a prediction. Estimate permissible calibration parameters on training trials, freeze them, and predict held-out wrench, strain, and motion time series. Use absolute errors, phase errors, sign accuracy, energy residuals, and coverage of prediction intervals.

Inspect residual structure. A consistent out-of-plane error suggests missing spatial dynamics; a timing-dependent shaft residual suggests missing modal physics; participant-specific force errors suggest allocation variability. Model revision follows a declared version and new holdout.

26.2.17 Decision Rules

Support is graded. A mechanism can pass geometry controls but fail human sign prediction; preload can alter stiffness but not performance; robustness can hold for a subgroup. Record each result separately. Do not collapse them into one “validated” label.

Predeclare what triggers rejection, narrowing, or redesign. A confidence interval crossing a practical threshold may mean inconclusive rather than no effect. Repeated confirmatory failure should demote the claim in the public evidence register.

26.2.18 Publication Regardless of Outcome

Register the study and commit to releasing qualified results whether supportive, null, or contradictory. Provide code, schemas, calibration, deviations, and limitations. Privacy-preserving data access can retain raw authority while sharing derived evidence and verification hashes.

An open resource earns trust when a negative result improves it. The model, figures, and lay explanation should update together so obsolete claims do not survive in an attractive PDF.

26.2.19 Reader Lab: Write the Failure Sentence First

Before analysis, complete this sentence for each primary claim: “We will narrow or reject the claim if ____ is observed under ____ qualified conditions.” Include effect direction, uncertainty, mediator, and negative controls. Post it with the registration and analysis commit.

Then create synthetic data representing support, null effect, sign reversal, sensor failure, and participant heterogeneity. Run the frozen pipeline and check that it produces the intended decisions without crashing or silently excluding unfavorable cases. Synthetic success qualifies analysis logic, not the mechanism.

After real data arrive, publish the registered result and deviations before exploratory refinements. If the primary claim fails but a subgroup pattern appears, label it exploratory and design a new held-out test. This sequence keeps failure scientifically productive.

26.2.20 Choosing a Minimal First Human Study

The full synchronized protocol is ambitious. A first decisive study could focus on the bilateral negative-couple prediction with instrumented grip, club motion, and shaft strain, plus bounded grip-separation

or geometry manipulation. It would not identify scapular strategy, but it could refute a central contact mechanism.

A second study could add transition stiffness and EMG to compare preload allocations. Staging reduces burden while preserving clear qualification. Each study should be designed around a discriminator rather than accumulating sensors without a primary question.

26.2.21 Stopping Rules and Governance

Define hardware-failure, discomfort, fatigue, and data-quality stopping rules. Record adverse events and protocol deviations. Protect participant identity and keep consent scope attached to every derived release. Scientific openness does not authorize disclosure of raw private data.

These controls belong inside the epic and evidence manifest, not in an appendix read only after collection. They make the proposed falsification program operational and responsible.

26.2.22 A Public Completion Checklist

Before calling an experiment ready, verify qualified hardware, frozen protocol, registered analysis, synthetic failure cases, participant-level power, identity and consent controls, calibrated time and frames, residual thresholds, adverse- event rules, and an outcome-independent publication plan. Each item links to an artifact and owner.

After collection, verify provenance, exclusions, code version, blinded primary result, deviations, uncertainty, and negative controls before interpretation. After publication, connect evidence status back to model and lay text. Completion is a closed trace, not only a paper submission.

26.2.23 If the Central Claim Fails

A failed negative-couple prediction would redirect attention to other contact, shaft, or control pathways. A failed preload mediation would remove justification for interpreting continuous loading through the dead-zone mechanism. A failed held-out policy would narrow optimization claims. None invalidates the general use of state, energy, and counterfactual analysis.

Writing these consequences in advance makes the theory genuinely revisable. It also reassures reviewers that the open resource is designed to learn rather than to sell a predetermined answer.

26.2.24 Chapter Checkpoint

The experimental program is credible when a reader can locate the registered prediction, calibration, data-quality gates, analysis, result, and consequence for the model. Every central mechanism has a feasible negative control and a result that changes its status.

This standard prevents “future work” from becoming an indefinite shield. Even a staged study can close one important uncertainty if it chooses a decisive observable and preserves qualification.

26.3 Where the Picture Breaks

The smoke alarm has a known target. Human strategies adapt to instrumentation and instructions. A manipulation intended to change preload may also change comfort, intent, or path. Manipulation

checks and mediation analysis are needed.

Preregistration cannot rescue poor measurement or an underspecified model. Exploratory findings remain useful when labeled and independently replicated.

Go Deeper: Read the complete [experimental falsification protocol](#), the [model-falsification matrix](#), and the [open qualification boundary](#).

Chapter 27

Open Evidence and the Reviewer Path

27.1 A Concrete Picture

A museum label can say a vase is ancient. A serious catalog adds provenance, material analysis, restoration history, photographs, and access rules. Reviewers can trace the claim instead of trusting presentation.

An open scientific resource needs the same layers. A polished PDF is the label, not the entire evidence. Source equations, scripts, parameters, generated data, figures, checksums, tests, and limitations let another person reconstruct the claim.

The Publication Is an Entry Point, Not the Evidence Boundary



A reviewer can walk backward from prose to reproducible evidence.

Figure 27.1: A Reviewer Path Runs From Claim to Figure, Data, Script, Test, Manifest, and Reproduction Result.

27.2 How the Mechanism Works

Each quantitative figure should point to machine-readable JSON or NPZ evidence. That evidence records parameters, conventions, software identity, outcomes, and residuals. A generating script rebuilds the figure. A test checks its invariants. A manifest lists hashes so reviewers know which artifact belongs to the release.

27.2.1 Determinism and Provenance

Random ensembles use fixed declared seeds and preserve sampled inputs. Build metadata records source commit and tool versions. Determinism does not mean the model is true; it means disagreements can be investigated rather than blamed on an unrecoverable run.

27.2.2 Live Links Are Part of Utility

This book links directly to technical source chapters, evidence directories, and cited literature. PDF URI annotations and HTML links are checked. A reader can move from a casual explanation to equations without searching a repository by filename.

27.2.3 Qualification Is More Than File Presence

The release validator checks hashes, required artifacts, PDF signature, link and outline counts, and generated-source freshness. Scientific tests check action- reaction, work-energy, constraints, limiting cases, counterfactual closure, and timestep behavior. Visual inspection catches clipped captions or unreadable plots that structural tests cannot.

27.2.4 A Reviewer Workbench

The reviewer path groups evidence by claim: interaction forces, hand-path work, two-hand couple, shaft, preload, spatial contact, uncertainty, and experiment. It states what each artifact supports and what remains unvalidated. Commands are copyable and fail closed on mismatches.

27.2.5 Open Does Not Mean Ungoverned

Code and synthetic evidence can be public. Human data may require controlled access. An open resource can publish schemas, derived deidentified results, analysis code, and provenance without exposing identities or licensed data.

Model Result: The current technical release is hash-bound and reproducible through a documented validator; paired figures and data expose the calculations.

Human Evidence: Release qualification does not add human evidence. It makes the absence and future path auditable.

Practical Interpretation: Trust should be able to travel backward from the sentence you read to the numbers, code, assumptions, and test that produced it.

27.2.6 The Reader Path Should Be Continuous

A casual reader begins with a figure and plain-language mechanism. A technical reader follows “Go Deeper” to equations and methods. A reviewer follows a figure to its source table, generator, model manifest, tests, and exact commit. These are not separate products. They are views of one evidence graph.

Links should target specific chapters, files, records, or DOI pages rather than generic repository roots. Stable anchors and release tags protect review from later navigation changes. The PDF must preserve clickable links and accessible bookmarks.

27.2.7 Canonical Source and Generated Artifacts

Text, data, figure scripts, and model configuration are canonical inputs. HTML, PDF, SVG, and rendered tables are generated outputs. Generated artifacts can be versioned for convenient review, but edits belong in canonical source. Build checks verify that public copies match source outputs byte for byte where that is the contract.

Every artifact should record or link to source commit, generation command, toolchain version, and content hash. A reviewer can then reproduce it or detect drift without trusting a filename such as “final.”

27.2.8 Evidence Manifests

A machine-readable manifest lists claim ID, evidence level, model rung, input files, outputs, units, frames, qualification status, limitations, and citations. Unknown or provisional fields fail closed. Synthetic data are labeled so they cannot be mistaken for participant evidence.

The manuscript cites claim IDs or linked artifacts. When new human evidence arrives, updating the manifest and affected passages becomes traceable. This is more maintainable than burying qualification in a long disclaimer.

27.2.9 Reproducible Environments

Lock runtime and dependencies, record operating-system assumptions, and provide one command for each primary analysis. Tests cover physics contracts, schema, citation integrity, title convention, rendering, and public artifact identity. Random processes use recorded seeds.

Reproduction should not depend on a proprietary graphical interface. Optional GUIs and workbenches call canonical libraries. A reviewer can use command-line tools and open formats to inspect every scientific result.

27.2.10 Figure Provenance

Each figure has a descriptive caption and alt text, paired vector formats for web and print, and source data. The plotting script freezes units, signs, colors, and accessibility choices. Multi-panel figures label model result, human evidence, and hypothesis where applicable.

Screenshots may document a three-dimensional model or interactive view, but they should link to the run and camera settings. A screenshot is illustration unless its data and generation path are auditable.

27.2.11 Live References

Bibliography entries should contain DOI, PubMed, publisher, or stable repository links where available. Claims in the lay text link to primary sources and to the technical treatment. Automated checks verify citation keys and URLs, while periodic review handles redirects or changed access.

Links do not replace a bibliography. The rendered PDF should contain complete reference metadata so it remains useful offline, while hyperlinks improve reviewer utility on phones and computers.

27.2.12 Issue and Epic Traceability

Each chapter and cross-cutting requirement has a subissue with explicit acceptance criteria. Pull requests link the issues, explain evidence and limitations, and include local and CI verification. The parent epic summarizes completion only after merged artifacts are publicly reachable.

Issue count is not proof of scientific quality. It is a way to prevent thin coverage, preserve decisions, and let later contributors see why a chapter or control exists. Duplicate or obsolete work is closed transparently.

27.2.13 Open Review Without Software Promotion

The product should be described as an open research resource. Examples may use repository implementations, but conclusions are expressed in physical quantities and portable contracts. Independent implementations are welcomed and their disagreements retained.

Avoid framing every result as a benefit of one package. A reviewer should be able to reproduce equations in another tool and still participate. Scientific value comes from testable definitions, evidence, and criticism—not platform adoption.

27.2.14 Qualification Status

Each release states which models are numerically verified, which predictions are synthetic, which human studies provide bounded context, and which hypotheses remain untested. Unresolved publication pins, missing calibration, or failed gates block promotion to a stronger status.

Status should appear near the relevant claim and in a consolidated register. Readers should never need to infer qualification from repository history.

27.2.15 Long-Term Preservation

Release source and artifacts under appropriate open licenses, archive stable versions, and preserve metadata. Large PDFs and graphics need storage policies that support review without weakening integrity checks. Content hashes and release assets can handle files larger than ordinary source limits.

Migration plans should keep links resolvable. If an external source disappears, retain its citation metadata and a lawful archival pointer where available.

27.2.16 A Reviewer Exercise

Choose one figure. Follow its link to data, rerun the generator, compare hash, inspect the model manifest, and trace the claim to a test and source. Then alter one parameter or negative control. A

healthy evidence product makes this path short and explains changed results.

The ability to disagree reproducibly is the central deliverable. A book that is easy to read and easy to challenge can serve both newcomers and specialists.

27.2.17 Reader Lab: Perform a Ten-Minute Trace

Open one chapter figure on a phone. Follow its caption link to the exact source data, inspect units and qualification, then open the generator and model manifest. The path should not require repository expertise. Back in the PDF, follow the primary citation and the deeper technical chapter. Record any dead end.

On a development machine, rerun the figure from the documented command. Compare output hash or numeric tolerance. Change one parameter and verify the artifact is visibly and semantically marked as exploratory. Then run the physics and publication gates.

If this path takes hours, the evidence may be open in principle but inaccessible in practice. Improving navigation, schemas, and small reproducible examples can advance review more than adding another framework layer.

27.2.18 Versioning a Living Scientific Book

Each release should state what changed in claims, evidence, model, and qualification. Stable chapter and claim identifiers let external reviews point to exact content. Corrections remain visible; old releases are archived rather than overwritten without trace.

Large artifacts can live in release assets or suitable large-file storage while source, manifests, and hashes remain in version control. The public URL should resolve without authentication and return the intended content type. A local PDF and deployed copy should be compared before completion is claimed.

27.2.19 Contribution Contract

Contributors can add a model rung, dataset, replication, or critique by satisfying shared schemas and tests. They need not adopt one engine or endorse the theory. New claims name falsifiers and evidence level; new figures include source and alt text; new human data retain consent and identity safeguards.

This contract makes disagreement composable. A contradictory implementation can sit beside baseline and change qualification through evidence rather than being treated as a software bug to hide.

27.2.20 A Minimum Release Bundle

The release should contain canonical chapter source, bibliography, figure data and generators, paired web/print vectors, PDF and HTML, model and run manifests, schemas, tests, exact commit, licenses, and qualification register. Public URLs are checked anonymously. The PDF is checked for bookmarks, live links, readable figures, and mobile-scale text.

Large artifacts should use an explicit storage route rather than an arbitrary small-file gate. Integrity comes from hashes, provenance, and reproducible generation, not from keeping scientific documents

under one megabyte. Source review remains practical because generated binaries are separated from canonical edits and compared by contract.

27.2.21 Keeping the Resource Scientifically Neutral

Lead with questions, definitions, and evidence. Mention implementation only when it enables reproduction. Use generic physical terms in conclusions and link to portable schemas. Compare independent engines and preserve disagreement.

Avoid page labels that describe the production request, promotional claims, or repetition of repository branding. The reader should experience a scientific book backed by open resources, not a brochure about the resources themselves.

27.2.22 The Maintenance Loop

New issue proposes a falsifiable change. Tests define the contract. Canonical source or engine changes. Local numerical and visual gates run. Protected review and CI merge the change. Public artifacts are verified. Evidence status and chapter links update. Obsolete issues and outputs are reconciled.

This loop is deliberately mundane. It is how a large living document remains trustworthy after the initial authoring effort.

27.2.23 Chapter Checkpoint

Open evidence is not a folder of files. It is a navigable, versioned chain from claim to source, model, run, data, figure, test, limitation, and public artifact. The chain must work for a phone reader and an independent technical reviewer.

When a link breaks, a model fails, or new evidence contradicts a claim, status updates at every view. Protected review and reproducible builds preserve the history. This is how a large publication stays scientifically neutral after its first release.

The ultimate test is whether an independent person can reproduce a result, change an assumption, disagree, and contribute the disagreement without needing permission from one software implementation.

The repository should also publish a compact reviewer index. For every major claim it names the accessible explanation, technical derivation, model rung, data output, figure, tests, qualification, and open issue. A static PDF can link to this evolving index while release tags preserve the reviewed version.

Automated link checks cannot prove a source supports a claim. Human review still verifies relevance, study design, and wording. Automation catches broken routes, missing keys, stale artifacts, and inconsistent titles so reviewers can spend time on scientific judgment.

27.3 Where the Picture Breaks

The museum catalog can preserve a stable object. Software dependencies and web links change. Exact hashes improve traceability but long-term preservation also needs archived releases and documented environments.

Reproducibility can reproduce a systematic mistake. Independent formulation, adversarial review, and new data remain necessary.

Go Deeper: Start with the [reviewer workbench](#), [open release qualification](#), and [data dictionary](#).

Chapter 28

A Practical Synthesis Without a Swing Prescription

28.1 A Concrete Picture

A weather map can explain pressure, wind, and uncertainty without telling every person whether to carry an umbrella. The decision depends on destination, tolerance for rain, and changing local conditions. A mechanics model can clarify the swing in the same way without declaring one motion correct for every golfer.

We can now assemble the picture: proximal work moves a linked system; constraints and inertia transmit and redirect effects; geometry changes leverage; two hands create visible and hidden wrench modes; negative torque can retain or stabilize; the shaft stores and returns energy; timing depends on state; robustness exposes tradeoffs; human evidence remains the final gate.

28.2 How the Mechanism Works

The useful framework is a sequence of questions.

First, choose the system and outcome. Are we explaining club kinetic energy, clubhead velocity, face/path, joint load, repeatability, or some combination?

Second, reconstruct the state and geometry. Where are the segments, how are they moving, and what elastic or activation states are loaded?

Third, close the instantaneous mechanics. Which forces, moments, constraints, gravity, and velocity terms balance acceleration? Are frames and reference points consistent?

Fourth, close work and energy over time. Which interfaces supply, transfer, store, return, or dissipate energy? Do positive and negative contributions cancel?

Fifth, intervene fairly. Hold state for pointwise attribution, integrate matched branches for persistence, and resimulate full schedules for strategy. Include negative controls.

The Framework Connects Mechanisms Without Selecting One Technique

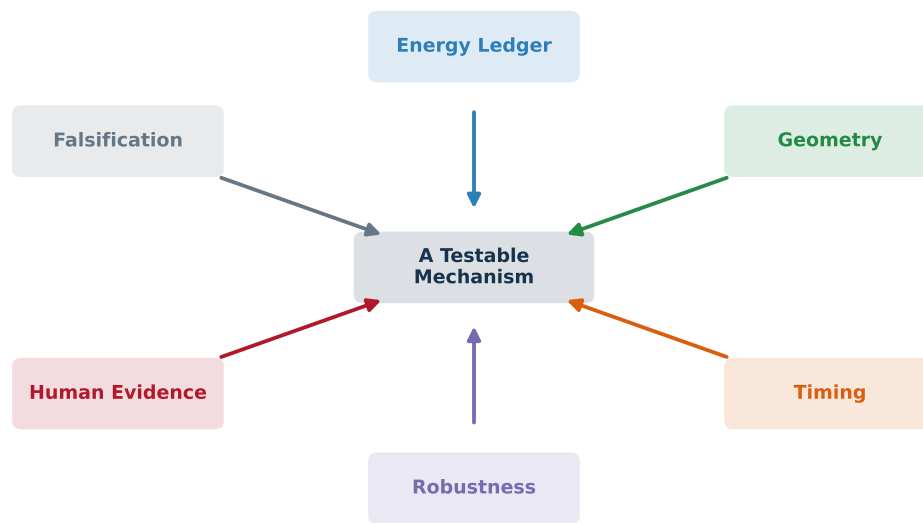


Figure 28.1: A Final Map Connects Energy Supply, Transfer, Storage, Geometry, Timing, Robustness, and Falsification Without Selecting One Technique.

Sixth, climb the model ladder. Does the mechanism survive two hands, compliance, a moving base, spatial motion, independent engines, and uncertainty?

Seventh, design the human test. Measure the variables that distinguish competing stories and preregister what would count against each.

28.2.1 What the Current Evidence Supports

The reduced models support a real mechanical possibility: club acceleration and hand-path work can be strongly influenced by interaction and state-dependent terms even with selected distal torque removed. Two-hand geometry can create a late negative club couple. Flexibility can modify timing while obeying energy closure. Arm- and wrist-dominant allocations can produce the same club task with different internal demands. Preload continuity helps only under a declared transmission family.

28.2.2 What Remains Open

The work does not establish a universal human technique, scapular mechanism, muscle preactivation strategy, injury benefit, or calibrated equipment effect. It does not identify one best allocation. The held-out model ensemble explicitly rejects a universal optimum across its outcomes.

28.2.3 Better Practical Questions

Instead of “should I hold lag?” ask which state variable and outcome the cue is supposed to change. Instead of “are the wrists passive?” ask whether direct wrist moment, impedance, and force-generated couple were measured separately. Instead of “is slack good?” ask which gap, stiffness, compliance, or

activation variable was observed. Instead of “did the hips transfer speed?” ask which boundary power and energy balance supports that statement.

These questions do not make instruction impossible. They make claims testable and participant-specific.

28.2.4 A Reader’s Next Experiment

Use the interactive workbench to compare clock and state triggers, remove a selected torque at one state, reverse a two-hand moment arm, or change shaft stiffness. Predict the result before running it. Then inspect forces, powers, work, energy closure, and falsifiers rather than only final speed.

Model Result: The project delivers a connected, openly reviewable set of mechanism tests across a common-observable ladder.

Human Evidence: The literature motivates and bounds the questions but does not yet validate the complete strategy mechanism.

Hypothesis: Some golfers may benefit from strategies that maintain useful proximal drive, manage distal resistance and preload, and permit state-dependent late transfer without force-transmission gaps. The opposite allocation remains a serious competing hypothesis.

Practical Interpretation: The framework is not a recipe. It is a way to ask better questions of a motion, a model, a sensor, or a coaching claim.

28.2.5 What the Framework Now Says

The swing is a coupled mechanical system. Proximal work, moving geometry, constraint reactions, direct distal control, bilateral hand allocation, shaft deformation, and ground interaction overlap. Clubhead speed is an outcome of their integrated trajectory. No single peak, torque sign, or body-part label contains the whole explanation.

The proximal-distal description remains useful when it means changing emphasis and energy pathways in a continuously linked system. It becomes misleading when turned into a strict relay or a universal timing script. The framework replaces that script with state, force, wrench, power, work, and counterfactual tests.

28.2.6 What the Models Establish

Verified reduced models demonstrate that interaction forces can carry and accelerate a distal link without direct distal drive; geometry controls the sign and leverage of that contribution; negative torque or contact couple can coexist with a positive later outcome; and multiple hand allocations can yield the same net club task. Flexible, moving-base, and spatial rungs test whether those mechanisms survive added physics.

These are model results under declared assumptions. Their strongest value is the observable predictions they generate. They do not establish a unique muscle strategy or prove that golfers exploit every simulated pathway.

28.2.7 The Two Strategy Extremes

One family emphasizes continuous arm/shoulder-girdle drive with wrist resistance from backswing through early downswing. Its proposed advantage is a loaded transition state with less transmission delay. The opposite family emphasizes greater wrist-created backswing club moment, arm resistance, then arm drive with wrist opposition in downswing. Its possible advantage is phase-specific use of different mechanical and biological resources.

Both can maintain contact and stiffness; both can create or avoid a measured force gap depending on implementation. Real strategies can lie between them. The meaningful comparison matches club task and measures bilateral wrench, stiffness, activation, work, load, and robustness.

28.2.8 A Decision Tree for the Researcher

If the question concerns linked inertia, begin with the double pendulum. If it concerns push–pull allocation, add two hands. If delay or “kick” is central, add shaft and transmission states. If ground or handle path is central, release the base. If out-of-plane couple matters, use spatial dynamics. If muscle claims are made, add biological measurements and a qualified musculoskeletal rung.

At each step, retain the smallest model that answers the question, common observables, negative controls, and closure tests. Add complexity only when a residual or hypothesis requires it.

28.2.9 A Decision Tree for the Evidence

If only motion is measured, report kinematics. With external wrenches and inertial assumptions, report net dynamics. With bilateral grip sensing, test hand allocation and contact couple. With shaft strain, test elastic pathways. With EMG and perturbation, constrain activation timing and effective stiffness. Do not climb the claim ladder without the corresponding evidence.

When evidence disagrees, localize the disagreement. A failed anatomical mapping does not automatically falsify the contact mechanics; a failed wrench prediction does. A model fitting speed but failing reactions needs revision.

28.2.10 What This Means for Creating Clubhead Speed

Within the mechanical framework, favorable conditions include sufficient whole- system work, a hand path and geometry that permit useful late club acceleration, control timing that avoids premature or incomplete release, and allocation that remains robust to execution error. An opposing moment can be useful if it shapes state and later power rather than simply wasting energy.

Those conditions are evaluation criteria, not body instructions. A specific golfer may meet them through different coordination. Speed must be considered with face/path, impact, internal load, and repeatability.

28.2.11 What “Manage the Negative Couple” Can Mean

Management is not necessarily conscious application of a negative wrist torque. It may mean arranging geometry and proximal motion so contact reactions have a favorable sign, maintaining enough engagement to avoid unwanted delay, and allowing direct control to regulate rather than fight the coupled motion. It may also mean avoiding excessive retention that leaves too little time.

The phrase should remain tied to measurable couple, power, timing, and outcomes. Without those, it is only a cue. The project does not recommend a cue from the current evidence.

28.2.12 A Practical Analysis Workflow

Begin with synchronized, qualified data. Reconstruct state and uncertainty. Close whole-system and interface ledgers. Plot force vectors and bilateral wrenches in declared frames. Run pointwise decompositions and matched-state forward branches. Compare model rungs and independent engines. Propagate parameter uncertainty and evaluate held-out participants.

Then classify each conclusion: mechanics identity, verified model result, human observation, hypothesis, or practical interpretation. Link it to the evidence graph. This workflow makes a sophisticated modeling product usable without pretending it is already a coaching authority.

28.2.13 Near-Term Completion Priorities

The highest-value next work is a qualified synchronized bilateral dataset, a spatial whole-body model with measured base and shaft state, participant-held-out prediction, and explicit arm/wrist allocation experiments. The workbench should expose counterfactuals, uncertainty, and failure regions with shared canonical engines. Documentation should update automatically from evidence manifests.

These priorities deepen falsifiability more than adding another attractive nominal animation. They also invite independent replication because the observables and contracts are software-neutral.

28.2.14 Questions a Reader Should Keep

What system and boundary are used? Which quantities were measured? Which were inferred or simulated? Does the ledger close? Is the comparison matched by state or task? Does geometry control the predicted sign? Can another allocation explain the same motion? Does the result survive uncertainty, higher model rungs, and held-out people? What finding would make the authors change their minds?

If those questions remain visible, the framework can grow without becoming a closed doctrine. It becomes an open map of mechanisms, evidence, and remaining uncertainty.

28.2.15 Final Boundary

Nothing in this synthesis establishes a universally superior swing, a prescribed scapular action, or a safe wrist-loading pattern. The research supports a more precise way to ask how transfer can occur and how competing explanations can be tested. Practical use requires qualified individual evidence and professional judgment appropriate to performance and health.

28.2.16 Reader Lab: Evaluate a New Claim

Suppose a new analysis says, “Trail-hand pull creates passive wrist lag and more speed.” Translate it into testable pieces. Define trail-hand force in a frame; define lag as state; define passive channel; identify power and work; specify the comparison and outcome. Ask whether arm, wrist, shaft, and geometry alternatives can reproduce it.

Then choose a model rung and negative controls. Predict bilateral wrench and shaft response. Match state for causal removal and club task for allocation comparison. Propagate uncertainty. Finally define a human measurement and a failure condition. Only after these steps should practical interpretation be considered.

This workflow does not drain intuition from the question. It converts intuition into an argument that another group can test.

28.2.17 The Desired End State of the Project

The complete product is a linked set of readable chapters, technical derivations, canonical simulation libraries, interactive experiments, qualified evidence, and reproducible publication. A newcomer can understand why a force vector matters; a specialist can audit equations; an experimentalist can see the next discriminating measurement; and an independent developer can implement the contracts elsewhere.

Completion does not mean the biological hypothesis is proven. It means the framework is sufficiently explicit, broad, and falsifiable that future data can change it cleanly. That is a more durable scientific objective than presenting one software package or swing strategy as finished truth.

28.3 Where the Picture Breaks

A synthesis map can create the illusion that every arrow has equal evidence. They do not. Structural mechanics, simulated model results, and human observations occupy different evidentiary levels. The claim labels must travel with any excerpt from this book.

The final danger is using scientific caution as vague neutrality. The models do make conditional predictions. They should be tested decisively, updated when wrong, and retained when they survive—not promoted prematurely and not dismissed because they are reduced.

Go Deeper: Read the monograph’s [discussion and conclusions](#), open the [interactive workbench](#), and use the [reviewer path](#) to challenge any claim that matters to you.

Chapter 29

Walk Through a Whole Swing Without Losing the Ledger

29.1 A Concrete Picture

Imagine standing beside a complicated relay race. The runners do not merely pass a baton from one hand to the next. They are connected by springs, they can push on the track, several of them can hold the baton at once, and the finish line is moving. If we watch only the baton, the race can look beautifully simple: one runner speeds up, then another, and finally the baton crosses the line. If we want to explain *why* it crossed quickly, however, we need a ledger. Who did work? Which connection redirected a force? Where was energy stored? Which apparent contribution was just a change of coordinates? Which conclusion survives when we rerun the race with one action removed?

A golf swing is not literally that relay, but the picture is useful because it forces us to keep several questions separate. The clubhead can gain speed while the hands slow. A wrist moment can be negative while the clubhead continues to accelerate. A force can be large while doing little work because it is nearly perpendicular to the point velocity. A shaft can store energy during one interval and return it during another. Two hands can create the same net club wrench with very different internal loads. None of those statements is a contradiction. These are mechanical possibilities under specified states and loads, not measurements of the imagined golfer. They become confusing when the ledger columns are mixed.

This chapter follows one imagined swing from preparation to impact. It does not claim that this is the unique swing of a skilled player. Instead, it shows how to carry the book's concepts through one continuous story without quietly changing the system boundary, reference point, model, or meaning of a word.

29.2 How the Mechanism Works

29.2.1 Before Motion: Write Down the Accounting Rules

Our imagined observer begins before the club moves. The chosen system is the golfer plus club when asking about external work from the ground. It becomes the club alone when asking what the hands

The Framework Connects Mechanisms Without Selecting One Technique

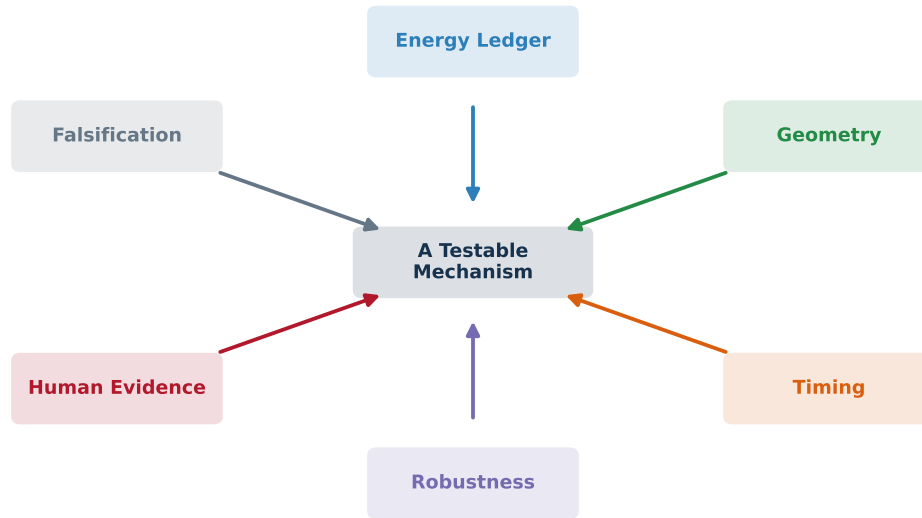


Figure 29.1: A Synthesis Map Connects State, Geometry, Forces, Power, Counterfactuals, Evidence, and Falsifiers Without Collapsing Them Into One Claim.

do to the club. It becomes the complete flexible club when asking about external grip work, but it may be divided into proximal and distal club segments when asking where shaft strain energy lives. Those are different ledgers. Changing between them is allowed; changing without saying so is not.

The observer also declares a laboratory frame, a club-attached frame, the grip center as the main wrench reference point, and positive angular directions. This sounds fussy until a force is transported from one hand to the grip center. The force remains the same physical interaction, but its reported moment changes with the reference point. For a rigid grip region in the same observer frame, the paired reduction is $P = \mathbf{F} \cdot \mathbf{v}_A + \mathbf{M}_A \cdot \omega$. For $\mathbf{r} = \mathbf{r}_B - \mathbf{r}_A$, use

$$\begin{aligned}\mathbf{v}_B &= \mathbf{v}_A + \omega \times \mathbf{r}, \\ \mathbf{M}_B &= \mathbf{M}_A - \mathbf{r} \times \mathbf{F}.\end{aligned}$$

The two evaluations give the same total power, although their force and moment parts change. This transports the same physical velocity field; changing to a moving observer is a different operation and generally changes power. A deforming grip cannot be reduced to one rigid twist without retaining the missing deformation power. Its individual contact forces need their actual material-point velocities. A persuasive diagram cannot repair inconsistent bookkeeping.

Finally, the observer lists the state variables available in the chosen model: segment angles and angular velocities for the simplest pendulum; pelvis and trunk motion for a moving-base model; shaft deflection and deflection rate for a flexible club; activation and series-force histories for a muscle-driven model. With specified future inputs or input laws, the retained state determines the model evolution. If an internal state is omitted, the same visible pose can hide different futures. Activation is a state, not an

energy quantity; an activation history does not itself measure elastic storage or metabolic cost. The state list determines which timing questions the model can answer.

Practical Interpretation: Before asking which body part “caused” speed, ask what objects are inside the accounting boundary, what instant or interval is being studied, and which stored states are being carried forward.

29.2.2 Preparation: Motion Creates a State, Not Merely a Position

During the imagined preparation, the golfer moves the club away from the ball. The visible result is a backswing pose, but the mechanical result is richer. Segments have velocities, tissues and the shaft may carry strain, muscles have activation histories, the two hand contacts carry some distribution of force, and the ground has exchanged impulse with the body. Two golfers can arrive at nearly the same photograph with different values in those hidden columns.

This is why “the top” is not a complete initial condition. A model initialized from a still pose and zero velocity has erased the route into that pose. Such a model can still answer a deliberately posed question, but it cannot be used as evidence that preparation history is irrelevant. The continuous-preparation experiments in the technical work exist precisely to avoid resetting an internal transmission state at the command transition.

Consider the two allocation extremes developed earlier. In one, a proximal push–pull contribution remains directionally loaded while the wrists provide an opposing contribution. In the other, the modeled roles reverse across the transition. Both command programs can match the same net desired moment before and after the change while taking different routes through internal transmission states. That does not establish the same transmitted moment or club trajectory throughout transition. The [archived allocation study](#) reports an advantage for persistent direction in nine positive-dead-zone cases, and equivalence of the chosen net error-impulse metric in three zero-width cases. The equivalence threshold is 10^{-10} N m s within that numerical study.

Removing a dead zone does not remove finite force-development time or make every internal trajectory equal. Two linear channels with the same time constant and matching summed commands can have the same summed output while their individual forces reverse at different times. The result is conditional on the transmission, initialization and metric. It neither identifies a human wrist dead zone nor selects a universal wrist or scapular strategy.

29.2.3 Transition: Replace the Magic Instant With a Neighborhood

The swing reverses direction over an interval, not at a mathematical point that contains the whole explanation. Near transition, some segment angular velocities approach zero while accelerations and internal forces need not. The club may still be moving one way as proximal segments begin accelerating the other way. A graph aligned to a named event can make this overlap visible, but the event label does not create the mechanism.

Our observer therefore records a transition neighborhood. The ledger contains configuration, velocity, applied generalized moments, constraint reactions, hand wrenches, shaft state, ground-reaction wrench, and the power of each declared interaction. If the observer wants to discuss preactivation, activation must be estimated from declared measurements or represented by a biological model; an EMG signal is not a direct measurement of every muscle force. If the observer wants to discuss slack, the word is

replaced by a quantity such as contact gap, tangent stiffness, low force, series strain, or force-development delay.

This disciplined language also makes opposing hypotheses stronger. Continuous loading might improve immediate transmission and perturbation resistance. A controlled unloading or role reversal might reduce internal loading, permit useful compliance, reshape the club path, or improve a participant's accuracy. The experiment must price both benefits and costs rather than declaring one side “connected” and the other “slack.”

29.2.4 Early Downswing: The Ground and Moving Base Join the Story

As the downswing begins, the golfer pushes on the ground and the ground pushes back. This ground wrench contributes to the golfer-plus-club momentum balance, together with gravity and any other external loads. Integrate all external forces for linear momentum and all external moments about a fixed inertial point for angular momentum. A moving angular-momentum reference requires the appropriate transport terms.

Impulse and work answer different questions. At an ideal stationary no-slip contact, material-point velocity is zero, so the contact force does zero work despite supplying impulse. A synthetic 100 N force maintained for 0.1 s has impulse 10 N s; at zero contact velocity its work is zero. Multiplying that force by center-of-mass velocity gives a different quantity used in center-of-mass kinetic accounting, not the contact's boundary power. The moving center of pressure is also not a material point whose migration speed can simply be multiplied by the ground force.

Real shoe/turf deformation, slip or a moving support can change this idealization. Their power requires the actual distributed tractions and contact velocities or a justified equivalent model. The ground can make a swing mechanically possible without serving as its source of mechanical energy. Muscle conversion of chemical energy and changes in gravitational or elastic storage remain separate entries.

Pelvis and trunk motion make the arm base move. In a prescribed-base pendulum, that motion is supplied as an input trajectory; the model does not pay the energetic or dynamic cost of producing it. In a finite-mass moving-base model, the base responds to applied forces, restoring terms, damping, and connected segment reactions. The second model can test whether a mechanism survives when the base is allowed to react. It is not automatically more truthful in every parameter, but it closes a question that the prescribed model leaves open.

The familiar proximal-to-distal pattern may now appear in segment speeds. Pelvis speed peaks, then trunk, arms, and club. This is a useful observation, not yet a causal decomposition. Overlapping accelerations, interaction forces, joint moments, and changing geometry can produce the ordered peaks. A peak-speed sequence alone cannot tell us which term did work or whether deliberately slowing one segment would improve the final outcome.

Human Evidence: Tinmark and colleagues studied 45 elite golfers and reported proximal-to-distal timing of peak angular-speed magnitudes for the pelvis, upper torso and hands across their partial/full-shot conditions (Tinmark et al. 2010). That result does not establish a universal pelvis–trunk–arm–club sequence or its cause. Marsan and colleagues compared seven angular-velocity definitions in 13 golfers' driver swings and found that the identified sequence depended strongly on the selected component (Marsan et al. 2019). Define the segments, measured quantity, processing and event rule before

comparing peak times; observations motivate mechanism tests rather than prescribe one control strategy.

29.2.5 Mid-Downswing: Geometry Changes What the Same Force Can Do

The arms and club now form a changing geometric machine. A force at the hands has a moment about a joint according to its moment arm. The same force vector can create a large turning effect at one configuration and almost none at another. Likewise, the same joint moment can produce different accelerations because the mass matrix and dynamic coupling terms change with configuration.

Suppose a hand force points mostly along the instantaneous hand path. Its power at that point may be large because force and velocity align. Suppose another force is almost perpendicular to the path. Its direct power is smaller, yet it may redirect the path, maintain a grip constraint, or create a couple with the other hand. Calling the first force “useful” and the second “wasted” would miss the constraint and geometry columns of the ledger.

The double pendulum makes the changing leverage visible. When the arm–club angle is large, proximal motion can create a different induced acceleration of the club than when the segments nearly align. The model’s off-diagonal inertial terms are not a mysterious extra force; they express how coordinates in a linked system share kinetic energy and acceleration. A counterfactual can then ask how the club acceleration changes at the same state when one declared control moment is set to zero. That pointwise result is not the same as rerunning the rest of the swing without the moment.

29.2.6 Delivery: Two Hands Can Produce Force and Couple Simultaneously

Near delivery, reducing the hands to a single point hides an important degree of freedom. The resultant hand force contributes to center-of-mass acceleration together with the other external forces. Its moment about the club center of mass also matters. For hand separation $\mathbf{d} = \mathbf{r}_R - \mathbf{r}_L$ and the midpoint G of the two hand contacts (not the club center of mass), the force-generated moment is

$$\mathbf{M}_G^{\text{force}} = \frac{1}{2} \mathbf{d} \times (\mathbf{F}_R - \mathbf{F}_L).$$

Equal and opposite forces form a pure couple, independent of reduction point. For a nonzero resultant, the reduced moment is point-dependent. To obtain the moment about the club center of mass C , add $(\mathbf{r}_G - \mathbf{r}_C) \times (\mathbf{F}_R + \mathbf{F}_L)$ to $\mathbf{M}_G^{\text{force}}$. Direct applied moments add to this wrench; they are not hidden inside the force moment. Many hand-load combinations can yield the same rigid-club wrench, while internal loads and compliant-contact power can differ.

This is the setting in which a negative grip-moment component may coexist with increasing clubhead speed. “Negative” refers to an oriented axis, and spatial moment power is the dot product $\mathbf{M} \cdot \boldsymbol{\omega}$, not one component’s sign. On a rigid club, hand power is the sum of the force and moment terms at one consistent point. At an anatomical rotational joint, the paired moment contribution instead uses the relative angular velocity of its two bodies. Neither net quantity uniquely identifies individual muscle work.

Gravity and other external loads can supply power while one component absorbs it. Elastic release can convert stored energy into kinetic energy within a flexible club; already acquired kinetic energy is part

of the state, not an additional power source. These distinctions explain coexistence without showing that an opposing moment improves a particular task. That requires an intervention and an objective.

Model Result: In the [archived forward two-hand study](#), a branch inherits the baseline state at 0.20 s and sets its applied ideal joint torques to zero. The force-generated moment about the study’s declared club reference remains negative at all stored samples over the next 0.05 s. Gravity, fixed-shoulder constraints and ideal bilateral hand contacts remain. This is archived model evidence, not a new run, a finding that human muscles are inactive, or identification of scapular retraction.

The archive separately reports near-machine-level wrench-power identities and nonzero integration work-energy residuals that decrease under step refinement. An algebraic identity closing tightly does not make the trajectory exact. Changing geometry, contact laws or actuator dynamics requires another scoped test; removing neural excitation with activation retained is a different intervention from removing ideal applied torque.

29.2.7 The Shaft: A Short Memory With Direction and Timing

The flexible shaft adds internal state. When it bends, it stores strain energy; when it straightens, it can return some of that energy. Damping dissipates a portion, and the deformation can alter clubhead orientation as well as speed. It is therefore misleading to describe the shaft only as a catapult that adds free energy. Its return comes from energy previously supplied to the club-body system, and its practical effect depends on timing, mode shape, damping, geometry, and impact conditions.

In our ledger, strain energy remains stored energy even when the shaft is partitioned. Choose material subdomains and a named cut section: each part has its own kinetic and strain energy, while traction power across the cut is an exchange term. Storage is measured in joules and power in watts; one does not turn into the other merely because the boundary moves on a diagram. A spatially moving cut additionally needs transport accounting.

For the whole flexible club, one useful convention is

$$\frac{d}{dt}(K + U_{\text{shaft}} + V_g) = P_{\text{hands}} - P_{\text{diss}},$$

with no other external power, gravity included once through V_g , and all retained translational, rotational and deformation kinetic energy in K . If U_{shaft} decreases, kinetic energy may rise without an extra external shaft-return term. The distribution of that rise among head, shaft and proximal parts depends on their interface powers; release need not all help the head. The shaft also makes two visually similar delivery poses dynamically different because bend and bend-rate states can differ.

This gives “late release” a more careful meaning. A delayed visible angle does not, by itself, reveal whether the shaft is storing energy, whether wrists are doing positive or negative work, or whether a constraint reaction is creating the observed acceleration. Those mechanisms need their own measurements and model variables.

29.2.8 Impact: Do Not Let One Number Swallow the Task

Clubhead speed immediately before impact is important, but it is not the only task variable. Face orientation, path, strike location, dynamic loft, shaft state, and variability all affect the ball outcome.

A simulated strategy that adds a small amount of peak speed while greatly increasing face or strike variability may not be better for the actual task.

The observer therefore reports a task vector rather than a victory number. That vector can include speed, path, face, strike, peak hand force, joint-load proxies, energy cost, and robustness to timing or parameter perturbations. The choice of weights is an objective, not a fact of mechanics. Different players and tasks can rationally occupy different points on the tradeoff surface.

At impact, the ledger also changes because the ball enters the interaction. The very short collision requires contact mechanics beyond the pre-impact swing model. The pre-impact state supplies initial conditions for that event; it does not by itself determine every launch outcome without a validated club–ball contact model.

29.2.9 Rewind the Swing With Counterfactual Scissors

Now the observer rewinds. First comes a pointwise decomposition: at selected states, set one declared control input to zero and recompute acceleration or constraint force without advancing time. This is a finite same-state comparison, not necessarily a derivative. For the illustrative nonlinear input map $a(u) = u^2$, changing $u = 2$ to zero changes a by -4 , while the local derivative predicts $2u\Delta u = -8$. In a fixed-mode control-affine model, the input derivative gives the finite change exactly; contact switches, saturation and actuator dynamics can invalidate that simplification. Recompute constraint reactions consistently.

Next comes a forward intervention: declare which command or policy changes, retain compatible configuration, velocity and internal state, and integrate. Input removal at each baseline state does not recreate this divergent path. Compare a declared diagnostic time or each branch’s own impact event, and report a branch that misses the ball. These answer different questions about consequences over time.

The observer distinguishes model controls from numerical verification. With bounded hand forces, collapsing separation removes the force moment about the common contact point; it does not remove a direct free moment or the resultant’s moment about a different point. Demanding a fixed nonzero couple as separation shrinks instead requires unbounded forces. Any geometry intervention must initialize a compatible state and check constraint rank.

Consistent coordinate representation and rigid-point transport should preserve the same observer’s total power. Timestep refinement checks numerical stability and convergence of the particular outcome; track constraint, equation and work-energy residuals with units and declared tolerances. A change under a real intervention can be the predicted effect, whereas failure of a representation identity indicates an implementation or convention problem. None of these checks alone validates the mechanism in a golfer.

Finally, the observer moves up the model ladder. Does the double-pendulum result survive a two-hand constrained model? Does it survive a moving base and flexible shaft? Does it survive spatial motion and parameter uncertainty? Do measured bilateral wrenches, kinematics, shaft strain, ground reactions, and activation timing show the predicted relationships in held-out participants? Adding complexity does not guarantee reduced uncertainty; it can introduce new unidentified parameters. Advance when a new tier answers the declared question and survives its verification and measurement tests. No earlier rung licenses skipping evidence needed for the eventual claim.

29.2.10 Read the Finished Ledger in Plain Language

At the end, the observer can say something careful and still useful. The linked geometry allows proximal motion, interaction forces, hand-force couples, direct moments, and shaft state to influence distal motion in overlapping ways. Ordered speed peaks are compatible with transfer but do not uniquely identify it. Negative torque can coexist with a positive distal outcome because sign, power, system boundary, and time interval are different questions. Continuous loading and role reversal are separable hypotheses whose predicted differences depend on stiffness, gaps, activation history, and the task being optimized. The present models make those statements testable; they do not select a universal human technique.

That conclusion is less dramatic than “energy flowed down the chain,” but it is far more informative. It tells us which quantities to measure, which model states not to reset, which counterfactuals to run, and what evidence could make the explanation fail.

29.2.11 Reader Lab: Build a One-Page Ledger

Choose one interval—transition, mid-downswing, delivery, or impact approach. Draw six columns labeled state, external wrench, internal exchange, stored energy, task outcome, and falsifier. Fill each column with quantities rather than body-part stories. If you write “hips transferred energy,” replace it with a measurable term: pelvis-to-trunk joint power, interaction work over the interval, or a counterfactual change in club kinetic energy. If you write “wrists resisted,” specify joint moment, angular velocity, and the sign of their product. If you write “no slack,” name the gap, stiffness, force, or activation variable.

Then ask what your one-page ledger cannot establish. That final blank space is not a weakness. It is the beginning of the next experiment.

29.2.12 Chapter Checkpoint

The whole-swing story remains coherent only when the accounting rules stay visible. Preparation creates state. Transition is an interval. The ground contributes impulse; its boundary work depends on actual contact motion. Geometry changes leverage. Two hands create both resultant force and couple. The shaft stores and returns energy. Impact adds a new contact problem. Counterfactuals separate local contributions from forward consequences. Human evidence is required before a model allocation is called a biological strategy.

29.3 Where the Picture Breaks

The relay picture suggests discrete handoffs, whereas a real swing contains overlapping power, simultaneous constraints, distributed tissues, and feedback. The walkthrough also combines lessons from several model tiers; no single executed model currently contains every element in one participant-calibrated simulation. A coherent ledger is not proof that every entry is experimentally accurate. Parameters, soft-tissue motion, muscle forces, contact behavior, and individual objectives remain uncertain. The synthesis earns a research program, not a swing prescription.

Go Deeper: Compare the complete [scientific monograph](#), the machine-readable [transmission robustness evidence](#), and the [reviewer workbench](#).

Chapter 30

A Curious Golfer and a Skeptical Reviewer Ask the Hard Questions

30.1 A Concrete Picture

Picture two readers at the same table. One is a curious golfer. This reader wants the mechanics to connect with sensations and visible motion: Why does the club seem to accelerate when the hands are not trying to fling it? Why might a negative torque help? Is “keeping tension” meaningful? The other is a skeptical reviewer. This reader keeps asking what was measured, what was assumed, which model produced the number, and what result would make the authors change their minds.

They are not opponents. The curious golfer prevents the explanation from hiding behind notation. The reviewer prevents the explanation from becoming a confident story that outruns the evidence. A mature open research resource needs both voices.

The conversation below is deliberately sustained. Each question is followed through mechanics, evidence, and limitations rather than answered with a slogan. The point is not to memorize a preferred response. It is to learn how to move from an intuitive question to a claim that can be checked.

The figure describes evidence provenance. The dialogue adds another obligation: definitions, assumptions, uncertainty, and rival explanations must be checked at every step. A reproducible result can still answer the wrong question.

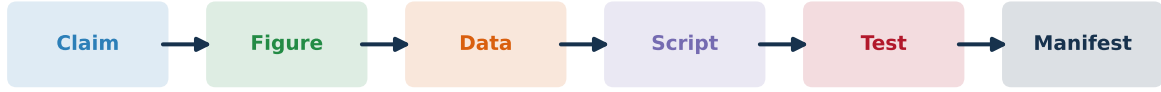
30.2 How the Mechanism Works

30.2.1 “Is Proximal-to-Distal Transfer Just a Fancy Name for Sequencing?”

The golfer begins with the visible pattern: pelvis, trunk, arms, and club often reach peak speed in that order. The reviewer agrees that this is a useful kinematic description, then separates it from the mechanism. A sequence tells us *when* selected maxima occur. Transfer asks how forces, moments, constraints, work, and stored energy change the motion of a declared subsystem.

Two swings can show similar peak ordering while differing in joint power, ground-reaction impulse, hand forces, shaft deformation, and robustness. Two other swings can differ in peak timing yet reach

The Publication Is an Entry Point, Not the Evidence Boundary



A reviewer can walk backward from prose to reproducible evidence.

Figure 30.1: A Reviewer Follows the Claim Through Its Figure, Data, Script, Test, and Manifest.

similar club states at impact. The sequence is therefore one observable among many. It becomes mechanically explanatory only when connected to a model that predicts the measured forces and energy changes and survives counterfactual tests.

The golfer’s useful takeaway is not that sequencing is unimportant. It is that “good sequence” should be unpacked. Does it mean a reproducible ordering of segment speeds, a particular phase relation, favorable hand-path geometry, low peak loading, or robust impact conditions? Each meaning suggests different measurements.

30.2.2 “Does Energy Literally Travel From the Hips Into the Club?”

The everyday phrase sounds like a parcel moving through a delivery network. Mechanical energy is a scalar accounting quantity, not a material substance. When we isolate the club, hand forces and moments can do work on it. When we isolate the trunk, joint interactions and external forces appear at its boundary. With a flexible shaft, strain energy can be stored internally and later returned. The numerical transfers depend on where the system boundary is drawn.

The reviewer asks for a power balance. For a rigid club, reduce all hand forces and directly applied moments to a resultant force $\mathbf{F}$ and moment $\mathbf{M}_A$ about a declared body point A . Their power on the club is

$$P_{h \rightarrow c} = \mathbf{F} \cdot \mathbf{v}_A + \mathbf{M}_A \cdot \boldsymbol{\omega}. \quad (30.1)$$

All vectors use the same axes and velocities are measured relative to the declared laboratory frame. The moment includes force moment arms about A ; one must not add those moments again as a second interaction. This is the wrench–velocity pairing explained in [Lynch and Park’s treatment of wrenches](#).

For a club subsystem with gravitational potential V_g and elastic storage U_s , a useful ledger is

$$\frac{d}{dt}(T + V_g + U_s) = P_{h \rightarrow c} + P_{\text{other}} - D, \quad D \geq 0. \quad (30.2)$$

Here T includes the modeled motion of the club's mass, D is dissipated mechanical power, and P_{other} includes external interactions other than the hands and gravity. Counting gravity through V_g means not adding gravitational power again on the right. A flexible club generally requires power at each moving contact and deformation degrees of freedom; one rigid-body twist cannot represent every contact velocity. Shaft energy return is a decrease in internal storage, not another external energy supply. Integrate this ledger over an interval and report its residual separately. For an impact, use the corresponding impulse/work or resolved-contact balance rather than assume that smooth-time power captures an instantaneous reset.

The golfer can still use a simple image: energy changes hands through boundary work and internal storage. The image stays honest only if we remember that several exchanges overlap and that energy is not tagged with the name of the body part that first supplied it.

30.2.3 “If the Hands Slow Down, How Can the Clubhead Speed Up?”

Speed belongs to a point or segment, not to the whole chain at once. The hands can reduce their tangential speed while the club rotates faster about them. Changing geometry can convert the motion of the proximal endpoint into angular motion of the distal segment. Constraint forces and joint moments determine how that conversion unfolds.

A skater analogy helps but is incomplete. Pulling mass inward changes rotational inertia in a nearly isolated system; a golf swing is externally forced, linked, and actively controlled. The full explanation requires the arm-club geometry, mass matrix, applied moments, constraint reactions, and hand-path motion.

The reviewer therefore rejects the statement “slowing the hands causes clubhead speed” unless a counterfactual defines the intervention. Were the hands slowed by removing force, applying an opposing force, changing the path, or changing the entire upstream motion? Those interventions produce different futures. Observed correlation between hand deceleration and club acceleration does not select one of them.

30.2.4 “How Can Negative Torque Possibly Help?”

Torque sign is defined about an oriented axis. In a planar model, a negative moment acting on a positively rotating club contributes negative power $\tau\omega$. In three dimensions, use $\mathbf{M} \cdot \boldsymbol{\omega}$; one negative component does not determine the full dot product. A joint's net moment instead pairs with its relative angular velocity, or with the matching generalized speed. These are different boundaries and different power accounts.

Momentum can sustain motion; it does not by itself cause acceleration. Acceleration follows from the complete force and moment balance, including the linked system's geometry and current velocity. For a constructed planar rotor with fixed inertia $I = 0.1 \text{ kg m}^2$ and $\omega = 4 \text{ rad/s}$, simultaneous moments of -2 and $+5 \text{ N m}$ give $\dot{\omega} = (-2 + 5)/0.1 = 30 \text{ rad/s}^2$. The opposing moment removes 8 W , the other supplies 20 W , and rotational kinetic energy increases at 12 W . The acceleration is caused by the net

moment. This example explains how an opposing contribution can coexist with speeding up, not why it is advantageous.

For a translating, rotating, deforming club, increasing total kinetic energy does not guarantee increasing clubhead speed or increasing angular speed. Those quantities require their own kinematics. Negative handle-moment power can coexist with positive hand-force power or release of previously stored strain energy, but the complete subsystem ledger decides the net result.

Negative torque may regulate release, alter face or path, manage shaft state, or reduce an overshoot. None of those possibilities makes negative torque automatically desirable. A negative moment applied too early, too strongly, or in the wrong configuration can reduce speed or disrupt the task.

The reviewer requests a signed plot of moment, angular velocity, moment power, resultant-force power, and net energy change. A plot of torque alone cannot support an energy claim. The golfer gains a better question: “What job is this negative moment doing during this interval, and what changes when it is removed in a matched counterfactual?”

30.2.5 “Is the Late Negative Couple Passive?”

The [archived forward two-arm study](#) reports a specific test: at 0.20 s, all six applied joint torques are set to zero and a new trajectory starts from exactly the achieved position and velocity. The force-generated moment about the declared club center remains negative for the tested 0.05 s. This is a forward branch, not a set of isolated zero-command calculations along the original motion.

It demonstrates feasibility for that planar rigid model, with fixed shoulders, gravity, and ideal bilateral hand constraints. The constraints enforce grip without a human contact-capacity model. The archived timestep and projection checks support a numerical model result; they do not establish a physiological strategy. Persistence also depends on the branch state: the 0.18 s cut starts with a nonnegative moment and becomes negative later, so the archive reports zero *immediate* negative persistence for that cut.

That does not mean the golfer’s body is passive. Muscles can remain active to maintain posture, hand contact, joint stability, and the state from which the constraint reaction emerges. Nor does it identify which muscles created the earlier state. “Passive mechanism” and “inactive person” are different claims.

The reviewer also demands a precise geometric control. The archive sets both grip offsets to zero at the club center, recomputes a compatible state, and reports zero force-generated moment there. More generally, shrinking the separation of two bounded forces eliminates their differential couple. It does not eliminate a directly applied free moment, nor the moment of a resultant force about a different reference point. Maintaining a fixed force-generated couple while its lever arm tends to zero would require unbounded forces.

30.2.6 “Do the Scapulae Create the Push–Pull Couple?”

Scapular protraction, retraction, rotation, and tilt change the shoulder socket and the geometry from which the arms act. They are plausible contributors to a bilateral push–pull pattern. Establishing that contribution requires a prediction linking measured scapular motion to the bilateral hand wrench; an anatomical possibility alone does not establish its importance in a swing.

The mapping is not unique. The same net club wrench can result from different combinations of trunk motion, scapular motion, shoulder and elbow moments, wrist moments, grip pressure, and muscle

co-contraction. Surface EMG reports electrical activation timing imperfectly; it does not directly report muscle force, hand-force direction, joint work, or a unique causal pathway.

The reviewer’s minimum evidence package includes bilateral grip wrenches, full-body and scapular kinematics of adequate quality, closed-chain inverse dynamics constrained by the internal wrench, forearm and shoulder-girdle EMG, and participant-level replication. Until then, “scapular action is a candidate mechanism” is supported more strongly than “scapular retraction creates the club couple.”

30.2.7 “If the Wrists Resist, Are They Passive?”

No. A joint can show little visible angular motion while muscles actively co-contrast, while passive tissues bear load, or while opposing moments nearly cancel. A negative wrist moment can involve active eccentric work, isometric activation at small angular velocity, elastic tissue loading, or a combination. The word “resist” describes a net mechanical relation only after the axis, moment, motion, and interval are defined.

This distinction matters for the two allocation extremes. The proximal-drive case in a reduced model is not equivalent to “arms active, wrists off.” The wrist contribution can remain negative and actively regulated. The wrist-led preparation case is not equivalent to “arms doing nothing.” The arms must still maintain geometry and contact. Biological redundancy prevents net joint moments from revealing a unique activation pattern.

The golfer may feel firmness, pressure, or stretch. Those sensations are useful reports, but the reviewer will not equate them directly with stiffness or torque without calibration. A sensation can motivate an experimental condition; it cannot substitute for the measurement.

30.2.8 “Is Slack Bad?”

The reviewer first removes the word from the question. Does slack mean a literal loss of grip contact, a low-force interval, tendon slack length, low tangent stiffness, a transmission dead zone, delayed activation, or simply a relaxed sensation? These quantities behave differently.

A contact gap prevents that contact from carrying compression until it closes; the effect on task precision depends on what other contacts can transmit. Elastic compliance can store energy, whereas damping dissipates it. Their effects on peak load, tracking, and disturbance response depend on direction, frequency, geometry, and control. Neither softness nor stiffness guarantees stability. A tendon has a slack-length parameter without implying that the entire movement feels loose.

There is direct evidence that force alone need not identify stiffness: [Höppner and colleagues](#) measured grip stiffness during perturbations in ten seated participants and found that voluntary co-contraction could change stiffness at similar grip force. Their posture-maintenance experiment does not establish a dynamic golf swing strategy. It supports measuring stiffness rather than inferring it from grip force or a cue to tense the hand.

The [archived transmission study](#) uses a phenomenological dead zone in two actuator channels. Its parameter sweep favors persistent loading direction in nine cases with positive dead-zone width; the three zero-width cases have equivalent net torque-error impulses within the declared 10^{-10} N m s tolerance. This comparison concerns that metric, matched commands, and initialization. It does not mean that every channel history becomes identical or that all real actuators lose their delay when a

gap is removed. The next experiment should estimate the actual gap, stiffness, or force-rise behavior instead of arguing over the word.

30.2.9 “Which Is Better: Arm-Led Preparation or Wrist-Led Preparation?”

The honest answer is that the current evidence does not establish a universal winner. In the matched-task model, proximal-dominant and wrist-dominant controls can produce the same club-level moment with different hand-force and generalized- torque demands. The allocation that minimizes one demand changes with geometry and need not minimize another.

Persistent proximal drive with wrist resistance may preserve a loaded transmission state through transition. Wrist-led preparation followed by role reversal may exploit compliance, reduce a proximal load, or suit another coordination objective. Intermediate allocations may dominate both extremes for a particular participant. Speed, accuracy, tissue loading, repeatability, and adaptability are not the same objective.

The reviewer asks for a preregistered tradeoff surface and held-out participants, not a group-average declaration. The golfer is invited to understand the options, not to treat the chapter as medical or coaching advice.

Hypothesis: Allocation strategies will differ most when the transmission has measurable state dependence and when the task includes perturbation or accuracy constraints, not merely an unperturbed peak-speed target.

30.2.10 “Does Preactivation Eliminate Slack?”

Preactivation can increase force readiness and effective stiffness, but it does not erase every kind of slack. It may take up a contact gap or develop series force before a rapid command. It may also increase internal load, energetic cost, and sensitivity to unwanted high-frequency disturbances. Its effect is directional and task dependent.

The reduced biological bridge carries internal activation and force states through a preparation-to-delivery change; a changed excitation command need not reset those states. Two programs with the same net target can therefore retain different internal histories. Yet a state visible inside a simulation is not necessarily identifiable from an external measurement.

The [archived biological comparison](#) also contains a crucial numerical limit. Its reversal-minus-persistent torque-error impulse decreases from approximately 0.00277 to 0.000514 N m s as the timestep is refined from 1 to 0.05 ms. The sign is unchanged over that grid, but the effect magnitude is not established as converged. Calling the effect merely “small” would hide this uncertainty. It is not a reliable physiological effect size, a performance benefit, or proof that one preparation history is superior.

The reviewer would measure pretransition activation, grip force, force-rise delay, stiffness proxies, and posttransition task error together. Activation without force and stiffness data is not enough. The golfer gets a balanced interpretation: readiness may matter, but “more activation everywhere” is not a scientifically supported prescription.

30.2.11 “Why Not Build One Huge Human Model and Settle It?”

A large model can represent more anatomy while introducing more parameters, uncertainty, and opportunities for compensation. If muscle paths, activation dynamics, contact properties, and participant scaling are poorly identified, the model may fit observed motion through many incompatible internal stories. Size is not the same as explanatory power.

The model ladder assigns each rung a question. A double pendulum exposes coupling and geometry. A two-hand constrained model exposes resultant and differential contact modes. A moving base tests whether prescribed support hid the mechanism. A flexible shaft carries strain state. A spatial model checks out-of-plane effects and frame conventions. A musculoskeletal model tests candidate biological allocations. Cross-engine agreement catches implementation errors but does not convert assumptions into measurements.

The reviewer prefers the smallest model capable of failing the claim being tested, followed by a declared fidelity gate. The golfer benefits because each result remains interpretable rather than disappearing inside a black box.

30.2.12 “Can Inverse Dynamics Tell Us Who Did the Work?”

Inverse dynamics estimates net generalized moments under a chosen model and measured kinematics. It is powerful, but a net joint moment can result from many muscle-force combinations. In a closed two-hand chain, unmeasured internal grip forces add another ambiguity. Filtering, inertial parameters, joint-center definitions, and soft-tissue artifact affect the result.

Joint power pairs a net generalized force with its power-conjugate speed. For an ideal rotational joint, use the moment dotted with the distal segment’s angular velocity relative to the proximal segment. For several coordinates, use $\tau^T \dot{q}$, with the appropriate speed mapping when coordinate derivatives are not angular-velocity components. This net power is not metabolic effort, individual muscle work, or energy that can be tagged and followed into one distal segment. Biarticular muscles, tendon storage, and constraints complicate that attribution.

The reviewer asks whether the internal wrench was measured, whether residuals close, how uncertainty was propagated, and whether alternative models change the conclusion. The golfer can still learn from joint-power curves, provided they are read as model-dependent net quantities rather than photographs of individual muscle action.

30.2.13 “What Would Actually Prove the Proposed Mechanism Wrong?”

This is the question that turns an explanation into a scientific program. First state the prediction’s scope and the size of a meaningful disagreement. An adequately measured absence of a predicted loading difference challenges the specified persistent-loading hypothesis only if the intended allocation was actually changed. A noisy or unsuccessful manipulation does not test it decisively. Failure to reject zero is not an equivalence test.

For the two-hand model, nonconverging power residuals or an unstable moment sign invalidate the reported numerical evidence until resolved. A nonzero moment from forces applied exactly at the reporting point violates the moment definition. A directly applied free moment is a different quantity. If the effect disappears when prescribed support is replaced with a responsive base, the result is

limited to the earlier support assumptions; its extension to the new model has failed, while the earlier mathematical example may remain valid.

For the scapular pathway, a precise failed prediction weakens that particular mechanism. An alternative allocation fitting the same observations equally well instead shows that those observations do not uniquely identify the mechanism. Both mechanisms may remain possible. A universal sequencing claim can be contradicted by a valid counterexample within its declared population and task. A shaft-energy prediction must survive comparison with measured phase and magnitude, including measurement and model uncertainty.

Negative results should be published with the same care as positive ones. A failed mechanism can leave useful invariants, validated measurement tools, and better-bounded alternatives. The reviewer is not asking for certainty; the reviewer is asking the authors to name the observations that would force a revision.

30.2.14 “Why Make the Evidence Open?”

An open paper is more useful when its figures, data summaries, scripts, event definitions, coordinate conventions, and residual gates can be inspected. Reviewers should be able to rerun a plot, change a parameter, and see whether a claim survives. Machine-readable manifests and hashes help distinguish a specific evidence snapshot from a later artifact.

Openness does not remove every barrier. Specialized engines, proprietary measurements, participant privacy, and computational cost can limit complete reproduction. The responsible response is to document those boundaries and publish the smallest sufficient evidence rather than implying that a static figure is the whole experiment.

For the golfer, open evidence means the explanation can improve without loyalty to a product or personality. For the reviewer, it means criticism can target a declared calculation instead of guessing how a result was made.

30.2.15 “What Can I Safely Take From This as a Golfer?”

You can take a way of asking questions. A swing creates a state, not merely a pose. Geometry changes what forces and moments can do. Two hands can create resultant and differential force modes; either can be zero in a particular state. Negative torque is not automatically harmful. Compliance can help or hurt depending on the task. Peak sequence, joint power, EMG, and clubhead speed answer different questions.

You should not take a universal instruction to pull with one scapula, push with one arm, hold a wrist angle, maximize tension, or deliberately apply negative torque. The models have not established the safety or superiority of those instructions for an individual. Technique changes that load wrists, elbows, shoulders, or the spine belong with qualified coaching and clinical judgment.

The deepest practical lesson is that a good strategy must be judged as a whole: speed, face and path control, strike, robustness, effort, and load. A mechanism that sounds elegant in isolation may be a poor trade for a particular person.

Practical Interpretation: Use the framework to make coaching claims more specific and testable. Do not turn a model variable directly into a body cue.

30.2.16 Reader Lab: Hold Both Voices at Once

Choose a familiar claim such as “keep the club loaded,” “fire the hips,” or “let the club release itself.” Let the curious golfer explain what the phrase feels like and what visible outcome it tries to produce. Then let the reviewer rewrite it using a state, force, moment, power, stiffness, or task variable. Name the model tier that could test it. Name one negative control. Name one human measurement. Finish with a sentence beginning, “I would change my mind if ...”

If the scientific rewrite no longer resembles the original claim, that is valuable information. The phrase may have been a useful personal cue without being a literal mechanism. Cues and mechanisms can coexist, but they should not borrow each other’s evidence.

30.2.17 A Full Conversation From First Impression to Testable Claim

The framework becomes easier to use when we slow down and follow one claim all the way through it. Imagine that a golfer watches a slow-motion swing and says, “The arms stopped, so their energy must have gone into the club.” That sentence feels persuasive because two visible events occur together: the hands appear to slow and the clubhead appears to speed up. It is a reasonable observation. It is not yet a complete mechanical explanation.

The curious golfer should begin by saying what was actually seen. Which point on the hands was tracked? Was “speed” a linear speed in the laboratory frame, an angular speed about the shoulder, or an angular speed relative to the club? Did the apparent slowing happen before, during, or after the clubhead speed-up? How much did the club’s direction change? A camera view can make a point look slow because it moves partly toward the camera. Even excellent motion capture needs a coordinate convention before “slowing” has one stable meaning.

Now the skeptical reviewer asks what “their energy” means. The arms have translational kinetic energy, rotational kinetic energy, gravitational potential energy, elastic energy in tissues, and interactions with the torso, hands, and club. Those accounts are not little colored parcels that can be watched crossing a joint. A segment can lose kinetic energy while muscles add positive work, because an even larger amount leaves through joint reactions. It can also gain kinetic energy while its net joint moment does negative power, because other links and constraints contribute more. The correct ledger needs the declared system boundary and every power crossing it.

For a first model, use the planar double pendulum. It is small enough that every term can be inspected. Reproduce the observed arm and club state at one instant, then calculate the full acceleration. Next set the selected arm or wrist moment to zero without changing the state and calculate again. The difference is the instantaneous acceleration influence of that intervention under the model, with its constraints and other inputs held fixed. Repeating it along the observed trajectory produces a history of same-state comparisons. Integrating an assigned power along that history is a trajectory-conditioned accounting quantity; it is not automatically the work or energy difference of a swing that actually evolved with the moment removed.

To answer the latter question, branch at a declared time, inherit the complete state, specify the remaining controls or feedback policy, and integrate both futures. Activation, contact, and elastic states must be inherited when the model has them. Each future uses its own velocities to calculate work.

A constructed one-dimensional example makes the distinction concrete. A 1 kg mass starts at rest and is pushed for 1 s by two constant 1 N forces. Its baseline velocity is $v_b(t) = (2 \text{ m/s}^2)t$. Each force does 1 J of work, and final kinetic energy is 2 J. Remove one force at the start: the remaining force produces $v_c(t) = (1 \text{ m/s}^2)t$, does 0.5 J, and leaves 0.5 J of kinetic energy. The forward energy difference is 1.5 J, whereas the removed force's work along the baseline was 1 J. The surviving force's work also changed because the trajectory changed. The same distinction matters in a coupled swing.

The reviewer now separates three checks. A change of coordinates is an **invariance check**: transform the same physical motion, force, and moment consistently and preserve the total power. Individual signed components can legitimately change sign when an axis is reversed. Refining the timestep or solver tolerance is a **numerical convergence check**. Removing gravity or reversing a physical moment arm is a **mechanical intervention**: it changes the system, so the outcome need not remain unchanged. A negative control is a deliberately chosen intervention predicted to remove the particular mechanism being tested, with other changes and limitations declared.

Even a change of reporting point requires care. For two points on the same rigid body, let $\mathbf{r} = \mathbf{r}_B - \mathbf{r}_A$. Then

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}, \quad \mathbf{M}_B = \mathbf{M}_A - \mathbf{r} \times \mathbf{F}. \quad (30.3)$$

Substitution in Equation 30.1 makes the two added scalar triple products cancel. Total power stays the same although the force and moment contributions change. This is a re-expression of one physical velocity field relative to the same observer. Changing to an observer moving relative to the laboratory changes measured velocities and can change power; it is not merely a change of axes. For a flexible shaft, velocities at separated points need not satisfy this rigid-body relation.

Suppose the double-pendulum result survives. The claim has earned one rung, not the entire ladder. Add the two-hand constraint model. The club now receives a resultant force, a force-generated couple from the separation of the hands, and any directly represented grip moment. Ask whether the late negative couple is still present and whether it closes the club's moment balance. Collapse the hand points together at the reporting point as a geometric control, resolving a compatible initial state for that geometry. Their force-generated moment there must be zero. For separated points at $\pm \mathbf{d}/2$ around their midpoint, the differential moment is

$$\mathbf{M}_{\text{diff}} = \frac{1}{2} \mathbf{d} \times (\mathbf{F}_R - \mathbf{F}_L). \quad (30.4)$$

Its magnitude is bounded by $\|\mathbf{d}\|(\|\mathbf{F}_R\| + \|\mathbf{F}_L\|)/2$, so it tends to zero with separation for bounded forces. Direct free moments remain separate. About another point, also include the resultant's moment arm; coincident hands away from that point can still exert a moment about it.

Next replace the rigid club with a flexible shaft. Some power can now enter strain energy rather than immediate rigid-body motion. Opposing handle-moment power can coexist with increasing shaft strain energy if other boundary work or decreasing kinetic energy supplies that storage. It does not itself prove that the shaft is loading. Energy extracted through the handle and energy dissipated within the shaft are separate terms. Only a strain-energy and dissipation ledger can distinguish those stories. The casual phrase “the shaft kicked” becomes several measurable questions: when did the shaft store

energy, how much returned before impact, in which bending or torsional mode, and how sensitive was the result to stiffness and damping?

Then let the base move. Prescribing the shoulder path can hide the reaction loads and work needed to create it. A moving-base model asks whether the same distal effect survives when the torso and support are allowed to respond. Finally move into three dimensions, where a moment that looks negative about one planar axis may coexist with positive work about another axis. At every rung, the reviewer asks the same question: did the mechanism survive, change meaning, or disappear when a previously hidden degree of freedom became available?

Human evidence enters after the model has produced distinct predictions. If two candidate strategies generate nearly identical club motion but different bilateral grip couples, measure the grip wrenches. If they predict different shaft histories, measure strain. If they predict a directionally persistent arm channel and a reversing wrist channel, estimate joint moments with the internal hand wrench constrained rather than guessed. EMG can help establish activation timing, but it cannot by itself supply the missing hand force or joint moment. The measurement package should be chosen because it separates the explanations, not because the instrument happens to be available.

If those measurements and checks succeed, the original sentence can be rewritten as the following reporting template, not as a result measured in this chapter: “During this interval, the measured hand point slowed in the laboratory frame while the clubhead accelerated. In the declared linked-segment model, arm-club constraint reactions were represented by the measured net grip wrench, which produced positive power on the club, while the modeled arm system lost kinetic energy. The energy balance closed within the stated residual, and the result survived the listed timestep, coordinate, and model-fidelity checks.” That sentence is longer than the slogan, but every phrase requires a specific piece of evidence. Count that grip interaction once, whether it is represented by contact reactions or an equivalent wrench. Simultaneous arm energy loss and positive grip power do not uniquely establish how much of the club’s gain originated in the arms: the arm subsystem also exchanges work with the torso and its own actuators.

The conclusion can still fail. The measured internal wrench may contradict the inverse-dynamics allocation. Shaft strain may show that energy returned after the useful impact window. A participant may create the same outcome with a different sequence. The result may vanish when shoulder motion is no longer prescribed. Those are not embarrassments. They identify exactly where a simple story stopped working and where a better one must begin.

30.2.18 A Full Conversation About “Keeping the System Loaded”

Consider a second familiar claim: “Keep tension in the system through the transition so there is no slack.” The curious golfer may be describing a very real sensation of connectedness. Perhaps grip pressure remains steady, the arms feel organized against the torso, and the club changes direction without a little uncontrolled wobble. That sensation is valuable as a report. It still contains several possible mechanisms.

The reviewer replaces “tension” with candidate measurements. Is there a nonzero resultant grip force, a nonzero differential hand couple, forearm muscle activation, scapular-muscle activation, shaft strain, tendon force, or high tangent stiffness about a joint? These variables can move independently. Two opposing muscle groups can raise stiffness while their net joint moment stays near zero. Two hand

forces can create a large couple with a small resultant force. A shaft can remain strained while the hands briefly reduce their active moment. “Loaded” therefore needs both a location and a quantity.

Now compare the two preparation extremes developed earlier. In the first, arm-and-shoulder channels create more of the backswing push–pull tendency while the wrists resist and regulate the club. The broad channel directions persist into early downswing. In the second, wrist moments create more of the backswing turning tendency while the arms resist; around transition the arms take over as the positive driver and the wrists oppose release. These are proposed channel allocations, not uniquely identified human muscle programs. The archived allocation sweep matches an instantaneous club control-moment target at a declared state. Matching an entire club path under evolving activation and contact constraints is a stronger requirement and needs a forward test.

In the declared dead-zone model, persistent direction has an advantage because it does not cross the modeled unloaded zone. Complete role reversal must travel through that zone before torque rebuilds. This is a useful conditional result, not proof that a human wrist contains backlash. Set the dead zone to zero in the archived sensitivity sweep and the two programs’ net torque-error impulses become equivalent within the declared tolerance. That control locates the difference within this transmission family; it does not establish a human wrist dead zone or a global optimum.

A biological system can preserve readiness during a net reversal without literal continuous torque. Antagonistic preactivation may keep short-range stiffness high. Tendon or shaft strain may preserve a loaded elastic state. Contact forces may remain nonzero while their moment about a selected axis changes sign. Conversely, a person may report continuous effort while the task-relevant force mode briefly approaches zero. Feel and mechanics may be related, but their mapping must be calibrated rather than assumed.

The two strategies also have costs that a peak-speed-only model will miss. Persistent co-contraction can raise joint compression, metabolic cost, fatigue, and sensitivity to unwanted high-frequency loading. A role reversal can use different muscle groups, exploit changing leverage, and permit useful reorientation. Some compliance can store energy, absorb disturbances, and protect tissue. Too much compliance or a true contact gap can delay force rise. There is no reason to expect one point on this tradeoff surface to be best for every participant and every objective.

A decisive experiment therefore matches the visible task while manipulating allocation. Measure bilateral grip wrenches, grip pressure, wrist and shaft kinematics, shaft strain, transition-aligned force rise, and suitable forearm and shoulder-girdle EMG. Estimate tangent stiffness with an ethical bounded perturbation or system-identification method. Compare not only clubhead speed but face and path variability, peak joint demand, repeatability, and response to a small disturbance. Hold participants out when selecting a strategy so the test does not turn a group average into a universal prescription.

Interpret the outcomes in stages. If the instructions do not change measured allocation, the manipulation failed. If allocation changes but the predicted gap or stiffness difference does not, the dead-zone explanation weakens. If the mediator changes but performance does not, the mechanism may be real yet unimportant for the task. If performance changes without the predicted mediator, another pathway deserves attention. If participants split into repeatable subgroups, the scientifically useful result may be a conditional map rather than a winner.

The careful practical translation is modest: a golfer and coach can distinguish continuous engagement from maximum effort and can ask which measurable mode a cue is intended to change. They should

not infer that more tension is always better, that wrists should be passive, or that one scapular action uniquely creates the club couple. The framework earns better questions and experiments; it does not yet earn a universal body instruction.

30.2.19 How to Carry the Method Into a New Question

When a new swing explanation appears, first write the observation without a causal verb. Next define the system boundary, coordinate frame, interval, and quantity. Draw the smallest model that contains the proposed mechanism and a negative control capable of removing it. Record the state before applying the counterfactual so that history is not silently reset. Close force, moment, power, and energy balances at the level required by the claim.

Then name at least one serious alternative. Increase model fidelity only where that alternative exploits something the smaller model omitted. Derive predictions that differ in measurable variables, including an internal variable when external motion is not identifying. Specify the residual, sensitivity, and participant-held-out criteria before looking for a preferred result.

Finally, translate the result twice. The scientific translation states the model, intervention, observable, uncertainty, and falsifier. The ordinary translation uses a concrete picture and says where that picture breaks. If the two translations disagree, revise them until neither borrows certainty from the other. That repeated movement between intuition and audit is how an accessible explanation can remain genuinely scientific rather than merely simple.

30.2.20 Chapter Checkpoint

The curious golfer asks questions that matter. The skeptical reviewer makes their meanings stable enough to test. Together they replace simple slogans with a linked chain: define the system and quantity, choose the smallest adequate model, preserve state history, measure the predicted observables, run negative controls, compare alternatives, and declare what would refute the claim. That chain is the real framework of the project.

30.3 Where the Picture Breaks

A dialogue can make disagreements look cleaner than they are. Real experiments face noisy measurements, participant variation, coupled objectives, missing data, and ethical limits. Some questions cannot be settled by one study, and a null result can reflect weak manipulation or inadequate measurement rather than absence of a mechanism. The two voices also do not replace domain experts in biomechanics, motor control, instrumentation, statistics, coaching, or clinical care. They provide a disciplined entry point for those collaborations.

Go Deeper: Audit the claims against the [complete scientific monograph](#), use the open [reviewer workbench](#), and inspect the project-level [falsification matrix](#).

Glossary

Acceleration: The rate at which velocity changes. It can change speed, direction, or both.

Ablation: A controlled model experiment in which one mechanism is removed or neutralized while the remaining conditions are held as comparable as possible.

Constraint: A rule limiting allowable motion, such as two linked segments sharing a joint or two hands remaining attached to one grip.

Coriolis Term: A cross-speed part of a motion equation under a declared coordinate convention. In the two-link example it requires simultaneous arm rotation and relative wrist motion; it is not a new external or muscle force.

Centripetal and Centrifugal: In an inertial description, centripetal names the inward requirement for curved motion. Centrifugal is the outward inertial interpretation in a rotating description. They are not two forces to add.

Counterfactual: A carefully stated “what if” comparison used to attribute a quantity or test whether a proposed mechanism persists.

Drift: The modeled zero-input part of the state derivative, or its acceleration component when that is the stated convention. Retained gravity, constraints, internal states, unselected inputs and prescribed boundary motion must be named. Zero declared input does not mean zero force or biological rest.

Energy: A scalar accounting quantity with specified forms, such as kinetic, gravitational, elastic or chemical energy. Values depend on the system and observer, and potential-energy zeros are conventional. Conservation of total energy does not require conservation of mechanical energy alone.

Falsifier: An observation or test result that would count against a claim.

Identifiability: Whether available observations can uniquely or usefully determine a model parameter, internal force, or mechanism.

Impulse: Force integrated over time. Angular impulse is moment integrated over time.

Interaction Force: A force transmitted at a connection between parts of a linked system.

Moment or Torque: The turning effect of a force or a direct rotational couple about a declared point and axis.

Power: The instantaneous rate of energy transfer or work, measured in watts.

Preload: A nonzero internal load already present before a command or event changes.

Robustness: Performance retained across a declared set of disturbances, uncertainties, participants, or model parameters.

State: The collection of variables needed by a model to predict its future with specified future inputs or input laws, commonly configuration, velocity, and internal states such as activation or deformation.

Squared-Speed Term: A velocity-dependent equation term containing one generalized speed squared. Its detailed allocation is coordinate-dependent.

Wrench: A combined force and moment expressed about a declared reference point and in a declared coordinate frame.

Work: The time integral of a declared mechanical power. Force power uses the material-point velocity; moment power uses the conjugate angular velocity. Joint power and external power on one body can therefore be different ledgers.

Pointwise Zero-Torque Counterfactual (ZTCF) Force: The generalized-force representation associated with the acceleration recomputed at a declared state after selected control removal. Its construction must name coordinates, mass mapping, retained loads and constraints. It is not an independently measured force, nor a universal tissue-force decomposition.

For the evidence-led technical treatment behind this tour, see the summary article [Proximal-to-Distal Energy Transfer in the Golf Swing](#), which connects the companion narrative to the monograph's evidence ledger.

References

- Ball, Kevin A., and Russell J. Best. 2007. “Different Centre of Pressure Patterns Within the Golf Stroke i: Cluster Analysis.” *Journal of Sports Sciences* 25 (7): 757–70. <https://doi.org/10.1080/02640410600874971>.
- Balzerson, Daniel, Joydeep Banerjee, and John McPhee. 2016. “A Three-Dimensional Forward Dynamic Model of the Golf Swing Optimized for Ball Carry Distance.” *Sports Engineering* 19: 237–50. <https://doi.org/10.1007/s12283-016-0197-7>.
- Betzler, Nils F., Stuart A. Monk, Eric S. Wallace, and Steve R. Otto. 2012. “Effects of Golf Shaft Stiffness on Strain, Clubhead Presentation and Wrist Kinematics.” *Sports Biomechanics* 11 (2): 223–38. <https://doi.org/10.1080/14763141.2012.681796>.
- Bourgain, Maxime, Philippe Rouch, Olivier Rouillon, Patricia Thoreux, and Christophe Sauret. 2022. “Golf Swing Biomechanics: A Systematic Review and Methodological Recommendations for Kinematics.” *Sports* 10 (6): 91. <https://doi.org/10.3390/sports10060091>.
- Choi, Hyeob, and Sukyung Park. 2020. “Three Dimensional Upper Limb Joint Kinetics of a Golf Swing with Measured Internal Grip Force.” *Sensors* 20 (13): 3672. <https://doi.org/10.3390/s20133672>.
- Featherstone, Roy. 2008. *Rigid Body Dynamics Algorithms*. Springer. <https://doi.org/10.1007/978-1-4899-7560-7>.
- Glazier, Paul S. 2011. “Movement Variability in the Golf Swing: Theoretical, Methodological, and Practical Issues.” *Research Quarterly for Exercise and Sport* 82 (2): 157–61. <https://doi.org/10.1080/002701367.2011.10599742>.
- Han, Ki Hoon, Christopher Como, Jemin Kim, et al. 2019. “Effects of the Golfer–Ground Interaction on Clubhead Speed in Skilled Male Golfers.” *Sports Biomechanics* 18 (2): 115–34. <https://doi.org/10.1080/14763141.2019.1586983>.
- Hume, Patria A., Justin Keogh, and Duncan Reid. 2005. “The Role of Biomechanics in Maximising Distance and Accuracy of Golf Shots.” *Sports Medicine* 35 (5): 429–49. <https://doi.org/10.2165/0007256-200535050-00005>.
- Jorgensen, Theodore P. 1999. *The Physics of Golf*. 2nd ed. Springer / AIP Press.

- Joyce, Christopher, Angus Burnett, Jodie Cochrane, and Kevin Ball. 2013. “Three-Dimensional Trunk Kinematics in Golf: Between-Club Differences and Relationships to Clubhead Speed.” *Sports Biomechanics* 12 (2): 108–20. <https://doi.org/10.1080/14763141.2012.728244>.
- Kao, John T., Marilyn Pink, Frank W. Jobe, and Jacquelin Perry. 1995. “Electromyographic Analysis of the Scapular Muscles During a Golf Swing.” *The American Journal of Sports Medicine* 23 (1): 19–23. <https://doi.org/10.1177/036354659502300104>.
- Koike, Sekiya. 2016. “Force and Moment Exerted by Each Hand on an Instrumented Golf Club.” *34th International Conference on Biomechanics in Sports*. <https://ojs.ub.uni-konstanz.de/cpa/article/view/6828>.
- Lazic, Stanley E. 2010. “The Problem of Pseudoreplication in Neuroscientific Studies: Is It Affecting Your Analysis?” *BMC Neuroscience* 11: 5. <https://doi.org/10.1186/1471-2202-11-5>.
- MacKenzie, Sasho J., Matthew McCourt, and Luc Champoux. 2020. “How Amateur Golfers Deliver Energy to the Driver.” *International Journal of Golf Science* 8 (1). <https://www.golfsciencejournal.org/article/12640-how-amateur-golfers-deliver-energy-to-the-driver>.
- MacKenzie, Sasho J., and Eric J. Springs. 2009. “A Three-Dimensional Forward Dynamics Model of the Golf Swing.” *Sports Engineering* 11 (4): 165–75. <https://doi.org/10.1007/s12283-009-0020-9>.
- Marsan, Thibault, Patricia Thoreux, Maxime Bourgain, Olivier Rouillon, Philippe Rouch, and Christophe Sauret. 2019. “Biomechanical Analysis of the Golf Swing: Methodological Effect of Angular Velocity Component on the Identification of the Kinematic Sequence.” *Acta of Bioengineering and Biomechanics* 21 (2): 115–20. <https://doi.org/10.5277/ABB-01318-2019-02>.
- McNally, William, and John McPhee. 2018. “Dynamic Optimization of the Golf Swing Using a Six Degree-of-Freedom Biomechanical Model.” *Proceedings* 2: 243. <https://doi.org/10.3390/proceedings2060243>.
- McPhee, John. 2022. “A Review of Dynamic Models and Measurements in Golf.” *Sports Engineering* 25. <https://doi.org/10.1007/s12283-022-00387-0>.
- Meister, David W., Amy L. Ladd, Erin E. Butler, et al. 2011. “Rotational Biomechanics of the Elite Golf Swing: Benchmarks for Amateurs.” *Journal of Applied Biomechanics* 27 (3): 242–51. <https://doi.org/10.1123/jab.27.3.242>.
- Milburn, Peter D. 1982. “Summation of Segmental Velocities in the Golf Swing.” *Medicine and Science in Sports and Exercise* 14 (1): 60–64. <https://pubmed.ncbi.nlm.nih.gov/7070260/>.
- Miura, Koryo. 2001. “Parametric Acceleration — the Effect of Inward Pull of the Golf Club at Impact Stage.” *Sports Engineering* 4 (2): 75–86. <https://doi.org/10.1046/j.1460-2687.2001.00071.x>.
- Morrison, Andrew, Denise McGrath, and Eric S. Wallace. 2016. “Motor Abundance and Control

- Structure in the Golf Swing.” *Human Movement Science* 46: 129–47. <https://doi.org/10.1016/j.humov.2016.01.009>.
- Nesbit, Steven M. 2005. “A Three Dimensional Kinematic and Kinetic Study of the Golf Swing.” *Journal of Sports Science and Medicine* 4 (4): 499–519. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3899667/>.
- Nesbit, Steven M., and Ryan McGinnis. 2009. “Kinematic Analyses of the Golf Swing Hub Path and Its Role in Golfer/Club Kinetic Transfers.” *Journal of Sports Science and Medicine* 8 (2): 235–46. <https://www.jssm.org/volume08/iss2/cap/jssm-08-235.pdf>.
- Nesbit, Steven M., and Monika Serrano. 2005. “Work and Power Analysis of the Golf Swing.” *Journal of Sports Science and Medicine* 4 (4): 520–33. <https://www.jssm.org/volume04/iss4/cap/jssm-04-520.pdf>.
- Putnam, Carol A. 1991. “A Segment Interaction Analysis of Proximal-to-Distal Sequential Segment Motion Patterns.” *Medicine and Science in Sports and Exercise* 23 (1): 130–44. <https://pubmed.ncbi.nlm.nih.gov/1997807/>.
- Putnam, Carol A. 1993. “Sequential Motions of Body Segments in Striking and Throwing Skills: Descriptions and Explanations.” *Journal of Biomechanics* 26 (Suppl. 1): 125–35. [https://doi.org/10.1016/0021-9290\(93\)90084-R](https://doi.org/10.1016/0021-9290(93)90084-R).
- Robertson, D. Gordon E., and David A. Winter. 1980. “Mechanical Energy Generation, Absorption and Transfer Amongst Segments During Walking.” *Journal of Biomechanics* 13 (10): 845–54. [https://doi.org/10.1016/0021-9290\(80\)90172-4](https://doi.org/10.1016/0021-9290(80)90172-4).
- Robinson, Patrick G., Howie J. Carson, Jim Richards, Andrew Murray, Andrew D. Duckworth, and Doug Campbell. 2023. “What Differences Exist Between the Lead and Trail Wrist in Extensor Carpi Ulnaris Activity and Golf Swing Joint Kinematics in Sub-Elite Golfers?” *Journal of Sports Sciences* 41 (17): 1596–604. <https://doi.org/10.1080/02640414.2023.2285121>.
- Rockafellar, R. Tyrrell, and Stanislav Uryasev. 2002. “Conditional Value-at-Risk for General Loss Distributions.” *Journal of Banking & Finance* 26 (7): 1443–71. [https://doi.org/10.1016/S0378-4266\(02\)00271-6](https://doi.org/10.1016/S0378-4266(02)00271-6).
- Sharp, Robin S. 2009. “On the Mechanics of the Golf Swing.” *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 465 (2102): 551–70. <https://doi.org/10.1098/rspa.2008.0304>.
- Sprigings, Eric J., and Sasho J. MacKenzie. 2002. “Examining the Delayed Release in the Golf Swing Using Computer Simulation.” *Sports Engineering* 5: 23–32. <https://doi.org/10.1046/j.1460-2687.2002.00094.x>.
- Sprigings, Eric J., and Robert J. Neal. 2000. “An Insight into the Importance of Wrist Torque in

- Driving the Golfball: A Simulation Study.” *Journal of Applied Biomechanics* 16 (4): 356–66. <https://doi.org/10.1123/jab.16.4.356>.
- Tinmark, Fredrik, John Hellström, Kjartan Halvorsen, and Alf Thorstensson. 2010. “Elite Golfers’ Kinematic Sequence in Full-Swing and Partial-Swing Shots.” *Sports Biomechanics* 9 (4): 236–44. <https://doi.org/10.1080/14763141.2010.535842>.
- Watson, Andrew, Andrew Murray, Alex Ehlert, et al. 2026. “Ground Reaction Force and Centre of Pressure During the Golf Swing and Associations with Clubhead Speed and Skill Level: A Systematic Review.” *Sports Medicine*, ahead of print. <https://doi.org/10.1007/s40279-025-02391-3>.
- Winter, David A. 2009. *Biomechanics and Motor Control of Human Movement*. 4th ed. John Wiley & Sons.
- Zajac, Felix E., and Michael E. Gordon. 1989. “Determining Muscle’s Force and Action in Multi-Articular Movement.” *Exercise and Sport Sciences Reviews* 17: 187–230. <https://pubmed.ncbi.nlm.nih.gov/2676547/>.
- Zheng, Naiquan, Steve W. Barrentine, Glenn S. Fleisig, and James R. Andrews. 2008. “Kinematic Analysis of Swing in Pro and Amateur Golfers.” *International Journal of Sports Medicine* 29 (6): 487–93. <https://doi.org/10.1055/s-2008-1038732>.